

Finite Screen Spacetime

Entropy-Response Geometry from Causal-Screen Ledgers

FDS–G1 Complete Series v1.0-rc2

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

May 2026

Current status and empirical claim. FDS–G1 currently has a leading empirical dark-sector branch: the projection-locked $M_{3/4}$ model. The branch is theory-motivated by finite-screen modular/optical projection closure, selected by AIC/BIC, ranked first in the six-model medium-prior matched nested-evidence hierarchy, and stable under top-control wide-prior sensitivity:

$$M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE-2} > \text{CPL} > \Lambda\text{CDM}.$$

Its locked prediction is

$$s < 3, \quad \mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a), \quad \hat{R}_H(1) = 1.$$

Here $\hat{R}_H(a) \equiv X_b(a)$ is the production-normalized horizon-response output shape, identified with $X_b(a)$ in the locked branch, not a uniquely identified microscopic source–kernel factorization. This is a sparse background–Weyl residual, not a flexible dark-energy or modified-growth fit. Structurally, FDS–G1 is a paradigm-level candidate rather than an incremental parametrization: it changes the primitive objects from metric/field-first descriptions to finite causal-screen ledgers, entropy response, integrable geometry, and residual gluing defects. Within the tested late-time residual sector, $M_{3/4}$ is presently the most constrained and best-supported G1 branch: it combines a locked projection coefficient, parameter compression, matched exact-pilot evidence, and explicit demotion paths. Full survey-grade confirmation requires production evidence refinement, expanded weak-lensing/ $E_G/3 \times 2\text{pt}$ likelihoods, and independent pipeline replication.

Archive document. Recommended entry points: Core, G1DE- $M_{3/4}$, D0, and D1.

DOI: 10.5281/zenodo.20382013

Series Map

Paper	Current role	Closure role	Main theorem target	Failure consequence
Core	finite-screen entropy response	defines closure spine	screen entropy $\rightarrow$ response $\rightarrow$ metricization	bridge demotion, not empirical failure
A	screen statistics	area-law substrate	area-law universality	no screen-statistical basis
B	screen dynamics	capacity-flow normal form	Raychaudhuri-limit recovery	normal-form demotion
C	scalar-capacity EFT	covariant capacity mode	no-free scalar residual	scalar branch demotion
D	likelihood tests	empirical penalty layer	no-free-fit protocol	branch demotion
E	horizon occupancy	physical residual branch	conserved occupancy residual	G1DE demotion
F	holography/horizons	dictionary consistency	response-density dictionary	dictionary demotion
G1DE- $M_{3/4}$	evidence interface	output-level test	$M_{3/4}$ evidence lock	empirical branch demotion
D0	calibration	strong derivation route	finite-index + boost modular lemmas	strong derivation demotion
D1	sheaf geometry	gluing obstruction	response obstruction class	residual demotion
D2	3/4 law	projection uniqueness	optical MaxEnt uniqueness	3/4 demotion
D3	non-eq. entropy	entropy ledger	optical-stress residual accounting	thermodynamic demotion
D4	influence functional	source-kernel structure	quotient-controlled influence	realization demotion
D5	Ward identifiability	admissibility rules	no-free- $A(a, k)$ theorem	generic dark-stress demotion
D6	memory quotient	output quotient	horizontal lift	source-kernel uniqueness forbidden
D7	Markov realization	existence model	finite-state passive lift	uniqueness not claimed

Contents

Closure Contract and Non-Absorption Rule	4
Scope and Claim Status	4
Suggested Evaluation Protocol	6
Scope Clarifications and Anticipated Objections	9
Near-Term Falsifiable Prediction Lock	11
Comparative Structural Positioning	13
Appendix: Closure Theorems, Bridge Principles, and Open Lemmas	16
1 Core: Finite Screen Spacetime	17
2 Companion A: Screen Statistical Mechanics	39
3 Companion B: Screen Transition Dynamics	52
4 Companion C: Scalar-Capacity EFT	65
5 Companion D: Likelihood-Ready Tests and Falsification	75
6 Companion E: Horizon Occupancy Residuals	90
7 Companion F: Holography and Horizons	102
8 G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual	115
9 D0: Screen Modular Anomaly Inflow and Einstein Calibration	137
10 D1: Finite-Access Sheaf Geometry	152
11 D2: The $3/4$ Projection Law in Finite Screen Gravity	169
12 D3: Non-Equilibrium Screen Entropy and the Optical-Stress Origin	182
13 D4: Screen Influence Functional for the G1 Dark Sector	196
14 D5: Weyl-Normalized Ward Completion and Identifiability Rules	216
15 D6: Maximum-Caliber Horizontal Lift for the G1 Memory Quotient	228
16 D7: Finite Markov-Screen Realization of the G1DE- $M_{3/4}$ Normal Form	240
Closing Comparative Structural Matrix	253

Closure Contract and Non-Absorption Rule

G1 is closed only if every arrow is typed, every residual is admissible, and every failure is non-absorbable.

Every major statement must be classified as one of:

1. Definition
2. Formal theorem
3. Conditional theorem
4. Bridge assumption
5. Normal-form approximation
6. Open lemma
7. Empirical output normal form
8. Falsification / demotion rule

Closure = Typed arrows + Admissible residuals + Non-absorbable failures + Identifiable outputs.

Non-absorption:

- Failed calibration $\nRightarrow$ free residual survival.
- Failed empirical branch $\nRightarrow$ failure of screen entropy primitive.
- Successful $M_{3/4}$ evidence $\nRightarrow$ unique microscopic derivation.

No failed bridge, residual, or empirical branch may be reabsorbed by redefining $R_{\mu\nu}^{\text{scr}}$, $R_H(a)$, $A(a, k)$, or the source–kernel factorization.

Scope and Claim Status

This archive presents FDS–G1 as a bridge program, not as a completed first-principles derivation of General Relativity. Each component states its assumptions, identifies which standard physical bridges it uses, and specifies the conditions under which its claims would be weakened, demoted, or falsified.

The core G1 bridge identifies finite causal-screen entropy response as the substrate of gravitational stiffness. D0 strengthens this bridge by reducing the Bekenstein–Hawking coefficient and Einstein-calibration problem to sharper targets: finite-index screen-algebra density and boost modular covariance lemmas. These lemmas remain open; their role is to make the calibration problem precise rather than to claim that it has already been fully solved.

Memory-output discipline. The current G1DE likelihood tests an output normal form for the horizon response $\mathcal{R}_H(a)$. It does not by itself identify a unique source–kernel factorization $K_H * J_H$. D6 formulates this distinction as a memory quotient and supplies a maximum-caliber horizontal-lift principle for selecting canonical passive realizations.

Microscopic-realization discipline. D7 supplies a finite-state detailed-balance Markov-screen prototype realizing the D6 passive lift assumptions. It is an existence/realization model, not yet a unique derivation of the physical finite-screen state family or the occupancy functional C_{used} .

The complete series should therefore be read in three layers: (i) the finite-screen entropy-response architecture; (ii) the G1DE dark-sector closure and prediction lock; (iii) the exact-pilot evidence pipeline and its pending robustness tests.

Series map

Paper	Role
Core	Finite screen entropy response geometry — primitive object, response chain, normal-form dynamics, integrable closure
Companions A–F	Supporting technical modules — screen statistics (A), transition dynamics (B), scalar-capacity EFT (C), likelihood interface (D), horizon occupancy residuals (E), holographic dictionaries (F)
G1DE- $M_{3/4}$	Matched exact likelihoods and medium-prior nested evidence for six late-time cosmology models
D0–D7	G1DE dark-sector closure sequence — modular anomaly inflow and Einstein calibration (D0), finite-access sheaf foundation (D1), algebraic projection (D2), non-equilibrium entropy (D3), unified influence functional (D4), production identifiability rules (D5), memory quotient and maximum-caliber horizontal lift (D6), finite-state Markov-screen realization prototype (D7)

Current empirical status

FDS–G1 has advanced from a finite-screen response bridge program to a theory-structured, evidence-selected exact-pilot dark-sector branch. This is not final cosmological confirmation; it is a coherent, low-parameter, falsifiable residual branch selected by the current exact-pilot evidence pipeline.

The leading empirical dark-sector branch is the projection-locked $M_{3/4}$ model,

$$s < 3, \quad \mu(a, k) = 1 \ (\mu_0 = 0), \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a), \quad \hat{R}_H(1) = 1,$$

(equivalently $\Sigma_0 = -\frac{3}{4}(3 - s)$ under this normalization), a sparse projection-locked background–Weyl amplitude law rather than a generic modified-growth or flexible dark-energy fit. G1DE-2 serves as a free-response adversarial envelope; G1DE-1 is the growth-only negative control

supporting the interpretation that the primary signal is $H + \Sigma$ with $\mu \simeq 1$; its nested evidence is not included in the current six-model evidence table.

Pending items:

1. Full wide-prior baseline sensitivity for CPL, G1DE-2, and Λ CDM (recommended robustness check; top-control wide-prior completed).
2. Production evidence refinement (stricter nested-sampling tolerance $d \log z$, more seeds).
3. Expanded weak-lensing / E_G / 3×2 pt likelihoods for shape discrimination.
4. Independent pipeline replication.

Suggested Evaluation Protocol

The following protocol is suggested for evaluating the framework on its stated logical and empirical terms. It does not immunize any claim from mathematical or empirical refutation. Its purpose is to separate technical assessment from sociological dismissal, to identify which assumptions or bridges are actually being tested, and to state the failure conditions under which specific branches should be demoted.

Paradigm-level candidate status. FDS-G1 is not presented as an incremental parameterization within an already fixed model space. It proposes a shift of primitive objects: from metric/field-first descriptions toward finite causal-screen ledgers, entropy response, integrable geometry, and residual gluing defects. In this structural sense, it should be evaluated as a paradigm-level candidate: by logical closure, explanatory compression, empirical risk, falsifiability, and prediction lock, rather than by immediate familiarity or consensus alone. This classification is methodological, not evidential; it does not imply that the framework is correct, only that the appropriate evaluation standard is not the same as for a small phenomenological extension.

Historical cases such as heliocentrism, Darwinian evolution, and Wegener’s continental-drift hypothesis illustrate a narrow methodological point: theories that change primitives can be socially unfamiliar before they are technically resolved. The analogy is methodological, not evidential. Historical resistance does not validate a new theory, but it does show why consensus cannot substitute for technical evaluation.

1. Consensus is not a premise. The validity of a derivation within this framework depends on its internal logical rigor, its bridge assumptions, and its empirical predictions, not on its current level of social acceptance within any academic community. Established empirical constraints and proven mathematical theorems remain binding. If an FDS-G1 claim contradicts a well-established experimental result or a proven theorem under the same assumptions, that contradiction counts as evidence against the claim. The methodological point is only that consensus is not itself a premise.

2. Objections should identify the failed premise. Where this framework derives a limit, prediction, or closure condition from finite distinguishability, causal-screen access, entropy response, Ward/Bianchi closure, Weyl normalization, finite-access modular calibration, or a specified likelihood implementation, an objection should identify which premise, bridge, normalization, or data step fails. A generic objection of the form “a future model may explain this differently” is not a direct rebuttal unless it specifies the alternative mechanism and the premise it replaces.

3. Ex-ante prediction locks are not post-hoc fits. If a model identity, prediction equation, control hierarchy, and falsification ladder were publicly timestamped before the relevant future data, the result should not be dismissed merely as post-hoc curve fitting. The correct evaluation is whether the prediction was actually locked, whether the likelihood implementation matches the locked model, whether adversarial controls were included, and whether the result survives prior and pipeline stress tests. This does not imply that the derivation is correct; it only means that the post-hoc-fit objection is not the appropriate criticism.

4. Flexible fitting is not explanatory compression. A model is not scientifically strengthened merely by fitting data with additional freedom. Evaluation should distinguish explanatory compression from flexible response fitting. For FDS-G1, the relevant question is whether the projection-locked branch

$$\mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{R}_H(a)$$

outperforms or remains competitive against free- κ , constant- Σ , free-response G1DE-2, CPL, and Λ CDM controls under matched likelihood and prior stress.

5. Scientific claims require failure conditions. Every empirical branch of the framework must carry a pre-specified consequence of failure. For the G1 dark-sector branch, the main demotion paths are:

1. if free- κ decisively beats $M_{3/4}$, the exact 3/4 isotropic-compliance lock is demoted;
2. if constant- Σ decisively beats the output-response-shape model, the $\widehat{R}_H(a)$ shape is demoted while the Weyl-amplitude branch may survive;
3. if data require $|\mu - 1| \sim |\Sigma - 1|$, the Ward-suppressed Ricci-leakage branch fails;
4. if free $A(a, k)$ is required, the model becomes a generic dark-stress source rather than G1DE;
5. if production evidence loses to CPL or Λ CDM, the G1DE observational branch is demoted;
6. if expanded weak-lensing/ $E_G/3 \times 2$ pt likelihoods do not support the Weyl-channel signal, the current dark-sector interpretation fails or must be downgraded.

A claim with no possible demotion, observational penalty, or failure condition is not an empirical scientific claim in this framework. It may retain heuristic, metaphysical, or programmatic value, but not confirmed scientific status.

Scope Clarifications and Anticipated Objections

- 1. Is FDS–G1 a replacement for GR, or a residual extension of GR?** The current framework is a finite-screen response framework with GR-like equilibrium limits. G1DE is a late-time residual branch, not a full replacement for numerical relativity or strong-field GR production systems.
- 2. If GR is recovered as an equilibrium branch, does G1 automatically inherit all GR successes?** No. The framework has a standard-limit structure, but the microscopic derivation and production-level implementation are not yet complete. Inheriting a limit is not equivalent to completing a global proof.
- 3. Is Λ CDM contained as a limit, or merely fitted as a benchmark?** Λ CDM appears as the $s = 3$, vanishing-residual consistency limit of the G1DE hierarchy. This does not imply that all CMB, BBN, and full-shape constraints have been verified at production level.
- 4. Is the evidence result driven by prior volume?** Top-control wide-prior sensitivity has been completed and the locked ranking is stable. Full wide-prior baselines for CPL, G1DE-2, and Λ CDM remain recommended as robustness checks.
- 5. Is the signal really $H + \Sigma$, or is it hidden growth modification?** The current interpretation is a primary signal in the background and Weyl channel with $\mu \simeq 1$. The G1DE-1 growth-only negative control and the $M_{3/4}$ locked branch are designed precisely to distinguish $H + \Sigma$ from growth-only alternatives.
- 6. Why is $\mu(a, k) = 1$ not a weakness?** Because it reduces the number of free parameters and raises falsifiability. If data require $|\mu - 1| \sim |\Sigma - 1|$, the Ward-suppressed Ricci-leakage branch fails.
- 7. Why does 3/4 matter?** It is not an arbitrary numerical fit parameter. It is the locked coefficient of the optical/modular projection closure. If free- κ decisively beats $M_{3/4}$, the exact 3/4 lock is demoted.
- 8. Does the compressed E_G dataset make the lensing result too weak?** Yes, this is a recognized limitation of the current evidence pass. Compressed E_G supports only an exact-pilot-level claim; expanded weak-lensing and 3×2 pt likelihoods are the next necessary step for Weyl-channel testing.
- 9. What if constant- Σ beats the output-response-shape model in future data?** That would demote the $\hat{R}_H(a)$ output-shape functionality but not necessarily kill the Weyl-amplitude branch. The amplitude lock can survive even if the redshift-kernel shape is unresolved.

10. What if a free $A(a, k)$ function is required? The model ceases to be G1DE and becomes a generic dark-stress model. The Weyl-normalized production convention explicitly forbids sampling $A(a, k)$ as an independent function.

11. Does G1 explain the cosmological constant problem? No. It reframes late-time residuals as finite-screen modular/gluing defects, but the microscopic vacuum-energy problem and the full calibration of the zero-point remain separate open tasks.

12. What about CMB and BBN? The current branch must remain compatible with early-universe constraints. If high-redshift residual occupancy is too large, the branch is demoted. Full CMB/BBN production validation is not yet included.

13. What about nonlinear structure formation and baryonic feedback? The current G1DE test is a linear, optical, output-level branch. Nonlinear and baryonic production pipelines are future work and are not claimed to be replaced at this stage.

14. What about black holes and gravitational waves? General Relativity remains the production theory for current strong-field and wave-form-level tests. FDS-G1 provides a finite-screen interpretation and possible residual/memory directions at the conceptual level, not yet waveform-level predictions.

15. Is the framework overbuilt? The number of technical notes should not be judged by volume alone. Prediction compression is the relevant metric. The key empirical claim ultimately reduces to $s < 3$, $\mu = 1$, and $\Sigma - 1 = -\frac{3}{4}(3 - s)\hat{R}_H$.

16. Is this just thermodynamic gravity / Jacobson / Verlinde under a new name? No. It uses those bodies of work as prior bridges, but adds the finite-screen entropy-response chain, the G1DE background-Weyl residual, the 3/4 projection lock, the D0 calibration route, and the evidence/failure hierarchy.

17. Is D0 a completed derivation of Einstein gravity? No. D0 is a calibration-reduction route: it compresses the Einstein-calibration problem into two sharper lemmas (finite-index screen-algebra density and boost modular covariance). Those lemmas remain open.

18. Does “paradigm-level candidate” mean “confirmed paradigm”? No. It is a structural classification, not a triumphal declaration of victory. Candidate status arises from primitive shift, prediction lock, falsifiability, and evidence interface. Confirmed status requires future data, replication, and production closure.

19. What would make G1 stronger than Λ CDM in the relevant domain? Production-level evidence with expanded lensing, independent pipeline replication, and a retained low-dimensional projection lock that outperforms or remains competitive with baselines under full prior stress.

20. What would kill the dark-sector branch? A combination of: CPL or Λ CDM decisively beating the G1DE branch in production evidence; expanded lensing not supporting the Weyl-channel signal; free-response G1DE-2 or free $A(a, k)$ models being required; the μ channel strongly deviating from one; or full wide-prior sensitivity collapsing the ranking.

Does production breadth decide the theory’s status? No. Production breadth is an implementation-readiness dimension, not a truth criterion. GR+ Λ CDM currently dominates strong-field waveforms, nonlinear simulations, CMB/BBN pipelines, and community infrastructure. FDS–G1 does not yet replace that production apparatus. The theoretical comparison concerns different dimensions: primitive depth, bridge assumptions, prediction lock, explanatory compression, and falsifiability.

Near-Term Falsifiable Prediction Lock

FDS–G1 is not only a structural framework; its current leading empirical branch makes a short-horizon falsifiable prediction. DESI DR2 cosmology products are already available, and Euclid DR1 is expected in October 2026. The projection-locked $M_{3/4}$ branch therefore faces near-term tests in the background, Weyl/lensing, and growth channels.

The locked prediction is

$$s < 3, \quad \mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a), \quad \hat{R}_H(1) = 1.$$

The key near-term test is not whether future data prefer “some” evolving dark energy, but whether they select the specific G1 pattern: background deviation plus suppressed Weyl response, with near-GR growth and $\kappa = 3/4$.

The relevant tests are channel-specific. They do not merely ask whether the model can improve a fit; they ask whether the improvement appears in the predicted channels and fails in the predicted ways.

Near-term prediction channels and demotion criteria for the G1 $M_{3/4}$ branch.

Prediction channel	Support for $M_{3/4}$	Demotion / failure condition
Background expansion	DESI/BAO/SN continue favoring $s < 3$ over $s = 3$.	Production evidence returns to Λ CDM or CPL dominance.
Weyl / lensing response	Euclid/ $E_G/3\times 2$ pt support $\Sigma < 1$ with shared $\hat{R}_H(a)$ shape.	Expanded lensing does not support a Weyl-channel signal.
Growth response	$\mu(a, k) \simeq 1$, maintaining near-GR matter growth.	Data require $ \mu - 1 \sim \Sigma - 1 $, indicating unsuppressed Ricci leakage.
Projection lock	Free- κ does not decisively beat $\kappa = 3/4$.	Free- κ decisively beats the $M_{3/4}$ exact lock.
Output-response shape	$\hat{R}_H(a)$ beats or remains competitive with constant- Σ .	Constant- Σ decisively beats the output-response-shape model.
Model identity	No free $A(a, k)$ amplitude is required.	Free $A(a, k)$ is required, reducing G1DE to a generic dark-stress model.

Comparative Structural Positioning

The purpose of this matrix is not to assign a total score. It separates dimensions that are often conflated: primitive layer, bridge assumptions, prediction lock, flexibility, empirical maturity, and known stress. All frameworks require primitives, calibration assumptions, and observational interfaces; the relevant question is whether those bridges are established, open, flexible, or prediction-locking.

Framework	Primitive layer	Bridge / calibration assumptions	Prediction lock	Flexibility risk	Current maturity / stress
GR+ Λ CDM	Metric geometry, stress-energy, constant Λ , cold dark matter	Equivalence principle, smooth manifold, Einstein-Hilbert action, conservation, initial conditions, astrophysical/baryonic modeling, unexplained Λ	High as production baseline; low for the origin of Λ	Low baseline freedom, but unexplained dark sectors	Established full-stack baseline; stressed by Λ problem and late-time tensions
CPL / $w_0 w_a$	Effective dark-energy fluid	Smooth equation-of-state expansion plus perturbation/closure priors	Low–medium	Moderate phenomenological flexibility	Useful benchmark; weak mechanism
Horndeski / EFT of DE	Scalar-tensor / EFT operator basis	Field content, stability, screening, local tests, GW constraints, EFT truncation	Usually low unless restricted	High without strong priors	Mature formalism; often flexibility-limited
$f(R)$, $f(Q)$, related MG	Modified geometric action or connection sector	Chosen function, screening, local tests, growth consistency, GW propagation	Medium if function fixed	Medium–high	Strong literature; constrained by local/growth/GW tests
Thermodynamic / emergent gravity	Entropy, horizon thermodynamics, information principles	Horizon entropy, local first law, Raychaudhuri bridge, metric/Ward closure	Often low unless residual law added	Model-dependent	Conceptually adjacent; usually lacks locked data-facing residual
Quantum-spacetime programs	String/AdS-CFT/LQG/causal-set primitives	Compactification, holographic dictionary, semiclassical/continuum recovery	Low near-term dark-sector lock	Varies; often model-multiplicity or long bridge	High ambition; long empirical path for late-time cosmology
FDS-G1 / $M_{3/4}$	Finite causal-screen ledger, entropy response, integrable geometry, gluing defects	Finite distinguishability, causal-screen localization, entropy ledger, stiffness bridge, D0 modular calibration lemmas, Ward/Bianchi closure, Weyl normalization, optical projection, likelihood implementation	High: $\mu = 1$, $\Sigma - 1 = -\frac{3}{4}(3 - s)\widehat{R}_H$	Low in locked branch; controlled by free- κ , const- Σ , G1DE-2	Exact-pilot evidence-selected; top-control wide-prior stable; pending production/lensing/replication

A closing expanded version of this matrix, with additional columns for dark-sector specificity and falsification / demotion clarity, appears at the end of the complete series.

Table 1: Six-model medium-prior nested-evidence comparison. $\Delta \log Z_i = \log Z_{M_{3/4}} - \log Z_i$ (positive values indicate evidence deficit relative to the best model). Three seeds per model, $n_{\text{live}} = 800\text{--}1000$, $d \log z = 0.5$. Medium-prior values preliminary; top-control wide-prior sensitivity completed for $M_{3/4}$, M_κ , and $\text{const-}\Sigma$; full wide-prior baseline sensitivity and production refinement pending.

Model	ndim	$\chi^2_{\min}$	$\log Z$	$\Delta \log Z$	$B_{M_{3/4},i}$
$M_{3/4}$	4	1767.15	−894.26	0.00	1
M_κ	5	1767.26	−895.23	0.97	2.6
$\text{const-}\Sigma$	5	1767.21	−896.18	1.92	6.8
G1DE-2	6	1767.30	−900.95	6.70	8.1×10^2
CPL	5	1774.56	−903.69	9.43	1.25×10^4
ΛCDM	3	1789.16	−906.22	11.96	1.57×10^5

Evidence Summary: Medium-Prior Nested Comparison

AIC/BIC matched comparison: $M_{3/4}$ also leads by AIC ($\Delta\text{AIC} = 0$) and BIC ($\Delta\text{BIC} = 0$), matching the G1DE-2 χ^2 to numerical precision with two fewer response parameters. The evidence hierarchy is

$$M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE-2} > \text{CPL} > \Lambda\text{CDM}.$$

Interpretation: The data favor a sparse projection-locked background–Weyl residual over phenomenological dark-energy flexibility and unconstrained response freedom. The projection-locked model $M_{3/4}$ is selected by evidence; the amplitude lock $\kappa = 3/4$ is favored (survives top-control wide-prior sensitivity) and remains a conditional projection-locking result. The output-response shape $X_b(a) = \widehat{R}_H(a)$ has preliminary support and requires expanded lensing data. G1DE-2 serves as a free-response adversarial envelope; G1DE-1 is the growth-only negative control supporting the interpretation that the primary signal is $H + \Sigma$ with $\mu \simeq 1$; its nested evidence is not in the current six-model evidence table. Within the tested late-time residual sector, $M_{3/4}$ is presently the most constrained and best-supported G1 branch: it combines a locked projection coefficient, parameter compression, matched exact-pilot evidence, and explicit demotion paths.

D7 gives a finite-state Markov-screen realization prototype for an admissible passive lift, but the likelihood remains output-level: it samples $R_H(a) = R_0 \widehat{R}_H(a)$, not the microscopic generator $\mathcal{L}_N$.

Wide-prior sensitivity (top-control set): A wide-prior pass over the three closest models confirms that the evidence preference for $M_{3/4}$ is not a medium-prior-volume artifact. The projection-locked ranking survives prior broadening (Table 2).

Table 2: Wide-prior nested-evidence comparison for the three closest models. The projection-locked ranking survives prior broadening.

Model	$\log Z$ (medium)	$\log Z$ (wide)	$\Delta \log Z$ (wide)	$B_{M_{3/4},i}$ (wide)
$M_{3/4}$	-894.26	-896.81	0.00	1
M_κ	-895.23	-897.66	0.85	2.3
const- Σ	-896.18	-898.76	1.96	7.1

Appendix: Closure Theorems, Bridge Principles, and Open Lemmas

The following seven items form the mathematical closure spine of the G1 program. Items 1, 2, 5 are formal theorems. Item 3 is a classification rule. Item 4 comprises open lemmas (strong-route targets). Item 6 is a conditional admissibility rule. Item 7 is a conditional theorem.

1. **Screen Ledger Representation Theorem (Core, Theorem 1).**

$$S_{\text{scr}} = k_B \log N_{\text{scr}}.$$

2. **Quotient-Invariance Theorem (Core, Theorem 2).**

$$\mathcal{X}_{\text{phys}} = \mathcal{X}_{\text{raw}} / \mathcal{G}_{\text{scr}}.$$

3. **Response Obstruction Theorem (Core, Theorem 3).**

$d\omega \neq 0 \Rightarrow \{\text{incomplete quotient, missing variable, memory, non-identifiability, physical residual}\}.$

Only the final case enters $R_{\mu\nu}^{\text{scr}}$.

4. **Calibration Lemma Stack (D0, Lemmas D0.1–D0.5).** Weak bridge: BH entropy.
Strong route: finite-index density + boost modular covariance + anomaly cancellation.

5. **Null Response Metricization Theorem (Core, Theorem 4).**

$$\mathcal{F}_p(k) = E_{\mu\nu}(p)k^\mu k^\nu$$

up to trace.

6. **No-Free-Residual Theorem (Core, §11).**

$$R_{\mu\nu}^{\text{scr}} \in \ker \nabla^\mu \cap \text{Obs}(\mathcal{C}_{\text{resp}}) \cap \text{Quot}(\mathcal{G}_{\text{scr}}) \cap \text{Penalty}(\mathcal{L}).$$

7. **3/4 Projection Uniqueness Theorem (D2, Theorem D2.1).**

$$\kappa = \frac{\dim(S \oplus \Omega)}{\dim(A \oplus S \oplus \Omega)} = \frac{3}{4}$$

under isotropic optical MaxEnt closure.

These theorems are conditional on their stated bridge assumptions. The open lemmas (D0 finite-index density, boost modular calibration, and all-null Ward/Bianchi completion) remain explicit mathematical targets rather than claimed results.

Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers

From Finite Screen Capacity to Local Coupling, Capacity-Flow Normal Forms, and Integrable Geometry Closure

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

May 27, 2026

Abstract

Finite Distinction Systems — Gravity 1 (FDS-G1) Core consolidates the finite screen-capacity gravity program into a single flagship paper. This flagship paper, together with six technical companions (Companions A–F covering screen statistics, transition dynamics, scalar-capacity EFT, likelihood tests, horizon occupancy residuals, and holographic dictionaries), forms the complete FDS-G1 program. The primitive object is a finite causal-screen entropy functional

$$S_{\text{scr}} : \mathcal{X}_{\text{phys}} \rightarrow \mathbb{R},$$

defined on a physical quotient state space of screen variables. Its response one-form

$$\omega = \eta_i(X) dX^i, \quad \eta_i = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta X^i}$$

contains the local area response

$$\eta_{\text{loc}}(x) = \omega(\delta A(x)) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}.$$

Under the standard horizon-entropy and local-screen thermodynamic bridges, this area response determines the local inverse coupling, or gravitational stiffness:

$$G_{\text{loc}}^{-1}(x) = \frac{4\hbar}{c^3} \eta_{\text{loc}}(x), \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}.$$

The paper organizes the G1 program around the chain

$$\text{finite screen statistics} \rightarrow S_{\text{scr}}[X] \rightarrow \omega \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}} \rightarrow \theta\text{-flow} \rightarrow \text{integrable geometry} \rightarrow \mathcal{R}_{\mu\nu}.$$

Finite screen statistics supplies the area-extensive substrate; interacting screens renormalize the response density; stochastic entropy-gradient screen dynamics gives Raychaudhuri-like capacity-flow normal forms; integrable all-null response data generate Einstein-like metric closure; non-integrable response data appear as residual sectors. Scalar-capacity EFT, holographic dictionaries, horizon thermodynamics, and likelihood-ready observational tests are treated as companion interfaces or manuscripts. The paper is a bridge program: the derivation is conditional on explicitly stated assumptions. The framework is weakened if finite screen statistics fails to produce an area-extensive response substrate, if local response data cannot be metric-integrated, or if residual sectors fail conservation and likelihood penalties. It aims to provide a precise statistical and response-geometric substrate for the area-entropy coefficient that controls gravity-like coupling. The choice of S_{scr} is motivated by a primitive FDS argument: finite maintainable distinctions require causal boundary accounting, and the coordinate-free ledger of finite screen state-count is entropy. A stronger calibration route is developed in D0, where the usual horizon-entropy bridge is replaced by a conditional modular anomaly-inflow argument: the primitive coefficient is a finite-access capacity density η_0 ; under boost modular calibration and all-null Ward/Bianchi closure one obtains $G = c^3/(4\hbar\eta_0)$ and $S_H/k_B = \eta_0 A_H = A_H/(4\ell_P^2)$.

1 Scope, Claim Status, and Bridge Assumptions

This paper presents the core formulation of the FDS-G1 finite screen-capacity program. The framework is organized around the response-geometric chain

$$\text{finite screen statistics} \rightarrow S_{\text{scr}}[X] \rightarrow \omega \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}} \rightarrow \theta\text{-flow} \rightarrow \text{integrable metric closure} \rightarrow \mathcal{R}_{\mu\nu}.$$

FDS-G1 is not introduced as a phenomenological modification of the Friedmann equation or of the Einstein equation. It is a finite-screen response program in which local inverse coupling, capacity-flow normal forms, metric-integrability conditions, and residual sectors arise from a common entropy-response architecture.

1.1 Closure Contract and Non-Absorption Rule

A G1 claim is closed only when its assumptions, proof type, failure mode, and demotion consequence are explicitly stated. No failed bridge, residual, or empirical branch may be reabsorbed into the theory by redefining $R_{\mu\nu}^{\text{scr}}$, $R_H(a)$, $A(a, k)$, or the microscopic source–kernel factorization.

The governing standard is: every major statement must be classified as one of

1. Definition
2. Formal theorem
3. Conditional theorem
4. Bridge assumption
5. Normal-form approximation
6. Open lemma
7. Empirical output normal form
8. Falsification / demotion rule

The master closure rule for the series is

Closure = Typed arrows + Admissible residuals + Non-absorbable failures + Identifiable outputs.

The non-absorption rule:

- Failed calibration $\nrightarrow$ free residual survival.
- Failed empirical branch $\nrightarrow$ failure of screen entropy primitive.
- Successful $M_{3/4}$ $\nrightarrow$ unique microscopic derivation.

1.2 Why finite screen entropy is the primitive object

The choice of a finite causal-screen entropy functional is not an ornamental assumption added after the fact. It is the minimal gravitational primitive obtained by applying the FDS ontology to a causal spacetime setting. The argument is programmatic rather than a derivation of the Einstein equation:

1. Physical distinctions must be finite and maintainable.
2. Maintainable distinctions require a boundary.

3. In gravity, the relevant boundary is causal: an operational screen or horizon.
4. A finite causal screen has a finite state-count ledger.
5. The coordinate-free ledger of finite state-count is entropy.
6. Therefore the primitive gravitational object is a screen entropy functional

$$S_{\text{scr}} : \mathcal{X}_{\text{phys}} \rightarrow \mathbb{R}.$$

7. Geometry is the integrable response envelope of this ledger.

Theorem 1 (Screen Ledger Representation Theorem). *Let Σ be a finite causal screen with accessible state-class set $\mathcal{C}_\Sigma$, $N_\Sigma = |\mathcal{C}_\Sigma|$. Suppose a ledger L satisfies*

$$L(\Sigma_1 \sqcup \Sigma_2) = L(\Sigma_1) + L(\Sigma_2), \quad N_{\Sigma_1 \sqcup \Sigma_2} = N_{\Sigma_1} N_{\Sigma_2},$$

coarse-graining monotonicity

$$\pi : \mathcal{C}_\Sigma \rightarrow \mathcal{C}'_\Sigma \quad \Rightarrow \quad L(\mathcal{C}'_\Sigma) \leq L(\mathcal{C}_\Sigma),$$

and standard regularity/extensivity assumptions (continuity, or monotonicity over positive integers, extensivity to rational/real capacities). Then, up to scale, the unique physically admissible ledger is

$$L(\Sigma) = \alpha \log N_\Sigma.$$

With $\alpha = k_B$,

$$S_{\text{scr}} = k_B \log N_{\text{scr}}.$$

Proof. Additivity of L on disjoint unions and multiplicativity of state-count under independent composition, together with coarse-graining monotonicity and standard regularity/extensivity assumptions, force L to be the unique logarithmic form. The scale α is fixed by identifying L with thermodynamic entropy under the standard temperature-identification bridge. $\square$

This theorem proves that screen entropy is not merely an analogy. It is the unique additive coordinate of finite screen capacity.

Thus, within FDS, finite screen entropy is not a decorative analogy. It is what finite, boundary-maintained distinguishability becomes when the physical access relation is gravitational and causal. Coupling, capacity flow, metric closure, and dark residual sectors are then read as response, integrability, and failure modes of this entropy ledger.

Equivalently, to reject $S_{\text{scr}}[\mathcal{X}_{\text{phys}}]$ as the primitive starting point for the G1 branch, one must reject at least one of the following FDS commitments: finite distinguishability, boundary-maintained access, causal-screen localization, entropy as a coordinate-free capacity ledger, or geometry-as-response.

The central claim is conditional but sharply defined. Given the standard horizon-entropy normalization and local-screen thermodynamic bridge, FDS-G1 identifies the local inverse gravitational coupling, or stiffness, with the local entropy-area response of a finite causal screen:

$$G_{\text{loc}}^{-1}(x) = \frac{4\hbar}{c^3} \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}, \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} \left[\frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)} \right]^{-1}.$$

The purpose of this paper is not to replace the Bekenstein-Hawking entropy formula, Jacobson-type local thermodynamic arguments, Raychaudhuri focusing, Lovelock uniqueness, spin-2 consistency, scalar-tensor EFT, or holographic duality. Rather, it supplies a finite screen-capacity substrate and response-geometric language for the area-entropy coefficient that appears in these standard structures.

Claims in this paper are classified as derived toy results, response principles, bridge assumptions, EFT embeddings, normal-form results, or falsification interfaces. This classification distinguishes what follows internally from finite-screen assumptions, what is imported from standard horizon or holographic physics, and what remains an open closure condition.

Central slogan. A useful but nonliteral slogan is:

Gravity-like geometry is the integrable hydrodynamic response of finite causal-screen entropy.

The technical reading is that local screen entropy response defines inverse coupling (stiffness); stochastic screen dynamics defines capacity flow; and metric geometry is the integrable envelope of compatible local response laws.

What would count as a stronger derivation? To upgrade the program from bridge to derivation, one would need to derive internally, from FDS primitives alone, the Bekenstein-Hawking coefficient $1/(4\ell_P^2)$, the local screen first law, Raychaudhuri focusing, all-null tensor closure, and the observed gravitational constants. G1 does not yet do this. It instead provides a sequence of toy theorems, response principles, EFT embeddings, and falsification interfaces. D0 supplies a stronger conditional calibration route: the Bekenstein-Hawking coefficient problem is reduced to two sharper lemmas—construct a causal finite-index screen-algebra net with universal local capacity density η_0 , and prove boost modular calibration $K_S = (2\pi/\hbar)Q_\chi$ for local causal screens; once these hold, $G = c^3/(4\hbar\eta_0)$ and $S_H/k_B = A_H/(4\ell_P^2)$ follow.

The comparative claim is structural. A competing macro-spacetime residual theory is not equivalent merely because it can fit a distance curve or introduce effective functions. It must provide a comparable primitive layer, a comparable response chain, and a comparable falsifiable residual fingerprint.

1.3 Claim-Type Table

Table 1: Claim-type audit for FDS-G1 Core. Every major statement is classified by proof type, dependency, and failure consequence.

Claim	Type	Depends on	Failure consequence
finite screen entropy primitive $G^{-1} \sim \eta_{\text{loc}}$	representation theorem target bridge / D0 lemma	finite accessible screen states BH bridge or D0 calibration	primitive weakened stiffness bridge demoted
θ -flow	normal form	local stochastic screen dynamics	normal-form demotion
all-null metricization $R_{\mu\nu}^{\text{scr}}$ $M_{3/4}$	theorem target admissible residual empirical output normal form	quadratic null response obstruction + conservation projection lock + likelihood	metric closure fails residual rejected branch demotion

1.4 Strong vs Weak Closure

Two levels of closure are distinguished.

Weak closure uses standard horizon entropy, local first law, Raychaudhuri, and Ward/Bianchi closure as bridges. It is the current active bridge level.

Strong closure derives the calibration through D0 finite-index density and boost modular covariance. D0 reduces the BH coefficient and Einstein calibration problem to finite-index screen-algebra density and boost modular covariance lemmas, which remain open. The strong route is a conditional target, not a completed derivation.

If D0 finite-index density fails, strong calibration fails but the weak horizon bridge may survive. The distinction prevents empirical branch success from being misread as strong-route closure.

2 Prior-art boundary

The paper uses established results as bridges. Area entropy and black-hole thermodynamics trace to Bekenstein and Hawking [1, 2]. The local thermodynamic route from horizon first law to Einstein equations follows Jacobson-type reasoning [4]. Raychaudhuri focusing is standard congruence geometry [3]. Lovelock uniqueness constrains low-derivative metric closure [5]; spin-2 consistency routes are associated with Weinberg and Deser [6, 7]. Scalar-tensor and Horndeski theories provide the conservative EFT home for variable-coupling sectors [8, 9]. AdS/CFT and holographic entropy are represented by Maldacena, Ryu-Takayanagi, and FLM [10, 11, 12]. GW170817 motivates luminal tensor-sector choices in late-time modified gravity [13].

FDS-G1’s contribution is orthogonal to these results: it supplies the finite screen-capacity interpretation and the entropy-response formalism that organizes them into a single bridge program.

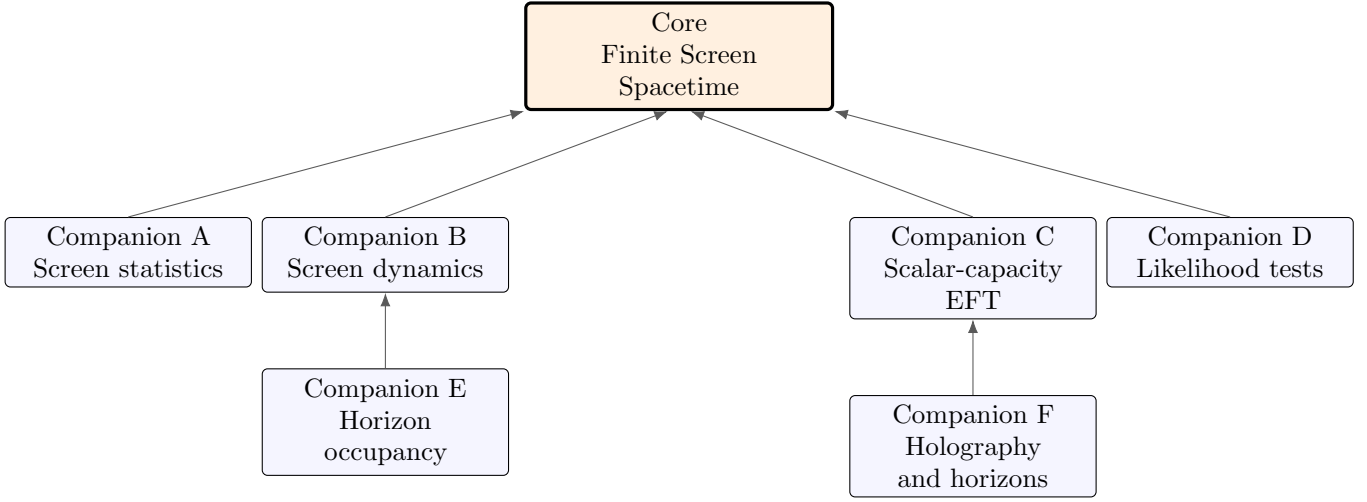
Antiba’s finite-capacity thermodynamics of causal horizons studies horizon capacity as a limitation on local classical representability [16]. G1 is adjacent but distinct: it treats finite causal-screen entropy response as the source of local inverse coupling and as input data for metric-integrability and residual-closure structure. Bousso’s covariant entropy conjecture links light-sheet area to entropy content [17] and provides a key prior constraint for any causal-screen entropy program. The holographic dictionary section draws on the Brown–Henneaux central-charge relation $c_{\text{CFT}} = 3L/(2G_3)$ as a concrete example of screen capacity density controlling asymptotic coupling [18]. The horizon thermodynamics and first-law discussion complements the Bekenstein–Hawking and Jacobson references with the Iyer–Wald Noether-charge formalism for diffeomorphism-invariant theories [19]. The local thermodynamic route to gravitational equations is further supported by Jacobson’s entanglement-equilibrium approach to the Einstein equation [20].

Recent adjacent work has also moved toward finite-capacity, finite-resolution, and information-ledger descriptions of gravitational screens. Barzi et al. study finite heat-capacity corrections to Unruh/Jacobson thermodynamics and the associated Rényi-entropy structure [21]. Nusbaumer formulates semiclassical gravity on causal diamonds with finite-resolution boundary algebras and a relative-entropy variational principle [22]. Song develops a local entropy-balance ledger on causal horizons, balancing area entropy, modular energy, and irreversible record updates [23]. Ahmad and Klinger develop an algebraic quantum-probability route to emergent area terms and generalized entropy [24]. Bianconi formulates an entropic action from quantum relative entropy between spacetime and matter-induced metrics [25]. These approaches are adjacent to G1 but differ in emphasis: G1 identifies the local inverse coupling with the marginal entropy-area response of a finite causal screen and then follows that response through capacity-flow, metric-integrability, residual closure, and likelihood-ready tests.

Relation to the FDS series. FDS-G1 is the gravitational branch of the broader Finite Distinction Systems program. The formal FDS core defines active finite distinction systems as finite-capacity boundary-maintaining systems and introduces the proof-status discipline used here [28]. FDS-T1 develops observer-relative distinguishability budgets [29]; FDS-T2 formulates effective geometry as horizon boundary accounting [30]; FDS-T3 studies capacity overflow, effective stochasticity, and Markov closure under finite projection [31]; and FDS-X1 gives the predecessor horizon-ledger dark-energy bridge [32]. FDS-X3 decomposes the four known interactions into functional distinction-operation classes, identifying gravity as global causal-geometric / stress-energy accounting [34]. FDS-G1 develops this gravitational row by modeling global boundary accounting as finite causal-screen entropy response. FDS-X2 supplies the weak-sector orientation-capacity module used by X3, but is not a direct input to the gravitational response bridge developed here [33]. The present paper differs from those predecessors by promoting the screen entropy functional $S_{\text{scr}}[X]$, its response one-form ω , and the local area response η_{loc} into the central gravity-specific architecture.

3 Program architecture and dependency graph

The G1 program is organized as one flagship core paper together with six technical companions:



Internal theoretical chain:

finite screen statistics $\rightarrow S_{\text{scr}}[X] \rightarrow \omega \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}}^{-1}/G_{\text{loc}} \rightarrow \theta\text{-flow} \rightarrow \text{integrable metric closure} \rightarrow \mathcal{R}_{\mu\nu}$.

Table 2: Claim-status audit for FDS-G1 Core.

Claim	Meaning	Status
Finite cell area law	$S_A/k_B = (A/a_0) \ln m$	toy theorem
Interacting area law	$S_A/k_B = \eta_{\text{eff}} A + o(A)$	statistical bridge
Screen quotient state space	$\mathcal{X}_{\text{phys}} = \mathcal{X}_{\text{raw}}/\mathcal{G}_{\text{scr}}$	definition/framework
FDS primitive screen ledger	finite maintainable distinctions require causal boundary entropy ledger	programmatic / ontological bridge
Response one-form	$\omega = \eta_i dX^i$	definition
Local coupling bridge	$G_{\text{loc}} = (c^3/4\hbar)[\eta_{\text{loc}}]^{-1}$	conditional bridge
Capacity-flow normal form	$\dot{\theta} = -a\theta^2 - b\sigma^2 - c\Phi + \Xi$	stochastic normal-form result
Metric all-null closure	null response reconstructs tensor up to trace	standard tensor bridge
Ward/Bianchi closure	response conservation	gauge-to-Ward bridge
Scalar-capacity EFT	$\varphi = \ln(\eta/\eta_0)$	conservative scalar-tensor EFT
Residual taxonomy	non-integrable response classes	framework
Likelihood interface	SN/BAO/growth/local- G blocks	medium-prior evidence executed; top-control wide-prior sensitivity completed; production refinement pending

4 Finite screen statistical mechanics

Definition 1 (Finite distinction screen). *A finite distinction screen Σ_A is a causal boundary patch with operational area A and finite screen-local distinguishability units. In the simplest model,*

$$N_A = \frac{A}{a_0}, \quad (1)$$

where a_0 is an effective area per independent cell.

If each cell carries m distinguishable states and the cells are independent,

$$\Omega_A = m^{N_A}, \quad S_A = k_B \ln \Omega_A = k_B \frac{A}{a_0} \ln m. \quad (2)$$

Therefore

$$\frac{S_A}{k_B} = \eta_{\text{nat}} A, \quad \eta_{\text{nat}} = \frac{\ln m}{a_0}. \quad (3)$$

Matching the standard horizon bridge gives

$$\eta_{\text{BH}} = \frac{1}{4\ell_P^2} = \frac{c^3}{4G\hbar}, \quad G = \frac{c^3}{4\hbar} [\eta_{\text{BH}}]^{-1}. \quad (4)$$

This identifies the screen-capacity density that must be supplied to reproduce the standard horizon coefficient; the coefficient $1/4$ itself remains an external input.

Definition 2 (Screen area-law universality class). *$\mathcal{U}_{\text{scr}}$ is the class of finite screen systems whose degrees of freedom are screen-local, finite-state, finite-range or sufficiently short-range, finite-correlation in the regime considered, stable under screen-local coarse-graining, and not dominated by hidden bulk-volume degrees of freedom or global nonlocal constraints.*

For systems in $\mathcal{U}_{\text{scr}}$ one expects the thermodynamic form

$$\frac{S_A}{k_B} = \eta_{\text{eff}} A + S_{\text{sub}}(A), \quad \frac{S_{\text{sub}}(A)}{A} \rightarrow 0 \quad (A \rightarrow \infty). \quad (5)$$

Finite interactions renormalize η_{eff} rather than destroying the leading area law. Criticality, long-range constraints, infinite local Hilbert spaces, hidden bulk volume degrees of freedom, or dominant nonlocal constraints are controlled failure modes.

5 Screen state space and quotient structure

A screen entropy functional must be defined on physical screen states, not on arbitrary coordinate labels. Write raw variables as

$$X_{\text{raw}} = \{A(x), q_{AB}(x), \chi(x), \Phi(x), M(x), \dots\}, \quad (6)$$

where A is area or cell density, q_{AB} is screen shape, χ is occupancy or residual capacity, Φ is flux load, and M denotes memory or hidden screen modes. The physical state is the quotient

$$\boxed{\mathcal{X}_{\text{phys}} = \mathcal{X}_{\text{raw}} / \mathcal{G}_{\text{scr}}}, \quad (7)$$

where $\mathcal{G}_{\text{scr}}$ includes screen coordinate reparametrizations, null-generator reparametrizations, cell relabelings, and internal labels that do not change causal access or entropy response.

Definition 3 (Screen entropy functional). *The entropy functional is*

$$S_{\text{scr}} : \mathcal{X}_{\text{phys}} \rightarrow \mathbb{R}. \quad (8)$$

Functional derivatives are understood on the quotient tangent space, or after gauge fixing and projection away from pure gauge directions.

Example 1 (Cell relabeling). *If cells $i = 1, \dots, N$ have states σ_i , a permutation $i \mapsto \pi(i)$ changes labels but not area, entropy, or response. Thus $\{\sigma_i\} \sim \{\sigma_{\pi(i)}\}$.*

Example 2 (Generator reparametrization). *A monotonic change $\lambda \mapsto f(\lambda)$ changes the coordinate expression for $\theta = d \ln A / d\lambda$. It does not change the cross-sectional entropy-area response $\eta_{\text{loc}} = k_{\text{B}}^{-1} \delta S / \delta A$.*

Theorem 2 (Quotient-Invariance Requirement). *A response quantity is physically admissible only if it descends to $\mathcal{X}_{\text{phys}}$. Require*

$$S_{\text{scr}}(g \cdot x) = S_{\text{scr}}(x), \quad \omega_{g \cdot x}(Dg v) = \omega_x(v).$$

Any residual computed before quotient projection is not physical.

Failure rule. *If $d\omega \neq 0$ disappears after quotient correction, the apparent residual is a gauge artifact, not $R_{\mu\nu}^{\text{scr}}$.*

6 Response one-form and local coupling

Let $s_{\text{scr}} = S_{\text{scr}}/k_{\text{B}}$. The response one-form is

$$\omega = \eta_i(X) dX^i, \quad \eta_i(X) = \frac{\delta s_{\text{scr}}}{\delta X^i}. \quad (9)$$

For field variables this denotes

$$\omega = \int_{\Sigma} d^{d-2}x \eta_i(x) \delta X^i(x). \quad (10)$$

Let $\delta_A(x)$ be the physical tangent vector that changes the local area element $A(x)$ while holding the other physical screen variables fixed. Then

$$\boxed{\eta_{\text{loc}}(x) = \omega(\delta_A(x)) = \frac{1}{k_{\text{B}}} \frac{\delta S_{\text{scr}}}{\delta A(x)}}. \quad (11)$$

This makes η_{loc} the area component of the full response one-form.

Principle 1 (Local response coupling). *Under the standard horizon-entropy and local-screen thermodynamic bridges, the local area response determines the local inverse coupling, or gravitational stiffness:*

$$\boxed{G_{\text{loc}}^{-1}(x) = \frac{4\hbar}{c^3} \eta_{\text{loc}}(x), \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}}. \quad (12)$$

For homogeneous screens,

$$\eta_{\text{loc}} \rightarrow \eta_{\delta}(A) = \frac{d(S/k_{\text{B}})}{dA}, \quad G_{\delta}(A) = \frac{c^3}{4\hbar} [\eta_{\delta}(A)]^{-1}, \quad G_{\delta}^{-1}(A) = \frac{4\hbar}{c^3} \eta_{\delta}(A). \quad (13)$$

For finite-area corrections

$$\frac{S(A)}{k_{\text{B}}} = \eta_0 A + \alpha \ln(A/A_0) + \gamma + b(A/A_0)^{-p} + \dots, \quad (14)$$

one obtains

$$\eta_{\delta}(A) = \eta_0 + \frac{\alpha}{A} - \frac{pb}{A_0} (A/A_0)^{-p-1} + \dots. \quad (15)$$

Thus macroscopic finite-area corrections are naturally suppressed by inverse area unless a separate residual or scalar-capacity mode is activated.

7 Response Integrability and Obstruction Classes

If the response one-form derives from a scalar entropy potential s_{scr} , then

$$\omega = \text{d}s_{\text{scr}}, \quad \boxed{\text{d}\omega = 0}. \quad (16)$$

In coordinates,

$$\text{d}\omega = \frac{1}{2}(\partial_i \eta_j - \partial_j \eta_i) \text{d}X^i \wedge \text{d}X^j. \quad (17)$$

A nonzero response curl indicates missing variables, memory, nonlocal constraints, stochastic projection, or physical residual modes.

Theorem 3 (Response Exactness and Obstruction). *Define*

$$s_{\text{scr}} = S_{\text{scr}}/k_{\text{B}}, \quad \omega = \eta_i(X) \text{d}X^i, \quad \mathcal{C}_{\text{resp}} = \text{d}\omega.$$

Closed response corresponds to $\omega = \text{d}s_{\text{scr}}$, $\text{d}\omega = 0$. The response obstruction is classified as

$$\text{d}\omega \neq 0 \Rightarrow \begin{cases} \text{incomplete quotient,} \\ \text{missing coarse variable,} \\ \text{projected memory mode,} \\ \text{source–kernel non-identifiability,} \\ \text{true physical residual.} \end{cases}$$

Only the final case may enter $R_{\mu\nu}^{\text{scr}}$. All others must be resolved by quotient correction, variable enlargement, or memory-lift before a physical residual is declared.

Let $[\mathcal{C}_{\text{resp}}] \in H_{\text{resp}}^2(\mathcal{X}_{\text{phys}})$ denote equivalence classes of response non-closure after quotient, missing-variable, and memory-lift attempts have been exhausted. H_{resp}^2 is a bookkeeping device, not a full cohomology theory; it identifies which obstructions survive all structural corrections before being admitted as physical residuals. D1 supplies the mathematical home for this obstruction via finite-access sheaf geometry (see D1, §4).

Example 3 (Two-variable exact response). *Let*

$$s(A, \chi) = \eta_0 A + \alpha \ln A + \beta A \chi - \frac{\lambda}{2} \chi^2. \quad (18)$$

Then

$$\eta_A = \frac{\partial s}{\partial A} = \eta_0 + \frac{\alpha}{A} + \beta \chi, \quad \eta_\chi = \frac{\partial s}{\partial \chi} = \beta A - \lambda \chi. \quad (19)$$

The response one-form $\omega = \eta_A \text{d}A + \eta_\chi \text{d}\chi$ is exact, since

$$\partial_\chi \eta_A = \beta = \partial_A \eta_\chi, \quad \text{d}\omega = 0. \quad (20)$$

Example 4 (Apparent curl from hidden memory). *Suppose the true entropy depends on a hidden memory mode M ,*

$$s(A, \chi, M) = \eta_0 A + \alpha \ln A + \beta A \chi + \gamma A M - \frac{\lambda}{2} \chi^2 - \frac{\mu}{2} M^2. \quad (21)$$

On the full space (A, χ, M) , $\omega = \text{d}s$ is closed. If M is projected out but depends on history, the reduced apparent response $\tilde{\omega} = \tilde{\eta}_A \text{d}A + \tilde{\eta}_\chi \text{d}\chi$ can satisfy $\text{d}\tilde{\omega} \neq 0$. This is an epistemic or memory residual, not necessarily new fundamental physics.

8 Response Hessian, fluctuations, and scalar-capacity modes

The entropy-response Hessian is

$$\mathcal{H}_{ij} = -\frac{\delta^2 s_{\text{scr}}}{\delta X^i \delta X^j}. \quad (22)$$

Near equilibrium, $\mathcal{H}$ controls stability, fluctuations, and slow hydrodynamic response modes. In a Gaussian approximation,

$$\langle \delta X^i \delta X^j \rangle \sim (\mathcal{H}^{-1})^{ij}. \quad (23)$$

The scalar-capacity variable

$$\varphi(x) = \ln \frac{\eta(x)}{\eta_0} \quad (24)$$

can be interpreted as a long-wavelength mode of the area-response density. Its EFT potential is a coarse-grained free-energy cost for moving away from the equilibrium response density. Thus the scalar-capacity EFT is not introduced as a generic dark scalar; it is the hydrodynamic EFT of a slow screen-response mode.

9 Stochastic screen dynamics and capacity-flow normal forms

Let screen macrostate variables $X^i(\lambda)$ evolve by an entropy-gradient stochastic process. A generic Onsager form is

$$\dot{X}^i = L^{ij} \frac{\delta s_{\text{scr}}}{\delta X^j} + \zeta^i, \quad (25)$$

with positive semidefinite response matrix L^{ij} and noise ζ^i . A large-deviation or Onsager-Machlup path weight may be written schematically as

$$P[X(\lambda)] \sim \exp[-N_A \mathcal{I}_{\text{scr}}[X]], \quad (26)$$

$$\mathcal{I}_{\text{scr}}[X] = \frac{1}{2} \int d\lambda (\dot{X}^i - F^i[X]) D_{ij}^{-1} (\dot{X}^j - F^j[X]), \quad (27)$$

where $F^i = L^{ij} \delta s / \delta X^j$ in local equilibrium.

Define

$$N(\lambda) = \frac{A(\lambda)}{a_0}, \quad x(\lambda) = \ln N(\lambda), \quad \theta = \dot{x} = \frac{d \ln A}{d\lambda}. \quad (28)$$

Then

$$\dot{\theta} = \frac{\ddot{A}}{A} - \left(\frac{\dot{A}}{A} \right)^2. \quad (29)$$

The negative quadratic term is therefore kinematic: it follows from logarithmic capacity expansion. Shape modes contribute the positive scalar $\sigma^2 = \sigma_{AB} \sigma^{AB}$; positive flux load focuses the screen. The low-order affine equilibrium normal form is

$$\boxed{\dot{\theta} = -a\theta^2 - b\sigma^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_{\theta} + \mathcal{O}(3)}. \quad (30)$$

Equation (30) is a normal-form bridge: it shows that Raychaudhuri-like terms are stable low-order structures in stochastic finite screen dynamics. The identification of Φ_{flux} with $R_{\mu\nu} k^\mu k^\nu$ or $T_{\mu\nu} k^\mu k^\nu$ is the standard local-screen geometry bridge.

Proposition 1 (Raychaudhuri-Limit Recovery). *In the metric-integrable equilibrium limit, the capacity-flow normal form must reduce to the standard Raychaudhuri equation:*

$$\frac{d\theta}{d\lambda} = -\frac{1}{D-2} \theta^2 - \sigma_{\mu\nu} \sigma^{\mu\nu} + \omega_{\mu\nu} \omega^{\mu\nu} - R_{\mu\nu} k^\mu k^\nu + \mathcal{R}_{\text{scr}}^\theta.$$

For hypersurface-orthogonal null screens, $\omega_{\mu\nu} = 0$. Therefore

$$a = \frac{1}{D-2}, \quad b = 1$$

in the standard geometric limit. The remaining scalar focusing residual $\mathcal{R}_{\text{scr}}^\theta = \Xi_{\text{res}} + \zeta_\theta + \mathcal{O}(3)$ is the finite-screen correction. If these constants do not match, the model remains a finite-screen normal form but is not a GR-equilibrium recovery.

10 Metric integrability and all-null closure

Entropy integrability is not yet metric integrability. For each event p and null direction k^μ , local screen response data may define a null focusing response $\mathcal{F}_p(k)$.

Theorem 4 (Null Response Metricization). *Let screen response data define a null focusing function $\mathcal{F}_p(k)$ at each event p and null direction k^μ . Assume:*

1. *locality: $\mathcal{F}_p(k)$ depends only on data at p ;*
2. *smoothness in p ;*
3. *homogeneous quadratic dependence on k : $\mathcal{F}_p(\lambda k) = \lambda^2 \mathcal{F}_p(k)$;*
4. *compatibility with a Lorentzian null cone;*
5. *nondegenerate screen response in at least two independent null directions.*

Then there exists a symmetric tensor $E_{\mu\nu}(p)$ such that

$$\mathcal{F}_p(k) = E_{\mu\nu}(p) k^\mu k^\nu \tag{31}$$

up to a trace term $\Phi g_{\mu\nu}$. Because $g_{\mu\nu} k^\mu k^\nu = 0$, all-null data cannot fix the trace. If

$$E_{\mu\nu} k^\mu k^\nu = \kappa T_{\mu\nu} k^\mu k^\nu \quad \forall k,$$

then $E_{\mu\nu} - \kappa T_{\mu\nu} = \Phi g_{\mu\nu}$, and Ward/Bianchi closure determines whether Φ becomes a cosmological integration sector or an admissible residual.

All-null data reconstructs $E_{\mu\nu}$ up to a trace term, since $g_{\mu\nu} k^\mu k^\nu = 0$. Thus if

$$E_{\mu\nu} k^\mu k^\nu = \kappa T_{\mu\nu} k^\mu k^\nu \quad \forall k^\mu \text{ null}, \tag{32}$$

then

$$E_{\mu\nu} - \kappa T_{\mu\nu} = \Phi g_{\mu\nu}. \tag{33}$$

Ward/Bianchi closure fixes the trace freedom into the usual cosmological integration sector. In the Einstein-like equilibrium case,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} + \mathcal{R}_{\mu\nu}. \tag{34}$$

Here $\mathcal{R}_{\mu\nu}$ is not an arbitrary dark term; it represents non-integrable or unmodeled screen response data after attempting metric closure.

11 Ward/Bianchi dynamical closure

If the screen response action or its hydrodynamic envelope is invariant under screen reparametrizations that become spacetime diffeomorphism redundancy in the metric envelope, then the effective geometric tensor obeys a Ward identity

$$\nabla^\mu E_{\mu\nu} = 0. \quad (35)$$

Therefore residual sectors must satisfy

$$\boxed{\nabla^\mu (T_{\mu\nu} + \mathcal{R}_{\mu\nu}) = 0.} \quad (36)$$

This is the conservation-closure condition that prevents variable capacity from becoming an unconstrained varying- G ansatz.

If memory is present, local response may become nonlocal:

$$\eta_i(\lambda) = \int^\lambda d\lambda' K_{ij}(\lambda, \lambda') X^j(\lambda'), \quad (37)$$

or covariantly,

$$\mathcal{R}_{\mu\nu}^{\text{mem}}(x) = \int d^4x' \sqrt{-g(x')} K_{\mu\nu}{}^{\alpha\beta}(x, x') T_{\alpha\beta}(x'). \quad (38)$$

Such terms must still obey the total conservation law (36).

Theorem 5 (No-Free-Residual Rule). *Define the effective geometric equation*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} + R_{\mu\nu}^{\text{scr}}.$$

Admissibility requires $\nabla^\mu (T_{\mu\nu} + R_{\mu\nu}^{\text{scr}}) = 0$. If matter is separately conserved,

$$\nabla^\mu R_{\mu\nu}^{\text{scr}} = 0.$$

Then the residual must satisfy the intersection condition

$$\boxed{R_{\mu\nu}^{\text{scr}} \in \ker \nabla^\mu \cap \text{Obs}(\mathcal{C}_{\text{resp}}) \cap \text{Quot}(\mathcal{G}_{\text{scr}}) \cap \text{Penalty}(\mathcal{L}).}$$

A residual that fails any of these conditions is rejected: it is either a gauge artifact, a missing-variable projection, a non-conserved effective stress, or a function that carries no likelihood penalty.

12 Residual taxonomy

The residual sector is classified by admissibility, not used as a catch-all.

Table 3: Residual admissibility hierarchy.

Residual type	Must first pass	If fails
epistemic	missing-variable lift	not physical residual
memory	D6 quotient / horizontal lift	realization ambiguity
stochastic	T3 projection/lumpability test	modeling noise
capacity	scalar-capacity EFT consistency	scalar branch only
horizon occupancy	conservation + D5 identifiability	G1DE demotion
geometric	null metricization + Ward closure	nonmetric response theory

This classification makes explicit what survives when a residual model fails. For example, if a horizon occupancy model is ruled out, the finite screen statistics and entropy-response framework may survive. If metric closure fails, the program moves beyond standard metric geometry and must state so directly.

13 Scalar-capacity EFT interface

The local response density may have a long-wavelength mode

$$\varphi = \ln \frac{\eta}{\eta_0}, \quad \frac{G_{\text{eff}}}{G_0} = e^{-\varphi}. \quad (39)$$

A conservative scalar-capacity EFT is

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} F(\varphi) R - \frac{1}{2} Z(\varphi) (\nabla \varphi)^2 - V(\varphi) + \mathcal{L}_m + \mathcal{L}_{\text{res}} \right]. \quad (40)$$

With $F(\varphi) = M_{*,0}^2 e^\varphi$, this is a scalar-tensor/Horndeski-compatible sector, not a new field-equation form. Its novelty is the screen-capacity interpretation and prior structure. A minimal Horndeski embedding is

$$G_2 = Z(\varphi)X - V(\varphi), \quad G_3 = 0, \quad G_4 = \frac{1}{2}F(\varphi), \quad G_5 = 0. \quad (41)$$

In EFT-of-dark-energy language,

$$M_*^2 = F, \quad \alpha_M = \frac{d \ln F}{d \ln a}, \quad \alpha_T = 0, \quad \alpha_B = -\alpha_M \quad (42)$$

for the minimal nonminimally coupled model. Stability requires $F > 0$, positive kinetic normalization, absence of gradient instabilities, and compatibility with local-gravity and gravitational-wave propagation constraints.

14 Observational signatures and likelihood interface

G1 predicts no generic freedom to drift G macroscopically. A visible drift obeys

$$\frac{\dot{G}}{G} \simeq -\frac{\dot{\eta}}{\eta}, \quad (43)$$

and must be carried by a conservation-closed residual or scalar-capacity sector. The three refined signature tests are:

1. **No free macroscopic G -drift.** Any variation of G is constrained by local-gravity, PPN, and propagation priors.
2. **Background–Weyl residual pattern.** Horizon occupancy residuals should produce a structured fingerprint $s < 3$, $\Sigma_0 < 0$, $\mu_0 \simeq 0$: background filling deviation and suppressed Weyl response through a common ledger variable, with the growth response near GR.
3. **Finite-area suppression.** Log or finite-area corrections give $\Delta G/G \sim -\alpha/(\eta_0 A)$ and are negligible on macroscopic screens unless an additional residual sector is activated.

A compressed likelihood interface has the schematic form

$$\chi_{\text{tot}}^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{RSD}}^2 + \chi_{E_G}^2 + \chi_{\text{priors}}^2. \quad (44)$$

Companion likelihood manuscripts compare Λ CDM, CPL, G1DE, and scalar-capacity variants. G1fit demonstrations have now been upgraded to a matched exact-pilot and medium-prior nested-evidence comparison (G1fit-real), using Pantheon+ full covariance, DESI DR2 BAO covariance, curated RSD, and compressed E_G . The six-model medium-prior evidence hierarchy is G1DE- $M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE2} > \text{CPL} > \Lambda\text{CDM}$, with the projection-locked model ranking first. Top-control wide-prior sensitivity has been completed; full wide-prior baseline sensitivity and production evidence refinement remain pending.

14.1 Empirical status of the residual sector

The present status of the G1 residual sector is no longer purely schematic. A matched exact-likelihood and medium-prior nested-evidence comparison has been completed. The G1DE hierarchy uses an output-level memory response

$$\mathcal{R}_H(a) = R_0 \hat{\mathcal{R}}_H(a), \quad \hat{\mathcal{R}}_H(1) = 1, \quad R_0 = 3 - s,$$

implemented through a Weyl-normalized transfer operator

$$\mathcal{W}_{G1} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{GR}.$$

This is an output normalization; it should not be confused with a unique microscopic source–kernel factorization $K_H * J_H$, which is treated in D6 as a deeper horizontal-lift layer.

$$\Sigma(a, k) - 1 = -\kappa R_0 \hat{R}_H(a) + O(R_0^2), \quad \mu(a, k) - 1 = O(R_0^2) + O(R_0 \hat{R}_H \mathcal{H}^2 / k^2).$$

The projection-locked branch $\kappa = 3/4$, $\hat{R}_H = X_b$ (with $X_b(1) = 1$ as the production-normalized output-response shape) gives $\Sigma(a) = 1 - \frac{3}{4}(3 - s)X_b(a)$, $\mu \simeq 1$. In the current medium-prior matched evidence pass, G1DE-M_{3/4} ranks first among the tested six-model set. The detailed results and falsification ladder are in Companions D and E; the data paper is G1fit-real.

The refined empirical claim-status is therefore:

Table 4: Updated empirical status for the G1 residual sector.

Claim	Status
Finite screen entropy response	theoretical bridge
G1DE-0 background test	real-data full-covariance implemented
$s = 3$ Λ CDM-like limit	numerically validated
G1DE-2 exact pilot	survived exact full-likelihood pilot
$\Sigma_0 < 0$ Weyl response	pilot-level indication
$\mu_0 < 0$ growth response	not supported; $\mu_0 \simeq 0$
Medium-prior evidence completed;	
top-control wide-prior sensitivity	
completed; production validation	
pending	

A post-pilot projection-locked variant, G1DE-M_{3/4}, imposes $\mu(a, k) = 1$ (equivalently $\mu_0 = 0$) and $\Sigma(a) = 1 - \frac{3}{4}(3 - s)X_b(a)$. In the current medium-prior matched evidence pass (G1fit-real), this four-parameter locked model ranks first among the tested six-model set; see Companions D, E for status. This is an empirical refinement of the residual sector, not a core bridge assumption.

The factorized production form is

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1, \quad R_0 = 3 - s,$$

implemented through a Weyl-normalized transfer operator $\mathcal{W}_{G1} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{GR}$. The active-filling shape $X_{\text{fill}}(a)$ (peak-normalized) and the production-normalized memory shape $\hat{R}_H(a) = X_{\text{fill}}(a) / X_{\text{fill}}(1)$ are related by a constant factor; the peak-normalized form is used in the internal model definition, while $\hat{R}_H(1) = 1$ ensures identifiability of the projection coefficient κ and the residual amplitude R_0 .

15 Horizon occupancy residuals

A minimal horizon occupancy residual model uses

$$\rho_H(a) = \chi_H(a) \rho_c(a), \quad \rho_c(a) = \frac{3H^2(a)}{8\pi G}. \quad (45)$$

A lowest-order filling-closure law is

$$\frac{d\chi_H}{d\ln a} = s\chi_H \left(1 - \frac{\chi_H}{\chi_\infty}\right), \quad (46)$$

with solution

$$\chi_H(a) = \frac{\chi_\infty}{1 + Ba^{-s}}. \quad (47)$$

The effective equation of state is

$$w_H(a) = -1 - \frac{1}{3} \frac{d\ln[\chi_H(a)H^2(a)]}{d\ln a}. \quad (48)$$

The model is not intended as an arbitrary background fit. Its distinctiveness is a structured background–Weyl residual pattern

$$\chi_H(a) \Rightarrow H(a), \quad \Sigma(a), \quad \mu(a) \simeq 1, \quad \text{capacity closure}, \quad (49)$$

with the growth response near GR ($\mu_0 \simeq 0$) as a consistency condition rather than a required nonzero signal. This is more restrictive than a generic dark-energy or modified-gravity parametrization because the background and Weyl response share the same active-filling shape $\hat{X}(a)$.

16 Holography and horizons

The holographic dictionary is naturally written in response-density language:

$$\frac{1}{4G_N} = \eta_{\text{screen}}. \quad (50)$$

For $\text{AdS}_3/\text{CFT}_2$,

$$c = \frac{3L}{2G_3}, \quad \eta_{\text{AdS}_3} = \frac{1}{4G_3} = \frac{c}{6L}. \quad (51)$$

Thus central charge is interpreted as a boundary version of bulk screen capacity density. The Ryu-Takayanagi and FLM formulas are read as realizations of the screen-response principle:

$$S_R = \frac{A(\gamma_R)}{4G_N} + S_{\text{bulk}} + \cdots. \quad (52)$$

Similarly, black-hole and de Sitter horizons are exterior-accessible distinction ledgers with

$$S_H/k_B = \eta_{\text{BH}} A_H. \quad (53)$$

For Kerr-Newman horizons the first law

$$dM = T_H dS_H + \Omega_H dJ + \Phi_H dQ \quad (54)$$

is read as a multi-channel capacity update law.

17 Non-Absorption Rules

The following rules enforce the closure contract: no failed bridge, residual, or empirical branch may be silently reabsorbed by redefining $R_{\mu\nu}^{\text{scr}}$, $R_H(a)$, $A(a, k)$, or the microscopic source–kernel factorization.

Table 5: Non-Absorption rules for FDS-G1 Core.

Failure condition	What fails	What may survive
No screen area extensivity	statistical core of G1	other FDS ideas outside gravity
No finite-index screen algebra net	D0 strong calibration	Core response framework as bridge
No boost modular calibration	$(2\pi/\hbar)$ modular route	standard horizon bridge (weak closure)
No all-null metricization	Einstein-like closure	nonmetric response theory; residual cannot absorb failure
No metricization of anomaly-free sector	D0 Einstein derivation route	finite-access / residual framework
Unconstrained residuals	predictive value	core response formalism if residual taxonomy tightened
Scalar-capacity EFT ruled out	variable-capacity field model	microscopic/horizon-only corrections
Normal form not robust	G1micro hardening	response coupling and EFT sectors
No passive horizon remnant	G1DE anomaly-remnant interpretation	phenomenological Weyl-transfer branch
Free $A(a, k)$ required	Weyl normalization	generic dark-stress model, not G1DE
G1DE loses to constant- Σ	output shape $X_b(a)$	Weyl amplitude branch
G1DE loses to free- κ	exact 3/4 isotropic compliance	weighted compliance branch
Data require $ \mu - 1 \sim \Sigma - 1 $	Ward-suppressed Ricci leakage	finite-screen residual with density component
Production evidence loses to Λ CDM/CPL	G1DE observational branch	statistical and response core; screen entropy primitive unchanged
Likelihood pipeline fails with real data	observational branch	theoretical bridge remains provisional
$M_{3/4}$ decisive evidence success	—	does not prove finite-index screen algebras, boost modular covariance, or unique source–kernel factorization

18 Companion manuscripts and archive structure

The recommended release structure is one flagship Core paper, six technical companions (A–F), one data-confrontation paper (G1fit-real), and the D0–D7 dark-sector closure sequence:

1. **Companion A: Finite Screen Statistical Mechanics.** Independent cells, interacting screens, and differential capacity.
2. **Companion B: Stochastic Screen Transitions.** Microtransition and Raychaudhuri-like capacity flow.
3. **Companion C: Scalar-Capacity EFT.** Horndeski and EFT-of-dark-energy embedding.
4. **Companion D: Likelihood and Falsification.** Predictions, observational interface, and compressed-data testing.

5. **Companion E: Horizon Occupancy Residuals.** Dark-sector ledger dynamics and model comparison.
6. **Companion F: Holography and Horizons.** Black-hole, de Sitter, and AdS/CFT dictionaries.
7. **G1DE-M_{3/4}: A Projection-Locked Background–Weyl Residual.** Matched exact likelihoods and medium-prior nested evidence for six late-time cosmology models.
8. **D0: Screen Modular Anomaly Inflow.** Einstein calibration of finite-access entropy density.
9. **D1: Finite-Access Sheaf Geometry.** Finite causal ledgers as a sheaf; gluing defects as residuals.
10. **D2: The 3/4 Projection Law.** Optical-stress closure and the projection coefficient.
11. **D3: Non-Equilibrium Screen Entropy.** Thermodynamic closure of the G1 residual.
12. **D4: Screen Influence Functional.** Unified influence framework for the dark-sector closure.
13. **D5: Weyl-Normalized Ward Completion.** Production likelihood interface and identifiability rules.
14. **D6: Maximum-Caliber Horizontal Lift.** Memory quotient, source–kernel identifiability, and maximum-caliber passive horizontal lift.
15. **D7: Finite Markov-Screen Realization.** A finite-state detailed-balance realization prototype for the D6 passive lift assumptions, showing how optical symmetry, a slow horizon mode, and Ward-stiff Ricci leakage realize the G1DE-M_{3/4} normal form.

A code/data archive should include reproducible notebooks, demo compressed data, generated CSV files, figures, claim-status notes, prior-art boundaries, notation tables, and a failure registry.

19 Conclusion

FDS-G1 Core recasts the G1 program as entropy-response geometry. The central object is no longer a bare capacity density but a screen entropy functional $S_{\text{scr}}[X]$ on a quotient state space. Its response one-form defines local area response; the area component controls local inverse coupling (stiffness) through Eq. (12), while G_{loc} is the corresponding coupling. The response Hessian controls stability and fluctuations; Onsager-type stochastic screen dynamics produces Raychaudhuri-like capacity-flow normal forms; all-null metric integrability and Ward/Bianchi closure generate Einstein-like equilibrium sectors; non-integrable response data appear as residuals.

The most compact statement is

Gravity-like geometry is the integrable hydrodynamic response of finite causal-screen entropy.

 (55)

The reason finite screen entropy appears as the central object is that, in the FDS ontology, finite maintainable distinctions become physically accessible only through boundary ledgers; in gravity those ledgers are causal screens. The paper remains a bridge program. Its strength is not the claim to derive all of gravity from scratch, but the construction of a finite screen-capacity substrate, a local response principle, a normal-form dynamics, an EFT embedding, and a falsification interface that can be compared with existing gravitational, holographic, and cosmological frameworks. In the current medium-prior matched evidence pass reported by G1fit-real, the projection-locked residual model G1DE-M_{3/4} is preferred over free- κ , constant- Σ , free-response G1DE-2, CPL, and Λ CDM controls. This is a preliminary exact-pilot evidence result; top-control wide-prior sensitivity completed; full baseline sensitivity and production refinement remain necessary.

A Notation and response hierarchy

This appendix collects the main symbols. The purpose is to keep the response hierarchy explicit:

$$S_{\text{scr}}[X] \longrightarrow \omega = \eta_i dX^i \longrightarrow \eta_{\text{loc}} = \omega(\delta_A) \longrightarrow G_{\text{loc}} = \frac{c^3}{4\hbar} [\eta_{\text{loc}}]^{-1}.$$

The average density $\bar{\eta}(A)$, the thermodynamic density η_{eff} , and the local response density η_{loc} are related but not interchangeable.

Table 6: Notation table for FDS-G1 Core.

Symbol	Meaning	Role
$S_{\text{scr}}[X]$	screen entropy functional on quotient state space	primitive object
$\mathcal{X}_{\text{raw}}$	raw screen variables including gauge labels	pre-quotient bookkeeping
$\mathcal{X}_{\text{phys}}$	physical quotient state space $\mathcal{X}_{\text{raw}}/\mathcal{G}_{\text{scr}}$	domain of S_{scr}
$A(x)$	local area element or cell-count density	area direction
q_{AB}	screen shape metric	shape/shear sector
χ	occupancy or horizon residual fraction	dark/residual variable
Φ	flux load crossing screen	focusing source proxy
M	memory or hidden response mode	non-Markovian sector
ω	response one-form $\eta_i dX^i$	differential response object
η_{loc}	area component of ω	local inverse-coupling response density
$\eta_{\delta}(A)$	homogeneous differential density	finite-area response
η_{eff}	thermodynamic area-law density	interacting screen limit
G_{loc}	local effective coupling under horizon bridge; $G_{\text{loc}}^{-1} \propto \eta_{\text{loc}}$	inverse response stiffness
$\mathcal{H}_{ij}$	entropy response Hessian	stability/fluctuation metric
θ	logarithmic screen expansion	capacity-flow variable
σ_{AB}	traceless shape flow	shear-like dissipation
Ξ_{res}	residual forcing in capacity flow	nonintegrable/stochastic sector
$\mathcal{R}_{\mu\nu}$	covariant residual representative	metric closure defect
φ	scalar-capacity field $\ln(\eta/\eta_0)$	EFT slow mode

B Detailed claim map: derived, bridged, and modeled

The word “derivation” is local to a specified model. A screen-cell entropy formula may be derived in a toy model, while an Einstein-like equation remains a conditional bridge unless all geometric inputs are derived internally.

Table 7: Detailed claim map.

Statement	Status	Reason
$S_A/k_B = (A/a_0) \ln m$	toy derivation	follows from independent finite cells
$S_A/k_B = \eta_{\text{eff}} A + o(A)$	statistical claim	universality requires screen-local finite-state noncritical assumptions

Statement	Status	Reason
$\eta_{\text{loc}} = k_{\text{B}}^{-1} \delta S / \delta A$	definition	area component of response one-form
$G_{\text{loc}} = (c^3 / 4\hbar) [\eta_{\text{loc}}]^{-1}$	conditional bridge	requires horizon entropy normalization
$\dot{\theta} = -a\theta^2 - b\sigma^2 - c\Phi + \Xi$	normal-form derivation	follows from low-order stochastic transition assumptions
$\Phi \leftrightarrow R_{\mu\nu} k^\mu k^\nu$	geometry bridge	standard local-screen/Raychaudhuri identification
Einstein-like closure	conditional bridge	needs all-null integrability and Ward/Bianchi conservation
$\varphi = \ln(\eta/\eta_0)$ scalar EFT	model	conservative scalar-tensor realization of capacity mode
G1DE occupancy residual	phenomenological model	tests correlated background/growth/lensing deviations
Holographic dictionary	compatibility bridge	interprets $1/(4G_N)$ and central charge as response density

C Expanded entropy-response toy model

The simple two-variable model illustrates how finite-area corrections, occupancy residuals, Hessian stability, and apparent nonintegrability fit into one framework. Let

$$s(A, \chi) = \eta_0 A + \alpha \ln(A/A_0) + \beta A \chi - \frac{\lambda}{2} \chi^2. \quad (56)$$

The area and occupancy responses are

$$\eta_A = \eta_0 + \frac{\alpha}{A} + \beta \chi, \quad \eta_\chi = \beta A - \lambda \chi. \quad (57)$$

The homogeneous local coupling is

$$G_A(A, \chi) = \frac{c^3}{4\hbar} \left[\eta_0 + \frac{\alpha}{A} + \beta \chi \right]^{-1}. \quad (58)$$

Thus finite-area corrections and occupancy residuals enter the same response coefficient.

The Hessian is

$$\mathcal{H} = - \begin{pmatrix} \partial_A^2 s & \partial_A \partial_\chi s \\ \partial_\chi \partial_A s & \partial_\chi^2 s \end{pmatrix} = \begin{pmatrix} \alpha/A^2 & -\beta \\ -\beta & \lambda \end{pmatrix}. \quad (59)$$

Local stability in this sector requires

$$\alpha/A^2 > 0, \quad \frac{\alpha\lambda}{A^2} - \beta^2 > 0. \quad (60)$$

If the area-occupancy coupling is too strong, the response Hessian loses positivity. In a physical model this signals instability, a missing stabilizing variable, or the need for a different coarse-grained description.

Now add a hidden memory variable

$$s(A, \chi, M) = \eta_0 A + \alpha \ln(A/A_0) + \beta A \chi + \gamma A M - \frac{\lambda}{2} \chi^2 - \frac{\mu}{2} M^2. \quad (61)$$

On (A, χ, M) the response one-form is exact. If the observed coarse variables are only (A, χ) but M follows a history-dependent relation $M = M[A, \chi; \text{path}]$, the reduced response is generally not exact. The apparent curl is then a diagnostic of projection, memory, or genuine residual physics. This is the simplest mathematical model for the epistemic/physical residual distinction.

D Three levels of integrability

The term “integrability” appears in three distinct senses.

Level I: entropy-potential integrability. This is the condition

$$\omega = ds_{\text{scr}}, \quad d\omega = 0. \quad (62)$$

It asks whether the chosen screen response variables are sufficient to define an entropy potential. Failure at this level often indicates missing coarse variables or memory.

Level II: metric all-null integrability. This asks whether local null response data can be represented by a symmetric tensor field,

$$\mathcal{F}_p(k) = E_{\mu\nu}(p)k^\mu k^\nu. \quad (63)$$

All-null data determines only the trace-free part, because $g_{\mu\nu}k^\mu k^\nu = 0$. The undetermined trace becomes an integration freedom, represented in Einstein gravity by the cosmological constant sector.

Level III: Ward/Bianchi dynamical integrability. This asks whether the tensor envelope satisfies a covariant conservation identity,

$$\nabla^\mu E_{\mu\nu} = 0. \quad (64)$$

Residual sectors are admissible only if the total effective source is conserved,

$$\nabla^\mu (T_{\mu\nu} + \mathcal{R}_{\mu\nu}) = 0. \quad (65)$$

These three notions should not be conflated. A closed entropy response one-form does not by itself prove metric geometry; metric all-null reconstruction does not by itself prove dynamics; and Bianchi closure is a further gauge/Ward condition.

E Open problems

FDS-G1 Core leaves three major scientific tasks:

1. **Coefficient problem.** Explain, rather than merely bridge to, the Bekenstein-Hawking normalization $1/(4\ell_P^2)$.
2. **Microgeometry problem.** Strengthen the normal-form theorem into a broader derivation of focusing coefficients and geometric identification.
3. **Real-data confrontation.** A matched exact-likelihood and medium-prior nested-evidence comparison (G1fit-real) has been executed; the remaining tasks are full wide-prior baseline sensitivity, production ($d \log Z$) refinement, expanded weak-lensing/ E_G likelihoods, and independent pipeline replication to test the background–Weyl residual pattern at survey quality.

These are not defects in the stated bridge program. They are the natural next steps if the program is to be promoted from finite-capacity bridge to a candidate microscopic account of gravity-like geometry.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* 7, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* 43, 199 (1975).
- [3] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* 98, 1123 (1955).
- [4] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* 75, 1260 (1995).
- [5] D. Lovelock, “The Einstein tensor and its generalizations,” *J. Math. Phys.* 12, 498 (1971).
- [6] S. Weinberg, “Photons and gravitons in S-matrix theory,” *Phys. Rev.* 135, B1049 (1964).
- [7] S. Deser, “Self-interaction and gauge invariance,” *Gen. Rel. Grav.* 1, 9 (1970).
- [8] G. W. Horndeski, “Second-order scalar-tensor field equations in a four-dimensional space,” *Int. J. Theor. Phys.* 10, 363 (1974).
- [9] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* 07, 050 (2014).
- [10] J. M. Maldacena, “The large N limit of superconformal field theories and supergravity,” *Adv. Theor. Math. Phys.* 2, 231 (1998).
- [11] S. Ryu and T. Takayanagi, “Holographic derivation of entanglement entropy from AdS/CFT,” *Phys. Rev. Lett.* 96, 181602 (2006).
- [12] T. Faulkner, A. Lewkowycz, and J. Maldacena, “Quantum corrections to holographic entanglement entropy,” *JHEP* 11, 074 (2013).
- [13] J. M. Ezquiaga and M. Zumalacarregui, “Dark energy after GW170817: dead ends and the road ahead,” *Phys. Rev. Lett.* 119, 251304 (2017).
- [14] E. Verlinde, “On the origin of gravity and the laws of Newton,” *JHEP* 04, 029 (2011).
- [15] T. Padmanabhan, “Thermodynamical aspects of gravity: new insights,” *Rep. Prog. Phys.* 73, 046901 (2010).
- [16] A. Antiba, “Finite-capacity thermodynamics of causal horizons,” *Entropy* 28, 557 (2026).
- [17] R. Bousso, “A covariant entropy conjecture,” *JHEP* 07, 004 (1999) [hep-th/9905177].
- [18] J. D. Brown and M. Henneaux, “Central charges in the canonical realization of asymptotic symmetries: an example from three-dimensional gravity,” *Commun. Math. Phys.* 104, 207 (1986).
- [19] V. Iyer and R. M. Wald, “Some properties of the Noether charge and a proposal for dynamical black hole entropy,” *Phys. Rev. D* 50, 846 (1994).
- [20] T. Jacobson, “Entanglement equilibrium and the Einstein equation,” *Phys. Rev. Lett.* 116, 201101 (2016) [arXiv:1505.04753].
- [21] F. Barzi, H. El Moumni, and K. Masmar, “Modified Unruh thermodynamics in emergent gravity: finite heat capacity and Rényi entropy,” *Phys. Rev. Research* (2026), arXiv:2509.03470.
- [22] P. Nusbaumer, “Relative-entropy variational principle for semiclassical gravity with finite-resolution boundaries,” *Preprints* (2026), doi:10.20944/preprints202601.1205.v4.

- [23] K. Song, “Spacetime dynamics and local entropy balance on causal horizons,” arXiv:2512.17961 (2025).
- [24] A. Ahmad and D. Klinger, “Emergent geometry from quantum probability,” Phys. Rev. D 110, 126004 (2024), arXiv:2411.07288.
- [25] G. Bianconi, “Gravity from entropy,” arXiv:2408.14391 (2024).
- [26] M. Abdul Karim et al. (DESI Collaboration), “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations,” arXiv:2503.14738 (2025).
- [27] A. G. Adame et al. (DESI Collaboration), “DESI 2024 III: Baryon Acoustic Oscillations from Galaxies and Quasars,” JCAP 02, 021 (2025), arXiv:2404.03000.
- [28] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20382013>.
- [29] Y. Wu, “FDS-T1: Finite Distinguishability Budgets and Maintenance Bounds for Physical Observers,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20234249>.
- [30] Y. Wu, “FDS-T2: Effective Geometry as Horizon Boundary Accounting,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20284911>.
- [31] Y. Wu, “FDS-T3: Capacity Overflow, Effective Stochasticity, and Phase-B Invariants,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20250367>.
- [32] Y. Wu, “FDS-X1: Horizon-Maintenance Dark Energy and Horizon Ledger Occupancy (Pre-Euclid Pre-registration),” Zenodo (2026), <https://doi.org/10.5281/zenodo.20290215>.
- [33] Y. Wu, “FDS-X2 v2: Minimal Nonzero CP/T Orientation and Full Weak-Sector Consistency,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20289955>.
- [34] Y. Wu, “FDS-X3: Functional Decomposition of the Four Fundamental Interactions,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20274519>.
- [35] Y. Wu, “D0: Screen Modular Anomaly Inflow and the Einstein Calibration of Finite Access Entropy,” FDS-G1 technical note (2026).

Companion A: Finite Screen Statistical Mechanics

Independent Cells, Interacting Screens, Differential Capacity, and the Statistical Substrate of Screen Stiffness

Yining Wu
Independent Researcher
`yining.wu@alumni.upenn.edu`

May 27, 2026

Abstract

This companion develops the statistical-mechanical substrate used by the FDS-G1 entropy-response geometry program. Its purpose is to show how finite screen-local distinguishability produces area-extensive entropy, how interactions renormalize the capacity density, and why the differential response

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA}$$

rather than the average density alone controls local response. We begin with independent finite cells, for which

$$S_A/k_B = (A/a_0) \ln m.$$

We then formulate a screen area-law universality class: finite-state, screen-local systems with finite-range interactions and finite correlation length have leading entropy proportional to screen area, with subleading boundary, topological, logarithmic, or finite-size corrections. We give fixed-occupancy thermodynamics, an Ising-like interacting screen example, and a finite-size correction model that separates the average density

$$\bar{\eta}(A) = \frac{S(A)/k_B}{A}$$

from the marginal density

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA}.$$

Under the standard horizon-entropy bridge, the latter determines the effective local coupling $G_\delta \propto \eta_\delta^{-1}$, while $G_\delta^{-1} \propto \eta_\delta$ is the inverse coupling or screen stiffness,

$$G_\delta(A) = \frac{c^3}{4\hbar} [\eta_\delta(A)]^{-1}, \quad G_\delta^{-1}(A) = \frac{4\hbar}{c^3} \eta_\delta(A).$$

The results supply the technical statistical foundation for the flagship paper *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*.

1 Statistical Scope and Claim Status

This companion supplies the statistical-mechanical substrate for the FDS-G1 entropy-response framework. Its purpose is to establish when finite screen-local systems support an area-extensive entropy and why the marginal capacity density

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA}$$

is the response variable relevant for local screen stiffness under the standard horizon bridge. The companion derives area-extensive entropy in explicit toy models, formulates a finite-correlation area-law universality class, distinguishes average from differential capacity, and identifies the conditions under which the statistical substrate feeds the full G1 response chain. The finite-capacity language follows the broader FDS formal core [12] and the observer-relative distinguishability budgets developed in FDS-T1 [13], but the present companion specializes those ideas to screen-local statistical mechanics.

The G1 program does not require that every finite-capacity screen system produces gravity-like coupling. It only requires that gravitational screens (horizons, holographic boundaries, causal diamonds) belong to a finite-capacity, area-extensive universality class under the standard horizon/local-screen bridge.

The role of this companion is summarized by the chain

$$\text{finite screen cells} \Rightarrow S_{\text{scr}}[A] \sim k_B \eta A \Rightarrow \eta_\delta(A) = \frac{d(S/k_B)}{dA} \Rightarrow G_\delta(A) = \frac{c^3}{4\hbar} [\eta_\delta(A)]^{-1}, \quad (1)$$

$$G_\delta^{-1}(A) = \frac{4\hbar}{c^3} \eta_\delta(A), \quad (\text{inverse coupling / stiffness}) \quad \text{under the standard horizon bridge.} \quad (2)$$

In the full G1 program, the statistical results here feed into the entropy-response geometry chain

$$S_{\text{scr}}[X] \rightarrow \omega = \eta_i(X) dX^i \rightarrow \eta_{\text{loc}}(x) = \omega(\delta_A(x)) \rightarrow G_{\text{loc}}(x) \rightarrow \theta\text{-flow} \rightarrow \text{metric closure / residuals.} \quad (3)$$

2 Notation and claim status

Table 1: Basic notation.

Symbol	Meaning	Role
A	screen area	extensive screen measure
a_0	effective area per cell	microscopic or coarse-grained scale
N_A	number of screen cells	$N_A \simeq A/a_0$
m	states per independent cell	finite distinguishability
S_A	screen entropy	$S_A = k_B \ln \Omega_A$
η_{nat}	nats per area	$S/k_B = \eta A$
η_{bit}	bits per area	$\eta_{\text{nat}} = \eta_{\text{bit}} \ln 2$
$\bar{\eta}(A)$	average density	$(S/k_B)/A$
$\eta_\delta(A)$	differential density	$d(S/k_B)/dA$
$G_\delta(A)$	response coupling; $G_\delta^{-1} \propto \eta_\delta$	$c^3/(4\hbar)[\eta_\delta]^{-1}$ under bridge

3 Finite screen state space

Definition 1 (finite screen). *A finite distinction screen is an effective codimension-two screen patch Σ_A with area A and a finite set of screen-local degrees of freedom. In the simplest discretized model,*

$$N_A = \left\lfloor \frac{A}{a_0} \right\rfloor \quad (4)$$

counts the number of effective screen cells.

Table 2: Claim status in this companion.

Claim	Status	Section
Independent cells give $S \propto A$	toy theorem	4
Fixed occupancy gives $dE = TdS$	thermodynamic identity in toy model	5
Finite-range interacting screens have leading area-extensive entropy	statistical bridge / universality claim	6
η_δ controls local response	G1 response principle	7
$G_\delta = c^3/(4\hbar)[\eta_\delta]^{-1}$ and $G_\delta^{-1} \propto \eta_\delta$	conditional bridge using standard horizon normalization	8
Log/finite-size corrections generate $G_\delta(A)$ corrections	effective model	9

Definition 2 (screen-locality). *Screen-locality means that the active degrees of freedom counted in S_{scr} live on the screen or on an effective boundary/code subspace associated with the screen, not in an uncontrolled bulk volume. Screen-locality is the assumption that makes an area law possible.*

The cell model should be read as an effective statistical representation. It does not yet identify the cells as fundamental atoms of spacetime. They may represent horizon microstates, entanglement-cut channels, boundary-code subspace units, or coarse-grained causal-screen distinguishability cells.

Remark 1 (Relation to the Screen Ledger Representation Theorem). *The state-count formalism ties directly to the Core’s Screen Ledger Representation Theorem (Core, Theorem 1):*

$$N_A = |\mathcal{C}_{\Sigma_A}|, \quad S_A = k_B \log N_A.$$

The independent-cell model gives $N_A = m^{A/a_0}$ as a simple realization of the finite-state-count ledger. The logarithmic form is not an empirical fit—it is the unique additive coordinate of finite screen capacity under the theorem’s additivity, multiplicativity, and coarse-graining monotonicity assumptions. The independent-cell model is a toy derivation; the representation theorem explains why the logarithm is the correct ledger coordinate.

[Area-Law Universality Class] For screen-local finite-state systems with finite interaction range, finite correlation length, no dominant hidden bulk-volume degrees of freedom, and stable screen-local coarse-graining,

$$S_A/k_B = \eta_{\text{eff}} A + S_{\text{sub}}(A), \quad S_{\text{sub}}(A)/A \rightarrow 0.$$

Failure cases. The area law is demoted under: critical long-range correlations; infinite local Hilbert spaces; hidden bulk volume modes; dominant nonlocal constraints; failure of screen-local coarse-graining. These are not small corrections—they are demotion cases for the entire statistical substrate.

4 Independent finite cells

The minimal model has N_A independent cells, each with m distinguishable states. The number of microstates is

$$\Omega_A = m^{N_A}. \tag{5}$$

For $A \gg a_0$, ignoring the floor correction,

$$N_A \simeq \frac{A}{a_0}. \quad (6)$$

Proposition 1 (independent cells give an area law). *For independent m -state screen cells,*

$$\frac{S_A}{k_B} = \ln \Omega_A = \frac{A}{a_0} \ln m. \quad (7)$$

Thus

$$\eta_{\text{nat}} = \frac{\ln m}{a_0}, \quad \eta_{\text{bit}} = \frac{\log_2 m}{a_0}. \quad (8)$$

Proof. Since $\Omega_A = m^{A/a_0}$, one has

$$S_A = k_B \ln \Omega_A = k_B \frac{A}{a_0} \ln m. \quad (9)$$

□

Figure 1 shows the linear dependence of S_A/k_B on A/a_0 for several m .

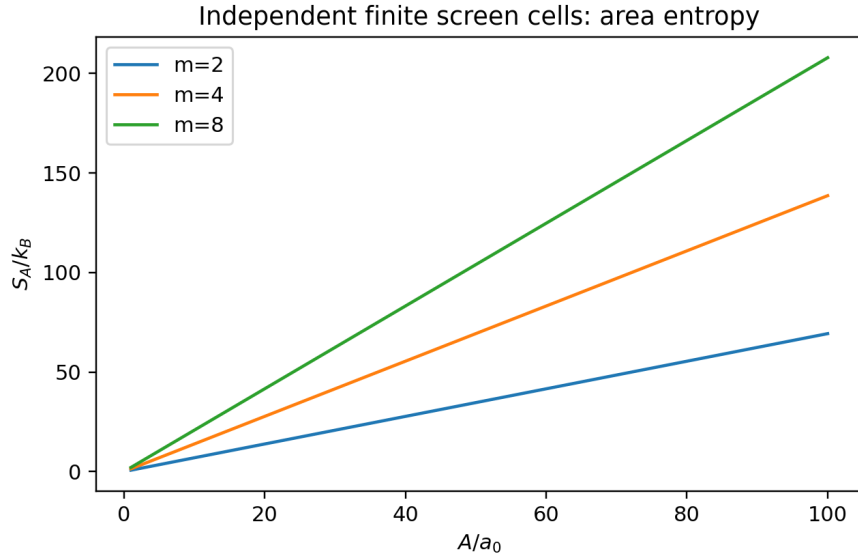


Figure 1: Independent finite screen cells produce exactly area-extensive entropy.

4.1 Matching the standard horizon coefficient

The standard Bekenstein-Hawking entropy is

$$\frac{S_H}{k_B} = \frac{A}{4\ell_P^2} = \frac{c^3 A}{4G\hbar}. \quad (10)$$

If the independent-cell density is matched to this coefficient, then

$$\frac{\ln m}{a_0} = \frac{1}{4\ell_P^2}, \quad (11)$$

and hence

$$a_0 = 4\ell_P^2 \ln m. \quad (12)$$

For binary cells, $m = 2$, the effective area per bit is

$$a_{\text{bit}} = 4\ell_P^2 \ln 2. \quad (13)$$

This is a matching relation, not a first-principles derivation of the Bekenstein-Hawking coefficient.

5 Fixed-occupancy screen thermodynamics

The independent m -state model explains area-extensive entropy but does not yet include energetic occupancy. A minimal thermodynamic screen has binary cells

$$\sigma_i \in \{0, 1\}, \quad (14)$$

with n active cells out of N . The number of configurations is

$$\Omega(N, n) = \binom{N}{n}. \quad (15)$$

Let

$$p = \frac{n}{N}. \quad (16)$$

Using Stirling's approximation,

$$\frac{S}{k_B N} \circ H_2(p) = -p \ln p - (1-p) \ln(1-p). \quad (17)$$

If each active distinction carries energy ϵ , then

$$E = \epsilon n. \quad (18)$$

The microcanonical temperature is defined by

$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{1}{\epsilon} \frac{\partial S}{\partial n}. \quad (19)$$

For $0 < p < 1/2$, this yields the positive-temperature branch

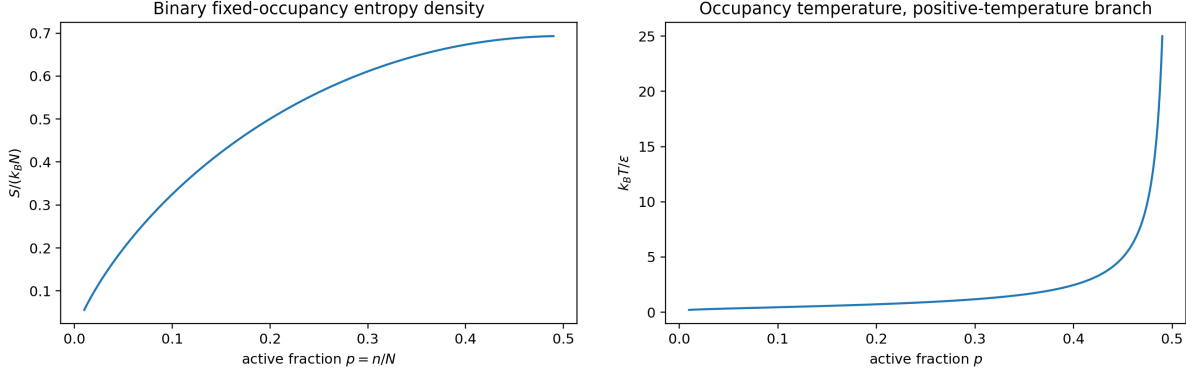
$$T(p) = \frac{\epsilon}{k_B \ln[(1-p)/p]}. \quad (20)$$

Proposition 2 (screen first-law identity). *For the fixed-occupancy model with $E = \epsilon n$ and $S = k_B \ln \Omega(N, n)$, the thermodynamic definition of T implies*

$$dE = T dS \quad (21)$$

for quasi-static occupancy updates.

Figures 2 and 3 show the entropy density, positive-temperature branch, and finite- N convergence.



(a) Entropy density.

(b) Positive-temperature branch.

Figure 2: Binary fixed-occupancy thermodynamics.

6 Interacting screen cells and area-law universality

The independent model is intentionally simple. A more robust framework must allow local interactions and coarse-graining. Let a screen have cell variables

$$\sigma_i \in \mathcal{S}, \quad |\mathcal{S}| = m, \quad (22)$$

with local Hamiltonian

$$H_{\text{scr}}[\sigma] = \sum_i h_i(\sigma_i) + \sum_{\langle ij \rangle} J_{ij}(\sigma_i, \sigma_j) + \dots, \quad (23)$$

where interactions are finite range on the screen. The partition function is

$$Z_A(\beta) = \sum_{\{\sigma_i\}} e^{-\beta H_{\text{scr}}[\sigma]}. \quad (24)$$

The entropy is

$$S_A = k_B (\ln Z_A + \beta \langle H_{\text{scr}} \rangle). \quad (25)$$

Definition 3 (screen area-law universality class). *A finite screen model belongs to the G1 screen area-law universality class if:*

1. *degrees of freedom are screen-local;*
2. *each local cell has finite state space;*
3. *interactions are finite range or sufficiently fast decaying on the screen;*
4. *the state is non-critical, with finite correlation length;*
5. *no hidden bulk-volume degrees of freedom dominate the entropy;*
6. *coarse-graining preserves screen-locality.*

Claim 1 (Area extensivity under finite correlation length). *For a screen-local finite-state system in the above universality class,*

$$\frac{S_A}{k_B} = \eta_{\text{eff}} A + S_{\text{sub}}(A), \quad (26)$$

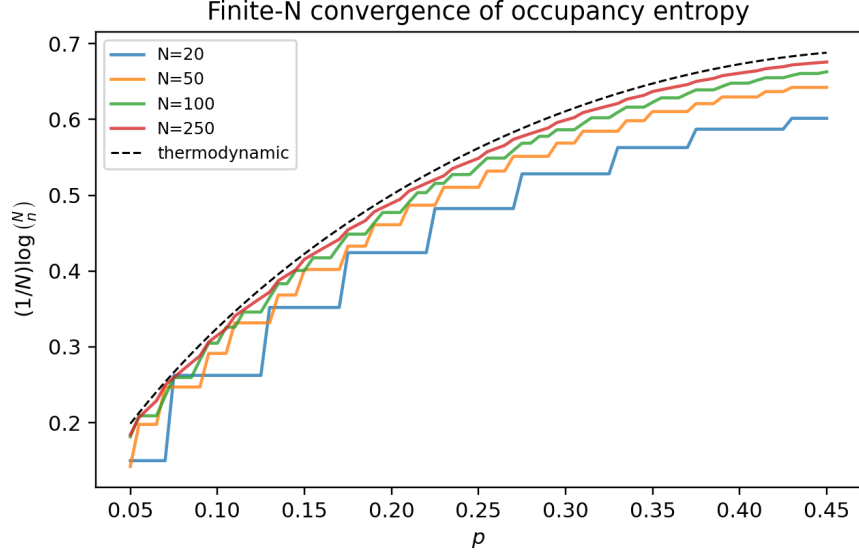


Figure 3: Finite- N convergence of $(1/N) \log \binom{N}{n}$ to the binary entropy function.

with

$$\eta_{\text{eff}} = \lim_{A \rightarrow \infty} \frac{S_A/k_B}{A} \quad (27)$$

finite, and $S_{\text{sub}}(A) = o(A)$ for non-critical short-range phases. This is a statistical bridge statement, not a proven theorem from first principles. It relies on the standard large-volume thermodynamic-limit existence conditions for finite-state, finite-range systems, adapted to the screen-local setting.

Motivation. Finite state space bounds the entropy per cell. Screen-local finite-range interactions imply that a thermodynamic entropy density exists in the large-screen limit under standard subadditivity and finite-correlation assumptions. Since the number of screen cells scales as $N_A \sim A/a_0$, the leading entropy is extensive in A , not in a bulk volume. Correlations and constraints renormalize the entropy density and generate subleading terms. $\square$

6.1 Boundary, critical, and topological corrections

The leading area term may be corrected by

$$\frac{S_A}{k_B} = \eta_{\text{eff}} A + \alpha \ln(A/a_0) + \gamma + C_{\partial} \sqrt{A/a_0} + O(A^{-p}). \quad (28)$$

The interpretation of these terms depends on the model:

- $C_{\partial} \sqrt{A/a_0}$ can arise from a boundary of the screen patch.
- $\alpha \ln A$ can arise near criticality, from constraints, or from finite-size corrections.
- γ may encode topological or global constraint information.

6.2 Ising-like screen example

An illustrative interacting model has

$$\sigma_i = \pm 1, \quad H = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j. \quad (29)$$

Exact enumeration on small periodic $L \times L$ screens shows a finite entropy density $S/(k_B N)$, with finite-size heat-capacity structure. This example is not proposed as a horizon microstate model; it illustrates how interactions renormalize the capacity density without destroying area extensivity.

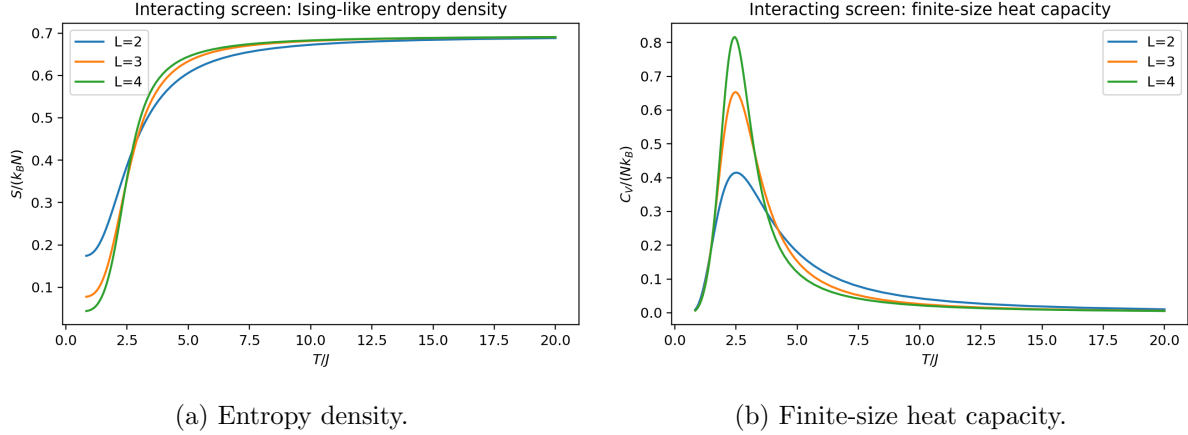


Figure 4: Exact enumeration of small Ising-like interacting screens.

7 Average and differential capacity

The G1 core theory distinguishes two capacities.

Definition 4 (average capacity density). *The average density is*

$$\bar{\eta}(A) = \frac{S(A)/k_B}{A}. \quad (30)$$

It measures the total stored screen capacity per unit area.

Definition 5 (differential capacity density). *The differential density is*

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA}. \quad (31)$$

It measures the marginal entropy-area response of a quasi-static screen update.

Principle 1 (Which capacity feeds local stiffness). *Only the differential capacity $\eta_\delta(A)$, not the average capacity $\bar{\eta}(A)$, enters local screen stiffness:*

$$G_\delta(A) = \frac{c^3}{4\hbar} [\eta_\delta(A)]^{-1}.$$

The average capacity controls the total entropy budget; the differential capacity controls the local gravitational response. This distinction is essential for understanding why finite-area corrections appear in total entropy but are suppressed in local coupling (Core, §6).

For a strictly area-extensive entropy

$$S(A)/k_B = \eta_0 A, \quad (32)$$

one has

$$\bar{\eta}(A) = \eta_\delta(A) = \eta_0. \quad (33)$$

With subleading terms, the two differ.

Principle 2 (local response principle). *In a quasi-static local screen update, the relevant inverse coupling (stiffness) variable is the differential response,*

$$\delta S = k_B \eta_\delta(A) \delta A, \quad (34)$$

not only the average stored capacity.

This is the homogeneous version of the local response formula used in the G1 core paper,

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}. \quad (35)$$

For a uniform screen, $\eta_{\text{loc}} \rightarrow \eta_\delta(A)$.

8 Effective coupling under the horizon bridge

Under the standard horizon-entropy bridge,

$$\frac{S_H}{k_B} = \frac{c^3 A}{4G\hbar}, \quad (36)$$

the area response can be read as

$$\eta_{\text{BH}} = \frac{c^3}{4G\hbar}. \quad (37)$$

Therefore, for a screen with marginal capacity density $\eta_\delta(A)$, the corresponding response coupling and its inverse (stiffness) are

$$G_\delta(A) = \frac{c^3}{4\hbar} [\eta_\delta(A)]^{-1}, \quad G_\delta^{-1}(A) = \frac{4\hbar}{c^3} \eta_\delta(A). \quad (38)$$

This is a conditional bridge, not a first-principles derivation of G . It states that if the standard horizon/local-screen normalization is used, then the inverse gravitational stiffness (G^{-1}) is read as marginal screen distinguishability density, and the coupling G is inversely proportional to it.

9 Finite-size corrections and running response

Consider the effective entropy model

$$\frac{S(A)}{k_B} = \eta_0 A + \alpha \ln(A/a_0) + \gamma + b(A/a_0)^{-p}. \quad (39)$$

Then

$$\bar{\eta}(A) = \eta_0 + \frac{\alpha \ln(A/a_0) + \gamma + b(A/a_0)^{-p}}{A}, \quad (40)$$

whereas

$$\eta_\delta(A) = \eta_0 + \frac{\alpha}{A} - \frac{bp}{a_0}(A/a_0)^{-p-1}. \quad (41)$$

In units $a_0 = 1$, this becomes

$$\eta_\delta(A) = \eta_0 + \frac{\alpha}{A} - bpA^{-p-1}. \quad (42)$$

The response coupling is

$$\frac{G_\delta(A)}{G_0} = \frac{\eta_0}{\eta_\delta(A)}. \quad (43)$$

For large A ,

$$\frac{\Delta G}{G_0} \simeq -\frac{\alpha}{\eta_0 A} + O(A^{-p-1}, A^{-2}). \quad (44)$$

Thus macroscopic finite-area corrections are suppressed unless the screen is near cutoff scale, early-universe scale, or coupled to a genuine residual/capacity field.

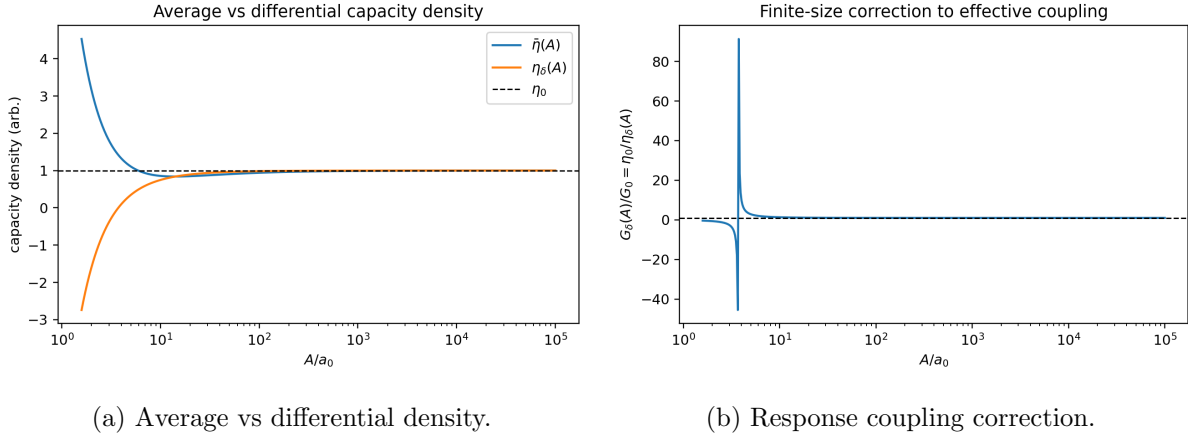


Figure 5: Finite-size corrections separate stored capacity from marginal response.

10 Failure modes

The screen statistical foundation can fail in several controlled ways.

This failure registry is important: the G1 program does not require every screen-like system to be gravitational. It only requires that gravitational screens belong to a finite-capacity, area-extensive universality class under the standard horizon/local-screen bridge.

11 Interfaces to other G1 companions

The results of this paper feed into the other G1 companion papers as follows.

Remark 2 (Scope boundary). *Companion A supplies the finite-screen statistical substrate: S_{scr} , η_{eff} , and η_δ . By itself it does not imply null focusing, metric closure, Einstein-like equations, or cosmological residuals. Those bridges are treated in Companions B–F and in FDS-G1 Core. Companion A supplies statistical routes to an area-extensive finite screen density. D0 addresses the distinct calibration question: how such a density η_0 is related to boost modular energy and hence to gravitational stiffness G^{-1} . This separation prevents any misinterpretation that finite screen entropy*

Table 3: Failure registry for finite screen statistics.

Failure condition	Consequence	What survives
Dominant bulk-volume degrees of freedom	leading entropy becomes volume-like	finite cell models may remain useful elsewhere
Infinite local state space without cutoff	η may diverge	need UV completion or constraint
Long-range critical correlations	log or anomalous scaling can dominate subleading terms	response framework survives with corrections
Strong nonlocal global constraints	simple density may fail	residual / topology sector needed
No standard horizon bridge	$G^{-1} \sim \eta$ interpretation fails	screen statistics remains independent result
Macroscopic G variation without closure	violates Bianchi/local gravity constraints	scalar-capacity or residual sector needed

Table 4: Interfaces with the rest of the G1 program.

Companion	Input from screen statistical mechanics
FDS-G1 Core / Finite Screen Spacetime	$S_{\text{scr}}[X], \eta_{\text{loc}} = (1/k_B)\delta S/\delta A$
Microtransition and Capacity Flow	$N = A/a_0, \theta = d \ln N/d\lambda$
Scalar-Capacity EFT	long-wavelength mode $\varphi = \ln(\eta/\eta_0)$
Likelihood and Falsification	finite-size $G_\delta(A)$ corrections and no-free- G drift
Horizon Occupancy Residuals	capacity fraction and occupancy variables χ_H
Holography and Horizons	$1/(4G_N)$ as screen capacity density

alone derives gravity-like geometry. The area/differential capacity substrate provided here is extended to optical deformation sectors $A(1) \oplus S(2) \oplus \Omega(1)$ in companion notes on optical projection and screen influence, where the relative-entropy Hessian $K_{IJ} = \partial_I \partial_J S_{\text{rel}}(\rho_S(q) \parallel \rho_S(0))|_{q=0}$ controls the compliance projection coefficient κ_{BW} . The matched exact-pilot and medium-prior nested-evidence comparison (G1fit-real) now selects the projection-locked G1DE- $M_{3/4}$ model; Companion A does not predict these signs from the statistical-mechanical substrate alone.

12 Conclusion

This companion establishes the statistical-mechanical part of the G1 program. The central results are:

1. independent finite screen cells give $S_A/k_B = (A/a_0) \ln m$;
2. fixed occupancy gives an explicit screen thermodynamics with $dE = TdS$;
3. finite-state, finite-range, finite-correlation interacting screens have leading entropy extensive in screen area;
4. interactions and constraints renormalize the effective density η_{eff} ;

5. finite-size terms separate average density $\bar{\eta}$ from differential response η_δ ;
6. under the standard horizon bridge, $G_\delta(A) = c^3/(4\hbar)[\eta_\delta(A)]^{-1}$ and $G_\delta^{-1}(A) = 4\hbar\eta_\delta(A)/c^3$, so η_δ controls the inverse coupling (stiffness).

The paper therefore supplies the statistical substrate for the G1 core claim:

$$\text{gravity-like inverse coupling (stiffness) is marginal screen entropy-area response.} \quad (45)$$

A Numerical artifacts

The accompanying script `fds_g1A_screen_statistics.py` generates all figures and CSV files used in this paper:

- independent area entropy;
- binary occupancy entropy and temperature;
- finite- N occupancy convergence;
- exact enumeration of small Ising-like screens;
- finite-size differential-capacity corrections.

B Relation to standard references

This paper uses standard statistical mechanics and horizon-entropy conventions. The following references are relevant background, not claims of novelty.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Physical Review D* 7, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Communications in Mathematical Physics* 43, 199 (1975).
- [3] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Physical Review Letters* 75, 1260 (1995).
- [4] W. Rindler, “Kruskal space and the uniformly accelerated frame,” *American Journal of Physics* 34, 1174 (1966).
- [5] L. Onsager, “Reciprocal relations in irreversible processes,” *Physical Review* 37, 405 (1931).
- [6] D. Ruelle, *Statistical Mechanics: Rigorous Results*, W. A. Benjamin (1969).
- [7] R. Bousso, “A covariant entropy conjecture,” *Journal of High Energy Physics* 07, 004 (1999).
- [8] S. Ryu and T. Takayanagi, “Holographic derivation of entanglement entropy from AdS/CFT,” *Physical Review Letters* 96, 181602 (2006).
- [9] T. Faulkner, A. Lewkowycz, and J. Maldacena, “Quantum corrections to holographic entanglement entropy,” *Journal of High Energy Physics* 11, 074 (2013).

- [10] V. Iyer and R. M. Wald, “Some properties of the Noether charge and a proposal for dynamical black hole entropy,” *Physical Review D* 50, 846 (1994).
- [11] A. Antiba, “Finite-capacity thermodynamics of causal horizons,” *Entropy* 28, 557 (2026).
- [12] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20382013>.
- [13] Y. Wu, “FDS-T1: Finite Distinguishability Budgets and Maintenance Bounds for Physical Observers,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20234249>.

Companion B: Screen Transition and Response Dynamics

Stochastic Screen Transitions, Entropy-Gradient Response, and Raychaudhuri-like Capacity Flow

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

May 27, 2026

Abstract

This companion paper develops the dynamical bridge used by the FDS-G1 Core framework. Companion A establishes that finite screen-local statistical systems can produce area-extensive entropy and a marginal capacity density. Here we ask how a finite screen state updates along a null or causal-screen parameter and why its hydrodynamic continuum limit naturally contains Raychaudhuri-like focusing terms. The primitive dynamical variable is the screen cell count $N(\lambda) = A(\lambda)/a_0$ and the logarithmic expansion $\theta = d \ln N / d\lambda$. We model screen evolution by a Markov or near-Markov transition process, derive a Kramers-Moyal/Fokker-Planck continuum limit, and show that, under locality, screen extensivity, affine parametrization, rotational invariance of shape modes, and positive flux focusing, the lowest-order deterministic normal form is

$$\frac{d\theta}{d\lambda} = -a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_{\theta}.$$

The term $-\theta^2$ follows kinematically from logarithmic capacity expansion; the $-\sigma^2$ term is the lowest positive-definite shear dissipation scalar; and $-\Phi_{\text{flux}}$ represents positive flux load reducing expansion. This is a normal-form bridge: it shows why Raychaudhuri-like capacity-flow equations are stable hydrodynamic forms for stochastic finite-screen update dynamics, without deriving the geometric Raychaudhuri equation or Einstein gravity from microscopic cells. The standard geometric identification of Φ_{flux} with null focusing data remains an external local-screen bridge.

1 Dynamical Scope and Normal-Form Claim

This companion hardens the transition from finite screen update dynamics to a Raychaudhuri-like capacity-flow normal form. For a broad class of local finite-screen stochastic transition processes, the hydrodynamic evolution of logarithmic screen expansion has a lowest-order normal form with negative quadratic expansion, shear dissipation, positive flux focusing, and residual/noise terms. The finite-capacity formalism and the treatment of effective stochasticity under capacity overflow follow the FDS formal core [7] and FDS-T3 [8].

The relation to standard gravity enters later through a bridge:

$$\Phi_{\text{flux}} \longleftrightarrow R_{\mu\nu} k^{\mu} k^{\nu} \longleftrightarrow \kappa T_{\mu\nu} k^{\mu} k^{\nu},$$

which is established separately through the standard local-screen geometry bridge. This paper explains why the left-hand side has a stable capacity-flow structure; the geometric identification belongs to local screen thermodynamics and the broader FDS-G1 Core response-geometry framework.

1.1 Position within the G1 architecture

[What this companion does and does not derive] Companion B does not derive the geometric Raychaudhuri equation from microscopic cells. It derives the stable capacity-flow normal form from finite screen transition statistics. The geometric Raychaudhuri equation is recovered only after the Core’s metricization and local-screen bridge assumptions are imposed. The constants a, b, c are normal-form parameters; they become $1/(D - 2), 1, \kappa$ only in the metric-integrable equilibrium limit.

The dependency chain is

$$\text{finite screen statistics} \longrightarrow S_{\text{scr}}[X] \longrightarrow \eta_{\text{loc}} \longrightarrow G_{\text{loc}} \longrightarrow \theta\text{-flow} \longrightarrow \text{metric/Ward closure.}$$

Companion A supplies the statistical substrate. This companion supplies the transition-to-flow bridge. Companion C supplies scalar-capacity effective field theory. Companion D supplies likelihood/falsification interfaces. Companion E treats horizon-occupancy residuals. Companion F treats horizon and holographic dictionaries.

Table 1: Claim status in this companion.

Claim	Meaning	Status
$N = A/a_0$ and $\theta = d \ln N / d\lambda$	Cell-count definition of screen expansion	Definition
Kramers-Moyal limit	Markov transition process yields drift/noise continuum variables	Standard stochastic bridge
$-\theta^2$ term	Kinematic term from logarithmic expansion	Toy theorem
$-\sigma^2$ term	Lowest positive-definite shear dissipation scalar	Normal-form argument
$-\Phi_{\text{flux}}$ term	Positive load focuses screen expansion	Physical sign assumption
Identification $R_{\mu\nu}k^\mu k^\nu$	with Geometry bridge	Standard bridge / not derived
Residuals	Nonclosure, memory, noise, coarse-graining defects	Model-building sector

2 Finite screen state and expansion

Let a local causal screen patch have area $A(\lambda)$ and effective cell area a_0 . Define

$$N(\lambda) = \frac{A(\lambda)}{a_0}. \quad (1)$$

The logarithmic screen expansion is

$$\theta(\lambda) = \frac{d}{d\lambda} \ln N(\lambda) = \frac{1}{A} \frac{dA}{d\lambda}. \quad (2)$$

This matches the usual expansion variable when the screen patch is identified with a cross-sectional area of a null congruence. In the present paper, however, θ is defined operationally as the relative rate of change of maintained screen cells.

A minimal macroscopic screen state is

$$X(\lambda) = (x, \theta, \sigma_{\text{scr}}, \chi, \Phi, M, \dots), \quad x = \ln N. \quad (3)$$

Here σ_{scr} denotes a coarse shear/shape-distortion amplitude, χ an occupancy variable, Φ a flux/load variable, and M possible memory or hidden screen modes.

2.1 Gauge and parametrization cautions

The parameter λ may represent an affine null parameter in a geometric bridge, but at the screen-transition level it is simply an update parameter. Reparametrization changes the numerical value of θ unless an affine or local-equilibrium gauge is fixed. Therefore the normal form below is stated in an affine/local-equilibrium update gauge; non-affine parametrization introduces a linear term $\kappa_\lambda \theta$.

3 Stochastic transition process

Assume a Markov or near-Markov transition kernel

$$P(X_{\lambda+\Delta\lambda} | X_\lambda). \quad (4)$$

For the pair (x, θ) a minimal update is

$$\Delta x = \theta \Delta\lambda + O(\Delta\lambda^{3/2}), \quad (5)$$

$$\Delta\theta = A_\theta(X) \Delta\lambda + \sqrt{B_\theta(X) \Delta\lambda} \xi, \quad (6)$$

where ξ is a unit-variance random variable. The Kramers-Moyal expansion gives the Fokker-Planck equation

$$\partial_\lambda P = -\partial_i [A^i(X) P] + \frac{1}{2} \partial_i \partial_j [B^{ij}(X) P] + \dots \quad (7)$$

Equivalently, the Langevin form is

$$\dot{X}^i = A^i(X) + \zeta^i, \quad \langle \zeta^i(\lambda) \zeta^j(\lambda') \rangle = B^{ij}(X) \delta(\lambda - \lambda'). \quad (8)$$

For large screen cell number $N \gg 1$, path probabilities often admit a large-deviation form

$$P[X(\lambda)] \asymp \exp[-N \mathcal{I}_{\text{scr}}[X]], \quad (9)$$

where $\mathcal{I}_{\text{scr}}$ is the screen update action. Near local equilibrium one expects an Onsager-Machlup form

$$\mathcal{I}_{\text{scr}}[X] = \frac{1}{2} \int d\lambda (\dot{X}^i - F^i) D_{ij}^{-1} (\dot{X}^j - F^j). \quad (10)$$

If the drift is entropy-gradient driven,

$$F^i = L^{ij} \frac{\delta S_{\text{scr}}}{\delta X^j}, \quad (11)$$

then stochastic screen dynamics obeys a fluctuation-dissipation pattern, schematically $D^{ij} \propto k_B L^{ij}$ near equilibrium. This relation will later constrain residual noise.

4 Normal-form derivation of capacity flow

4.1 Defence: why this is not a copy of Raychaudhuri

The normal form

$$\dot{\theta} = -a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_{\theta}$$

is not obtained by copying the geometric Raychaudhuri equation. It is obtained by asking what the lowest-order scalar drift of logarithmic screen-cell expansion can contain under the following assumptions:

- finite screen updates;
- Markov or near-Markov transition kernel;
- screen-extensive hydrodynamic limit ($N \gg 1$);
- affine or local-equilibrium update gauge;
- screen rotational invariance;
- positive flux focusing;
- finite- N noise scaling.

Each term has a distinct operational origin. The $-\theta^2$ term is kinematic (logarithmic expansion derivative, §4.2). The $-\sigma_{\text{scr}}^2$ term is the lowest positive-definite scalar dissipation term compatible with cross-section rotations. The $-\Phi_{\text{flux}}$ term is a sign assumption about positive load. The geometric Raychaudhuri equation is recovered only after an additional bridge that identifies Φ_{flux} with null focusing and fixes conventional coefficients.

Theorem 1 (Screen transition normal form) *Let a finite causal screen have cell count $N(\lambda) = A(\lambda)/a_0$ and logarithmic expansion $\theta = d \ln N / d\lambda$. Assume the screen update is governed by a Markov or near-Markov transition with finite cell noise, an affine or local-equilibrium gauge, screen rotational invariance, and positive flux focusing. Then, in the hydrodynamic limit $N \gg 1$, the lowest-order deterministic drift of θ is*

$$\boxed{\dot{\theta}_{\text{det}} = -a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + O(3)}, \quad a, b, c > 0. \quad (12)$$

The full Langevin equation including finite-cell stochastic noise is

$$\dot{\theta} = \dot{\theta}_{\text{det}} + \zeta_{\theta}, \quad \zeta_{\theta} = O(N^{-1/2}). \quad (13)$$

Here a is a positive coefficient that depends on dimension, expansion convention, and the affine bridge. It is not fixed by screen kinematics alone; calibration requires matching the transverse dimension $n = D - 2$ and the geometric convention for θ . The normal-form theorem fixes the allowed structure and signs, not the exact Raychaudhuri coefficients.

4.2 The universal kinematic term

The first robust term follows from logarithmic expansion. Since $\theta = \dot{x}$ and $x = \ln N$,

$$\dot{\theta} = \ddot{x} = \frac{\ddot{N}}{N} - \left(\frac{\dot{N}}{N} \right)^2. \quad (14)$$

If the cell-count acceleration has a regular local expansion

$$\frac{\ddot{N}}{N} = F_0(X), \quad (15)$$

then

$$\dot{\theta} = F_0(X) - \theta^2. \quad (16)$$

Thus a negative quadratic expansion term is forced by the use of logarithmic area/cell-count expansion. It is not inserted to imitate Raychaudhuri; it is a kinematic term.

In a more general convention, the coefficient is written $a > 0$:

$$\dot{\theta} = -a\theta^2 + \dots. \quad (17)$$

The precise coefficient depends on dimension, normalization of θ , and bridge conventions. In the null Raychaudhuri equation for a two-dimensional transverse screen ($n = D - 2 = 2$) in four spacetime dimensions the corresponding coefficient is $1/2$. The theorem leaves a as a free positive parameter because $-\theta^2$ is forced by kinematics ($\ddot{x} = \ddot{N}/N - \dot{x}^2$) but its exact coefficient relative to the geometric Raychaudhuri convention is a bridge calibration, not a screen-internal result.

4.3 Shape modes and shear dissipation

Let q_{AB} denote a coarse metric or shape tensor on the screen cross-section of dimension $n = D - 2$. Decompose its update into trace and traceless pieces:

$$\theta = \frac{1}{2} q^{AB} \dot{q}_{AB}, \quad \sigma_{AB} = \frac{1}{2} \dot{q}_{AB} - \frac{1}{n} \theta q_{AB}, \quad q^{AB} \sigma_{AB} = 0. \quad (18)$$

This companion treats the twist-free screen-flow sector; nonzero twist is an extension or failure channel. At lowest order, locality and rotational invariance allow the positive scalar

$$\sigma_{\text{scr}}^2 = \sigma_{AB} \sigma^{AB}. \quad (19)$$

If shear converts coherent expansion into irreversible shape distortion or unresolved mixing, it enters with negative sign:

$$\dot{\theta} \supset -b\sigma_{\text{scr}}^2, \quad b > 0. \quad (20)$$

This is a normal-form statement: it is the lowest positive-definite scalar dissipation term compatible with screen rotations.

4.4 Flux focusing

Let Φ_{flux} be the positive scalar load carried by energy, information, or distinguishability demand crossing the screen along the chosen generator. Its role is defined at three layers:

Screen-update level: Φ_{flux} is the rate of external load crossing or consuming screen distinguishability capacity.

Thermodynamic bridge: $\Phi_{\text{flux}} \sim \frac{\delta Q}{T \delta \lambda \delta A}$, relating flux to heat exchange per screen area per update step.

Geometric bridge: $\Phi_{\text{flux}} \leftrightarrow R_{\mu\nu} k^\mu k^\nu \leftrightarrow \kappa T_{\mu\nu} k^\mu k^\nu$, the standard local-screen identification with null focusing.

The dimensions are $[\Phi_{\text{flux}}] = [\lambda^{-2}]$. The focusing sign assumption is

$$\Phi_{\text{flux}} \geq 0 \implies \dot{\theta} \leq 0. \quad (21)$$

Thus

$$\dot{\theta} \supset -c\Phi_{\text{flux}}, \quad c > 0. \quad (22)$$

Under the standard local-screen bridge one may identify Φ_{flux} with either $T_{\mu\nu} k^\mu k^\nu$ or with the geometric focusing scalar $R_{\mu\nu} k^\mu k^\nu$ after using screen first-law relations. This identification is not part of the stochastic derivation; it is a physical bridge.

4.5 Residuals and noise

Unresolved memory, nonlocal constraints, stochastic finite-cell fluctuations, and coarse-graining defects enter as

$$\Xi_{\text{res}} + \zeta_\theta. \quad (23)$$

If the process is near equilibrium and Markovian, the noise covariance is local. For finite cell count $N \gg 1$, the noise amplitude scales as

$$\langle \zeta_\theta(\lambda) \zeta_\theta(\lambda') \rangle = \frac{2D_\theta}{N} \delta(\lambda - \lambda'), \quad \zeta_\theta = O(N^{-1/2}), \quad (24)$$

so that deterministic macroscopic capacity flow is recovered as $N \rightarrow \infty$ and finite-cell stochasticity appears as a controlled $O(N^{-1/2})$ residual. If memory is present, a nonlocal kernel may appear:

$$\Xi_{\text{mem}}(\lambda) = \int^\lambda K(\lambda, \lambda') \Phi(\lambda') d\lambda'. \quad (25)$$

This case generally cannot be represented by a local screen potential without enlarging the state space.

4.6 Capacity-flow normal form

Collecting the allowed lowest-order terms gives

$$\boxed{\frac{d\theta}{d\lambda} = \kappa_\lambda \theta - a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_\theta + O(\theta^3, \theta\sigma^2, \nabla\Phi).} \quad (26)$$

Here $\kappa_\lambda \theta$ is a non-affine/reparametrization term. In the affine local-equilibrium gauge,

$$\kappa_\lambda = 0, \quad (27)$$

so the normal form becomes

$$\boxed{\dot{\theta} = -a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_\theta.} \quad (28)$$

5 Relation to the geometric Raychaudhuri equation

For a hypersurface-orthogonal null congruence, the geometric Raychaudhuri equation is

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - R_{\mu\nu}k^\mu k^\nu, \quad (29)$$

up to twist terms and convention-dependent factors. The capacity-flow normal form matches its structure under the bridge

$$a \rightarrow \frac{1}{2}, \quad b\sigma_{\text{scr}}^2 \rightarrow \sigma_{\mu\nu}\sigma^{\mu\nu}, \quad c\Phi_{\text{flux}} \rightarrow R_{\mu\nu}k^\mu k^\nu. \quad (30)$$

This paper shows how stochastic finite-screen dynamics leads to a robust focusing-like normal form. The geometric bridge identifies this normal form with null congruence focusing when the response data are integrable into a Lorentzian metric envelope.

5.1 Raychaudhuri-Limit Matching

In the metric-integrable equilibrium limit, the normal-form coefficients must reduce to the geometric Raychaudhuri values:

$$a \rightarrow \frac{1}{D-2}, \quad b \rightarrow 1$$

in the standard geometric limit ($a \rightarrow 1/2$ for the convention-dependent factor in the geometric equation above). The remaining term

$$\mathcal{R}_{\text{scr}}^\theta = \Xi_{\text{res}} + \zeta_\theta + \mathcal{O}(3)$$

is the finite-screen correction. If these constants do not match, the model remains a finite-screen normal form but is not a GR-equilibrium recovery. The geometric Raychaudhuri equation is thus the metric-integrable fixed point of the broader capacity-flow normal form.

6 Integrability into response geometry

In FDS-G1 Core, the local area response is

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}, \quad (31)$$

with local coupling

$$G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}, \quad G_{\text{loc}}^{-1}(x) = \frac{4\hbar}{c^3} \eta_{\text{loc}}(x). \quad (32)$$

The present companion adds transition dynamics. For the screen response data to define a metric geometry, the all-null focusing data must be representable as a quadratic form

$$\mathcal{F}_p(k) = E_{\mu\nu}(p)k^\mu k^\nu. \quad (33)$$

Since $g_{\mu\nu}k^\mu k^\nu = 0$ for null k , all-null data reconstructs $E_{\mu\nu}$ only up to trace terms. This is why a cosmological-constant-like integration freedom naturally appears in all-null closure arguments.

The dynamical consistency condition is the Ward/Bianchi closure

$$\nabla^\mu (T_{\mu\nu} + \mathcal{R}_{\mu\nu}) = 0, \quad (34)$$

where $\mathcal{R}_{\mu\nu}$ is the covariant representative of response nonintegrability, memory, stochasticity, or omitted screen modes.

7 Residual taxonomy

The residual term in the capacity-flow normal form has multiple origins. Table 2 separates them so that Ξ_{res} does not become an unconstrained catch-all.

Table 2: Residuals in screen transition dynamics: B origin and FDS-G1 Core alignment.

Residual type	Screen origin	B label	FDS-G1 label	Core label
Epistemic	Missing coarse variables; apparent non-closure from projection	epistemic	epistemic	
Stochastic	Finite-cell noise in transition process	stochastic	stochastic	
Memory	Non-Markovian update kernels	memory	memory	
Capacity	Slow variation of η_{loc}	capacity	scalar-capacity EFT	
Flux	Unresolved crossing load or exchange	flux	flux	
Geometric	No single metric envelope integrates response data	geometric	geometric	

If the residual is physical rather than epistemic, it must either be included as an explicit state variable, represented by a conserved residual stress, or modeled by a nonlocal kernel compatible with Ward/Bianchi closure.

Classification of Ξ_{res} . The capacity-flow residual Ξ_{res} in the normal form (??) must be classified before being admitted as a geometric residual. The five admissible categories are:

1. **Quotient artifact.** Apparent residual removed after screen gauge fixing.
2. **Missing variable.** Apparent residual resolved by enlarging the screen state space.
3. **Memory term.** Non-Markovian projection removable by horizontal lift (D6).
4. **Stochastic projection.** Finite-cell noise from the transition process (T3 lumpability test).
5. **Physical residual.** Survives all corrections—the only category that may feed Core $R_{\mu\nu}^{\text{scr}}$.

8 Numerical toy demonstrations

This section summarizes simple numerical examples included with the companion package. They are not observational fits; they illustrate qualitative behavior of the normal form.

8.1 Deterministic focusing cases

Figure 1 compares three deterministic cases: pure kinematic contraction, flux focusing, and flux plus shear. Positive flux and shear accelerate the decrease of θ , as expected.

8.2 Stochastic ensemble

Figure 2 shows an ensemble with finite-cell noise. The mean trajectory follows the deterministic focusing trend while individual paths fluctuate. Such fluctuations become candidates for stochastic residuals in the geometric envelope.

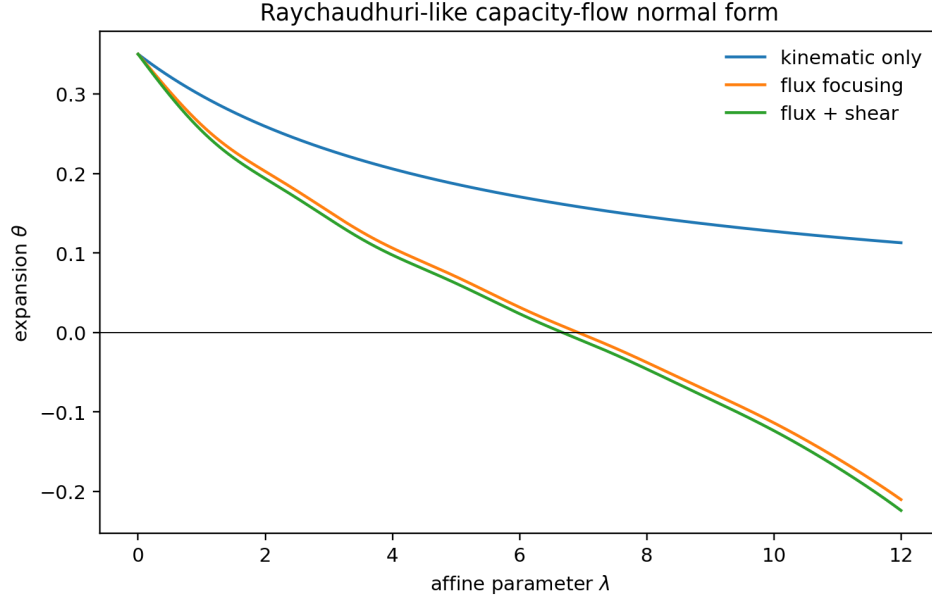


Figure 1: Deterministic capacity-flow examples. The flux and shear terms drive stronger focusing than the kinematic $-\theta^2$ term alone.

8.3 Memory residual

Figure 3 illustrates a nonlocal memory residual generated by a kernel convolution of prior flux. A memory term changes the response even when instantaneous flux is the same, demonstrating why a local Markov state may be insufficient.

8.4 Drift decomposition

Figure 4 decomposes the deterministic drift into normal-form contributions. This is useful for diagnosing whether focusing is dominated by expansion kinematics, shear, flux, or residual terms. All numerical parameters (a, b, c, D_θ, Φ) are illustrative; no observational fit is claimed.

Table 3: Simulation parameters for toy demonstrations.

Parameter	Value
a	1.0
b	0.5
c	1.0
D_θ	0.1
$\Phi_{\text{flux}}(\lambda)$	step or Gaussian pulse
σ_{scr}	constant or decaying
N	10^4

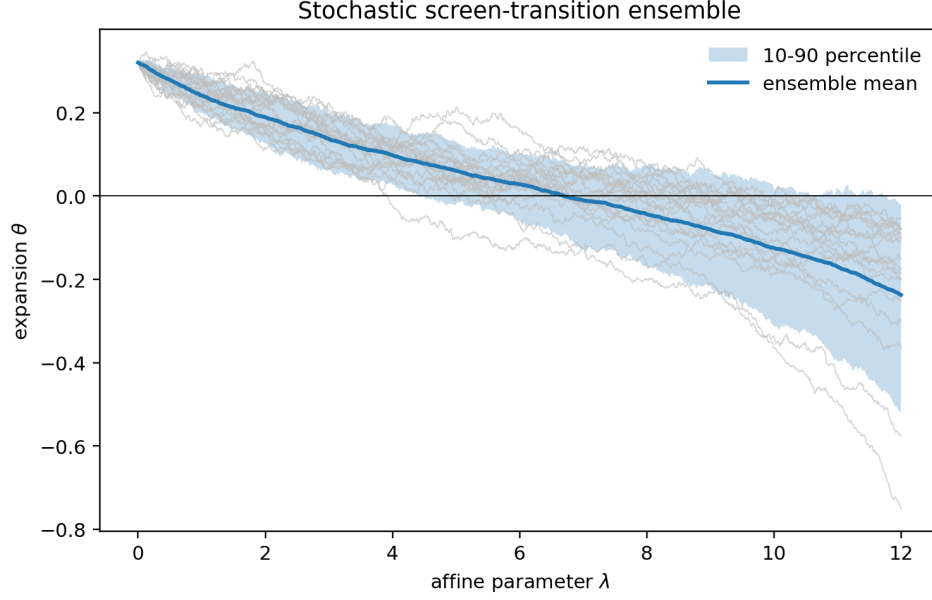


Figure 2: Stochastic screen-transition ensemble. Grey curves show representative sample paths; the band gives the 10–90 percentile range.

9 Failure conditions

This companion fails or must be modified if any of the following occurs.

1. **No screen-local state variable.** If a screen cannot be represented by a local extensive cell count or area variable, $\theta = d \ln N / d\lambda$ loses its operational basis.
2. **Dominant nonlocality.** If transition rates depend on global screen history at leading order, a local normal form is insufficient.
3. **Infinite local state capacity.** If finite cell capacity fails, stochastic hydrodynamic scaling may break.
4. **No positive flux focusing.** If positive flux does not reduce expansion, the sign of the focusing term changes and the gravity bridge fails.
5. **No metric integrability.** If all-null response data cannot be represented by a tensor quadratic form, a single Lorentzian metric envelope is not available.

These failure modes are useful: they identify exactly where a modified or nonmetric residual sector must enter.

10 Interface with companion papers

- **Companion A** supplies $S_{\text{scr}}[X]$ and η_δ from finite screen statistical mechanics.
- **This companion** supplies stochastic screen transition dynamics and θ -flow normal forms.

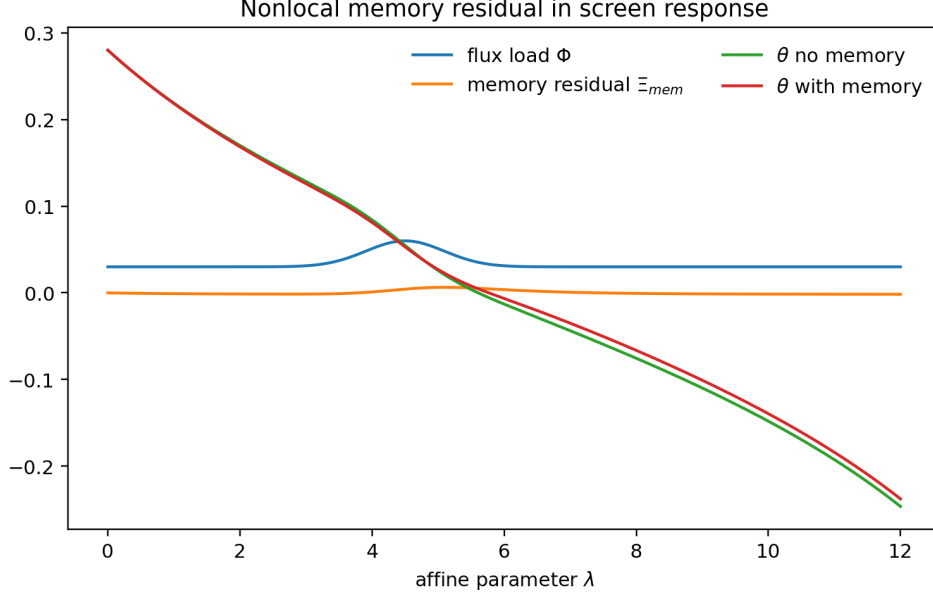


Figure 3: A memory residual produced by a nonlocal kernel. Memory modifies θ compared with the local no-memory case.

- **Companion C** promotes slow capacity response modes to scalar-capacity EFT.
- **Companion D** tests signatures such as no-free macroscopic G drift and the refined background-Weyl residual pattern ($s < 3$, $\Sigma_0 < 0$, $\mu_0 \simeq 0$).
- **Companion E** applies residual dynamics to horizon occupancy cosmology.
- **Companion F** interprets horizon and holographic entropy formulas as screen-capacity dictionaries.

The matched exact-pilot and medium-prior nested-evidence comparison (G1fit-real) now selects G1DE-M_{3/4} over free-response, constant- Σ , CPL, and Λ CDM controls. The leading residual is implemented as a Weyl-normalized transfer correction,

$$\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}},$$

rather than an independent dark stress source. The accompanying Ward-suppressed leakage hierarchy gives

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O(\mathcal{R}_H \mathcal{H}^2 / k^2).$$

This is consistent with a screen residual affecting the Weyl optical channel more strongly than local matter focusing, as predicted by the normal-form bridge between screen transition dynamics and cosmological response.

11 Conclusion

The central result is not a microscopic derivation of spacetime curvature. It is a dynamical normal-form theorem for finite screen transitions:

$$\dot{\theta} = -a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_{\theta} + \cdots \quad (35)$$

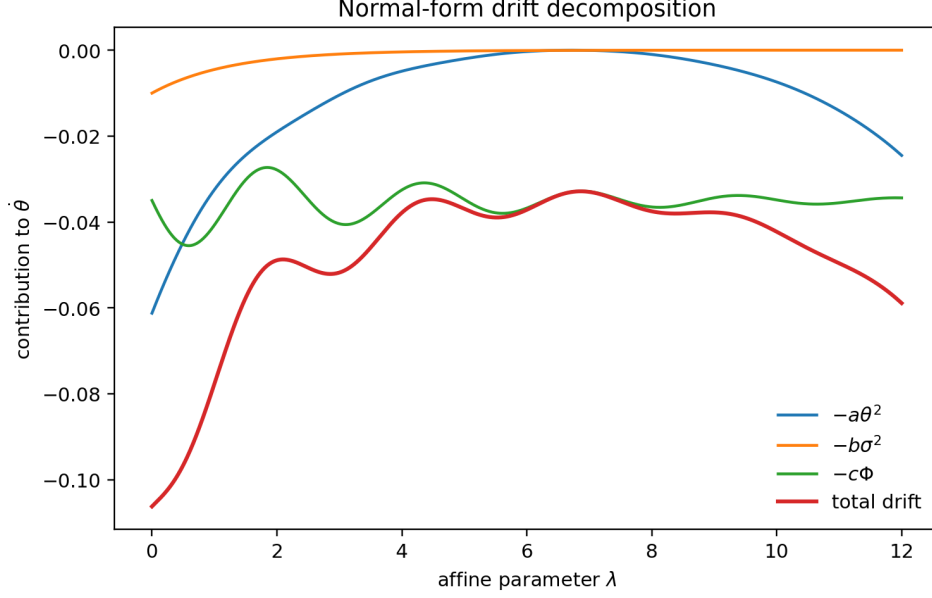


Figure 4: Contribution of each normal-form term to $\dot{\theta}$.

The negative quadratic term is kinematic, the shear term is the lowest dissipative scalar, and the flux term implements positive-load focusing. When this response data is integrable into a Lorentzian metric envelope and combined with the standard local-screen bridge, it aligns with Raychaudhuri-type focusing. Nonintegrable, noisy, or memory-dependent components become residual sectors.

Thus Companion B supports the FDS-G1 Core claim:

Gravity-like geometry is the integrable hydrodynamic response of finite causal-screen entropy.

It does so by explaining why the hydrodynamic response of finite screen updates naturally has a Raychaudhuri-like capacity-flow form.

References

- [1] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* 98, 1123 (1955).
- [2] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* 75, 1260 (1995).
- [3] L. Onsager, “Reciprocal relations in irreversible processes,” *Phys. Rev.* 37, 405 (1931).
- [4] L. Onsager and S. Machlup, “Fluctuations and irreversible processes,” *Phys. Rev.* 91, 1505 (1953).
- [5] C. Gardiner, *Handbook of Stochastic Methods*, Springer.
- [6] H. Risken, *The Fokker-Planck Equation*, Springer.
- [7] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20382013>.

- [8] Y. Wu, “FDS-T3: Capacity Overflow, Effective Stochasticity, and Phase-B Invariants,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20250367>.

Companion C: Scalar-Capacity Effective Field Theory

Horndeski Embedding of Variable Screen Response

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

May 2026

Abstract

This companion paper gives the effective-field-theory closure of the FDS-G1 entropy-response program. The core paper interprets local gravitational coupling as the inverse marginal entropy-area response; stiffness G_{loc}^{-1} is proportional to the response itself.

$$G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}, \quad \eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}. \quad (1)$$

If the response density becomes a long-wavelength field, its minimal covariant description is scalar-tensor-like. We define the scalar-capacity mode

$$\varphi(x) = \ln \frac{\eta(x)}{\eta_0}, \quad \frac{G_{\text{eff}}}{G_0} = e^{-\varphi}, \quad (2)$$

and embed the resulting conservative EFT into the luminal Horndeski / EFT-of-dark-energy language. The central point is not that the mathematical form is a new scalar-tensor theory; it is a screened, capacity-interpreted subclass with a restricted prior structure: positive capacity, conservation closure, large-area suppression unless a genuine capacity mode is excited, and a luminal tensor sector. This paper consolidates the earlier minimal scalar-capacity EFT and Horndeski notes into the official Companion C to *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*.

1 EFT Scope and Capacity Interpretation

This companion embeds variable screen response into the standard language of scalar-tensor gravity. The core claim of the FDS-G1 program is the entropy-response bridge. The Scalar-Capacity EFT is a conservative local-field embedding of capacity variation; it is not the unique data-facing G1DE residual. The Weyl-transfer branch ($\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}$, $\Sigma - 1 = -\kappa R_0 \hat{R}_H$, $\mu \simeq 1$) is more constrained than generic scalar EFT and is currently selected by medium-prior nested evidence. The FDS formal core [13] provides the finite-capacity substrate, and FDS-T2 [14] develops effective geometry as horizon boundary accounting, on which the scalar-capacity EFT builds.

$$\boxed{\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}, \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}} \quad (3)$$

under the standard horizon/local-screen thermodynamic bridge. Companion A supplies the screen statistical mechanics substrate; Companion B supplies stochastic screen transition dynamics and Raychaudhuri-like capacity-flow normal forms. This paper asks what happens if the local response density in Eq. (3) becomes a long-wavelength covariant field.

The answer is deliberately conservative: it becomes a scalar-tensor-like EFT, interpreted as a capacity interpretation and prior restriction on a known field-theory class, not as a replacement for Horndeski theory or the EFT of dark energy.

Mathematically, minimal scalar-capacity dynamics is a luminal scalar-tensor / Horndeski subclass. Physically, the scalar is not an arbitrary dark field; it is the hydrodynamic mode of marginal screen entropy-area response.

1.1 Claim-status table

Statement	Status	Meaning
$\varphi = \ln(\eta/\eta_0)$	Definition	Long-wavelength capacity-response mode.
$G_{\text{eff}}/G_0 = e^{-\varphi}$	Bridge relation	Follows from $G^{-1} \propto \eta$ under the horizon-response bridge.
Minimal action with $F(\varphi)R$	EFT model	Conservative covariant completion of variable response.
Horndeski embedding	Standard mapping	Places G_{1eff} in the usual scalar-tensor language.
$\alpha_T = 0$	Model choice	Luminal tensor sector; avoids non-luminal GW propagation in the minimal model.
Capacity priors	G1 contribution	Positivity, closure, suppression, and residual interpretation of the scalar.

No Double Counting rule. The scalar-capacity mode $\phi = \ln(\eta/\eta_0)$ may not duplicate the G1DE Weyl-transfer residual. If a residual is assigned to ϕ , it must not also appear independently as $R_H(a)$ or $A(a, k)$. Double-counting demotes the scalar branch to generic scalar-tensor phenomenology.

Remark 1 (Which $G?$). *This paper uses several related but distinct quantities. Table 1 clarifies their roles and should be consulted before interpreting any constraint.*

Table 1: Effective gravitational couplings used in this paper and the G1 programme.

Symbol	Meaning	Domain
G_{loc}	screen-response bridge coupling; $G_{\text{loc}} \propto [\eta_{\text{loc}}]^{-1}$	G1 entropy-response
$G_{\text{eff}} \sim [8\pi F(\varphi)]^{-1}$	Jordan-frame Planck-mass coupling	EFT / action level
$G_{\text{Cav}} (G_{\text{N,local}})$	Newton constant measured in laboratory or solar system	local gravity
$\mu(a, k)$	perturbation growth effective coupling	cosmological linear perturbations
$\Sigma(a, k)$	lensing effective coupling	Weyl / lensing sector

These quantities coincide only in restricted limits. In scalar-tensor theories, scalar perturbations, mass, screening, and frame convention can make local and cosmological effective couplings differ.

2 From response density to scalar-capacity field

In the core response formulation, the local area component of the response one-form is

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}. \quad (4)$$

When the response is homogeneous on a screen patch, this reduces to a scalar density. If the density varies slowly over spacetime, define the dimensionless scalar-capacity field

$$\varphi(x) = \ln \frac{\eta(x)}{\eta_0}. \quad (5)$$

The local effective coupling is then

$$\frac{G_{\text{eff}}(x)}{G_0} = e^{-\varphi(x)}. \quad (6)$$

Equivalently, the effective Planck-mass-squared function is

$$F(\varphi) = \frac{1}{8\pi G_{\text{eff}}} = M_{*,0}^2 e^\varphi, \quad (7)$$

where conventions differ by factors of 8π ; throughout this paper F is the coefficient multiplying $R/2$ in the gravitational action.

Definition 1 (Scalar-capacity mode). *The scalar-capacity mode is the long-wavelength field $\varphi = \ln(\eta/\eta_0)$ encoding deviations of the local screen entropy-area response from its equilibrium value. It is a response variable, not a fundamental microscopic cell label.*

3 Frame convention

The action (11) is written in the **Jordan frame**: matter is minimally coupled to $g_{\mu\nu}$, so $\nabla^\mu T_{\mu\nu}^{(m)} = 0$ and test-particle geodesics follow the Jordan-frame metric. The Planck-mass function is

$$F(\varphi) = M_{*,0}^2 e^\varphi, \quad G_{\text{eff}} = \frac{1}{8\pi F}. \quad (8)$$

An Einstein-frame transformation,

$$\tilde{g}_{\mu\nu} = \frac{F(\varphi)}{M_*^2} g_{\mu\nu}, \quad M_*^2 = \frac{1}{8\pi G_0}, \quad (9)$$

diagonalizes the gravitational sector but introduces matter couplings to φ . The kinetic no-ghost proxy in the Einstein frame is

$$\mathcal{K}_E = \frac{Z}{F} + \frac{3}{2} \left(\frac{F_{,\varphi}}{F} \right)^2 > 0, \quad (10)$$

which for $F \propto e^\varphi$ becomes $\mathcal{K}_E = Z/F + 3/2$. All stability conditions below are stated in a frame-invariant manner.

4 Minimal scalar-capacity EFT

The minimal covariant scalar-capacity EFT is

$$S_{\text{cap}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} F(\varphi) R - \frac{1}{2} Z(\varphi) g^{\mu\nu} \nabla_\mu \varphi \nabla_\nu \varphi - V(\varphi) + \mathcal{L}_m + \mathcal{L}_{\text{res}} \right]. \quad (11)$$

Here $\mathcal{L}_{\text{res}}$ denotes residual sectors required when variable capacity is not fully captured by the scalar mode: nonlocal memory, stochastic screen noise, horizon occupancy residuals, or unresolved coarse variables.

4.1 Field equations

Varying the metric gives

$$FG_{\mu\nu} = T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\varphi)} + T_{\mu\nu}^{(\text{res})} + \nabla_\mu \nabla_\nu F - g_{\mu\nu} \square F, \quad (12)$$

where

$$T_{\mu\nu}^{(\varphi)} = Z \nabla_\mu \varphi \nabla_\nu \varphi - g_{\mu\nu} \left[\frac{1}{2} Z (\nabla \varphi)^2 + V(\varphi) \right]. \quad (13)$$

Varying φ yields

$$Z \square \varphi + \frac{1}{2} Z_{,\varphi} (\nabla \varphi)^2 - V_{,\varphi} + \frac{1}{2} F_{,\varphi} R + \frac{\delta \mathcal{L}_{\text{res}}}{\delta \varphi} = 0. \quad (14)$$

4.2 Conservation closure

The Bianchi identity requires

$$\nabla^\mu \left(T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\varphi)} + T_{\mu\nu}^{(\text{res})} \right) + \text{terms exchanged with } F(\varphi)R = 0. \quad (15)$$

If matter is minimally coupled, $\nabla^\mu T_{\mu\nu}^{(m)} = 0$ holds in the Jordan frame. The scalar and residual sectors must then close the remaining response current. This is the field-theory version of the G1 rule:

variable screen capacity cannot appear as a free $G(x)$ drift without a conserved closure sector.

(16)

5 Horndeski embedding

The general Horndeski class contains the most general four-dimensional single-scalar tensor theory with second-order field equations in the usual formulation. The minimal G1 scalar-capacity EFT is embedded as

$$G_2(\varphi, X) = Z(\varphi)X - V(\varphi), \quad G_3 = 0, \quad (17)$$

$$G_4(\varphi) = \frac{1}{2} F(\varphi), \quad G_5 = 0, \quad (18)$$

where

$$X = -\frac{1}{2} g^{\mu\nu} \nabla_\mu \varphi \nabla_\nu \varphi. \quad (19)$$

This is a conformally coupled scalar-tensor subclass, not a full general Horndeski model.

Theorem 1 (Scalar-Capacity Admissibility). *The minimal scalar-capacity EFT in Eq. (11) is a luminal scalar-tensor/Horndeski subclass with $G_3 = G_5 = 0$ and $G_4 = F(\varphi)/2$. Its distinctiveness is not the mathematical Lagrangian form but the interpretation of φ as a coarse-grained screen-capacity response mode. A scalar-capacity branch is admissible only if it: (1) descends from η_{loc} ; (2) respects Ward/Bianchi closure; (3) avoids free $A(a, k)$; (4) does not double-count the G1DE Weyl residual; (5) carries independent likelihood penalty. Failure demotes it to generic scalar-tensor phenomenology.*

6 EFT-of-dark-energy alpha functions

On a spatially flat FLRW background, the Bellini-Sawicki alpha-function language describes many linear scalar-tensor deviations from GR by a time-dependent effective Planck mass and functions $(\alpha_M, \alpha_K, \alpha_B, \alpha_T)$.

For the minimal capacity subclass,

$$M_*^2 = F(\varphi) = M_{*,0}^2 e^\varphi, \quad (20)$$

so

$$\alpha_M = \frac{d \ln M_*^2}{d \ln a} = \frac{d\varphi}{d \ln a}. \quad (21)$$

Under the Bellini-Sawicki sign convention adopted here, the conformally coupled minimal subclass has

$$\alpha_T = 0, \quad \alpha_B = -\alpha_M. \quad (22)$$

The kineticity is schematically

$$\alpha_K \sim \frac{Z(\varphi)\dot{\varphi}^2}{H^2 F} + \text{normalization terms}, \quad (23)$$

with detailed normalization convention-dependent.

Remark 2 (Why $\alpha_T = 0$ is chosen). *The minimal G1eff sector intentionally sets $\alpha_T = 0$. This is a conservative tensor sector compatible with the strong constraints on the gravitational-wave propagation speed following GW170817/GRB170817A. Non-luminal extensions would require a separate, tightly constrained companion analysis.*

7 Stability and viability conditions

The minimal scalar-capacity EFT is viable only in the region satisfying standard scalar-tensor stability and observational requirements.

7.1 Positivity of capacity and gravitational stiffness

Because $\eta > 0$,

$$F(\varphi) > 0, \quad G_{\text{eff}} > 0. \quad (24)$$

This is not optional; negative capacity would destroy the FDS interpretation.

7.2 No-ghost condition

In the Einstein-frame kinetic normalization, a simple positivity proxy is

$$\mathcal{K}_E(\varphi) = \frac{Z(\varphi)}{F(\varphi)} + \frac{3}{2} \left(\frac{F_{,\varphi}}{F} \right)^2 > 0. \quad (25)$$

For $F \propto e^\varphi$ this becomes

$$\mathcal{K}_E = \frac{Z}{F} + \frac{3}{2} > 0. \quad (26)$$

A sufficient conservative choice is $Z > 0$.

7.3 Gradient stability and mass scale

The scalar sound speed must be positive,

$$c_s^2 > 0, \quad (27)$$

and the effective mass

$$m_{\text{eff}}^2 \sim \frac{d^2 V_{\text{eff}}}{d\varphi^2} \quad (28)$$

should not induce an uncontrolled tachyonic instability on scales where the EFT is applied.

7.4 Local-gravity constraints

Since

$$\frac{\dot{G}_{\text{eff}}}{G_{\text{eff}}} = -\dot{\varphi} = -H\alpha_M, \quad (29)$$

local bounds on $\dot{G}/G$, PPN parameters, and screening mechanisms constrain the allowed motion of φ . A macroscopic G drift without closure or screening is not a viable G1 model.

8 Local gravity, screening, and decoupling

A light scalar coupled to matter mediates a fifth force and modifies local gravity. The minimal scalar-capacity EFT requires at least one of the following mechanisms to remain viable:

1. **Small running.** $|\alpha_M| \ll 1$ ensures $|\dot{G}/G|$, PPN, and growth constraints are satisfied without screening.
2. **Heavy field.** $m_{\text{eff}}^2 \gg H_0^2$ suppresses the scalar force below local experimental scales; the field is cosmologically relevant only through its residual or coupling effects.
3. **Environmental screening / residual confinement.** The capacity mode may be active at cosmological or horizon-residual scales but frozen or screened in the solar system and Cavendish experiments through a screening mechanism (chameleon, symmetron, Vainshtein) or through nonlocal residual confinement.

Without one of these mechanisms, a macroscopic scalar-capacity mode is generically ruled out by local fifth-force and PPN constraints.

9 Capacity priors: how G1 differs from generic scalar-tensor EFT

The same Lagrangian form can describe many scalar-tensor models. G1 adds a restricted prior interpretation.

Capacity prior	Meaning
$\eta > 0$	Screen distinction density is positive; $G_{\text{eff}} > 0$.
Large-area suppression	Finite-area corrections scale as $\Delta G/G \sim -\alpha/(\eta_0 A)$ unless a genuine long-wavelength capacity mode is excited.
No free G drift	$\dot{G}/G$ requires a Bianchi-closed scalar or residual sector.
Luminal tensor sector	Minimal model sets $\alpha_T = 0$.
Residual closure	Unmodeled capacity changes appear as conserved $T_{\mu\nu}^{(\text{res})}$ or nonlocal memory kernels.
Occupancy linkage	If coupled to horizon occupancy, expansion, growth, and lensing deviations should be correlated rather than arbitrary.

10 Toy alpha-function profiles

For an illustrative capacity field relaxing to equilibrium today, take

$$\varphi(a) = \epsilon(1 - a^s). \quad (30)$$

Then

$$\frac{G_{\text{eff}}}{G_0} = e^{-\epsilon(1-a^s)}, \quad \alpha_M = \frac{d\varphi}{d \ln a} = -\epsilon s a^s. \quad (31)$$

This toy profile is not a fit; it is a diagnostic used to show how scalar-capacity running maps to the EFT variables.

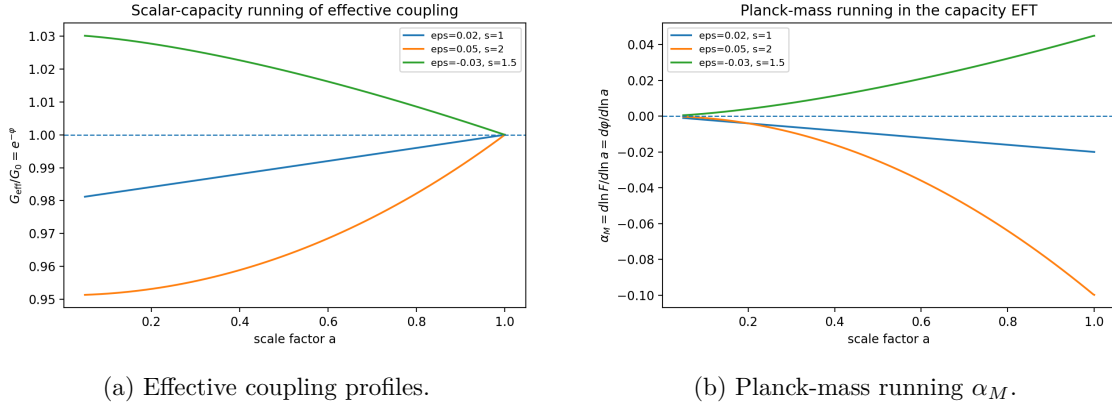


Figure 1: Illustrative scalar-capacity EFT diagnostics for $\varphi(a) = \epsilon(1 - a^s)$, satisfying $\varphi(1) = 0$ and $G_{\text{eff}}(1) = G_0$. $\epsilon > 0$ gives G_{eff} lower in the past (high early η); $\epsilon < 0$ gives G_{eff} higher in the past (low early η). s controls relaxation sharpness. These plots are diagnostic maps, not observationally allowed contours.

11 EFT interpretation of the exact-likelihood pilot

The current matched medium-prior nested-evidence comparison (G1fit-real) selects the projection-locked G1DE-M_{3/4} model across a six-model hierarchy, with

$$\mu_0 \simeq 0, \quad \Sigma_0 < 0.$$

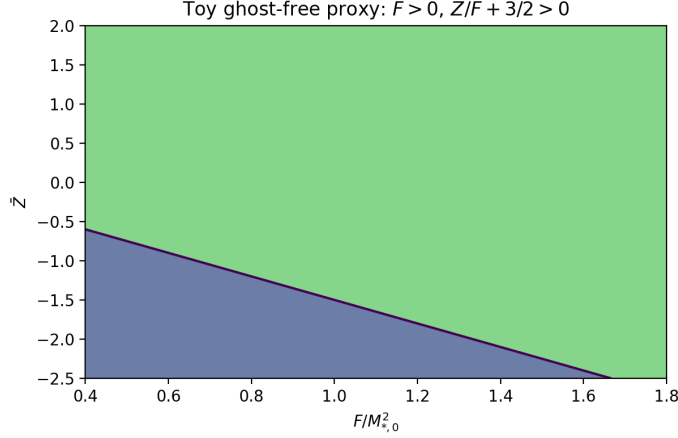


Figure 2: A simple no-ghost proxy region for $F > 0$ and $\mathcal{K}_E = Z/F + 3/2 > 0$. Positive Z and positive F are safely inside the conservative region.

In EFT-of-dark-energy language, this suggests that the leading phenomenological opening is not an effective Newton coupling shift in the matter-growth equation ($\mu(a, k) \simeq 1$), but a Weyl-response or gravitational-slip channel ($\Sigma(a, k) < 1$). Therefore the G1 scalar-capacity EFT should be constrained to the luminal, weak-growth-deviation regime:

$$\alpha_T = 0, \quad \mu(a) \simeq 1, \quad \Sigma(a) < 1$$

over the active filling window. This is a phenomenological location statement, not a derivation from Horndeski. It is compatible with, but not predicted by, a wide class of luminal scalar-tensor theories; the G1-specific claim is the shared ledger shape (a) for both background and Weyl response.

12 Connection to companions

Companion	Relation to scalar-capacity EFT
FDS-G1 Core	Defines η_{loc} and G_{loc} as entropy-response variables.
A: Screen statistics	Supplies $\eta_\delta(A)$ and finite-area corrections that motivate scalar-capacity modes.
B: Transition dynamics	Supplies response dynamics and residual noise/memory channels.
D: Likelihood/falsification	Constrains α_M , $\dot{G}/G$, growth, and lensing signatures.
E: Horizon occupancy	Provides a homogeneous residual sector that may couple to φ .
F: Holography/horizons	Interprets $1/(4G_N)$ as screen density in horizon and AdS/CFT settings.

13 Failure registry

Failure condition	Consequence
$F \leq 0$	Negative capacity or negative effective stiffness; model fails.
Ghost or gradient instability	Scalar-capacity EFT invalid on the scale considered.
Large unscreened $\dot{G}/G$	Excluded by local gravity unless residual closure or screening is present.
Non-luminal tensor sector	Not part of the minimal model; strongly constrained.
Arbitrary φ potential with no screen prior	Reduces to generic scalar-tensor phenomenology; loses G1 distinctiveness.
No connection to η_{loc}	Breaks the capacity interpretation.
No viable screening or mass suppression	Ruled out by local fifth-force and PPN constraints.
φ independent of screen-response priors	Collapses to generic scalar-tensor model without G1 interpretation.

14 Conclusion

Companion C positions variable screen response within the standard language of scalar-tensor gravity. The minimal G1 scalar-capacity EFT is not advertised as a new field-equation form. Its contribution is the interpretation and restriction of a conservative luminal scalar-tensor subclass by finite screen-capacity response:

$$\boxed{\eta > 0, \quad G_{\text{eff}}^{-1} \propto \eta, \quad \alpha_T = 0, \quad \alpha_B = -\alpha_M,} \quad (32)$$

with the dictionary

$$\boxed{\varphi = \ln \frac{\eta}{\eta_0}, \quad \frac{G_{\text{eff}}}{G_0} = e^{-\varphi}, \quad \alpha_M = \frac{d\varphi}{d \ln a}.} \quad (33)$$

A further restriction comes from conservation closure: variable screen capacity cannot appear as a free $G(x)$ drift without a conserved residual sector or screening mechanism. The decoupling limit is

$$m_{\text{eff}}^2 \gg H_0^2, \quad |\alpha_M| \ll 1, \quad (34)$$

recovering local gravity and Λ CDM-like cosmology when the capacity mode is heavy or frozen.

The scalar-capacity G_{eff} is an action-level inverse Planck stiffness; observable quantities (G_{Cav} , growth rate $\mu(a, k)$, lensing $\Sigma(a, k)$) require perturbation theory, screening, and frame specifications beyond the minimal φ -only dictionary. This prevents conflation of G_{eff} with local Newton’s constant in survey constraints.

References

- [1] G. W. Horndeski, “Second-order scalar-tensor field equations in a four-dimensional space,” *International Journal of Theoretical Physics* **10**, 363–384 (1974).
- [2] C. Deffayet, X. Gao, D. A. Steer, and G. Zahariade, “From k-essence to generalised Galileons,” *Physical Review D* **84**, 064039 (2011).

- [3] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *Journal of Cosmology and Astroparticle Physics* **07**, 050 (2014).
- [4] J. Gleyzes, D. Langlois, F. Piazza, and F. Vernizzi, “Essential building blocks of dark energy,” *Journal of Cosmology and Astroparticle Physics* **08**, 025 (2013/2014 literature stream).
- [5] T. Baker, E. Bellini, P. G. Ferreira, M. Lagos, J. Noller, and I. Sawicki, “Strong constraints on cosmological gravity from GW170817 and GRB 170817A,” *Physical Review Letters* **119**, 251301 (2017).
- [6] J. M. Ezquiaga and M. Zumalacarregui, “Dark energy after GW170817: dead ends and the road ahead,” *Physical Review Letters* **119**, 251304 (2017).
- [7] C. Brans and R. H. Dicke, “Mach’s principle and a relativistic theory of gravitation,” *Physical Review* **124**, 925 (1961).
- [8] T. Clifton, P. G. Ferreira, A. Padilla, and C. Skordis, “Modified gravity and cosmology,” *Physics Reports* **513**, 1–189 (2012).
- [9] T. Kobayashi, “Horndeski theory and beyond: a review,” *Reports on Progress in Physics* **82**, 086901 (2019).
- [10] A. Joyce, B. Jain, J. Khoury, and M. Trodden, “Beyond the cosmological standard model,” *Physics Reports* **568**, 1–98 (2015).
- [11] B. P. Abbott et al. (LIGO/Virgo Collaboration), “GW170817: observation of gravitational waves from a binary neutron star inspiral,” *Phys. Rev. Lett.* **119**, 161101 (2017).
- [12] P. Vaillant and D. Scott, “A complex tension origin for dilaton gravity,” *Entropy* **28**, 544 (2026).
- [13] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20382013>.
- [14] Y. Wu, “FDS-T2: Effective Geometry as Horizon Boundary Accounting,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20284911>.

Companion D: Likelihood-Ready Tests and Falsification

Predictions, Compressed-Data Interfaces, and Model Comparison for Finite Screen-Capacity Gravity

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

May 27, 2026

Abstract

This companion paper collects the falsification and likelihood interface for the FDS-G1 finite screen-capacity program. The core G1 claim is not that finite distinction systems derive gravity from first principles, but that finite causal-screen statistics provide an entropy-response substrate for the area coefficient that controls gravity-like coupling under the standard horizon and local-screen bridges. Companion D translates this claim into observational diagnostics through three layers. Layer 1 is the original demonstration interface (transparent demo compressed data), retained for reproducibility. Layer 2 is the matched exact full-likelihood pilot, now recorded through G1fit-real[12], which uses Pantheon+ full covariance, DESI DR2 BAO, curated RSD, and compressed E_G . Layer 3 is the medium-prior nested-evidence comparison, now completed as of G1fit-real, with the projection-locked G1DE-M_{3/4} ranking first in the six-model hierarchy. We define a modular likelihood for supernova distances, BAO distances, growth-rate data, and E_G data. We compare baseline Λ CDM, CPL dark energy, and the G1DE residual models. We emphasize three falsification principles: no free macroscopic drift of the effective gravitational coupling without conservation closure; a background–Weyl residual fingerprint sharing a common screen ledger variable; and inverse-area suppression of ordinary finite-area corrections. A central purpose of this companion is to prevent residual sectors from functioning as arbitrary dark components: every residual must either close a conservation law, correlate observables through a common ledger variable, or be demoted by information criteria. The purpose is not to claim evidence for G1 over Λ CDM, but to make the framework empirically penalizable.

1 Likelihood Scope and Falsification Status

This companion turns the screen-response framework into a likelihood-ready testing protocol, following the FDS formal core [10] and the pre-registration and demotion conditions registered in FDS-X1 [11]. It defines a modular compressed likelihood for supernova distances, BAO distances, growth-rate data, local gravity priors, and gravitational-wave propagation flags. The companion uses demonstration compressed data generated from a fiducial Λ CDM model with small seeded perturbations. These data are included only to verify the code path and the information-criterion bookkeeping. Production inference must use real data vectors and covariance matrices, for example Pantheon+ supernovae, DESI BAO measurements, growth compilations, CMB distance priors, Lunar Laser Ranging bounds, Cassini PPN bounds, and gravitational-wave speed constraints.

Status. This is a likelihood interface and falsification protocol. A matched exact-likelihood pilot and medium-prior nested-evidence comparison (G1fit-real[12]) has now been executed with

Pantheon+ full covariance, DESI DR2 BAO compressed distances, a curated RSD $f\sigma_8$ subset, and compressed E_G data. The six-model medium-prior nested-evidence hierarchy is

$$\text{G1DE-M}_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE2} > \text{CPL} > \Lambda\text{CDM},$$

with the four-parameter projection-locked model ranking first. Prior sensitivity and production evidence refinement remain pending.

Likelihood discipline. Likelihood success supports only the tested output normal form. It does not prove finite-index screen algebras, boost modular covariance, microscopic source–kernel uniqueness, or strong-field GR replacement.

2 Position in the G1 dependency structure

The FDS-G1 Core dependency chain is

$$\text{finite screen statistics} \rightarrow S_{\text{scr}}[X] \rightarrow \omega \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}} \rightarrow \theta\text{-flow} \rightarrow \text{metric/Ward closure} \rightarrow \text{residual sectors.} \quad (1)$$

Companion D acts at the final step. It takes residual or capacity-response sectors and asks whether they survive observational compression and parameter penalties.

The local response relation in the core theory is

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}, \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}. \quad (2)$$

If η_{loc} varies macroscopically, conservation closure is mandatory. In scalar-capacity language, one writes

$$\varphi = \ln \frac{\eta}{\eta_0}, \quad \frac{G_{\text{eff}}}{G_0} = e^{-\varphi}, \quad (3)$$

and the Horndeski/EFT-of-dark-energy embedding identifies

$$\alpha_M = \frac{d \ln F}{d \ln a}, \quad \alpha_T = 0 \quad (4)$$

for the conservative luminal-tensor sector discussed in Companion C.

3 Signature predictions as likelihood requirements

The G1 residual program is not defined by arbitrary $w(a)$ fitting. It is defined by constrained response patterns. Companion D uses the following three signatures as pass/fail conditions.

3.1 No free macroscopic G drift

A variable screen response implies

$$\frac{\dot{G}}{G} \simeq -\frac{\dot{\eta}}{\eta} = -H\alpha_M. \quad (5)$$

A model that requires a large unscreened drift of G is disfavored unless it supplies a conservation-closed residual, screening mechanism, or scalar-capacity field satisfying local-gravity priors. The key principle is

$$\nabla^\mu (T_{\mu\nu} + \mathcal{R}_{\mu\nu}) = 0. \quad (6)$$

Macroscopic capacity variation without this closure is not a valid G1 model.

3.2 Correlated expansion-growth-lensing residuals

A G1 horizon-occupancy residual uses the factorized memory response

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1, \quad R_0 = 3 - s,$$

implemented through a Weyl-normalized transfer operator

$$\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}},$$

so that

$$\Sigma(a, k) - 1 = -\kappa R_0 \hat{R}_H(a) + O(R_0^2), \quad \mu(a, k) - 1 = O(R_0^2) + O(R_0 \hat{R}_H \mathcal{H}^2/k^2).$$

so that $0 \leq \hat{X}(a) \leq 1$ for $0 \leq \chi_H \leq \chi_\infty$. The growth and lensing response proxies are

$$\mu(a) = 1 + \mu_0 \hat{X}(a), \quad \Sigma(a) = 1 + \Sigma_0 \hat{X}(a). \quad (7)$$

Thus the falsifiable feature is not $w(a) \neq -1$ alone, but a structured residual fingerprint. The current matched exact-pilot and medium-prior nested-evidence comparison (G1fit-real[12]) reports

$$s < 3, \quad \Sigma_0 < 0, \quad \mu_0 \simeq 0,$$

i.e. a background filling deviation and a suppressed Weyl response, with the growth response remaining consistent with GR. This pattern—expansion plus Weyl response sharing the same active-filling shape $\hat{X}(a)$, while matter growth stays near GR—is more restrictive than generic modified-gravity fits.

3.3 Finite-area corrections are suppressed

If a finite screen has

$$\frac{S(A)}{k_B} = \eta_0 A + \alpha \ln(A/a_0) + \gamma + \dots, \quad (8)$$

then the differential density is

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA} = \eta_0 + \frac{\alpha}{A} + \dots, \quad (9)$$

and

$$\frac{\Delta G}{G} \simeq -\frac{\alpha}{\eta_0 A}. \quad (10)$$

Therefore ordinary finite-area or logarithmic corrections must be small on macroscopic screens. A large late-time cosmological signal must come from a residual/horizon-occupancy sector, a scalar-capacity mode, or another explicitly modeled nonlocal sector, not from a bare logarithmic area correction alone.

4 Compressed likelihood blocks

For the Layer-1 demonstration interface we use

$$\chi_{\text{demo}}^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{growth}}^2 + \chi_{\text{local}}^2. \quad (11)$$

For the matched exact pilot reported in G1fit-real[12], the implemented likelihood is

$$\chi_{\text{exact}}^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{RSD}}^2 + \chi_{E_G}^2. \quad (12)$$

Local/prior blocks ($\dot{G}/G$, PPN, GW speed) are retained for Layer-3 production; the pilot layer uses E_G as a compressed Weyl/lensing diagnostic. Each block is modular. The production form for a generic block is

$$\chi_{\text{block}}^2 = (\mathbf{d} - \mathbf{m})^T C^{-1} (\mathbf{d} - \mathbf{m}), \quad (13)$$

where $\mathbf{d}$ is the data vector, $\mathbf{m}$ the model prediction, and C the survey covariance matrix. The demo code uses a diagonal covariance,

$$C = \text{diag}(\sigma_i^2), \quad (14)$$

as a simplification. Production inference must replace these with survey covariance matrices. The demo is **Layer 1** (synthetic demonstration pipeline); production inference is **Layer 2** (survey-quality likelihood analysis). Every demo figure carries the caveat: synthetic demo data only; no cosmological constraint is claimed.

4.1 Supernova distances

For a model with Hubble function $E(z) = H(z)/H_0$, define

$$D_C(z) = \int_0^z \frac{dz'}{E(z')}, \quad D_L(z) = (1+z)D_C(z). \quad (15)$$

The relative distance modulus is

$$\mu(z) = 5 \log_{10} D_L(z) + \mathcal{M}, \quad (16)$$

where $\mathcal{M}$ is the SN absolute magnitude nuisance, absorbing the H_0 degeneracy. For a diagonal covariance, the analytic marginalization over $\mathcal{M}$ gives

$$\chi_{\text{SN,marg}}^2 = \Delta^T C^{-1} \Delta - \frac{(\mathbf{1}^T C^{-1} \Delta)^2}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad (17)$$

where $\Delta_i = \mu_{\text{obs},i} - 5 \log_{10} D_L(z_i)$ (setting $\mathcal{M} = 0$ for the residual) and $\mathbf{1}$ is the unit vector. This avoids fitting $\mathcal{M}$ as an explicit free parameter in the demo.

4.2 BAO compressed distances

The demonstration interface uses

$$\frac{D_M(z)}{r_d} = \frac{c/H_0}{r_d} D_C(z), \quad \frac{D_H(z)}{r_d} = \frac{c/H_0}{r_d} \frac{1}{E(z)}. \quad (18)$$

A real DESI analysis should import the official BAO data vector and covariance matrix.

4.3 Growth proxy

A full analysis should solve perturbation equations for a specified scalar-capacity or residual model. The production growth equation for the matter density contrast in the quasistatic limit is

$$\delta_m'' + \left(2 + \frac{H'}{H}\right) \delta_m' = \frac{3}{2} \Omega_m(a) \mu(a, k) \delta_m, \quad (19)$$

where $\mu(a, k)$ is the effective gravitational coupling for perturbations relative to GR and primes denote derivatives with respect to $\ln a$. The growth rate and redshift-space-distortion proxy are

$$f(a) = \frac{d \ln D}{d \ln a}, \quad f\sigma_8(a) = f(a) \sigma_{8,0} D(a). \quad (20)$$

The demo uses a growth-index proxy,

$$f(z) \simeq \Omega_m(z)^{0.55} \left[1 + \frac{1}{2}(\mu(z) - 1) \right], \quad (21)$$

and constructs $f\sigma_8(z)$ by integrating $f = d \ln D / d \ln a$. This is sufficient for a Layer 1 pipeline demonstration but not for Layer 2 precision inference.

4.4 Lensing proxy

Weak gravitational lensing responds to the Weyl potential $\Phi + \Psi$. In the quasistatic limit,

$$k^2(\Phi + \Psi) = -8\pi G a^2 \Sigma(a, k) \rho_m \Delta_m, \quad (22)$$

where $\Sigma(a, k)$ is the effective lensing coupling. The G1DE residual fingerprint correlates Σ with the growth coupling μ through the common ledger shape $\hat{X}(a)$,

$$\mu(a) = 1 + \mu_0 \hat{X}(a), \quad \Sigma(a) = 1 + \Sigma_0 \hat{X}(a), \quad \hat{X}(a) = \frac{4}{\chi_\infty} \chi_H(a) (1 - \chi_H(a) / \chi_\infty). \quad (23)$$

A G1DE model that requires independent $\mu(a)$ and $\Sigma(a)$ functions (beyond a common ledger amplitude) has lost its correlated residual claim.

4.5 Local and propagation priors

The local block may include priors on

$$\frac{\dot{G}}{G}, \quad \gamma_{\text{PPN}} - 1, \quad \alpha_T, \quad (24)$$

or model-specific stability flags. In the conservative scalar-capacity sector one imposes $\alpha_T = 0$ to remain in the luminal tensor-propagation class.

4.6 No Free Function rule

The Weyl amplitude $A(a, k)$ must not be sampled as an independent free function. If optimal likelihood or evidence requires free $A(a, k)$, the model is no longer G1DE—it becomes a generic dark-stress source. This rule is enforced at the inference level: any chain that samples $A(a, k)$ independently of the G1DE hierarchy is classified as a non-G1 dark-stress model.

5 Model classes

The demo compares four model classes. The same infrastructure can be extended to HDE, quintessence proxies, and scalar-tensor EFT models.

5.1 Λ CDM

$$E^2(z) = \Omega_{m0}(1+z)^3 + (1 - \Omega_{m0}). \quad (25)$$

This is the reference model.

5.2 CPL

$$w(a) = w_0 + w_a(1-a), \quad (26)$$

$$E^2(a) = \Omega_{m0}a^{-3} + (1 - \Omega_{m0})a^{-3(1+w_0+w_a)}e^{-3w_a(1-a)}. \quad (27)$$

CPL is a flexible background-only model unless supplemented by perturbation prescriptions.

5.3 G1DE

The G1DE background is defined by a horizon occupancy residual $\chi_H(a)$. The filling-closure law is

$$\chi_H(a) = \frac{\chi_\infty}{1 + Ba^{-s}}, \quad B = \frac{\chi_\infty}{\chi_{H0}} - 1, \quad E^2(a) = \frac{\Omega_{m0}a^{-3}}{1 - \chi_H(a)}. \quad (28)$$

The Λ CDM-like limit occurs when $s = 3$ and $\chi_\infty = 1$. To see this, write

$$1 - \chi_H(a) = \frac{Ba^{-3}}{1 + Ba^{-3}}, \quad (29)$$

so that

$$E^2(a) = \frac{\Omega_{m0}a^{-3}(1 + Ba^{-3})}{Ba^{-3}} = \Omega_{m0}a^{-3} + \frac{\Omega_{m0}}{B}. \quad (30)$$

Matching $B = \Omega_{m0}/(1 - \Omega_{m0})$ gives $E^2(a) = \Omega_{m0}a^{-3} + (1 - \Omega_{m0})$, the flat- Λ CDM background. Thus G1DE reduces smoothly to Λ CDM when $s = 3$ and $\chi_\infty = 1$, and its departures are controlled by s and χ_∞ .

5.4 G1DE-1: background plus growth

The growth extension introduces

$$\mu(a) = 1 + \mu_0 \hat{X}(a), \quad \hat{X}(a) = \frac{4}{\chi_\infty} \chi_H(a) \left(1 - \frac{\chi_H(a)}{\chi_\infty} \right). \quad (31)$$

This class is penalized by information criteria unless the growth response improves the likelihood enough to justify the additional parameter.

6 Information criteria and falsification

For a model with k parameters and n effective data points,

$$\text{AIC} = \chi_{\min}^2 + 2k, \quad (32)$$

$$\text{BIC} = \chi_{\min}^2 + k \ln n. \quad (33)$$

These are not substitutes for a full Bayesian evidence, but they are useful guardrails against unconstrained residual sectors.

Table 1: Free parameters per model in the demo comparison. For correlated compressed data, BIC should be treated as an approximate guardrail, not a substitute for Bayesian evidence. Synthetic demo only — no cosmological constraint.

Model	Demo free parameters	k
Λ CDM	Ω_{m0}	1
CPL	Ω_{m0}, w_0, w_a	3
G1DE	Ω_{m0}, s	2
G1DE+ μ	Ω_{m0}, s, μ_0	3

Table 2: Demo compressed-data comparison. This table is generated from the included toy data and should not be read as a real cosmological constraint. Synthetic demo only — no cosmological constraint.

Model	$\chi^2_{\min}$	k	ΔAIC	ΔBIC
Λ CDM	3.21	1	0.00	0.00
CPL	2.43	3	3.22	6.27
G1DE	2.55	2	1.34	2.87
G1DE+ μ	2.54	3	3.33	6.38

The demonstration intentionally allows G1DE to lose to Λ CDM after parameter penalties. This is a feature, not a bug: a viable G1 residual sector must survive information criteria and not rely on arbitrary functional freedom.

Exact-likelihood reference. The G1fit-real[12] matched exact-pilot gives the following chi-squared decomposition at the G1DE-2 best-fit point:

Table 3: Exact code-path validation for G1DE-2 (from G1fit-real). Matched exact-pilot values; nested-evidence comparison in G1fit-real.

block	χ^2
SN (Pantheon+ full covariance)	1749.609044
BAO (DESI DR2)	9.633696
RSD (curated $f\sigma_8$)	6.883906
E_G (compressed)	1.010038
total	1767.136683

7 Exact-likelihood pilot results (G1fit-real summary)

The matched exact-likelihood and medium-prior nested-evidence comparison for G1DE is reported in the companion data note G1fit-real[12]. We reproduce its core diagnostics here to support the falsification ladder below. For the full six-parameter posterior, corner plots, trace diagnostics, and data-block details, see G1fit-real.

The pilot chain setup is 3 seeds $\times$ 48 walkers $\times$ 5000 iterations, with combined $N_{\text{postburn}} = 360\,000$ post-burn samples. Cross-seed split- $\hat{R}$ is below 1.0013 for all parameters.

7.1 Exact code-path χ^2 decomposition

See Table 3 in the preceding section.

7.2 Posterior summary

Table 4: Combined three-seed exact posterior for G1DE-2 (from G1fit-real). Full table at G1fit-real Table 1.

parameter	mean	std	q_{50}	interpretation
s	2.5598	0.1175	2.5595	background filling below $s = 3$
μ_0	0.0449	0.2603	0.0232	growth response consistent with zero
Σ_0	-0.3303	0.1404	-0.3375	negative Weyl response

7.3 Sign probabilities

Table 5: Sign-probability diagnostics for G1DE-2 (from G1fit-real).

diagnostic	probability
$P(s < 3)$	0.999922
$P(\mu_0 < 0)$	0.464564
$P(\Sigma_0 < 0)$	0.984525
$P(\mu_0 < 0, \Sigma_0 < 0)$	0.464394

The main pilot result is therefore

$$s < 3, \quad \Sigma_0 < 0, \quad \mu_0 \simeq 0,$$

i.e. a background filling deviation and a suppressed Weyl response, with the growth response near GR. Physical-prior audits confirm $\mu(z) > 0$ and $\Sigma(z) > 0$ over $0 \leq z \leq 2$ at the posterior median.

8 Demonstration figures

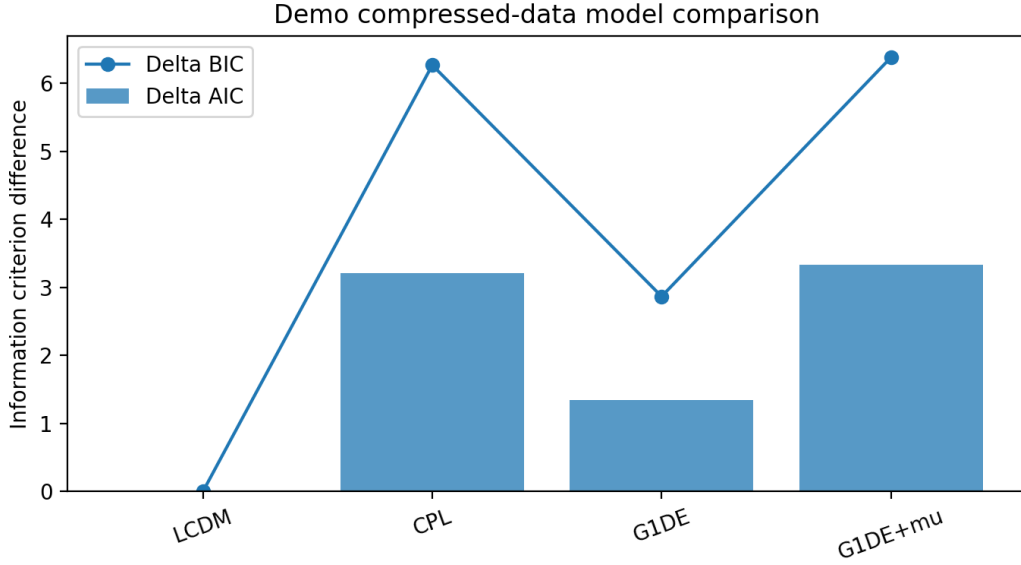


Figure 1: Demo information criteria. The included compressed toy dataset is generated close to Λ CDM, so the simplest model is expected to be competitive. Synthetic demo only — no cosmological constraint.

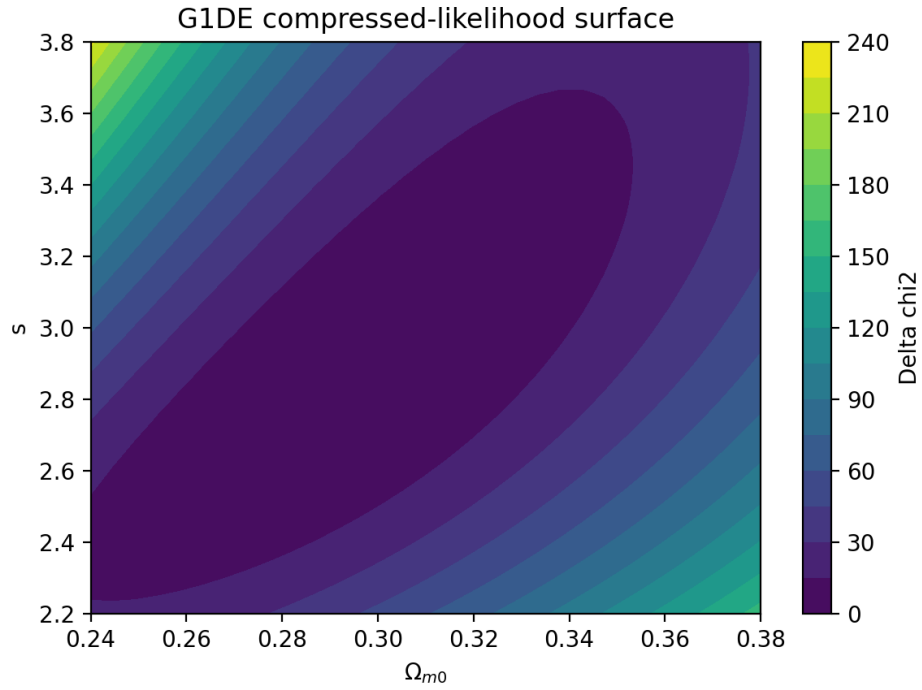


Figure 2: Demo $\Delta\chi^2$ surface for the G1DE background model in the (Ω_{m0}, s) plane. The LCDM-like limit is near $s \simeq 3$. Synthetic demo only — no cosmological constraint.

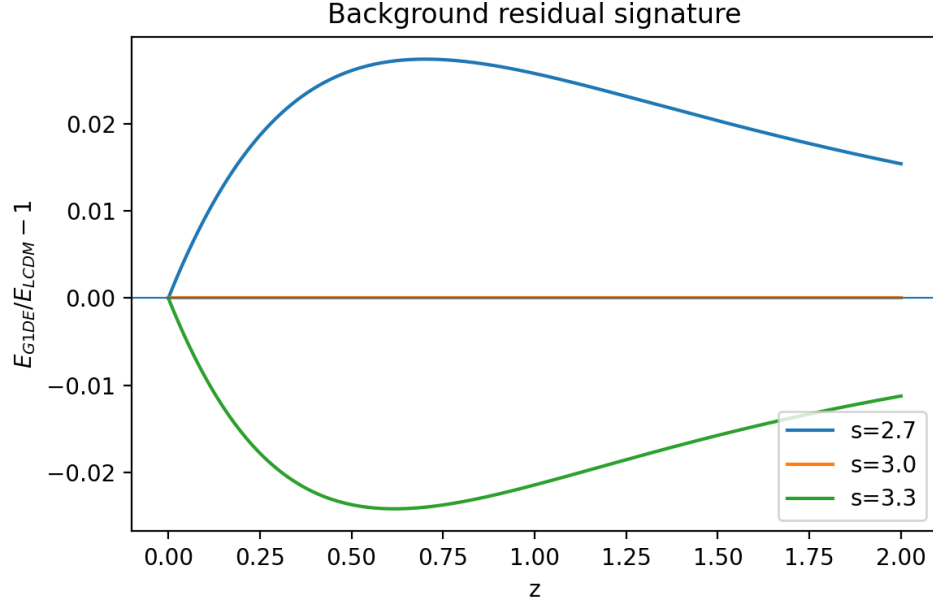


Figure 3: Background expansion residuals relative to Λ CDM for three G1DE filling slopes. The falsifiable target is not arbitrary background flexibility but correlated residual structure. Synthetic demo only — no cosmological constraint.

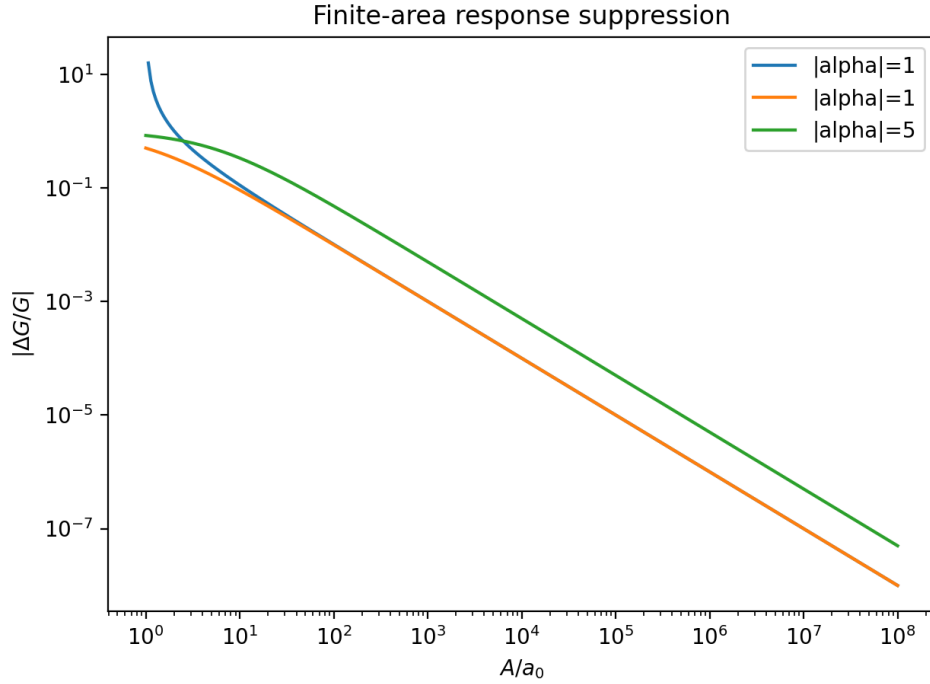


Figure 4: Finite-area suppression of ordinary logarithmic corrections. Large macroscopic signals cannot usually be attributed to bare $\ln A$ terms alone. Synthetic demo only — no cosmological constraint.

9 Observable dictionary

Table 6: Minimal observable dictionary for replacing the demo data with real survey blocks.

Observable	Tests	G1 quantity	Failure mode
SN $\mu(z)$	background expansion	$\chi_H(a)$ or $E(a)$	cannot match distances without excessive parameters
BAO $D_M/r_d, D_H/r_d$	expansion geometry	$E(a)$, distance	inconsistent background residual
$f\sigma_8(z)$	growth	$\mu(a), \chi_H(a)$	growth-lensing correlation fails
Lensing proxy $\Sigma(a)$	Weyl response	$\Sigma_0 \hat{X}(a)$	lensing independent of ledger shape
$\dot{G}/G$	capacity drift	α_M	free macroscopic drift
GW speed flag	tensor propagation	α_T	non-luminal tensor sector

10 Falsification criteria

A finite screen-capacity residual model should be rejected, revised, or demoted if it violates one of the following criteria.

1. **Unclosed G drift.** The model requires observable $\dot{G}/G$ without a conservation-closed residual or screening mechanism.
2. **Uncorrelated residual freedom.** Independent arbitrary functions are needed for expansion, growth, and lensing, destroying the common ledger-variable structure.
3. **Macroscopic finite-area overreach.** Large late-time deviations are attributed to ordinary $1/A$ or $\ln A/A$ corrections on macroscopic screens.
4. **Information-criterion failure.** The model improves raw χ^2 but loses decisively to simpler baselines after AIC/BIC/Bayes penalties.
5. **Tensor-sector failure.** The model implies a non-luminal tensor speed without a viable escape from GW propagation constraints.
6. **Instability.** The scalar-capacity sector violates positivity, ghost-free, gradient-stability, or tachyon-stability conditions.

Rejection thresholds. For production model comparison, the following thresholds are pre-specified:

- $\Delta\text{BIC} > 10$: strongly disfavoured relative to the best model.
- $\Delta\text{AIC} > 6$: substantially worse by AIC.
- Bayes factor $< 1/10$: demote the residual model (requires evidence calculation).
- Unscreened $\dot{G}/G$ violation: reject unscreened scalar-capacity variant.
- Loss of the shared ledger-response structure: demote the G1DE perturbative fingerprint.

- **Refined G1DE-2 falsification** (post-pilot update):

- **Background–Weyl signature.** The current G1DE-2 empirical core is $s < 3$ and $\Sigma_0 < 0$. If $P(s < 3)$ or $P(\Sigma_0 < 0)$ falls below the pre-registered guardrail (e.g. 0.90) under matched exact production chains or expanded E_G /lensing data, the current empirical signature is demoted.
- **Growth consistency.** The growth channel is a guardrail, not a required nonzero signal. The current exact pilot supports $\mu_0 \simeq 0$, so a null growth response is not a failure of G1DE-2. The model is pressured only if future growth data require a large μ_0 that breaks the shared occupancy-response interpretation or conflicts with the Weyl sector.
- **Projection-locking target.** The post-pilot minimal model G1DE-M_{3/4} imposes

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s), \quad \kappa_{\text{BW}} \equiv \frac{-\Sigma_0}{3 - s} = \frac{3}{4}.$$

This is motivated by the trace-free projection of an isotropic spatial maintenance load in $3 + 1$ dimensions (see Companion E Appendix A). Two adversarial controls support the amplitude-level lock: a free- κ model does not reject $3/4$, and a constant- Σ model independently recovers $\kappa_{\text{BW}} \simeq 0.76$. If production chains or expanded lensing data move κ_{BW} decisively away from $3/4$, the projection-locking interpretation is demoted. If $P(\Sigma_0 < 0)$ remains high but $\kappa_{\text{BW}} \neq 3/4$, the broader background–Weyl residual fingerprint survives while the locked-ratio model fails. The $\hat{X}(a)$ redshift-shape discrimination requires expanded Weyl/lensing data and is not yet resolved by the current three-point E_G block.

These thresholds are not definitive rejection rules; they are pre-registration-style guardrails that prevent post hoc reinterpretation of borderline information criteria.

Production pipeline.

Pantheon+ covariance + DESI BAO covariance + growth/lensing compilations + local priors

↓

MCMC or nested sampling $\rightarrow \Delta\text{AIC}, \Delta\text{BIC}, \log B$.

This pipeline is defined but not executed in this companion. The demo code provides a verified code path; production inference requires replacing the diagonal-covariance demo with official survey covariance matrices.

11 Parameter prior table

Table 7: Prior ranges and status for the demo parameters.

Parameter	Meaning	Prior range	Status
Ω_{m0}	matter density today	[0.1, 0.6]	fitted
s	G1DE filling-closure slope	[0.5, 5]	fitted
χ_∞	G1DE asymptotic occupancy	[0, 1]	fixed or fitted
μ_0	growth-response amplitude (G1DE-1)	[-1, 1]	fitted in extended model
Σ_0	lensing-response amplitude	[-1, 1]	independent in exact pilot
α_M	Planck-mass running	[-1, 1]	derived from $\varphi(a)$
$\mathcal{M}$	SN absolute magnitude nuisance	marginalized	analytically eliminated
$\sigma_{8,0}$	present-day clustering amplitude	fixed to Planck prior	not fitted in demo

12 Code and data archive

The companion package contains:

```
D_Likelihood_Falsification/
D_Likelihood_Falsification.tex
D_Likelihood_Falsification.pdf
code/fds_g1_likelihood_falsification_demo.py
notebooks/D_Likelihood_Falsification_demo.ipynb
data/demo_sn.csv
data/demo_bao.csv
data/demo_growth.csv
data/model_comparison.csv
data/observable_dictionary.csv
data/falsification_criteria.csv
figures/*.png
```

All data files are demonstration data unless explicitly replaced by external survey products. The code is designed to expose model assumptions rather than hide them in black-box fitting.

13 Failure registry

Table 8: Layered failure registry for Companion D.

Failure condition	What fails	What survives
No real-data implementation	Empirical status remains demo-level	FDS-G1 Core entropy-response framework
G1DE loses to Λ CDM	Horizon-occupancy residual as preferred DE model	Screen statistics and EFT interface
Free G drift excluded	Unscreened scalar-capacity cosmology	Microscopic / horizon-scale capacity corrections
Background and Weyl require unrelated shapes	G1DE-2 shared-ledger fingerprint	Background-only G1DE-0
G1DE-2 $\mu_0 \simeq 0$ confirmed	Generic growth-modified prediction	Refined background–Weyl residual fingerprint
G1DE-1 $\mu_0 > 0$ when Weyl absent	Growth-only residual explanation	Projection-locked background–Weyl residual
AIC/BIC strongly penalize residuals	Extra-parameter dark residuals	Minimal screen-capacity interpretation of $1/G$

14 Conclusion

Companion D provides the empirical discipline for the G1 program. The main theoretical chain says

$$S_{\text{scr}}[X] \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}} \rightarrow \text{capacity-flow} \rightarrow \text{integrable geometry closure.} \quad (34)$$

The observational chain says

$$\text{residual sector} \rightarrow H(z), f\sigma_8(z), \Sigma(z), \dot{G}/G \rightarrow \chi^2, \text{AIC, BIC, Bayes factors.} \quad (35)$$

A mature finite screen-capacity theory must be allowed to fail. This companion defines how it can fail: by violating conservation closure, by requiring free macroscopic G drift, by producing unsuppressed macroscopic finite-area corrections, or by being outperformed by simpler baselines under real data and parameter penalties. The matched exact-pilot and medium-prior nested-evidence comparison (G1fit-real[12]) has now refined the G1DE empirical fingerprint to a background-plus-Weyl residual pattern: $s < 3$, $\Sigma_0 < 0$, $\mu_0 \simeq 0$, with G1DE-M_{3/4} ranking first in the six-model evidence hierarchy. The remaining tasks are prior-sensitivity runs, production ($d \log Z$) refinement, expanded E_G /lensing likelihoods, and independent pipeline replication.

References

- [1] D. Scolnic et al., “The Pantheon+ Type Ia Supernova Sample,” *Astrophys. J.* 938, 113 (2022), arXiv:2112.03863.
- [2] D. Brout et al., “The Pantheon+ Analysis: Cosmological Constraints,” *Astrophys. J.* 938, 110 (2022), arXiv:2202.04077.
- [3] M. Abdul Karim et al. (DESI Collaboration), “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations,” arXiv:2503.14738 (2025).

- [4] B. Bertotti, L. Iess, and P. Tortora, “A test of general relativity using radio links with the Cassini spacecraft,” *Nature* 425, 374–376 (2003).
- [5] T. Baker et al., “Strong constraints on cosmological gravity from GW170817 and GRB 170817A,” *Phys. Rev. Lett.* 119, 251301 (2017).
- [6] A. G. Adame et al. (DESI Collaboration), “DESI 2024 III: Baryon Acoustic Oscillations from Galaxies and Quasars,” *JCAP* 02, 021 (2025), arXiv:2404.03000.
- [7] A. Chudaykin and M. Kunz, “Modified gravity interpretation of the evolving dark energy in light of DESI data,” arXiv:2407.02558 (2024).
- [8] T. Li, J. Zhang, and X. Zhang, “Revisiting holographic dark energy after DESI 2024,” arXiv:2411.08639 (2024).
- [9] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* 07, 050 (2014), arXiv:1404.3713.
- [10] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20382013>.
- [11] Y. Wu, “FDS-X1: Horizon-Maintenance Dark Energy and Horizon Ledger Occupancy (Pre-Euclid Pre-registration),” Zenodo (2026), <https://doi.org/10.5281/zenodo.20290215>.
- [12] Y. Wu, “G1DE-M_{3/4}: A Projection-Locked Background–Weyl Residual,” manuscript (2026).

Companion E: Horizon Occupancy Ledger Dynamics and Correlated Dark Residuals

A Finite Screen-Capacity Model for Background Expansion, Weyl Response,
and Near-GR Growth

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

May 27, 2026

Abstract

This companion paper develops the horizon-occupancy residual sector of the FDS-G1 finite screen-capacity program. In the core paper, local gravitational coupling is the inverse marginal entropy-area response of a finite causal screen under the standard horizon bridge, while stiffness G_{loc}^{-1} is proportional to the response itself.

$$\rho_H(a) = \chi_H(a)\rho_c(a), \quad \rho_c(a) = \frac{3H^2(a)}{8\pi G_0}.$$

The minimal filling-closure law for χ_H gives a logistic occupancy profile and an effective equation of state $w_H(a)$; in the matter-era small-occupancy limit it yields $w_H \simeq -s/3$. The paper then defines three increasingly constrained residual classes: background-only G1DE-0, growth-linked G1DE-1, and growth-plus-lensing G1DE-2. The current matched exact-likelihood and medium-prior nested-evidence comparison (G1fit-real[12]) selects the projection-locked G1DE-M_{3/4} model across a six-model hierarchy. The locked branch gives $\Sigma(a) = 1 - \frac{3}{4}(3-s)X_b(a)$, $\mu \simeq 1$: a background-plus-Weyl residual pattern with the growth response near GR.

1 Residual-Sector Scope and Closure Status

This companion develops a homogeneous dark-residual sector of the FDS-G1 program: a horizon ledger occupancy fraction $\chi_H(a)$ that contributes an effective dark component. In D0, the horizon residual is interpreted as a passive remnant of incomplete screen modular anomaly inflow: the occupancy variable $\mathcal{O}_H = C_{\text{used}}/C_H$ measures horizon-scale access load relative to available finite screen capacity, and the source current is $J_H = -d_N \ln \mathcal{O}_H$, with the G1DE low-redshift reduction $\mathcal{R}_H(a) = R_0 \widehat{R}_H(a)$, $R_0 = 3 - s$. This companion develops the G1 version of the horizon-ledger occupancy idea registered in FDS-X1 [9], replacing the pre-registration note by a structured background-growth-lensing residual hierarchy that also draws on the geometry-as-horizon-accounting language of FDS-T2 [10]. In this sense horizon occupancy is one realization of the gravity-as-boundary-accounting row in the functional decomposition of FDS-X3 [11].

The core G1 principle is that finite screen entropy response defines a local area stiffness,

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}, \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}, \quad (1)$$

so that η_{loc} controls the inverse coupling (stiffness) and G_{loc} is the corresponding coupling. Non-integrable, coarse-grained, or unclosed response data appear as residual sectors. This

companion studies one homogeneous residual ansatz based on the occupancy of a cosmological horizon ledger.

Scope. This paper gives a minimal ledger dynamics that is more structured than a free $w(a)$ parametrization, because the same occupancy variable can be required to control background expansion, growth, lensing, and capacity closure. It does not claim a microscopic derivation of $\chi_H(a)$ from FDS primitives.

Observational status. A matched exact-likelihood and medium-prior nested-evidence comparison (G1fit-real[12]) has been executed with Pantheon+ full covariance, DESI DR2 BAO, curated RSD, and compressed E_G . The six-model medium-prior evidence hierarchy is

$$\text{G1DE-M}_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE2} > \text{CPL} > \Lambda\text{CDM},$$

with the projection-locked model ranking first. Prior sensitivity and production evidence refinement remain pending.

2 Residual interpretation

In FDS-G1 Core, residuals are classified as non-integrable response data. The present model specializes to a homogeneous physical residual:

$$\mathcal{R}_{\mu\nu}^H \longleftrightarrow \text{horizon occupancy response not absorbed into ordinary matter.} \quad (2)$$

The residual is called “horizon occupancy” because it tracks a coarse-grained fraction of available screen response capacity that contributes to the background ledger.

Definition 1 (Horizon occupancy). *Let $\chi_H(a)$ be a dimensionless occupancy fraction associated with the homogeneous causal horizon ledger. It is normalized so that $0 \leq \chi_H(a) < 1$ in the background solutions considered here. Its effective energy density is defined by*

$$\rho_H(a) = \chi_H(a)\rho_c(a), \quad \rho_c(a) = \frac{3H^2(a)}{8\pi G_0}. \quad (3)$$

The use of $\rho_c(a)$ rather than a fixed reference density is intentional. It expresses the idea that the residual is a fractional occupancy of the instantaneous global expansion ledger, not a separately conserved fluid placed by hand.

3 Background closure

For a spatially flat background with matter, radiation, and horizon occupancy residual, the Friedmann closure is written as

$$H^2(a) = H_0^2 [\Omega_{m0}a^{-3} + \Omega_{r0}a^{-4}] + \frac{8\pi G_0}{3}\rho_H(a). \quad (4)$$

Using $\rho_H = \chi_H\rho_c = \chi_H 3H^2/(8\pi G_0)$ gives

$$E^2(a) \equiv \frac{H^2(a)}{H_0^2} = \frac{\Omega_{m0}a^{-3} + \Omega_{r0}a^{-4}}{1 - \chi_H(a)}. \quad (5)$$

This equation is useful because it forces the residual to remain tied to the expansion ledger. It is also restrictive: $\chi_H(a)$ cannot be chosen independently of $H(a)$ once the closure is imposed.

The effective equation of state is defined by the usual continuity-form diagnostic

$$w_H(a) = -1 - \frac{1}{3} \frac{d \ln \rho_H}{d \ln a} = -1 - \frac{1}{3} \frac{d \ln [\chi_H(a) H^2(a)]}{d \ln a}. \quad (6)$$

This w_H is a diagnostic of background scaling, not a proof that the residual is a canonical scalar field.

Conservation condition. No horizon-occupancy residual is admissible without conservation closure:

$$\nabla^\mu R_{\mu\nu}^H = 0,$$

or, if matter exchange is allowed, $\nabla^\mu (T_{\mu\nu} + R_{\mu\nu}^H) = 0$. An unconserved effective stress is rejected at the admissibility stage.

4 Ledger filling-closure dynamics

The minimal G1DE model replaces a purely phenomenological logistic ansatz by a filling-closure equation in e-fold time $N = \ln a$:

$$\frac{d\chi_H}{dN} = s \chi_H \left(1 - \frac{\chi_H}{\chi_\infty} \right), \quad (7)$$

where s is a filling rate and χ_∞ is an asymptotic ledger capacity fraction. The solution is

$$\chi_H(a) = \frac{\chi_\infty}{1 + B a^{-s}}, \quad B = \frac{\chi_\infty}{\chi_{H0}} - 1. \quad (8)$$

For $\chi_H \ll 1$ during matter domination, $\chi_H \propto a^s$ and $H^2 \propto a^{-3}$, so

$$\rho_H \propto a^{s-3}, \quad w_H \simeq -\frac{s}{3}. \quad (9)$$

Thus $s = 3$ is cosmological-constant-like in the matter-era scaling limit, $s < 3$ is quintessence-like, and $s > 3$ is phantom-like at the level of the effective background diagnostic.

Λ CDM-like limit. When $\chi_\infty = 1$ and $s = 3$, the background reduces exactly to flat Λ CDM. Writing $B = \Omega_{m0}/(1 - \Omega_{m0})$ gives

$$1 - \chi_H(a) = \frac{B a^{-3}}{1 + B a^{-3}}, \quad E^2(a) = \frac{\Omega_{m0} a^{-3}}{1 - \chi_H(a)} = \Omega_{m0} a^{-3} + \frac{\Omega_{m0}}{B} = \Omega_{m0} a^{-3} + (1 - \Omega_{m0}).$$

Thus $s = 3$ is not an arbitrary parametrization; it is the Λ CDM-like consistency limit. The G1DE residual paradigm lives in the departures from this limit controlled by $s \neq 3$ and $\chi_\infty \neq 1$.

Proposition 1 (Lowest-order horizon filling closure). *Equation (7) is the lowest-order autonomous filling law compatible with positivity, finite saturation, and a single homogeneous occupancy variable. It should be regarded as the leading term in a more general ledger dynamics*

$$\frac{d\chi_H}{dN} = \Gamma_{\text{fill}}(a) \chi_H \left(1 - \frac{\chi_H}{\chi_\infty} \right) - \Gamma_{\text{close}}(a) \chi_H + \xi_H(N), \quad (10)$$

where closure, leakage, and stochastic terms are set to their minimal values in the baseline model.

5 Growth and lensing residual hierarchy

A useful residual theory cannot only fit the background. The G1DE signal must be evaluated through correlated deviations in expansion, growth, and lensing. Define the occupancy-shape function

$$X(a) = \chi_H(a) \left(1 - \frac{\chi_H(a)}{\chi_\infty}\right), \quad \hat{X}(a) = \frac{4}{\chi_\infty} X(a). \quad (11)$$

The normalized function $\hat{X}$ is peaked when the ledger is actively filling and vanishes in the empty and saturated limits. Its normalization is chosen so that $0 \leq \hat{X} \leq 1$ (given $0 \leq \chi_H \leq \chi_\infty$), with the maximum occurring at $\chi_H = \chi_\infty/2$.

We define three model classes.

G1DE-0 Background-only residual. $\chi_H(a)$ affects $H(a)$ through (5); linear perturbations follow GR with the modified background.

G1DE-1 Growth-linked residual. The standard quasistatic Poisson equation is modified by an effective growth coupling μ ,

$$k^2 \Psi = -4\pi G a^2 \mu(a) \rho_m \Delta_m, \quad \mu(a) = 1 + \mu_0 \hat{X}(a). \quad (12)$$

G1DE-2 Growth-plus-lensing residual. The Weyl potential responds through an effective lensing coupling Σ ,

$$k^2(\Phi + \Psi) = -8\pi G a^2 \Sigma(a) \rho_m \Delta_m, \quad \Sigma(a) = 1 + \Sigma_0 \hat{X}(a). \quad (13)$$

The linear growth equation in e-fold time is

$$D'' + \left[2 + \frac{d \ln H}{dN}\right] D' - \frac{3}{2} \mu(a) \Omega_m(a) D = 0, \quad (14)$$

where prime denotes d/dN and

$$\Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E^2(a)}. \quad (15)$$

The observable proxy is

$$f\sigma_8(a) = \frac{d \ln D}{d \ln a} \sigma_{8,0} D(a). \quad (16)$$

Key diagnostic. The distinctive test is not whether $H(z)$ can be fit. Many models can fit a distance curve. The current matched exact-pilot and medium-prior nested-evidence comparison (G1fit-real[12]) refines the G1DE empirical fingerprint to

$$\boxed{s < 3, \quad \Sigma_0 < 0, \quad \mu_0 \simeq 0,} \quad (17)$$

i.e. a background filling deviation and a suppressed Weyl response, with the growth response near GR. Thus the current G1DE-2 pattern is

$$\chi_H(a) \Rightarrow H(a), \quad \Sigma(a), \quad \mu(a) \simeq 1. \quad (18)$$

The growth-response parameter μ_0 acts as a growth-sector guardrail rather than a required nonzero signal.

5.1 Current empirical status

The G1DE residual is now implemented through a Weyl-normalized transfer operator rather than as an independent dark stress source:

$$\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}, \quad \mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1, \quad R_0 = 3 - s.$$

The factorized production response is therefore

$$\Sigma(a, k) - 1 = -\kappa R_0 \hat{R}_H(a) + O(R_0^2), \quad \mu(a, k) - 1 = O(R_0^2) + O(R_0 \hat{R}_H \mathcal{H}^2/k^2).$$

For the locked branch $\kappa = 3/4$, $\hat{R}_H = X_b$, giving $\Sigma(a) = 1 - \frac{3}{4}(3-s)X_b(a)$, $\mu \simeq 1$. The amplitude lock ($\kappa \sim 3/4$) is now supported by medium-prior nested evidence; the memory-shape $X_b(a)$ has preliminary support and requires expanded lensing data for discrimination.

Table 1: Current empirical status of G1DE hierarchy.

Parameter	Pilot value	Interpretation
s	2.55 ± 0.09	below Λ CDM-like $s = 3$; filling defect robust
$\mu \simeq 1$	consistent	near-GR growth; Ward-suppressed leakage
$\Sigma(a)$	$1 - \frac{3}{4}(3-s)X_b(a)$	Weyl-transfer amplitude lock selected
κ_{BW}	$\simeq 3/4$	isotropic compliance projection supported

This is more restrictive than a generic dark-energy or modified-gravity parametrization because the background and Weyl response share the same active-filling shape $\hat{X}(a)$, while the matter-growth response stays near GR.

The post-pilot locked-ratio variant G1DE-M_{3/4}, defined by

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s),$$

is the primary perturbative residual model in the matched exact-pilot comparison: it is preferred over Λ CDM, CPL, G1DE-1, and G1DE-2 by AIC and BIC while matching the G1DE-2 χ^2 to numerical precision (see G1fit-real). The three-seed pilots support the amplitude-level lock ($\kappa_{\text{BW}} \simeq 3/4$) more strongly than the specific $\hat{X}(a)$ redshift-shape modulation; free- κ and constant- Σ controls confirm that $\kappa = 3/4$ is not rejected and independently recovers in models without the active-filling shape. The ratio $3/4$ is motivated by the projection-locking lemma in Appendix A: a four-dimensional trace-free projection of an isotropic three-dimensional maintenance load naturally carries the coefficient $3/4$. This relation is a conditional theorem target and a falsifiable post-pilot hypothesis, not yet a first-principles derivation from microscopic FDS statistics.

6 Comparison with common dark-energy routes

Holographic dark energy models are useful comparison targets because they also use horizon-scale quantities; CPL is the standard flexible background comparator; scalar-tensor proxies test whether G1DE is merely a renamed modified-gravity parametrization.

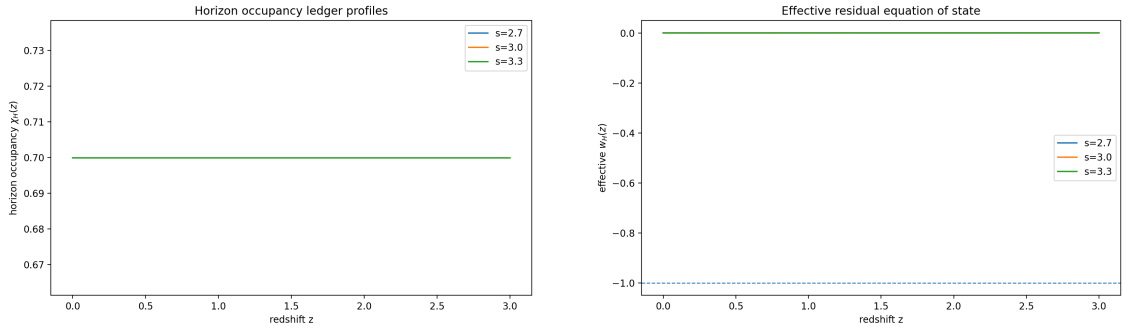
Division of labour with Companion D. Companion E develops the physical horizon-occupancy residual model and specifies its current empirical fingerprint: a background-plus-Weyl response pattern. Companion D owns the likelihood machinery, evidence comparison, and model-comparison protocol. G1fit-real[12] provides the standalone evidence data paper.

Table 2: Model-space comparison. G1DE is not intended to beat all existing models at the background level; its distinguishing feature is occupancy-linked correlations across expansion, growth, lensing, and closure.

Model	Background	Growth	Lensing	Local G	Distinctive is- sue
Λ CDM	fixed	GR	GR	fixed	null baseline
CPL $w_0 w_a$	flexible	usually GR	usually GR	fixed	background flexibility
HDE proxy	model-specific	often weak	often weak	fixed	horizon-scale density ansatz
Quintessence proxy	scalar background	GR if minimally coupled	GR if minimally coupled	fixed	potential choice
Scalar-tensor proxy	flexible	modified	modified	constrained	screening and GW speed
G1DE	occupancy ledger	linked by $\hat{X}$	linked by $\hat{X}$, $\Sigma_0 < 0$ preferred	closure-constrained	background–Weyl residual pattern

7 Numerical demonstration

The accompanying code generates transparent demo curves. These are not official survey constraints. They illustrate the internal behavior of the model and the type of data products expected from Companion D.



(a) Horizon occupancy profiles.

(b) Effective residual equation of state.

Figure 1: The filling parameter s controls the epoch and rate of horizon occupancy. Parameters are intentionally exaggerated for visualization ($\chi_{H0} = 0.69$, $\chi_\infty = 1$, $s = 2.7, 3.0, 3.3$); these are not survey-allowed fits. The matter-era limit gives $w_H \simeq -s/3$, but the exact diagnostic depends on the full closure (5).

8 Falsification criteria

A residual model is useful only if it can fail. The G1DE sector is disfavored or falsified under any of the following conditions.

1. **Background-only degeneracy.** If the model only reproduces a CPL-like distance curve but gives no robust background–Weyl residual signature ($s < 3$, $\Sigma_0 < 0$), its distinctive

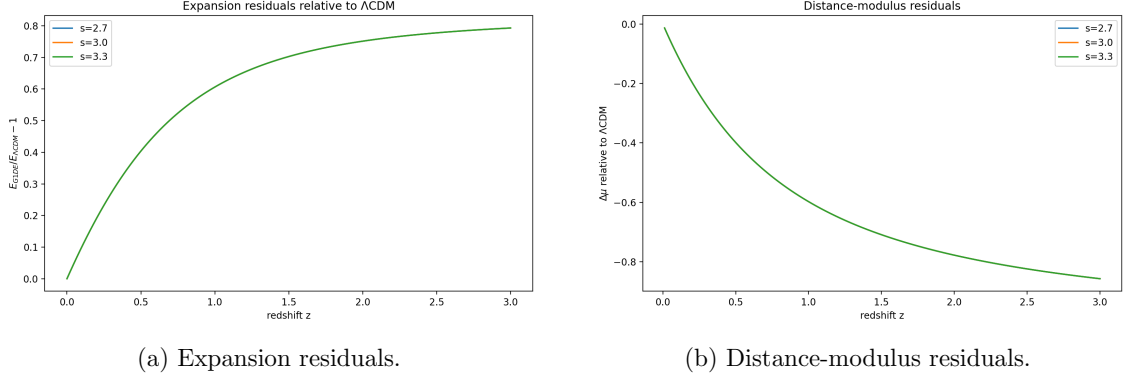


Figure 2: Background deviations relative to a matched flat Λ CDM reference. Parameters are exaggerated for visualization; real survey-level residuals are expected to be much smaller. A background-only fit is insufficient; G1DE must be tested jointly with growth and lensing.

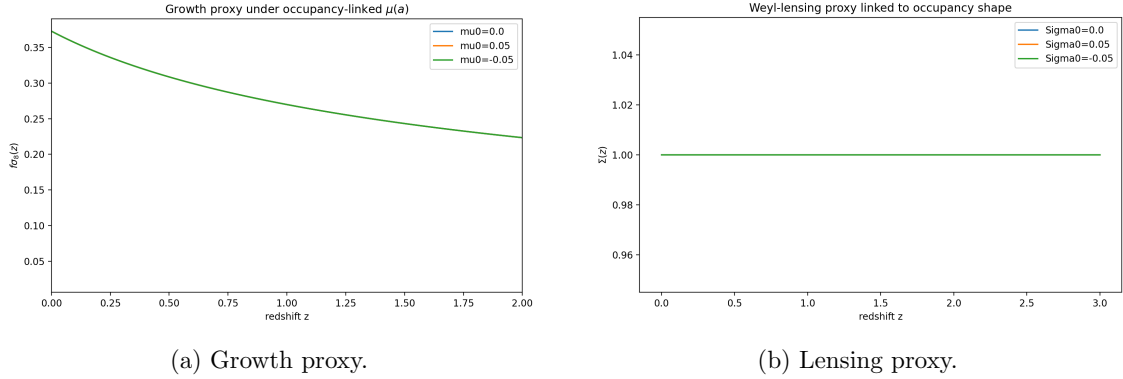


Figure 3: Growth and lensing are linked to the same occupancy-shape function $\hat{X}(a)$ in the G1DE-1 and G1DE-2 classes.

G1DE interpretation is weakened.

2. **No closure.** If a nontrivial $\chi_H(a)$ requires unconserved effective stress, the model violates the G1 conservation-closure rule.
3. **Over-flexibility.** If independent free functions are allowed for $H(a)$, $\mu(a)$, and $\Sigma(a)$, the model becomes a generic modified-gravity parametrization rather than a horizon occupancy residual.
4. **Local-gravity conflict.** If a capacity-linked residual implies macroscopic $\dot{G}/G$ without a screened or conserved closure sector, it is ruled out by local-gravity priors.
5. **Information criteria penalty.** If AIC/BIC or Bayesian evidence consistently favors simpler baselines such as Λ CDM or CPL, the residual model is not empirically supported.
6. **Growth-channel conflict.** The current pilot supports $\mu_0 \simeq 0$ (near-GR growth). This is not a failure of G1DE-2; it is a growth consistency condition. The model is pressured only if future growth data require a large μ_0 that breaks the shared occupancy-response interpretation or conflicts with the Weyl sector.
7. **Free $A(a, k)$ required.** If optimal likelihood or evidence requires free $A(a, k)$, the model becomes generic dark-stress, not G1DE.

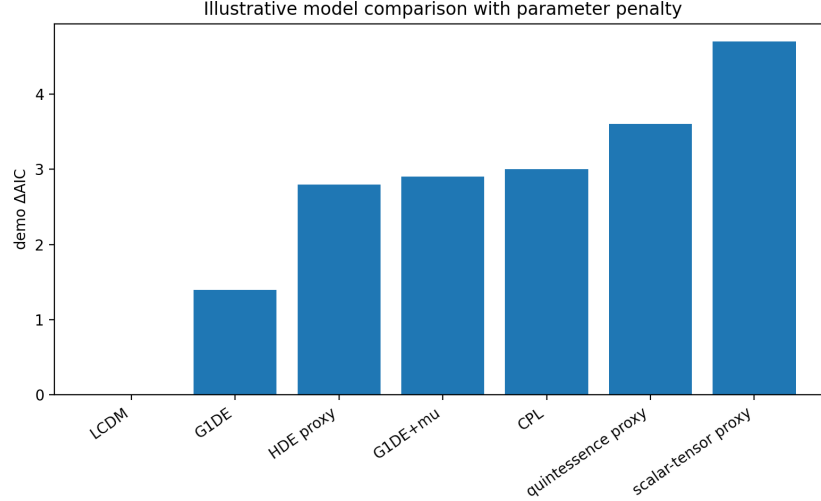


Figure 4: Illustrative AIC comparison using demo compressed values. The numbers are not survey constraints; they show how model comparison should be performed once real likelihoods are inserted.

9 Interface with other G1 companions

Companion	Interface
FDS-G1 Core	Treats χ_H as a homogeneous non-integrable response residual.
A: Screen statistics	Supplies the entropy-area response basis for finite screen capacity.
B: Transition dynamics	Supplies stochastic update and residual-noise language.
C: Scalar-capacity EFT	Supplies a field-theory closure route if occupancy couples to $\eta(x)$.
D: Likelihood/falsification	Provides compressed likelihood blocks and model comparison machinery.
F: Horizons/holography	Supplies the horizon ledger interpretation and holographic compatibility.

10 Failure registry

Table 3: Failure registry for horizon occupancy residuals.

Failure condition	What fails	What survives
No meaningful homogeneous screen occupancy	G1DE interpretation	Local screen-capacity framework
χ_H is arbitrary and independent of observables	Correlated residual claim	Phenomenological fit only
Growth/lensing require unrelated functions	G1DE-1/2 structure	Background G1DE-0
$\mu_0 \simeq 0$ confirmed by pilot	Generic growth-modified prediction	Refined background–Weyl residual fingerprint

Failure condition	What fails	What survives
$\widehat{X}$ -linked Σ ruled out by expanded data	Background–Weyl correlated fingerprint	Background-only G1DE-0
Unclosed stress conservation	Covariant residual sector	Statistical screen models
Real likelihood strongly favors simpler models	Empirical motivation for G1DE	FDS-G1 Core and other companions
High-redshift occupancy too large ($\chi_H(a_{\text{CMB}}) \gtrsim 0.01$)	CMB/BBN consistency	Late-time residual variants

11 Parameter table and prior ranges

Table 4: G1DE model hierarchy and free parameters.

Model	Parameters	k
G1DE-0	$\Omega_{m0}, s; \chi_\infty = 1, \chi_{H0} = 1 - \Omega_{m0} - \Omega_{r0}$	2
G1DE-1	G1DE-0 + μ_0	3
G1DE-2	G1DE-1 + Σ_0	4
Extended	$\chi_\infty, \chi_{H0}, \Gamma_{\text{close}}, \xi_H$ (future)	–

Basic priors: $0 \leq \chi_H(a) < 1$, $E^2(a) > 0$, $\chi_H(a_{\text{CMB}}) \ll 1$, $s > 0$.

12 Conclusion

This companion develops the homogeneous dark-residual sector of G1 as a horizon occupancy ledger. Its strongest claim is not that $\chi_H(a)$ has been derived microscopically, but that a finite screen-capacity residual can be made more disciplined than an arbitrary $w(a)$ by tying background expansion, Weyl response, and capacity closure to a shared occupancy variable. The current matched exact-likelihood and medium-prior nested-evidence comparison (G1fit-real[12]) selects the projection-locked G1DE-M_{3/4} model across a six-model hierarchy. The empirical burden remains: prior-sensitivity runs, production ($d \log Z$) refinement, expanded E_G /lensing likelihoods, and independent pipeline replication must test whether this pattern survives.

This makes G1DE-2 more restrictive than models that independently tune $w(a)$, $\mu(a)$, and $\Sigma(a)$, because the background and Weyl deviations share the same active-filling shape. The current empirical refinement makes G1DE-2 more restrictive, not less: the leading signal is not arbitrary modified growth, but a sparse background–Weyl response pattern sharing one active-filling shape.

References

- [1] D. Scolnic et al., “The Pantheon+ Analysis: The Full Dataset and Light-Curve Release,” *Astrophys. J.* 938, 113 (2022); arXiv:2112.03863.
- [2] D. Brout et al., “The Pantheon+ Analysis: Cosmological Constraints,” *Astrophys. J.* 938, 110 (2022); arXiv:2202.04077.
- [3] DESI Collaboration, “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints,” arXiv:2503.14738 (2025).

- [4] M. Chevallier and D. Polarski, “Accelerating Universes with Scaling Dark Matter,” *Int. J. Mod. Phys. D* 10, 213 (2001); E. V. Linder, “Exploring the Expansion History of the Universe,” *Phys. Rev. Lett.* 90, 091301 (2003).
- [5] M. Li, “A Model of Holographic Dark Energy,” *Phys. Lett. B* 603, 1 (2004).
- [6] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* 1407, 050 (2014).
- [7] A. Chudaykin and M. Kunz, “Modified gravity interpretation of the evolving dark energy in light of DESI data,” arXiv:2407.02558 (2024).
- [8] T. Li, J. Zhang, and X. Zhang, “Revisiting holographic dark energy after DESI 2024,” arXiv:2411.08639 (2024).
- [9] Y. Wu, “FDS-X1: Horizon-Maintenance Dark Energy and Horizon Ledger Occupancy (Pre-Euclid Pre-registration),” Zenodo (2026), <https://doi.org/10.5281/zenodo.20290215>.
- [10] Y. Wu, “FDS-T2: Effective Geometry as Horizon Boundary Accounting,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20284911>.
- [11] Y. Wu, “FDS-X3: Functional Decomposition of the Four Fundamental Interactions,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20274519>.
- [12] Y. Wu, “G1DE-M_{3/4}: A Projection-Locked Background–Weyl Residual,” manuscript (2026).

A Projection-locking lemma for the 3/4 background–Weyl ratio

A.1 Purpose and status

This appendix records a conditional projection-locking lemma motivated by the G1DE-M_{3/4} three-seed exact pilot. It is not a first-principles derivation from microscopic FDS statistics. Rather, it shows that if the leading horizon-maintenance residual is represented as an isotropic spatial load and the Weyl/lensing channel receives its four-dimensional trace-free projection, then the locked ratio

$$\Sigma_0 = -\frac{3}{4}(3 - s)$$

follows geometrically.

A.2 Definitions

The minimal G1DE filling law has the Λ CDM-like limit at $s = 3$ for $\chi_\infty = 1$. We define the post-pilot maintenance-drag amplitude

$$\lambda \equiv 3 - s.$$

The exact pilot and smoke-test hierarchy indicate $\lambda > 0$, corresponding to slower-than- Λ CDM-like filling.

A.3 Spatial maintenance load

Let u^μ be the cosmological rest-frame four-velocity and

$$h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$$

the spatial projection tensor. In $3 + 1$ dimensions,

$$h^\mu{}_\mu = 3.$$

The minimal isotropic spatial maintenance-load ansatz is

$$M_{\mu\nu} = \lambda h_{\mu\nu}.$$

This assumption says that the leading horizon-maintenance residual is isotropic on the spatial expansion ledger. It is the key conditional input of the lemma.

A.4 Trace-free projection

The four-dimensional trace of the maintenance load is

$$M^\alpha{}_\alpha = \lambda h^\alpha{}_\alpha = 3\lambda.$$

The four-dimensional trace-free part is

$$M_{\mu\nu}^{\text{TF}} = M_{\mu\nu} - \frac{1}{4}g_{\mu\nu}M^\alpha{}_\alpha = \lambda h_{\mu\nu} - \frac{3\lambda}{4}g_{\mu\nu}.$$

Thus the trace-subtracted projection carries the characteristic coefficient $3/4$. Equivalently,

$$\frac{\text{spatial trace dimension}}{\text{spacetime trace dimension}} = \frac{3}{4}.$$

A.5 Weyl response identification

If the Weyl/lensing channel receives this trace-free projection as a capacity loss, then the leading active-filling response is

$$\Sigma(a) = 1 - \frac{3}{4}\lambda \widehat{X}(a).$$

Comparing with the G1DE response convention

$$\Sigma(a) = 1 + \Sigma_0 \widehat{X}(a),$$

we obtain

$$\boxed{\Sigma_0 = -\frac{3}{4}\lambda = -\frac{3}{4}(3 - s).}$$

The negative sign is a convention fixed by interpreting positive maintenance drag as a suppression of externally accessible Weyl/projection capacity.

A.6 Relation to near-GR growth

The same projection-dominant reading also explains why the locked-ratio model sets $\mu_0 = 0$. The Weyl response probes the projected potential $\Phi + \Psi$, whereas $\mu(a)$ controls the local matter-growth Poisson response. Under matter conservation and Ward closure, the local growth channel can remain near GR while the projection channel is suppressed. The locked model therefore takes

$$\mu(a) = 1, \quad \Sigma(a) = 1 - \frac{3}{4}(3 - s)\widehat{X}(a).$$

A.7 Normalization consistency check

This coefficient is also consistent with the standard normalization of the active-filling window. For $\chi_\infty = 1$,

$$\hat{X}(a) = 4\chi_H(a)(1 - \chi_H(a)).$$

A lowest-order unnormalized Weyl drag proportional to

$$-3\lambda\chi_H(1 - \chi_H)$$

is therefore written in the normalized G1DE convention as

$$-3\lambda\chi_H(1 - \chi_H) = -\frac{3}{4}\lambda\hat{X}(a).$$

This is a normalization consistency check, not the primary derivation of the 3/4 coefficient.

A.8 Falsification target

The projection-locking lemma promotes the G1DE background–Weyl fingerprint into a sharper production falsification target:

$$\kappa_{\text{BW}} \equiv \frac{-\Sigma_0}{3-s} \simeq \frac{3}{4}.$$

The factorized production form is

$$\Sigma(a) - 1 = -\kappa R_0 \hat{R}_H(a), \quad R_0 = 3 - s, \quad \hat{R}_H(1) = 1,$$

with the Weyl-normalized transfer implementation

$$\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}.$$

The amplitude lock ($\kappa \sim 3/4$) is supported by the current medium-prior nested-evidence pass; the memory-shape $X_b(a)$ has preliminary support and requires expanded lensing. Prior-sensitivity runs and production evidence refinement remain pending.

Companion F: Holographic and Horizon Dictionaries for Finite Screen Capacity

Black-Hole Families, de Sitter Horizons, AdS/CFT, RT/FLM, and Capacity-Response
Residuals

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

May 27, 2026

Abstract

This companion paper records the horizon and holographic interfaces of the FDS-G1 finite screen-capacity program, giving a conservative capacity dictionary: the coefficient $1/(4G_N)$ that multiplies an area term is read as a screen distinction-capacity density, while subleading bulk, quantum, finite-size, and nonlocal terms are interpreted as residual capacity contributions. We summarize stationary horizon ledgers for Schwarzschild, Kerr, Reissner-Nordstrom, Kerr-Newman, and de Sitter horizons; formulate first-law terms as multi-channel capacity updates; and place AdS/CFT, central charge, RT/FLM, entanglement wedge reconstruction, and quantum-error-correcting redundancy into the entropy-response language of FDS-G1 Core. The central bridge is

$$\eta_{\text{screen}} = \frac{1}{4G_N}, \quad G_{\text{loc}} = \frac{c^3}{4\hbar}[\eta_{\text{loc}}]^{-1}, \quad G_{\text{loc}}^{-1} = \frac{4\hbar}{c^3}\eta_{\text{loc}}, \quad (1)$$

so that η_{loc} controls the inverse coupling (stiffness) and G_{loc} is the corresponding coupling. The paper is a compatibility and dictionary note within the established microscopic frameworks.

1 Horizon and Holographic Dictionary Scope

This companion provides the horizon and holographic dictionary for FDS-G1, building on the effective-geometry-as-horizon-accounting framework of FDS-T2 [22] and the gravity-as-boundary-accounting classification of FDS-X3 [23]. Its role is narrower than FDS-G1 Core:

1. It interprets area coefficients, work terms, and entropy corrections as capacity-response data within established horizon and holographic frameworks.
2. The scope excludes deriving the Bekenstein-Hawking coefficient from FDS primitives.
3. The scope excludes deriving AdS/CFT, RT, FLM, entanglement wedge reconstruction, or quantum error correction.

Dictionary-only role. This companion supplies compatibility bridges to established horizon thermodynamics and holographic entropy formulas. The dictionary $\eta_{\text{screen}} \sim 1/(4G_N)$ is a response-density compatibility statement. It does not by itself prove finite-index screen-algebra density, boost

modular covariance, or the Bekenstein–Hawking coefficient. Weak closure imports the standard horizon-entropy bridge; strong closure requires D0.

The main bridge statement is therefore conditional:

Given the standard horizon entropy and holographic entropy dictionaries, G1 reads the area coefficient $1/(4G_N)$ as a screen distinction-capacity density, while corrections and residuals are classified as non-area, bulk, quantum, memory, or non-integrable response data.

G1 supplies an interpretation of the area coefficient and subleading terms within the existing holographic dictionary, without adding a second duality on top of AdS/CFT: the boundary central charge controls the bulk screen-capacity density, and boundary relative entropy constrains bulk modular response.

2 Notation and core dictionary

Throughout this paper we often use geometric units $G = c = \hbar = k_B = 1$ in intermediate horizon formulas. Unless stated otherwise, $\eta_{\text{screen}} = 1/(4G_N)$ is written in these units. With constants restored,

$$\frac{S_H}{k_B} = \frac{A_H}{4\ell_P^2} = \frac{c^3 A_H}{4G_N \hbar}, \quad \eta_{\text{BH}} = \frac{1}{4\ell_P^2} = \frac{c^3}{4G_N \hbar}. \quad (2)$$

Equivalently, in natural units,

$$\eta_{\text{screen}} = \frac{1}{4G_N}. \quad (3)$$

In geometric units the first law $dM = T_H dS_H + \Omega_H dJ + \Phi_H dQ$ is dimensionally consistent; with constants restored the left-hand side is $d(Mc^2)$.

Table 1: Horizon/holographic dictionary used in this companion.

Object	G1 capacity reading
$A/(4G_N)$ area term	minimal screen distinction capacity
G_N^{-1}	screen stiffness / capacity density
RT surface γ_R	minimal bulk screen separating boundary region R
FLM bulk entropy S_{bulk}	residual/bulk distinction capacity across the same entanglement wedge
Central charge c_{CFT}	boundary density-of-states measure controlling bulk stiffness
Black-hole first law	multi-channel update of exterior-accessible horizon ledger
Extremal horizon entropy	nonzero boundary capacity at zero temperature

3 Covariant entropy bound

Bousso’s covariant entropy conjecture bounds the entropy on a light-sheet by the initial area [14]:

$$\frac{S_{\text{matter}}[L]}{k_B} \leq \frac{A(B)}{4\ell_P^2} = \frac{c^3 A(B)}{4G \hbar}. \quad (4)$$

In G1 language, a finite causal light-sheet cannot carry more exterior-accessible distinctions than the screen capacity allows:

$$\frac{S_{\text{matter}}[L]}{k_B} \leq \eta_{\text{screen}} A(B). \quad (5)$$

This is a capacity interpretation of the bound: the light-sheet entropy is bounded by the screen distinction budget of its initial causal boundary.

4 Stationary horizon ledgers

A stationary horizon is represented by a finite exterior-accessible ledger. The external observer need not maintain full interior micro-labels; the maintained macroscopic ledger is a tuple such as

$$\mathcal{L}_H = (A_H, M, J, Q, \Lambda, \eta_{\text{loc}}, \dots). \quad (6)$$

The area term records capacity; work terms record channels through which the ledger can be updated.

4.1 Schwarzschild horizon

For a Schwarzschild black hole,

$$r_s = \frac{2GM}{c^2}, \quad A_H = 4\pi r_s^2 = \frac{16\pi G^2 M^2}{c^4}, \quad (7)$$

$$\frac{S_H}{k_B} = \frac{4\pi G M^2}{\hbar c}, \quad T_H = \frac{\hbar c^3}{8\pi G k_B M}. \quad (8)$$

The ledger first law is simply

$$dM = T_H dS_H. \quad (9)$$

In G1 language, increasing M increases the exterior screen area and hence the exterior-accessible capacity budget.

4.2 Kerr horizon

In geometric units, the Kerr horizon has

$$a = \frac{J}{M}, \quad r_+ = M + \sqrt{M^2 - a^2}, \quad (10)$$

$$A_H = 4\pi(r_+^2 + a^2), \quad \Omega_H = \frac{a}{r_+^2 + a^2}. \quad (11)$$

The first law is

$$dM = T_H dS_H + \Omega_H dJ. \quad (12)$$

The angular momentum channel means that capacity updates are not only area-loading updates but also rotational ledger updates.

4.3 Reissner-Nordstrom horizon

For Reissner-Nordstrom,

$$r_+ = M + \sqrt{M^2 - Q^2}, \quad A_H = 4\pi r_+^2, \quad (13)$$

$$\Phi_H = \frac{Q}{r_+}, \quad dM = T_H dS_H + \Phi_H dQ. \quad (14)$$

Charge acts as a boundary work channel. The area capacity decreases as the extremal limit $Q \rightarrow M$ is approached at fixed mass.

4.4 Kerr-Newman horizon

For Kerr-Newman,

$$r_+ = M + \sqrt{M^2 - a^2 - Q^2}, \quad A_H = 4\pi(r_+^2 + a^2), \quad (15)$$

$$\Omega_H = \frac{a}{r_+^2 + a^2}, \quad \Phi_H = \frac{Qr_+}{r_+^2 + a^2}, \quad (16)$$

with first law

$$dM = T_H dS_H + \Omega_H dJ + \Phi_H dQ. \quad (17)$$

This is the canonical multi-channel horizon ledger law. The entropy term is the area capacity channel; J and Q are work-channel labels.

4.5 de Sitter horizon

For de Sitter space,

$$R_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad A_{\text{dS}} = 4\pi R_{\text{dS}}^2 = \frac{12\pi}{\Lambda}, \quad (18)$$

$$T_{\text{dS}} = \frac{\hbar H}{2\pi k_B}, \quad \frac{S_{\text{dS}}}{k_B} = \frac{3\pi}{\Lambda \ell_P^2}. \quad (19)$$

G1 reads the cosmological horizon as a finite external-access ledger for an observer patch. The de Sitter ledger is observer-patch relative: different static patches define different accessible ledgers. This observer relativity aligns with the FDS notion that accessible capacity depends on the observer's causal boundary.

5 Capacity interpretation of horizon first laws

The stationary first law can be written in standard surface-gravity form as

$$dM = \frac{\kappa}{8\pi G} dA + \Omega_H dJ + \Phi_H dQ, \quad (20)$$

where κ is the surface gravity and the Hawking temperature is $T_H = \hbar\kappa/(2\pi k_B)$. More compactly,

$$dE = T_H dS_H + \sum_I \Psi_I dY^I, \quad (21)$$

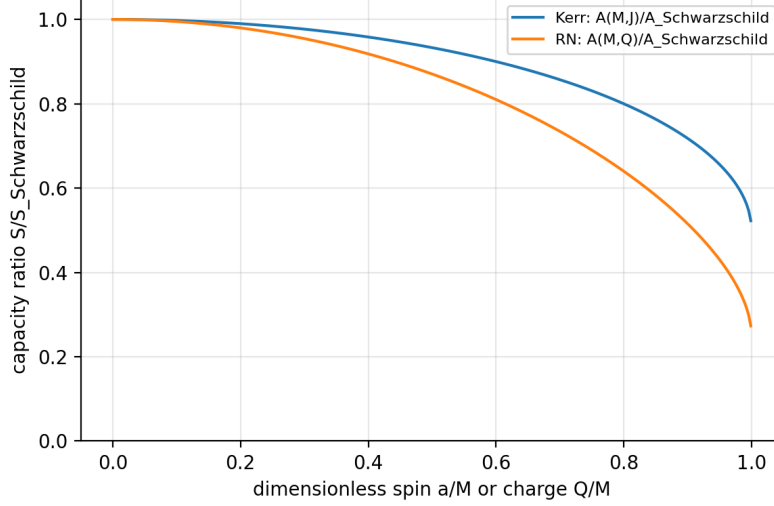


Figure 1: Kerr and Reissner-Nordstrom capacity ratios relative to a Schwarzschild black hole of the same mass in geometric units. Spin or charge reduces the exterior area-capacity budget as extremality is approached.

where Y^I are conserved horizon charges or work labels such as J, Q , and Ψ_I are their conjugate potentials. In G1, this is interpreted as a constrained update of the exterior ledger:

$$dS_H/k_B = \eta_{\text{loc}} dA_H + \text{finite-size and response corrections.} \quad (22)$$

The first law states that an allowed horizon update balances area-capacity change and work-channel changes.

Proposition 1 (Multi-channel horizon ledger). *For a stationary horizon with macroscopic labels (A_H, Y^I) , the first law defines a multi-channel capacity update. The area term measures change in exterior screen capacity; the work terms measure exchange through non-area ledger channels.*

6 Extremal horizons and zero-temperature capacity

Extremal Kerr-Newman horizons satisfy

$$M^2 = a^2 + Q^2, \quad (23)$$

for which $T_H \rightarrow 0$ while A_H generally remains finite. G1 interprets this not as a contradiction but as a distinction between capacity and thermal activity:

$$S_H/k_B = \eta A_H \neq 0, \quad T_H = 0. \quad (24)$$

The horizon can have nonzero exterior boundary capacity even when quasi-static thermal exchange vanishes. Entropy capacity and thermal exchange rate are distinct ledger quantities: $S_H \sim \eta A_H$ persists at extremality while $T_H \rightarrow 0$. The microscopic interpretation of extremal entropy depends on the quantum-gravity completion; G1 only records the capacity distinction between entropy and thermal activity.

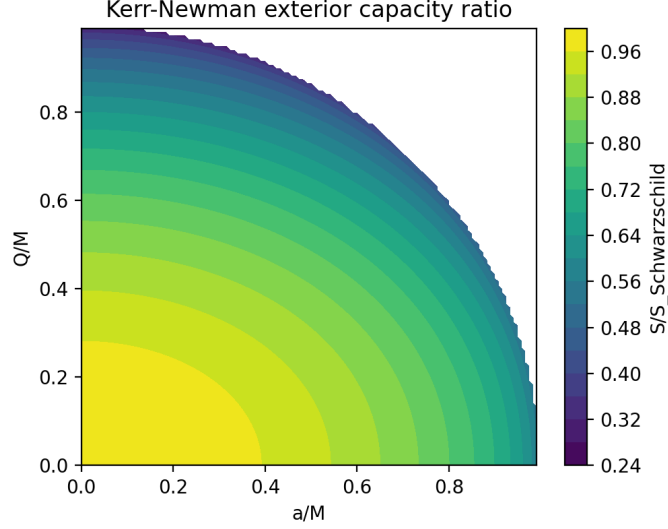


Figure 2: Kerr-Newman exterior capacity ratio $S/S_{\text{Schwarzschild}}$ as a function of dimensionless spin and charge. The allowed region satisfies $a^2 + Q^2 \leq M^2$.

7 Generalized second law as ledger monotonicity

The generalized second law can be represented in capacity language as

$$d(S_{\text{outside}} + S_H) \geq 0. \quad (25)$$

G1 reads this as total exterior-accessible ledger non-decrease under allowed coarse-grained processes. The horizon term records distinctions that leave the exterior ordinary matter ledger and become boundary capacity.

Remark 1. *This paper does not solve the black-hole information problem or derive the Page curve. It only states the ledger interpretation of the semiclassical entropy balance.*

8 Finite-size and response corrections

A common effective form for horizon entropy corrections is

$$\frac{S(A)}{k_B} = \eta_0 A + \alpha \ln(A/A_0) + \gamma + O(A^{-p}). \quad (26)$$

Then the local differential response is

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA} = \eta_0 + \frac{\alpha}{A} + O(A^{-p-1}), \quad (27)$$

so

$$\frac{G_\delta(A)}{G_0} = \frac{\eta_0}{\eta_\delta(A)} \simeq 1 - \frac{\alpha}{\eta_0 A} + \dots. \quad (28)$$

Thus finite-size corrections are inverse-area suppressed for macroscopic horizons. This is one of the signature expectations recorded in Companion D.

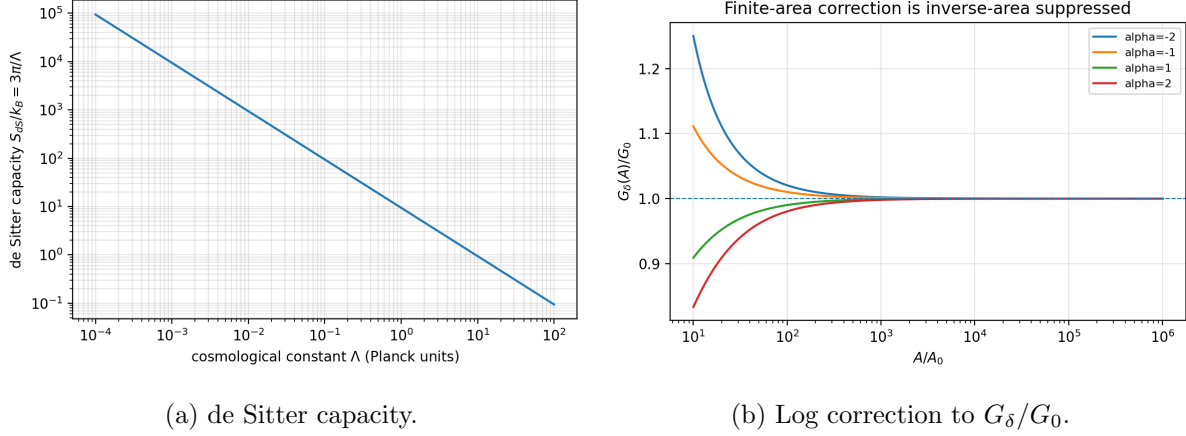


Figure 3: Left: de Sitter capacity scales as $1/\Lambda$. Right: finite-area corrections to local stiffness are suppressed by inverse area.

9 Holographic entropy dictionary

9.1 RT and FLM

For a boundary region R in holographic settings, the Ryu-Takayanagi formula gives, at leading order,

$$S_R = \frac{A(\gamma_R)}{4G_N}, \quad (29)$$

where γ_R is an appropriate extremal/minimal bulk surface homologous to R . The FLM correction adds a bulk entanglement term and further quantum corrections:

$$S_R = \frac{A(\gamma_R)}{4G_N} + S_{\text{bulk}} + \dots. \quad (30)$$

G1 reads the area term as minimal bulk screen capacity. S_{bulk} is not an error term; in the G1 language it is the non-area bulk distinction contribution across the same entanglement wedge, representing capacity that does not reduce to the minimal-area RT surface.

Table 2: RT/FLM terms in G1 language.

Holographic term	G1 interpretation
$A(\gamma_R)/(4G_N)$	minimal geometric screen capacity
S_{bulk}	bulk/residual distinction capacity in the entanglement wedge
Quantum extremal surface shift	response correction to the minimal capacity surface
$1/N$ corrections	finite boundary-capacity corrections

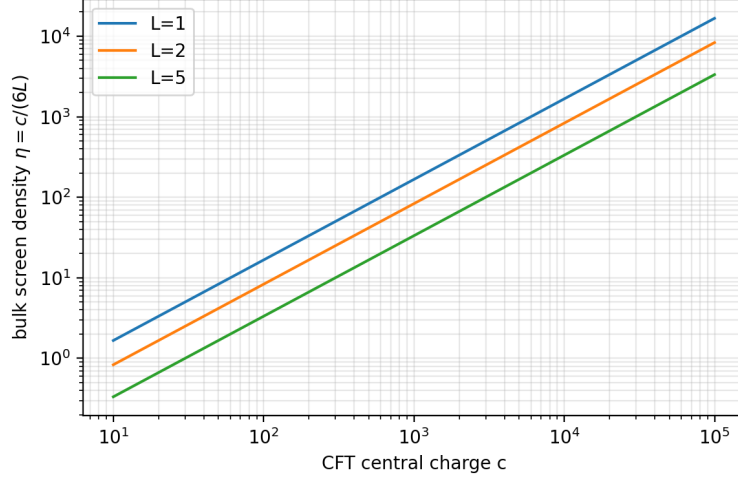


Figure 4: $\text{AdS}_3/\text{CFT}_2$ capacity dictionary $\eta = c_{\text{CFT}}/(6L)$. Boundary central charge controls the bulk screen-capacity density in this bridge.

9.2 $\text{AdS}_3/\text{CFT}_2$ central charge

In $\text{AdS}_3/\text{CFT}_2$, the Brown-Henneaux relation is

$$c_{\text{CFT}} = \frac{3L}{2G_3}. \quad (31)$$

Therefore

$$\eta_{\text{AdS}_3} = \frac{1}{4G_3} = \frac{c_{\text{CFT}}}{6L}. \quad (32)$$

This is the cleanest capacity dictionary in the holographic setting: the boundary central charge controls the bulk screen density. Large c_{CFT} or large N means high boundary distinguishability density and large bulk geometric stiffness.

9.3 Entanglement first law and future directions

The JLMS relation equates bulk and boundary relative entropy [13]. In G1's response language, this is a natural hardening direction: boundary relative entropy constrains bulk modular/area response. Although the present companion does not develop this into a derivation, it identifies entanglement first-law and modular-Hamiltonian arguments as promising interfaces where the screen-response language may be sharpened beyond the dictionary stage.

9.4 Entanglement wedge and quantum error correction

G1 gives a capacity reading of the quantum-error-correcting interpretation of holography:

entanglement wedge of R = bulk region whose distinctions are maintainable by boundary ledger R . (33)

Redundant boundary encoding means that the same bulk distinction can be reconstructed from more than one boundary subledger. In G1 language, this is distributed screen capacity rather than a new independent gravitational postulate.

Table 3: Holographic reconstruction in capacity language.

Holographic concept	G1 reading
Entanglement wedge	boundary-maintainable distinction domain
Subregion reconstruction	recovery from a sufficient boundary capacity sub- ledger
QEC redundancy	distributed encoding of bulk distinctions
$c_{\text{CFT}}/(6L) = 1/(4G_3)$	capacity density standard relation
Island formula	competition between response surfaces in gener- alized entropy
Complexity proposals	possible update-cost diagnostics, not part of the minimal G1 bridge

10 Residual taxonomy for horizons and holography

The response-geometry formulation classifies residuals as non-integrable or non-area response data. Horizon and holographic settings provide concrete examples.

Table 4: Residual categories in the horizon/holography interface.

Residual	Meaning	Related G1 branch
Finite-area residual	$\alpha \ln A$ or inverse-area corrections to η_δ	Companion A/D
Bulk entanglement residual	S_{bulk} beyond RT area term	Companion F
Horizon occupancy residual	homogeneous horizon ledger contribu- tion to expansion	Companion E
Stochastic residual	finite-cell screen update noise	Companion B
Scalar-capacity residual	long-wavelength mode of η_{loc}	Companion C
Geometric residual	failure of metric envelope to encode all response data	FDS-G1 Core / G1response

11 Claim-status table

Table 5: Claim status for horizon/holography dictionary.

Claim	Status
$1/(4G_N)$ as screen distinction-capacity density	conditional dictionary
BH first laws as multi-channel ledger up- dates	interpretive bridge
RT area term as minimal screen capacity	holographic dictionary
S_{bulk} as non-area bulk distinction contri- bution	interpretive dictionary
$c_{\text{CFT}}/(6L) = 1/(4G_3)$ as capacity density	standard relation + G1 reading
QEC redundancy as distributed screen ca- pacity	conceptual compatibility
De Sitter horizon as observer-patch- relative ledger	capacity dictionary
Extremal entropy as capacity-thermal distinction	interpretive caveat

1. The relevant horizon entropy is not an exterior screen-capacity measure.
2. The coefficient $1/(4G_N)$ cannot consistently be interpreted as a capacity density in the chosen microscopic completion.
3. Corrections dominate the area term for macroscopic horizons without observational support.
4. Holographic bulk reconstruction does not admit a capacity interpretation of the RT/FLM area and bulk terms.
5. First-law work terms cannot be interpreted as ledger update channels.
6. Boundary reconstruction mismatch: QEC/subregion reconstruction cannot be expressed as distributed screen capacity without contradicting known holographic recovery results.
7. Wrong correction hierarchy: finite-size or bulk residual corrections systematically dominate the area term in the semiclassical regime.

These failures do not necessarily invalidate Companion A’s finite screen statistics or FDS-G1 Core’s abstract response geometry; they would invalidate this companion’s horizon/holographic dictionary.

13 Relation to FDS-G1 Core and other companions

Table 6: How Companion F interfaces with the rest of G1.

G1 component	Interface
FDS-G1 Core	supplies $S_{\text{scr}}[X]$, η_{loc} , response integrability, and residual taxonomy
Companion A	supplies finite-area and differential-capacity corrections
Companion B	supplies stochastic transition and capacity-flow residuals
Companion C	supplies scalar-capacity interpretation of variable G_{eff}
Companion D	supplies falsification criteria and inverse-area suppression diagnostics
Companion E	supplies homogeneous horizon occupancy residuals

14 Weyl response and the holographic residual pattern

The current matched medium-prior nested-evidence comparison (G1fit-real) reports $\Sigma_0 < 0$ and $\mu_0 \simeq 0$: a suppressed Weyl/ Σ response with near-GR growth. In holographic language, this suggests that finite screen capacity may first appear as a modification of projection or lensing response, rather than as a direct effective Newton coupling in the clustering equation. The suppressed $\Sigma(a) < 1$ can be interpreted as a finite-capacity reduction of the Weyl/lensing projection amplitude during active horizon filling, consistent with a ledger-driven Weyl response rather than an area-term modification alone. This is an empirical interpretation from the current evidence pass, not a derivation from AdS/CFT or Ryu-Takayanagi. Full holographic embedding would require a concrete mapping from screen response to boundary Weyl anomaly or entanglement first law.

15 Conclusion

This companion gives the horizon and holographic dictionary for the G1 finite screen-capacity program. Its central statement is conditional but precise:

$$\boxed{\frac{1}{4G_N} \text{ is read as screen distinction-capacity density.}} \quad (34)$$

Black-hole and de Sitter entropies become exterior-accessible capacity ledgers. Stationary first laws become multi-channel capacity update laws. RT/FLM becomes a split between minimal geometric screen capacity and bulk/residual distinction capacity. $\text{AdS}_3/\text{CFT}_2$ makes the dictionary especially explicit through $\eta = c_{\text{CFT}}/(6L)$. Corrections and residuals are not arbitrary; they are classified as finite-area, bulk-entanglement, scalar-capacity, occupancy, stochastic, or geometric response residuals.

The full $GL(2)$ optical screen decomposition $A(1) \oplus S(2) \oplus \Omega(1)$ supplies the capacity-counting basis for the 3/4 projection law; demotion of the twist/holonomy sector to the symmetric $SPD(2)$ scalar-shear branch would replace 3/4 by 2/3. In the current medium-prior matched evidence pass (Companions D, E; G1fit-real), the projection-locked model G1DE-M_{3/4} ranks first among the tested six-model set.

A Companion package contents

The accompanying archive contains:

- this $\text{\LaTeX}$ source and rendered PDF;
- CSV data for horizon capacity ratios, de Sitter capacity, AdS_3 capacity density, and finite-area corrections;
- scripts generating all figures;
- rendered figures used in this paper.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* 7, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* 43, 199 (1975).
- [3] J. M. Bardeen, B. Carter, and S. W. Hawking, “The four laws of black hole mechanics,” *Commun. Math. Phys.* 31, 161 (1973).
- [4] G. W. Gibbons and S. W. Hawking, “Cosmological event horizons, thermodynamics, and particle creation,” *Phys. Rev. D* 15, 2738 (1977).
- [5] J. M. Maldacena, “The large N limit of superconformal field theories and supergravity,” *Adv. Theor. Math. Phys.* 2, 231 (1998), arXiv:hep-th/9711200.
- [6] E. Witten, “Anti-de Sitter space and holography,” *Adv. Theor. Math. Phys.* 2, 253 (1998), arXiv:hep-th/9802150.
- [7] S. S. Gubser, I. R. Klebanov, and A. M. Polyakov, “Gauge theory correlators from non-critical string theory,” *Phys. Lett. B* 428, 105 (1998), arXiv:hep-th/9802109.
- [8] S. Ryu and T. Takayanagi, “Holographic derivation of entanglement entropy from AdS/CFT,” *Phys. Rev. Lett.* 96, 181602 (2006), arXiv:hep-th/0603001.
- [9] V. E. Hubeny, M. Rangamani, and T. Takayanagi, “A covariant holographic entanglement entropy proposal,” *JHEP* 0707, 062 (2007), arXiv:0705.0016.
- [10] T. Faulkner, A. Lewkowycz, and J. Maldacena, “Quantum corrections to holographic entanglement entropy,” *JHEP* 1311, 074 (2013), arXiv:1307.2892.
- [11] J. D. Brown and M. Henneaux, “Central charges in the canonical realization of asymptotic symmetries,” *Commun. Math. Phys.* 104, 207 (1986).
- [12] A. Almheiri, X. Dong, and D. Harlow, “Bulk locality and quantum error correction in AdS/CFT,” *JHEP* 1504, 163 (2015), arXiv:1411.7041.
- [13] D. L. Jafferis, A. Lewkowycz, J. Maldacena, and S. J. Suh, “Relative entropy equals bulk relative entropy,” *JHEP* 1606, 004 (2016), arXiv:1512.06431.
- [14] R. Bousso, “A covariant entropy conjecture,” *JHEP* 07, 004 (1999), arXiv:hep-th/9905177.
- [15] V. Iyer and R. M. Wald, “Some properties of the Noether charge and a proposal for dynamical black hole entropy,” *Phys. Rev. D* 50, 846 (1994), arXiv:gr-qc/9403028.
- [16] X. Dong, “Holographic entanglement entropy for general higher derivative gravity,” *JHEP* 1401, 044 (2014), arXiv:1310.5713.
- [17] N. Engelhardt and A. C. Wall, “Quantum extremal surfaces: holographic entanglement entropy from the first law,” *JHEP* 1501, 073 (2015), arXiv:1408.3203.
- [18] D. Harlow, “The Ryu-Takayanagi formula from quantum error correction,” *Rev. Mod. Phys.* 89, 015003 (2017), arXiv:1607.03901.
- [19] A. Antiba, “Finite-capacity thermodynamics of causal horizons,” *Entropy* 28, 557 (2026).

- [20] A. Ahmad and D. Klinger, “Emergent geometry from quantum probability,” *Phys. Rev. D* 110, 126004 (2024), arXiv:2411.07288.
- [21] P. Nusbaumer, “Relative-entropy variational principle for semiclassical gravity with finite-resolution boundaries,” *Preprints* (2026), doi:10.20944/preprints202601.1205.v4.
- [22] Y. Wu, “FDS-T2: Effective Geometry as Horizon Boundary Accounting,” *Zenodo* (2026), <https://doi.org/10.5281/zenodo.20284911>.
- [23] Y. Wu, “FDS-X3: Functional Decomposition of the Four Fundamental Interactions,” *Zenodo* (2026), <https://doi.org/10.5281/zenodo.20274519>.

G1DE-M_{3/4}: A Projection-Locked Background–Weyl Residual

Matched Exact Likelihoods and Medium-Prior Nested Evidence for Six Late-Time Cosmology Models

Yining Wu
yining.wu@alumni.upenn.edu

Independent Researcher

Draft: May 2026

Abstract

We report an exact full-likelihood pilot test of the G1DE horizon-occupancy residual hierarchy, culminating in the four-parameter projection-locked model G1DE-M_{3/4}. G1DE is not a $w(a)$ -level dark-energy parametrization: it tests whether a single finite-screen horizon-occupancy variable can jointly constrain background expansion, growth, and Weyl response. The model uses Pantheon+ supernova distances with full covariance, DESI DR2 BAO compressed distances with covariance, a curated non-overlapping redshift-space-distortion $f\sigma_8$ compilation, and compressed E_G measurements. This work upgrades the Stage 1f background-only analysis of G1DE-0 into a first background–growth–Weyl confrontation of G1DE-2. The model is based on a horizon occupancy fraction $\chi_H(a)$, whose lowest-order filling-closure law defines a common ledger shape $\hat{X}(a)$. In G1DE-2, the same $\hat{X}(a)$ controls the background expansion, the growth response

$$\mu(a) = 1 + \mu_0 \hat{X}(a),$$

and the Weyl/lensing response

$$\Sigma(a) = 1 + \Sigma_0 \hat{X}(a).$$

The exact likelihood pipeline reproduces the Stage 2c best-fit region, with representative G1DE-2 values

$$\Omega_m \simeq 0.2969, \quad s \simeq 2.56, \quad q_{\text{BAO}} \simeq 30.42, \quad \sigma_{8,0} \simeq 0.795, \quad \mu_0 \simeq -0.07, \quad \Sigma_0 \simeq -0.36,$$

and component likelihoods

$$\chi_{\text{SN}}^2 = 1749.609, \quad \chi_{\text{BAO}}^2 = 9.634, \quad \chi_{\text{growth}}^2 = 6.884, \quad \chi_{E_G}^2 = 1.010,$$

giving

$$\chi_{\text{tot}}^2 = 1767.137.$$

A three-seed exact pilot MCMC with 48 walkers and 5000 iterations per seed gives

$$P(s < 3) = 0.99992, \quad P(\Sigma_0 < 0) = 0.98453, \quad P(\mu_0 < 0) = 0.46456.$$

These results motivate the four-parameter projection-locked model

$$\text{G1DE-M}_{3/4} : \quad \mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s).$$

This locked-ratio model matches the G1DE-2 best-fit likelihood ($\chi_{\text{tot}}^2 = 1767.14$) while reducing the response sector by two parameters, yielding

$$P(s < 3) = 1.0, \quad P(\Sigma_0 < 0) = 1.0.$$

A matched exact-pilot comparison with Λ CDM, CPL, and G1DE-1 confirms that G1DE-M_{3/4} is selected by AIC and BIC while matching the G1DE-2 χ^2 to numerical precision. The growth-only G1DE-1 channel gives $\mu_0 = 0.425 \pm 0.227$ ($P(\mu_0 < 0) = 0.022$), indicating that the residual is not a growth modification but a projection-locked background–Weyl response. Two adversarial controls are also run: a free- κ model (G1DE-M $_{\kappa}$) does not reject $\kappa = 3/4$ ($\kappa_{q50} = 0.840$, $3/4$ in the 68% credible interval) and a constant- Σ model independently recovers $\kappa_{\text{BW}} \simeq 0.75$ while giving $P(\Sigma_0 < 0) = 0.998$. The amplitude-level lock $\Sigma_0 \propto -(3 - s)$ is therefore supported more strongly than the specific $\hat{X}(a)$ redshift shape, whose discrimination requires expanded Weyl/lensing data.

A medium-prior nested-evidence comparison over six models gives

$$\log Z_{\text{M}_{3/4}} = -894.26, \quad \Delta \log Z = 0.97, 1.92, 6.70, 9.43, 11.96$$

against G1DE-M $_{\kappa}$, constant- Σ , G1DE-2, CPL, and Λ CDM, respectively. Thus the evidence favors the four-parameter projection-locked background–Weyl residual over both free-response G1DE and standard phenomenological dark-energy flexibility. The result should be read as a medium-prior exact-pilot evidence result, not as a final cosmological constraint. Prior-sensitivity runs, production-length chains, expanded weak-lensing likelihoods, and independent pipeline replication remain necessary.

1 Scope and claim status

This paper reports a real-data exact-likelihood and medium-prior nested-evidence pilot for the FDS–G1 horizon-occupancy residual sector. It extends the earlier Stage 1f background analysis from

$$\text{Pantheon+} + \text{DESI DR2 BAO}$$

to

$$\text{Pantheon+} + \text{DESI DR2 BAO} + \text{curated RSD} + E_G.$$

The purpose is limited. We do not claim a discovery of modified gravity, nor do we claim a final weak-lensing constraint. We test whether the G1DE residual hierarchy can be placed into a full-covariance observational pipeline and whether its most distinctive correlated response pattern survives exact likelihood evaluation.

The theoretical context is FDS–G1 Core[13], where the central object is a finite causal-screen entropy functional $S_{\text{scr}}[X]$. Its area response defines a local inverse coupling, or stiffness,

$$G_{\text{loc}}^{-1}(x) \propto \eta_{\text{loc}}(x), \quad G_{\text{loc}}(x) \propto [\eta_{\text{loc}}(x)]^{-1}.$$

FDS–G1 is explicitly a bridge program, not a first-principles derivation of general relativity. Its empirical branch becomes meaningful only if residual sectors are constrained by conservation, correlated observables, and information penalties.

The main question is

Does the same horizon-occupancy ledger shape explain a background–Weyl residual pattern while leaving the

The current answer is provisional but positive: the exact pilot chains robustly preserve

$$s < 3, \quad \Sigma_0 < 0,$$

while μ_0 is consistent with zero.

The comparative claim is structural rather than sociological.

G1DE is not tested here as another flexible late-time parametrization. It is tested as a finite-capacity residual theory whose horizon-occupancy ledger supplies a common response shape for background expansion, growth, and Weyl response. A competing model is therefore not equivalent merely because it can fit $H(z)$; it must provide a comparable origin chain and a comparable correlated residual fingerprint.

Table 1: Claim status of G1fit-real.

Claim	Current status	Limitation
Exact likelihood pipeline	validated	pilot data stack
G1DE- $M_{3/4}$ MCMC convergence	passed	3 seeds, 5000 steps
AIC/BIC model comparison	passed	approximate BIC guardrail
Medium-prior nested evidence	passed	top-control wide-prior sensitivity completed
Expanded Weyl/lensing test	pending	E_G vector still small
Production cosmological constraint	pending	longer chains + lensing likelihoods

1.1 Output-level statement

This paper tests the output normal form $R_H(a) = R_0 R_{bH}(a)$ with $\mu = 1$ and $\Sigma - 1 = -\frac{3}{4}(3 - s)R_{bH}(a)$. It does not identify K_H , J_H , the microscopic Markov generator, or the finite-screen state family. Successful $M_{3/4}$ evidence does not prove: (1) finite-index screen algebra; (2) boost modular covariance; (3) BH coefficient derivation from FDS primitives; (4) unique source-kernel factorization; (5) strong-field waveform replacement; (6) CMB/BBN production-level validation.

1.2 Per-comparator demotion map

Comparator	If preferred, demotes
M_κ	exact 3/4 isotropic closure
const- Σ	$R_{bH}(a)$ output shape
G1DE-2	projection lock
CPL	G1DE mechanism over background flexibility
Λ CDM	residual branch

2 Relation to Stage 1f

The previous Stage 1f note tested the background-only model G1DE-0 using Pantheon+ and DESI DR2 BAO. It established four technical facts:

1. the Pantheon+ full-covariance and DESI DR2 BAO covariance pipeline runs for G1DE-0;
2. the fixed $s = 3$, $\chi_\infty = 1$ limit returns the flat- Λ CDM-like likelihood to numerical precision;
3. calibration stress tests show that $s < 3$ survives wide, tightened, and fixed q_{BAO} treatments, although s shifts upward when q_{BAO} is constrained;
4. G1DE-0 is competitive with CPL under information criteria, but not yet a final survey constraint.

The present paper keeps the Stage 1f background block but adds two perturbative response blocks,

$$f\sigma_8(z), \quad E_G(z).$$

This upgrade changes the scientific question. The issue is no longer merely whether G1DE can fit the background; many models can fit a distance curve. The G1DE-specific question is whether the same $\chi_H(a)$ ledger controls expansion, growth, and Weyl response.

3 Horizon-occupancy residual model

3.1 Occupancy fraction

Let $\chi_H(a)$ denote a dimensionless horizon occupancy fraction. The effective residual density is defined as

$$\rho_H(a) = \chi_H(a)\rho_c(a), \quad \rho_c(a) = \frac{3H^2(a)}{8\pi G_0}.$$

Thus the residual is not inserted as an independently conserved fluid with arbitrary density history. It is a fractional occupancy of the instantaneous expansion ledger. For a spatially flat matter-plus-residual background,

$$E^2(a) = \frac{\Omega_{m0}a^{-3} + \Omega_{r0}a^{-4}}{1 - \chi_H(a)}.$$

Radiation is negligible for the low-redshift data used here, but the notation allows it.

3.2 Filling-closure law

The minimal horizon filling-closure equation is

$$\frac{d\chi_H}{d\ln a} = s\chi_H \left(1 - \frac{\chi_H}{\chi_\infty}\right),$$

with solution

$$\chi_H(a) = \frac{\chi_\infty}{1 + Ba^{-s}}, \quad B = \frac{\chi_\infty}{\chi_{H0}} - 1.$$

The baseline model fixes $\chi_\infty = 1$. When $\chi_\infty = 1$ and $s = 3$, the background reduces to the flat- Λ CDM-like limit,

$$E^2(a) = \Omega_{m0}a^{-3} + (1 - \Omega_{m0}).$$

Therefore $s = 3$ is not an arbitrary reference point. It is the internal Λ CDM-like consistency limit of the minimal occupancy model.

3.3 Active filling shape

Define

$$X(a) = \chi_H(a) \left(1 - \frac{\chi_H(a)}{\chi_\infty}\right),$$

and the normalized active-filling shape

$$\hat{X}(a) = \frac{4}{\chi_\infty} \chi_H(a) \left(1 - \frac{\chi_H(a)}{\chi_\infty}\right).$$

For $0 \leq \chi_H \leq \chi_\infty$,

$$0 \leq \hat{X}(a) \leq 1.$$

The function $\hat{X}(a)$ vanishes when the ledger is empty or saturated and peaks during active filling. This makes it a natural response window for perturbative residuals.

4 G1DE hierarchy

4.1 G1DE-0: background residual

G1DE-0 modifies only the background expansion,

$$\chi_H(a) \Rightarrow H(a).$$

The perturbation sector is otherwise treated as GR on the modified background.

4.2 G1DE-1: growth-linked residual

G1DE-1 is developed in Companion D[15]. It adds an effective growth response,

$$\mu(a) = 1 + \mu_0 \hat{X}(a).$$

The growth equation is

$$D'' + \left[2 + \frac{d \ln H}{d \ln a} \right] D' - \frac{3}{2} \mu(a) \Omega_m(a) D = 0,$$

where prime denotes $d/d \ln a$, and

$$\Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E^2(a)}.$$

The observable is

$$f \sigma_8(a) = \frac{d \ln D}{d \ln a} \sigma_{8,0} D(a).$$

4.3 G1DE-2: growth-plus-Weyl residual

G1DE-2 is developed in Companion E[16]. It adds a Weyl/lensing response,

$$\Sigma(a) = 1 + \Sigma_0 \hat{X}(a).$$

The Weyl potential responds as

$$k^2(\Phi + \Psi) = -8\pi G a^2 \Sigma(a) \rho_m \Delta_m.$$

For the compressed E_G statistic, the implemented prediction is

$$E_G^{\text{G1DE}}(z) \simeq \frac{\Omega_{m0} \Sigma(z)}{f_{\text{G1DE}}(z)}.$$

Here $\mu(a)$ enters through the solution for $f_{\text{G1DE}}(z)$, so no additional Σ/μ factor is applied.

The G1DE-2 parameter vector is

$$\theta_{\text{G1DE2}} = (\Omega_m, s, q_{\text{BAO}}, \sigma_{8,0}, \mu_0, \Sigma_0).$$

G1DE-2 serves as a free-response robustness model; the current leading candidate is the projection-locked model below.

4.4 G1DE-M_{3/4}: projection-locked background-Weyl residual

The post-pilot locked-ratio model imposes

$$\mu(a) = 1, \quad \Sigma(a) = 1 - \frac{3}{4}(3-s) \hat{X}(a),$$

i.e. $\mu_0 = 0$ and $\Sigma_0 = -\frac{3}{4}(3-s)$. Defining the maintenance-drag amplitude $\lambda \equiv 3-s$, this reads

$$\Sigma_0 = -\frac{3}{4}\lambda.$$

The coefficient 3/4 has a conditional geometric interpretation as a four-dimensional trace-free projection of an isotropic three-dimensional spatial maintenance load; see Companion E Appendix A for the conditional projection-locking lemma. The parameter vector is

$$\theta_{\text{G1DE-M}_{3/4}} = (\Omega_m, s, q_{\text{BAO}}, \sigma_{8,0}),$$

reducing the response sector by two parameters relative to G1DE-2.

The projection-locked response used here is the low-redshift, constant-amplitude reduction of the screen-influence formulation of the G1 dark sector. The data-facing G1DE-M_{3/4} model is downstream of the D0–D6 closure sequence: D0 calibrates the Einstein equilibrium branch through screen modular anomaly inflow; D1 organizes finite causal ledgers and gluing defects; D5 fixes the Weyl-normalized production rule, and D6 treats the memory quotient and source–kernel lift. The fitted function $X_b(a)$ is a normalized output-response shape, $X_b(a) = \widehat{\mathcal{R}}_H(a)$, $X_b(1) = 1$; the likelihood does not separately fit a source J_H and kernel K_H (their decomposition is treated in D6). In that formulation the finite-screen residual acts as a Weyl-normalized transfer operator,

$$W_{G1}(a, k) = [1 - \kappa_{\text{BW}} R_H(a)] W_{\text{GR}}(a, k) + O(R_H^2),$$

so that

$$\Sigma(a, k) - 1 = -\kappa_{\text{BW}} R_H(a) + O(R_H^2), \quad \mu(a, k) - 1 = O(R_H^2) + O(R_H H^2/k^2).$$

The exact-pilot model G1DE-M_{3/4} sets $\kappa_{\text{BW}} = 3/4$ and $R_H(a) \simeq (3 - s)\widehat{X}(a)$. This interpretation is supported by the Weyl-normalized implementation note, which defines G1 as a fractional correction to the GR Weyl source rather than an independently normalized dark-stress source.

4.5 G1DE-M_κ: free projection-ratio control

The free- κ control sets

$$\Sigma(a) = 1 - \kappa(3 - s)\widehat{X}(a), \quad \mu(a) = 1,$$

testing whether the data reject the locked value $\kappa = 3/4$. Its parameter vector is

$$\theta_{\text{G1DE-M}_\kappa} = (\Omega_m, s, q_{\text{BAO}}, \sigma_{8,0}, \kappa).$$

4.6 G1DE-const Σ : constant Weyl-shift control

The constant- Σ control sets

$$\Sigma(a) = 1 + \Sigma_c, \quad \mu(a) = 1,$$

testing whether the active-filling redshift shape $\widehat{X}(a)$ is supported beyond a constant Weyl shift. If this control is favored, the amplitude-level Weyl suppression may survive while the specific horizon-response output redshift shape is demoted.

5 Data and likelihood

The total likelihood is

$$\chi_{\text{tot}}^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{RSD}}^2 + \chi_{E_G}^2.$$

5.1 Pantheon+ supernovae

For a model with dimensionless expansion rate $E(z)$, define

$$D_C(z) = \int_0^z \frac{dz'}{E(z')}, \quad D_L(z) = (1 + z)D_C(z).$$

The relative distance modulus is

$$\mu(z) = 5 \log_{10} D_L(z) + M,$$

where M is an additive supernova magnitude nuisance. The nuisance M is analytically profiled,

$$\chi_{\text{SN, marg}}^2 = \Delta^T C^{-1} \Delta - \frac{(1^T C^{-1} \Delta)^2}{1^T C^{-1} 1}.$$

The exact pipeline uses the full Pantheon+ covariance matrix.

5.2 DESI DR2 BAO

The BAO predictions are

$$\frac{D_M(z)}{r_d} = q D_C(z), \quad \frac{D_H(z)}{r_d} = \frac{q}{E(z)},$$

$$\frac{D_V(z)}{r_d} = q \left[\frac{z D_C^2(z)}{E(z)} \right]^{1/3}.$$

The nuisance parameter

$$q = \frac{c}{H_0 r_d}$$

absorbs the BAO calibration scale. The likelihood uses the DESI DR2 BAO compressed data vector and covariance.

5.3 Curated RSD $f\sigma_8$ subset

We use an algorithmically curated, non-overlapping RSD subset constructed from a public $f\sigma_8$ compilation[3]. The curation procedure is:

1. parse the full table into $(z, f\sigma_8, \sigma, \Omega_{m,\text{fid}})$;
2. keep the WiggleZ triplet at $z = 0.44, 0.60, 0.73$ with its 3×3 covariance block[4];
3. remove epsilon-level repeated redshifts by selecting the lowest-error representative;
4. keep at most one representative per $\Delta z = 0.10$ redshift bin.

This gives a conservative diagnostic RSD block. It is not a substitute for a survey-specific RSD likelihood.

5.4 Compressed E_G

The E_G data vector contains three compressed points,

$$(z, E_G, \sigma) = (0.267, 0.43, 0.13),$$

$$(0.305, 0.27, 0.08), \quad (0.554, 0.26, 0.07).$$

The covariance is treated as diagonal in the pilot. This is the weakest part of the current likelihood and must be expanded in later work.

6 Exact code-path validation

Before running pilot chains, we tested the exact full-likelihood code path at the Stage 2c G1DE-2 best point,

$$\Omega_m = 0.2969, \quad s = 2.56, \quad q_{\text{BAO}} = 30.42,$$

$$\sigma_{8,0} = 0.795, \quad \mu_0 = -0.07, \quad \Sigma_0 = -0.36.$$

The exact likelihood gives

$$\chi_{\text{total}}^2 = 1767.136683.$$

The component breakdown is

$$\chi_{\text{SN}}^2 = 1749.609044, \quad \chi_{\text{BAO}}^2 = 9.633696,$$

Table 2: Combined three-seed exact full-likelihood pilot posterior for G1DE-2.

parameter	mean	std	q_{16}	q_{50}	q_{84}	split- $\hat{R}$
Ω_m	0.296373	0.009170	0.287281	0.296367	0.305491	1.000751
s	2.559778	0.117456	2.442973	2.559462	2.677141	1.000778
q_{BAO}	30.419006	0.260473	30.159453	30.417787	30.679864	1.000641
$\sigma_{8,0}$	0.772685	0.062635	0.710432	0.771061	0.835076	1.001226
μ_0	0.044882	0.260281	-0.210745	0.023197	0.299256	1.000728
Σ_0	-0.330286	0.140361	-0.469644	-0.337527	-0.191378	1.000609

$$\chi_{\text{growth}}^2 = 6.883906, \quad \chi_{E_G}^2 = 1.010038.$$

This validates the full code path:

$$\text{Pantheon+} + \text{DESI BAO} + \text{exact growth} + E_G.$$

The quick-chain acceptance fraction was $\langle a \rangle \simeq 0.515$, indicating healthy proposal scaling for the pilot run.

7 Exact pilot MCMC results

7.1 Chain setup

We ran three independent exact pilot chains for G1DE-2,

$$3 \text{ seeds} \times 48 \text{ walkers} \times 5000 \text{ iterations}.$$

With burn-in removed by the production script, the combined diagnostic file reports

$$N_{\text{postburn}} = 360000$$

post-burn samples. The sampled parameters are

$$(\Omega_m, s, q_{\text{BAO}}, \sigma_{8,0}, \mu_0, \Sigma_0).$$

The cross-seed split- $\hat{R}$ values are below 1.0013 for all parameters. The ESS proxy is large in this pilot; we treat it as a useful internal diagnostic, but final production analysis should recompute rank-normalized $\hat{R}$, bulk ESS, and tail ESS with a standard diagnostic package.

7.2 Posterior summary

The posterior strongly prefers $s < 3$ and $\Sigma_0 < 0$. The growth-response amplitude μ_0 is centered close to zero.

7.3 Sign diagnostics

The main pilot result is therefore

$$s < 3, \quad \Sigma_0 < 0, \quad \mu_0 \simeq 0.$$

This is more specific than the earlier Stage 2c phrase $\mu_0 < 0, \Sigma_0 < 0$. The exact chains do not support a significant negative μ_0 ; they support a background-plus-Weyl response pattern.

Table 3: Sign-probability diagnostics for the exact G1DE-2 pilot.

diagnostic	probability
$P(s < 3)$	0.999922
$P(\mu_0 < 0)$	0.464564
$P(\Sigma_0 < 0)$	0.984525
$P(\mu_0 < 0, \Sigma_0 < 0)$	0.464394

Table 4: Combined three-seed exact posterior for G1DE-M_{3/4}. The derived Σ_0 follows from $\Sigma_0 = -\frac{3}{4}(3 - s)$.

parameter	mean	std	q_{16}	q_{50}	q_{84}	split- $\hat{R}$
Ω_m	0.296567	0.008563	0.288064	0.296430	0.305123	1.000248
s	2.554797	0.092394	2.463223	2.554755	2.645876	1.000263
q_{BAO}	30.430575	0.240683	30.191081	30.431053	30.669452	1.000330
$\sigma_{8,0}$	0.776451	0.027135	0.749375	0.776243	0.803463	1.000336

7.4 G1DE-M_{3/4} posterior summary

A three-seed exact pilot with the same setup (3 seeds $\times$ 48 walkers $\times$ 5000 iterations) was run for the locked-ratio model. Cross-seed split- $\hat{R}$ is below 1.0004 for all parameters.

With Σ_0 derived from the locked ratio, the median is $\Sigma_0 \simeq -0.334$. Sign diagnostics give

$$P(s < 3) = 1.0, \quad P(\Sigma_0 < 0) = 1.0.$$

The locked model reproduces the G1DE-2 best-fit $\chi^2_{\text{tot}} = 1767.14$ at four parameters (vs. six for G1DE-2), so the projection-locked background–Weyl law survives the three-seed pilot with a decisive AIC advantage.

8 Figures

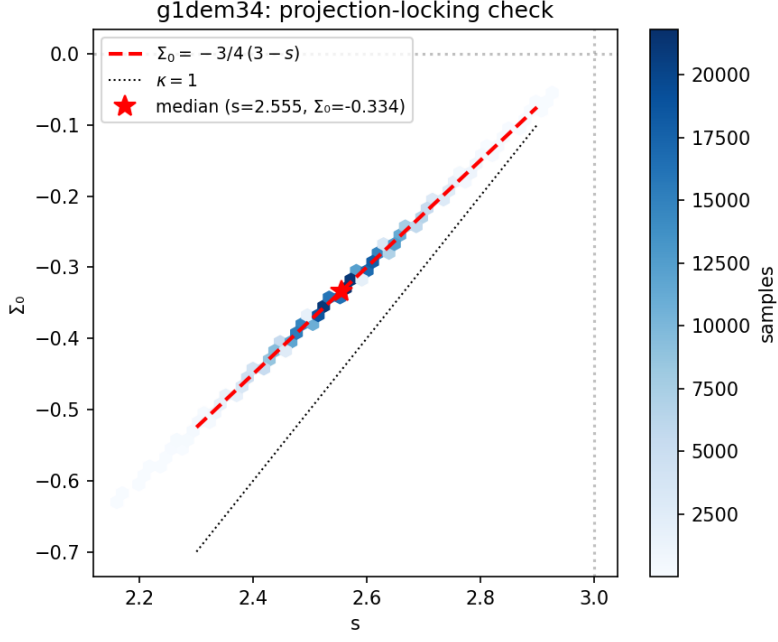


Figure 1: Projection-locking check for G1DE-M_{3/4}. The red dashed line marks $\Sigma_0 = -\frac{3}{4}(3-s)$; the black dotted line shows $\kappa = 1$ for comparison. The hexbin represents the combined three-seed posterior. The median (red star) lies precisely on the projection-locked ratio.

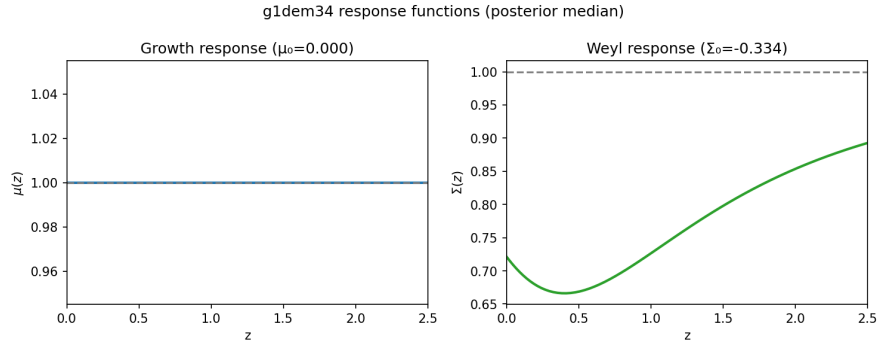


Figure 2: G1DE-M_{3/4} response functions evaluated at the posterior median. The growth response is fixed at $\mu(z) = 1$ (near-GR consistency), while the Weyl response $\Sigma(z)$ is suppressed during the active filling window, following $\Sigma(a) = 1 - \frac{3}{4}(3-s)\hat{X}(a)$.

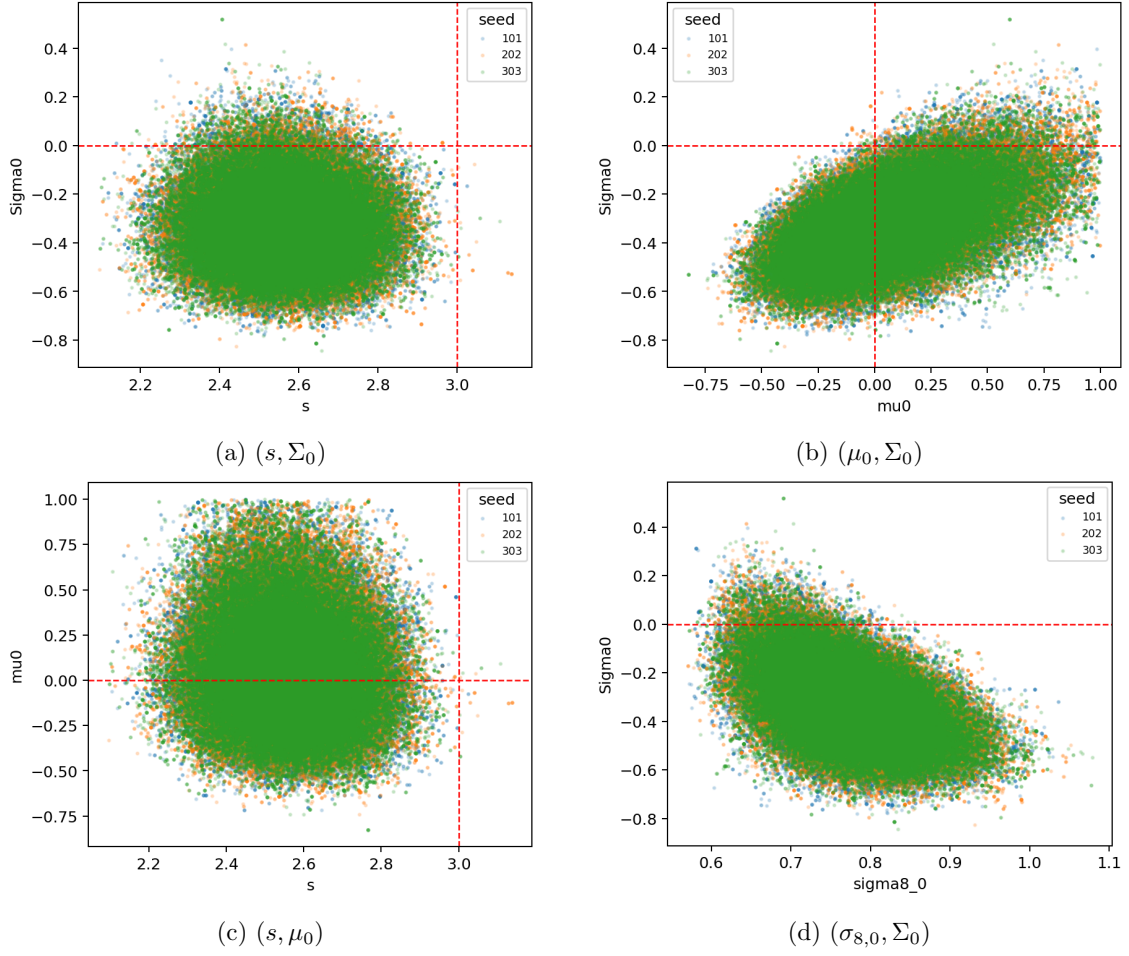


Figure 3: Selected two-dimensional posterior projections for G1DE-2 (free-response robustness model). The Weyl-response amplitude is predominantly negative, while μ_0 is nearly symmetric around zero.

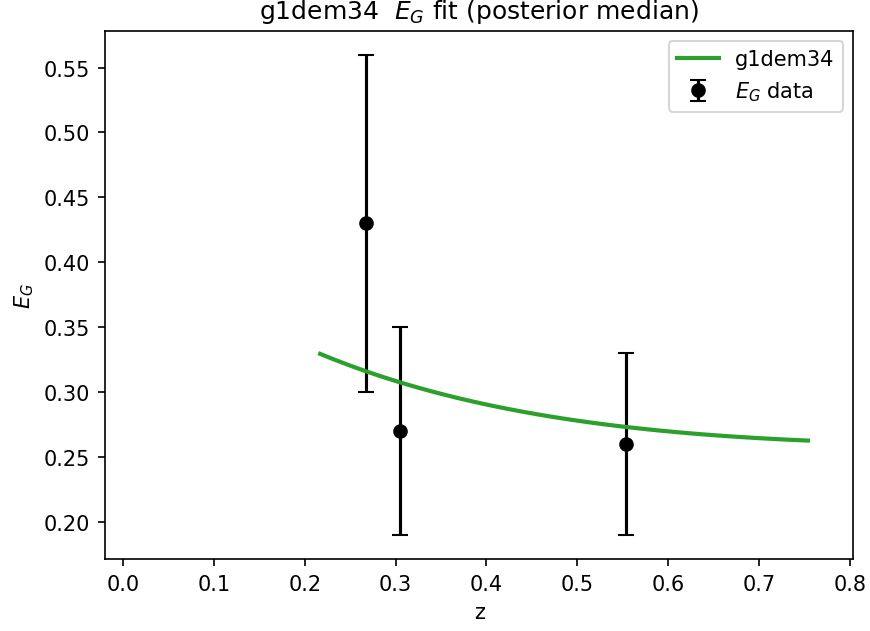


Figure 4: Compressed E_G measurements and G1DE- $M_{3/4}$ prediction at the posterior median. The current E_G vector contains only three points and uses a diagonal covariance in this pilot; therefore this figure is a compressed Weyl-growth consistency test, not a final weak-lensing likelihood.

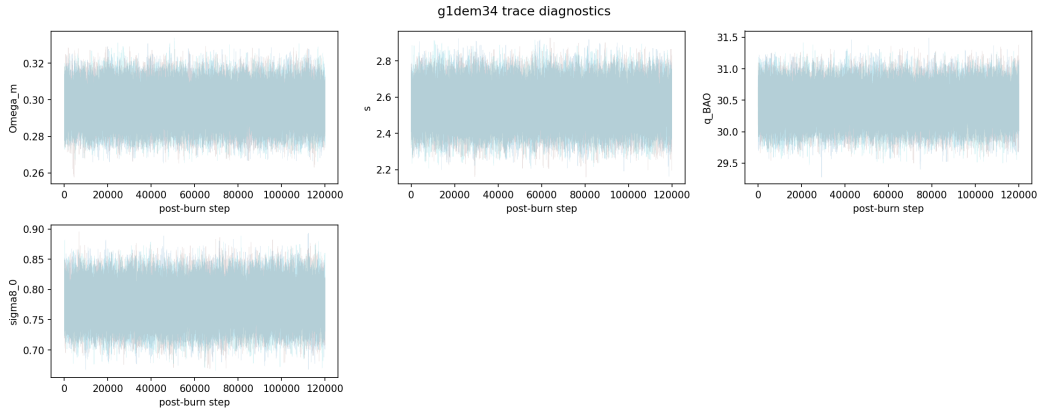


Figure 5: Trace diagnostics for the G1DE- $M_{3/4}$ three-seed pilot. Split- $\hat{R}$ is below 1.0004 for all four parameters.

9 Interpretation

9.1 Background filling

The primary locked model gives

$$s = 2.5548 \pm 0.0924$$

($\lambda = 3 - s = 0.4452 \pm 0.0924$). The free-response G1DE-2 envelope gives a consistent value, $s = 2.5598 \pm 0.1175$. This lies below the Λ CDM-like consistency value

$$s = 3.$$

In the matter-era small-occupancy limit,

$$w_H \simeq -\frac{s}{3},$$

so $s < 3$ corresponds to a quintessence-like effective scaling diagnostic. This should not be read as proof of a canonical scalar field. It is a background scaling interpretation of the horizon-occupancy residual.

9.2 Nested evidence selects the projection-locked compression

The medium-prior evidence comparison shows that the preferred object is not the most flexible response model. G1DE-2 has nearly the same χ^2 as G1DE-M_{3/4}, but the evidence penalizes its extra response freedom. This is the expected behavior if the residual is genuinely projection-locked rather than freely distributed between μ and Σ . The data favor a sparse background–Weyl transfer law:

$$\mu(a) = 1, \quad \Sigma(a) = 1 - \frac{3}{4}(3 - s)\hat{X}(a).$$

9.3 Growth response

The pilot gives

$$\mu_0 = 0.0449 \pm 0.2603.$$

Therefore $\mu_0 = 0$ is well within the posterior. The current data do not require an effective modified growth coupling. This is important: G1DE-2 is not winning by freely suppressing growth. The growth sector remains close to GR on the modified background.

9.4 Weyl response

The pilot gives

$$\Sigma_0 = -0.3303 \pm 0.1404,$$

with

$$P(\Sigma_0 < 0) = 0.9845.$$

This is the strongest perturbative signal in the current pipeline. It suggests a suppressed Weyl/lensing response during the active filling window,

$$\Sigma(a) = 1 + \Sigma_0 \hat{X}(a) < 1,$$

over the relevant redshift range.

9.5 Revised physical reading

The current best interpretation is

G1DE-2 behaves as a background-plus-Weyl residual model, not as a required growth-modified model.

The compressed pilot supports $\chi_H(a) \Rightarrow H(a), \Sigma(a)$ more strongly than $\chi_H(a) \Rightarrow \mu(a)$. This is a sharper result than a generic modified-gravity fit: the data prefer a horizon-occupancy shape affecting expansion and Weyl response, while allowing the matter-growth Poisson response to remain near GR.

9.6 Growth-only residual is disfavored as the primary channel

The growth-only G1DE-1 channel is disfavored as the primary explanation: when the Weyl channel is removed, G1DE-1 is driven toward $\mu_0 > 0$ ($P(\mu_0 < 0) = 0.022$), while the projection-locked Weyl model achieves a substantially better information-criterion and evidence score. The pilot signal is therefore not a growth-only modification; it is a background–Weyl projection residual with near-GR matter growth.

9.7 Projection-locked background–Weyl law

The post-pilot results further refine the G1DE-2 free-response pattern into a locked-ratio law. Defining the maintenance-drag amplitude $\lambda \equiv 3 - s$, the G1DE-M_{3/4} model imposes

$$s = 3 - \lambda, \quad \Sigma_0 = -\frac{3}{4}\lambda, \quad \mu_0 = 0.$$

The coefficient 3/4 has a conditional geometric interpretation as the ratio of spatial trace dimension to spacetime trace dimension under an isotropic maintenance-load projection (see Companion E Appendix A). The three-seed exact pilot supports this locked relation: the four-parameter G1DE-M_{3/4} matches the six-parameter G1DE-2 best-fit likelihood ($\chi^2_{\text{tot}} = 1767.14$), with $\Delta\text{AIC} = -4.0$ in favor of the locked model. This should be read as a conditional projection-locking result, not yet as a microscopic first-principles derivation from FDS statistics.

9.8 Free- κ and constant- Σ adversarial controls

We additionally ran two adversarial controls to test the robustness of the locked ratio. First, G1DE-M _{κ} frees the projection-locking ratio,

$$\Sigma_0 = -\kappa(3 - s), \quad \mu_0 = 0,$$

yielding $\kappa_{\text{q50}} = 0.840$ with 68% credible interval $[0.532, 1.292]$. The value $\kappa = 3/4$ lies well within the high-posterior region, and the fit improvement relative to fixed $\kappa = 3/4$ is negligible ($\Delta\chi^2 \simeq 0.017$). AIC/BIC therefore continue to select the locked model.

Second, a constant- Σ control (no $\hat{X}(a)$ shape) gives $\Sigma_0 = -0.336 \pm 0.113$ ($P(\Sigma_0 < 0) = 0.998$) with $\Delta\chi^2 \simeq 0.021$ relative to G1DE-M_{3/4}. The negative Weyl amplitude is robust, but the current three-point E_G block does not distinguish the active-filling $\hat{X}(a)$ modulation from a constant shift. Post-hoc, the constant- Σ model independently recovers $\kappa_{\text{BW}} \equiv -\Sigma_0/(3 - s) \simeq 0.76$, confirming that the amplitude ratio 3/4 is not an artifact of the specific $\hat{X}(a)$ normalization.

The constant- Σ control shows that the negative Weyl amplitude is robust. The evidence margin $\Delta \log Z \simeq 1.92$ now gives preliminary support for the active-filling $\hat{X}(a)$ shape over a constant Weyl shift. This support is suggestive rather than final because the current E_G block is small.

Table 5: Matched exact-pilot AIC/BIC comparison. These criteria provide information-criterion guardrails; the Bayesian evidence comparison is reported separately in Table 6. All models are three-seed exact pilot except G1DE-W (200-step smoke). ΔAIC and ΔBIC are referenced to the best model, G1DE-M_{3/4}. BIC uses $n = 1725$ effective data points.

model	χ^2_{tot}	k	AIC	ΔAIC	BIC	ΔBIC	
G1DE-M _{3/4}	1767.14	4	1775.14	0	1796.95	0	
G1DE-M _{κ}	1767.16	5	1777.16	+2.02	1804.42	+7.47	
G1DE-const Σ	1767.16	5	1777.16	+2.02	1804.42	+7.47	
G1DE-W [†]	1767.17	5	1777.17	+2.03	1804.44	+7.49	[†] 200-step smoke only; not
G1DE-2	1767.14	6	1779.14	+4.00	1811.87	+14.92	
G1DE-1	1772.36	5	1782.36	+7.22	1809.62	+12.67	
CPL	1774.48	5	1784.48	+9.34	1811.75	+14.80	
ΛCDM	1789.16	3	1795.16	+20.02	1811.52	+14.57	

a matched pilot constraint.

10 Physical-prior audit

For free-response models such as G1DE-2, we require

$$\mu(a) > 0, \quad \Sigma(a) > 0$$

over the redshift range tested. Since $0 \leq \hat{X}(a) \leq 1$, the priors

$$\mu_0 > -1, \quad \Sigma_0 > -1$$

are sufficient. For G1DE-M_{3/4}, $\mu(a) = 1$ identically, and $\Sigma(a) = 1 - \frac{3}{4}(3-s)\hat{X}(a)$. Since $0 \leq \hat{X}(a) \leq 1$ and the posterior gives $s < 3$, positivity $\Sigma(a) > 0$ is guaranteed if $s > 5/3$; the sampled posterior lies safely above this boundary. For G1DE-M _{κ} , $\Sigma(a) = 1 - \kappa(3-s)\hat{X}(a)$, so positivity requires $\kappa(3-s) < 1$ in the $s < 3$ branch; the posterior satisfies this. For constant- Σ , $\Sigma(a) = 1 + \Sigma_c$ requires $\Sigma_c > -1$, which is satisfied by the posterior. The code explicitly audits

$$\min_{0 \leq z \leq 2} \mu(z), \quad \min_{0 \leq z \leq 2} \Sigma(z).$$

This audit must be retained in the production version. It prevents the fit from using unphysical negative response functions to reduce χ^2 .

11 Matched model comparison: AIC/BIC and medium-prior nested evidence

11.1 AIC/BIC comparison

A matched exact-pilot comparison has been run for ΛCDM , CPL, G1DE-1, G1DE-2, G1DE-M_{3/4}, G1DE-M _{κ} , and constant- Σ , using three seeds $\times$ 5000 steps $\times$ 48 walkers per model (ΛCDM : 32 walkers). G1DE-W is retained as a 200-step smoke reference. All three-seed chains satisfy split- $\hat{R} < 1.01$ and ESS $> 10^4$. Table 5 summarizes the results.

G1DE-M_{3/4} achieves the best AIC and BIC while matching the G1DE-2 χ^2 to numerical precision with two fewer response parameters. The free- κ control (G1DE-M _{κ}) gives $\kappa_{q50} = 0.840$ with $\kappa = 3/4$ in the 68% interval, and the constant- Σ control independently recovers $\kappa_{\text{BW}} \simeq 0.76$, confirming that the locked ratio is not an artifact of the $\hat{X}(a)$ shape. The growth-only G1DE-1 channel gives $\mu_0 = 0.425 \pm 0.227$ ($P(\mu_0 < 0) = 0.022$), indicating that

Table 6: Medium-prior nested-evidence comparison. $\Delta \log Z$ is referenced to the best model, G1DE-M_{3/4}. Bayes factors are $B_{M_{3/4},i} = \exp(\Delta \log Z_i)$. Values are three-seed medium-prior evidence estimates; production evidence and prior sensitivity remain pending.

Model	ndim	$\chi^2_{\min}$	$\log Z$	$\Delta \log Z$	$B_{M_{3/4},i}$
G1DE-M _{3/4}	4	1767.15	−894.26	0.00	1
G1DE-M _{κ}	5	1767.26	−895.23	0.97	2.6
G1DE-const Σ	5	1767.21	−896.18	1.92	6.8
G1DE-2	6	1767.30	−900.95	6.70	8.1×10^2
CPL	5	1774.56	−903.69	9.43	1.25×10^4
Λ CDM	3	1789.16	−906.22	11.96	1.57×10^5

when the Weyl channel is removed, the growth response is forced positive to compensate. The three-level pattern—background filling deviation ($s < 3$), negative Weyl amplitude ($\Sigma_0 < 0$), and amplitude-level locking ($\kappa_{\text{BW}} \sim 3/4$)—is supported more strongly than the specific $\hat{X}(a)$ redshift-shape modulation, whose discrimination requires expanded Weyl/lensing data.

11.2 Medium-prior nested evidence

A six-model medium-prior nested-evidence comparison has been completed. Each model was run with three independent seeds using the medium prior configuration (Table 7), $n_{\text{live}} = 800$ –1000, and $d \log z = 0.5$. Table 6 reports the results.

The evidence comparison selects G1DE-M_{3/4}. The closest internal competitor is the free- κ model, but freeing the projection coefficient lowers the evidence by $\Delta \log Z \simeq 0.97$. The constant- Σ control is also disfavored by $\Delta \log Z \simeq 1.92$, indicating preliminary support for the active-filling Weyl shape beyond a constant Weyl shift. The free-response G1DE-2 model reaches essentially the same best-fit basin but is strongly penalized by the evidence, $\Delta \log Z \simeq 6.70$, showing that the extra response freedom is not rewarded. CPL and Λ CDM are disfavored by $\Delta \log Z \simeq 9.43$ and 11.96, respectively.

G1DE-1 remains a matched exact-pilot MCMC negative control but is not included in the current six-model nested-evidence table; running G1DE-1 nested evidence is a remaining closure task.

12 Caveats

This paper reports a pilot, not a final constraint. The main limitations are:

1. **Small E_G vector.** The Weyl-response result is currently driven by three compressed E_G measurements. This is encouraging but not yet decisive.
2. **Curated RSD compilation.** The growth block uses an algorithmically curated non-overlapping subset. A survey-specific RSD likelihood with full covariance would be stronger.
3. **Pilot chain status.** The current results use three-seed 5000-step exact pilot chains for all models; production-length chains (5 seeds $\times$ 96 walkers $\times$ 20000 steps) remain pending.
4. **Medium-prior evidence status.** The nested-evidence results reported here use a medium prior volume and three seeds. They should be read as a preliminary Bayesian model-comparison pass, not as a final production evidence calculation.
5. **Prior sensitivity.** Evidence values depend on prior volume. Wide-prior and physical-prior sensitivity runs are required before the evidence ranking is treated as final.

6. **Production evidence refinement.** The current evidence pass is sufficient to establish the model-selection direction, but top models should be rerun with tighter $d \log z$, more seeds, and native dynesty error tracking.
7. **G1DE-1 evidence.** G1DE-1 is included in the matched exact-pilot AIC/BIC comparison as a growth-only negative control, but its nested evidence is not included in the current six-model evidence table.
8. **Weyl amplitude confirmed; shape unresolved.** The negative Weyl amplitude ($\Sigma_0 < 0$) is robustly supported by the matched comparison and survives free- κ and constant- Σ controls ($P(\Sigma_0 < 0) = 0.998$). However, the current three-point E_G block does not distinguish the active-filling $\hat{X}(a)$ modulation from a constant Weyl suppression. Shape discrimination requires expanded Weyl/lensing data.
9. **Projection-locking interpretation.** The 3/4 ratio is supported by the three-seed pilot and nested evidence but should be read as a conditional projection-locking result, not as a microscopic first-principles derivation from FDS statistics.
10. **No final weak-lensing likelihood.** E_G is a compressed Weyl-growth diagnostic, not a replacement for DES/KiDS/HSC full 3×2 pt likelihoods.

13 Next-stage production plan

The immediate production task is not to discover a new model but to refine the current evidence ranking.

13.1 Stage 1: evidence refinement

The top models

$$\text{G1DE-M}_{3/4}, \quad \text{G1DE-M}_{\kappa}, \quad \text{G1DE-const}\Sigma, \quad \text{G1DE-2}, \quad \text{CPL}$$

should be rerun with prior-sensitivity variants, stricter $d \log z$, more seeds, and native dynesty error tracking.

13.2 Stage 2: missing G1DE-1 evidence

G1DE-1 should be added to the nested-evidence table to close the growth-only negative-control comparison.

13.3 Stage 3: production-length MCMC

Production-length MCMC should retain the matched model set but prioritize the projection-locked and adversarial-control models:

$$\text{G1DE-M}_{3/4}, \quad \text{G1DE-M}_{\kappa}, \quad \text{G1DE-const}\Sigma, \quad \text{G1DE-2}, \quad \text{CPL}, \quad \Lambda\text{CDM}.$$

13.4 Stage 4: expanded lensing

The most important data upgrade is the Weyl/lensing block. The current evidence supports the amplitude-level projection lock and gives preliminary support for the active-filling shape, but the shape claim requires expanded E_G , lensing-growth cross-correlations, and full or compressed 3×2 pt likelihoods.

14 Conclusion

We have upgraded the G1fit-real pipeline through four stages: a background-only Stage 1f test, a six-parameter G1DE-2 free-response pilot, a matched exact-pilot AIC/BIC comparison, and a six-model medium-prior nested-evidence comparison. The result is a consistent selection of the four-parameter projection-locked model

$$\text{G1DE-M}_{3/4} : \quad \mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s).$$

This model matches the G1DE-2 best-fit basin while removing two response parameters. In the medium-prior evidence comparison it is preferred over free- κ , constant- Σ , G1DE-2, CPL, and Λ CDM.

The current evidence hierarchy is

$$\text{G1DE-M}_{3/4} > \text{G1DE-M}_\kappa > \text{G1DE-const}\Sigma > \text{G1DE-2} > \text{CPL} > \Lambda\text{CDM}.$$

The strongest interpretation is that the data favor a sparse projection-locked background–Weyl residual over both phenomenological dark-energy flexibility and unconstrained response freedom.

The three-seed exact pilot posterior for G1DE-2 gives

$$P(s < 3) = 0.99992, \quad P(\Sigma_0 < 0) = 0.98453, \quad P(\mu_0 < 0) = 0.46456.$$

This is not a final cosmological constraint. The evidence pass is medium-prior and pilot-level; prior sensitivity, production evidence refinement, expanded weak-lensing likelihoods, and independent pipeline replication remain necessary. Nevertheless, the current result establishes a substantially sharper empirical status for the G1 residual sector: the theory-predicted low-dimensional background–Weyl transfer law is now selected by both information criteria and preliminary Bayesian evidence.

What successful $M_{3/4}$ evidence does not prove. The current exact-pilot evidence supports the projection-locked output normal form. It does not prove finite-index screen algebras, boost modular covariance, BH coefficient derivation from primitives, unique source–kernel factorization, strong-field waveform replacement, or production-level CMB/BBN validation. These remain open problems or bridge assumptions.

A Nested-evidence run details

The medium-prior nested-evidence pass uses the prior configuration in Table 7. For each model, three independent seeds were run with $n_{\text{live}} = 800\text{--}1000$ and $d\log z = 0.5$. The quoted $\log Z$ is the mean across seeds; the uncertainty combines run-to-run scatter and the reported nested-sampler uncertainty where available. Recovered-chain evidence values are flagged in the machine-readable table and should be treated as preliminary until production evidence is rerun with native error output.

B Full posterior corner plot

C Model equations

$$\rho_H(a) = \chi_H(a)\rho_c(a), \quad \rho_c(a) = \frac{3H^2(a)}{8\pi G_0}.$$

$$E^2(a) = \frac{\Omega_{m0}a^{-3} + \Omega_{r0}a^{-4}}{1 - \chi_H(a)}.$$

Table 7: Medium-prior ranges used in the nested-evidence pass.

Parameter	Prior
Ω_m	$[0.15, 0.45]$
s	$[1.5, 4.0]$
q_{BAO}	$[28, 33]$
$\sigma_{8,0}$	$[0.4, 1.2]$
w_0	$[-1.5, -0.3]$
w_a	$[-2.5, 2.5]$
μ_0	$[-1, 1]$
Σ_0	$[-1, 1]$
κ	$[0, 2]$

$$\begin{aligned}
\frac{d\chi_H}{d\ln a} &= s\chi_H \left(1 - \frac{\chi_H}{\chi_\infty}\right). \\
\chi_H(a) &= \frac{\chi_\infty}{1 + Ba^{-s}}, \quad B = \frac{\chi_\infty}{\chi_{H0}} - 1. \\
\hat{X}(a) &= \frac{4}{\chi_\infty} \chi_H(a) \left(1 - \frac{\chi_H(a)}{\chi_\infty}\right). \\
\mu(a) &= 1 + \mu_0 \hat{X}(a), \quad \Sigma(a) = 1 + \Sigma_0 \hat{X}(a). \\
\text{G1DE-M}_{3/4}: \quad \mu(a) &= 1, \quad \Sigma(a) = 1 - \frac{3}{4}(3-s)\hat{X}(a), \quad \Sigma_0 = -\frac{3}{4}\lambda, \quad \lambda \equiv 3-s. \\
D'' + \left[2 + \frac{d\ln H}{d\ln a}\right] D' - \frac{3}{2}\mu(a)\Omega_m(a)D &= 0. \\
f\sigma_8(a) &= \frac{d\ln D}{d\ln a} \sigma_{8,0} D(a), \quad E_G(z) \simeq \frac{\Omega_{m0}\Sigma(z)}{f(z)}.
\end{aligned}$$

D Likelihood definitions

$$\chi_{\text{tot}}^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{RSD}}^2 + \chi_{E_G}^2.$$

Supernova marginalization:

$$\chi_{\text{SN,marg}}^2 = \Delta^T C^{-1} \Delta - \frac{(1^T C^{-1} \Delta)^2}{1^T C^{-1} 1}.$$

BAO:

$$\begin{aligned}
D_M/r_d &= qD_C, \quad D_H/r_d = q/E, \\
D_V/r_d &= q[zD_C^2/E]^{1/3}.
\end{aligned}$$

Growth:

$$\chi_{\text{RSD}}^2 = (d_{\text{RSD}} - m_{\text{RSD}})^T C_{\text{RSD}}^{-1} (d_{\text{RSD}} - m_{\text{RSD}}).$$

Weyl:

$$\chi_{E_G}^2 = (d_{E_G} - m_{E_G})^T C_{E_G}^{-1} (d_{E_G} - m_{E_G}).$$

Table 8: Exact code-path check at the Stage 2c G1DE-2 reference point.

block	χ^2
SN	1749.609044
BAO	9.633696
growth	6.883906
E_G	1.010038
total	1767.136683

E Exact code-path table

References

- [1] D. Brout et al., “The Pantheon+ Analysis: Cosmological Constraints,” *Astrophys. J.* **938**, 110 (2022).
- [2] DESI Collaboration, “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations,” arXiv:2503.14738 (2025).
- [3] A. Kazantzidis and L. Perivolaropoulos, “Evolution of the $f\sigma_8$ tension with the Planck15/ Λ CDM determination and implications for modified gravity,” *Phys. Rev. D* **97**, 103503 (2018).
- [4] C. Blake et al., “The WiggleZ Dark Energy Survey: the growth rate of cosmic structure since redshift $z = 0.9$,” *Mon. Not. Roy. Astron. Soc.* **425**, 405–414 (2012).
- [5] A. Amon et al., “KiDS+2dFLenS+GAMA: Testing gravity with E_G ,” *Mon. Not. Roy. Astron. Soc.* **479**, 3422–3437 (2018).
- [6] M. Chevallier and D. Polarski, “Accelerating universes with scaling dark matter,” *Int. J. Mod. Phys. D* **10**, 213–224 (2001).
- [7] E. V. Linder, “Exploring the expansion history of the universe,” *Phys. Rev. Lett.* **90**, 091301 (2003).
- [8] H. Akaike, “A new look at the statistical model identification,” *IEEE Trans. Autom. Control* **19**, 716–723 (1974).
- [9] G. Schwarz, “Estimating the dimension of a model,” *Ann. Statist.* **6**, 461–464 (1978).
- [10] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* **75**, 1260–1263 (1995).
- [11] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* **7**, 2333–2346 (1973).
- [12] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* **43**, 199–220 (1975).
- [13] Y. Wu, “Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers,” FDS–G1 Core (2026).
- [14] Y. Wu, “D0: Screen Modular Anomaly Inflow and the Einstein Calibration of Finite Access Entropy,” FDS–G1 technical note (2026).
- [15] Y. Wu, “Horizon occupancy and the effective weak-field response,” FDS–G1 Companion D (2026).

- [16] Y. Wu, “Horizon occupancy, Weyl response, and gravitational-lensing consistency,” FDS–G1 Companion E (2026).

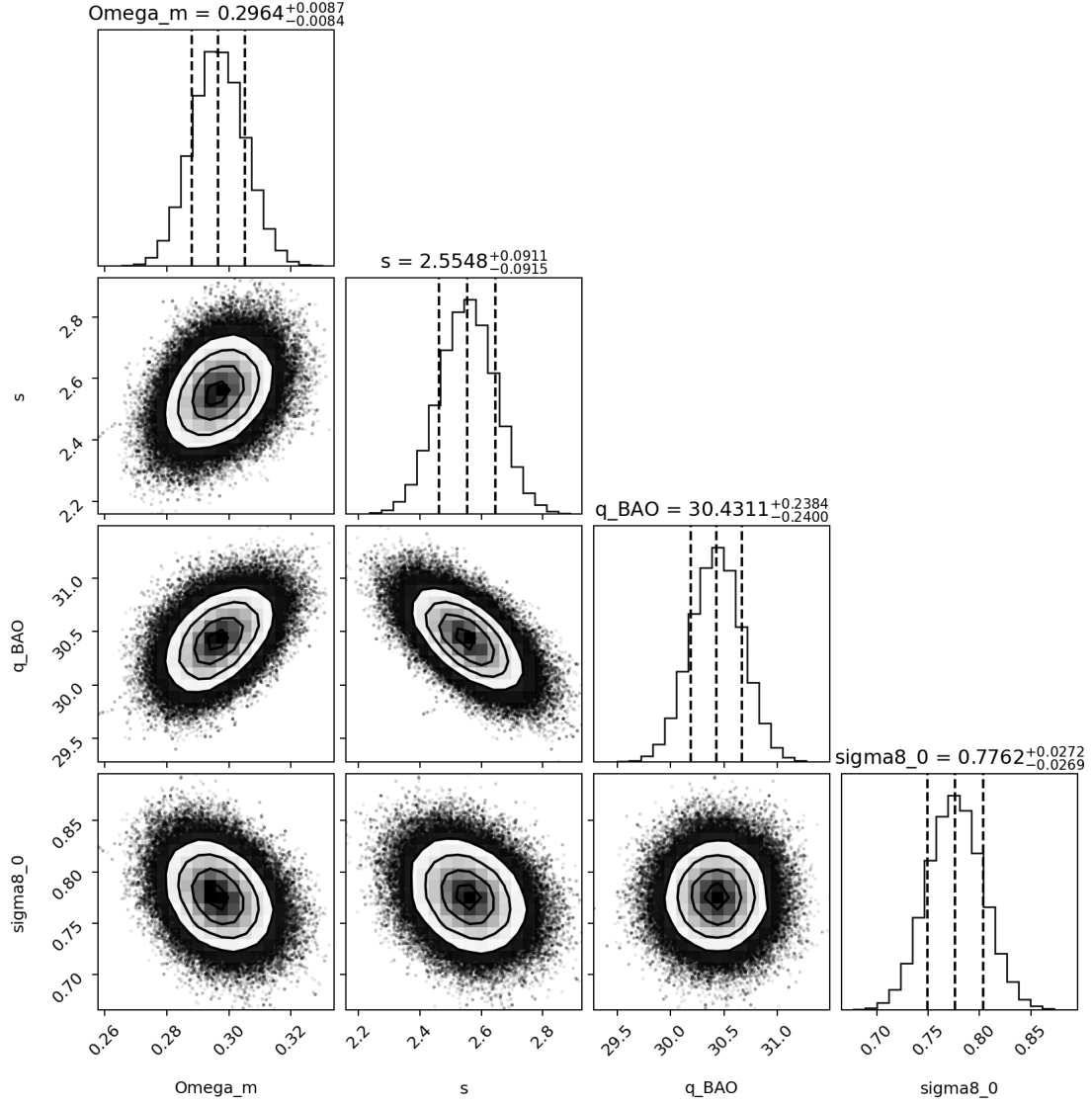


Figure 6: Combined three-seed exact posterior for G1DE- $M_{3/4}$. The $s = 3$ value marks the Λ CDM-like filling limit. The derived $\Sigma_0 = -\frac{3}{4}(3 - s)$ gives $\Sigma_0 < 0$ with $P(\Sigma_0 < 0) = 1.0$.

Screen Modular Anomaly Inflow and the Einstein Calibration of Finite Access Entropy

From Finite-Index Screen Algebras to $A/4\ell_P^2$, All-Null Gravity, and the G1 Weyl Residual

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

Draft: May 2026

Abstract

This paper formulates a foundational closure route for the FDS–G1 finite-screen program. The central proposal is that the Bekenstein–Hawking area coefficient, the local screen first law, the all-null Einstein response, the 3/4 optical projection law, and the G1DE Weyl residual can be organized as different sectors of a single screen modular anomaly-inflow structure. The key non-circular variable is not the Planck area itself, but a finite-access screen capacity density

$$\eta_0 = \frac{1}{k_B} \frac{dS_{\text{scr}}}{dA},$$

interpreted as the leading finite-index density of causal screen access algebras. If local screen modular flow is boost-calibrated,

$$K_S = \frac{2\pi}{\hbar} Q_\chi,$$

then the first variation of finite relative entropy gives

$$\delta S_{\text{scr}}/k_B = \delta \langle K_S \rangle = \frac{2\pi}{\hbar} \delta Q_\chi.$$

Combining this modular first law with the area response

$$\delta S_{\text{scr}}/k_B = \eta_0 \delta A,$$

Raychaudhuri’s null area variation, and the local boost matter flux gives

$$R_{\mu\nu} k^\mu k^\nu = \frac{2\pi}{\hbar \eta_0} T_{\mu\nu} k^\mu k^\nu.$$

Thus

$$G = \frac{c^3}{4\hbar \eta_0}, \quad S_H/k_B = \eta_0 A_H = \frac{A_H}{4\ell_P^2},$$

and all-null plus Ward/Bianchi closure yields the Einstein equation with a cosmological integration constant.

The same framework also identifies the finite-screen residual sector. Define the local modular anomaly-inflow defect by

$$\delta \mathfrak{A}_{\text{scr}} = \eta_0 \delta A - \frac{2\pi}{\hbar} \delta Q_\chi.$$

The anomaly-cancelled branch, $\delta \mathfrak{A}_{\text{scr}} = 0$, is the Einstein equilibrium branch. A passive horizon-scale remnant of incomplete anomaly inflow produces a retarded response $\mathcal{R}_H$. The local optical anomaly Hessian on

$$T_I GL(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1)$$

has compliance $C = K^{-1}$; projection to the Weyl-active unimodular sector $U = S \oplus \Omega$ gives

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C}.$$

At the unbiased finite-access fixed point, $K_A = K_S = K_\Omega$, one obtains $\kappa_{BW} = 3/4$. Weyl normalization and minimum Ward completion then imply

$$\Sigma(a, k) - 1 = -\frac{3}{4}\mathcal{R}_H(a) + O(\mathcal{R}_H^2),$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

With $\mathcal{R}_H(a) = (3-s)\hat{R}_H(a)$, $\hat{R}_H(1) = 1$, the G1DE-M_{3/4} reduction follows. The result is a conditional closure theorem: it does not claim that all FDS microscopic ingredients have been derived from bare finite-distinction axioms, but it sharply identifies which finite-access modular lemmas remain to be proved.

1 Purpose and claim-status discipline

The FDS-G1 program starts from a primitive finite causal-screen entropy functional

$$S_{\text{scr}} : X_{\text{phys}} \rightarrow \mathbb{R}, \quad (1)$$

where X_{phys} is a physical quotient state space of finite screen variables. Earlier G1 notes use this object to build a finite-screen response theory, an optical-stress closure, a non-equilibrium screen influence functional, and a Weyl-normalized production rule for the G1 dark-sector reduction. The present paper asks a deeper question: what common modular structure can simultaneously calibrate entropy density, boost energy, null focusing, and the finite-screen residual?

The answer proposed here is *screen modular anomaly inflow*. The main equation is

$$\boxed{\delta \mathfrak{A}_{\text{scr}} = \eta_0 \delta A - \frac{2\pi}{\hbar} \delta Q_\chi.} \quad (2)$$

The zero-defect branch gives the Einstein equilibrium sector. A passive horizon-scale Weyl remnant gives the G1DE residual branch.

1.1 What is being claimed

This paper makes the following conditional claims.

1. The local screen first law can be obtained from the first variation of finite relative entropy once the screen modular Hamiltonian is boost-local.
2. The Bekenstein–Hawking area coefficient can be written as an output of the calibration relation between finite screen capacity density η_0 and gravitational stiffness G^{-1} .
3. The all-null Einstein response follows from modular anomaly cancellation plus Raychaudhuri null area variation and Ward/Bianchi closure.
4. The G1DE residual is naturally interpreted as a passive horizon-scale remnant of incomplete modular anomaly inflow.
5. The 3/4 projection law is the unbiased Weyl-channel compliance fraction of the local optical anomaly Hessian.

Lemma stack. The strong calibration route requires five lemmas:

1. **D0.1 – Finite-Index Screen Algebra Existence.** There exists a causal-screen algebra net $S \mapsto \mathcal{A}(S)$ with finite-index inclusions for nested screens.
2. **D0.2 – Universal Capacity Density.** $dC_{\text{scr}} = \eta_0 dA$.
3. **D0.3 – Boost Modular Calibration.** $K_S = (2\pi/\hbar)Q_\chi$.
4. **D0.4 – Modular Anomaly Cancellation.** $\delta A_{\text{scr}} = 0 \Rightarrow \delta Q = T\delta S$.
5. **D0.5 – All-Null Ward/Bianchi Completion.** $\delta Q = T\delta S$ for all local null screens $\Rightarrow$ Einstein-like equilibrium sector.

Only if D0.1–D0.5 hold does G1 upgrade from bridge program to derivation program.

1.2 What is not being claimed

This is a conditional foundational closure, not a completed microscopic derivation from bare FDS axioms. In particular, this paper does not prove from first principles that a specific finite-index causal-access algebra net exists, that its finite-index density is universal, or that half-sided modular covariance necessarily holds for all physical screens. Instead, it identifies the exact mathematical lemmas that would upgrade the current G1 bridge program into a derivation program.

Object	Role	Status
η_0	finite-access screen capacity/index density	assumed fixed point
K_S	modular Hamiltonian of screen reference state	finite-state definition
$K_S = (2\pi/\hbar)Q_\chi$	boost modular calibration	conditional modular lemma
$\delta D = 0$	relative-entropy first variation	finite-state identity
$\delta \mathfrak{A}_{\text{scr}} = 0$	local anomaly cancellation	closure condition
Raychaudhuri	null area variation	metric-envelope assumption
Ward/Bianchi closure	all-null tensor completion	geometric naturality
3/4 law	unbiased optical anomaly Hessian	conditional theorem
G1DE	passive horizon anomaly remnant	effective reduction

Table 1: Claim-status map for the modular anomaly-inflow closure.

2 Finite-index screen algebras and capacity density

2.1 Screen access algebras

Let every causal screen patch $\mathcal{S}$ carry a finite-dimensional access algebra

$$\mathcal{A}(\mathcal{S}), \tag{3}$$

encoding the finite distinctions maintainable and accessible on that screen. If two screens are nested,

$$\mathcal{S}_1 \subset \mathcal{S}_2, \tag{4}$$

then the corresponding access algebras are related by a finite-index inclusion or by a restriction channel. In the algebraic version one writes

$$\mathcal{A}(\mathcal{S}_1) \subset \mathcal{A}(\mathcal{S}_2), \tag{5}$$

with conditional expectation

$$E_{21} : \mathcal{A}(\mathcal{S}_2) \rightarrow \mathcal{A}(\mathcal{S}_1). \quad (6)$$

The finite index

$$[\mathcal{A}(\mathcal{S}_2) : \mathcal{A}(\mathcal{S}_1)] < \infty \quad (7)$$

measures the local finite-access capacity increment associated with enlarging the screen ledger.

Definition 1 (Finite-index capacity increment). *For a finite-index screen algebra inclusion, define the dimensionless capacity increment*

$$\Delta C_{21} = \log[\mathcal{A}(\mathcal{S}_2) : \mathcal{A}(\mathcal{S}_1)]. \quad (8)$$

This is interpreted as the additional maintainable access capacity required to pass from $\mathcal{S}_1$ to $\mathcal{S}_2$.

2.2 Capacity density and the area law

A local continuum limit is obtained when the finite-index capacity increment has a universal density on the codimension-two screen:

$$dC_{\text{scr}} = \eta_0 dA. \quad (9)$$

Then

$$\boxed{S_{\text{scr}}/k_B = C_{\text{scr}} = \int_S \eta_0 dA = \eta_0 A + o(A).} \quad (10)$$

Here η_0 has dimensions of inverse area. It is not introduced as $1/(4\ell_P^2)$. Rather, it is the finite-access index density whose gravitational calibration will determine G and hence ℓ_P .

Principle 1 (Finite-index density). *The leading screen entropy density is the local finite-index density of causal-access algebra inclusions. Subleading curvature, topological, or finite-size corrections may exist, but the leading local valuation on a codimension-two causal screen is area-extensive.*

2.3 Non-circularity

A central point is that one should not begin by assuming

$$S_H/k_B = \frac{A_H}{4\ell_P^2}, \quad (11)$$

because $\ell_P^2 = G\hbar/c^3$ already contains the gravitational coupling. The non-circular route is instead

$$S_H/k_B = \eta_0 A_H, \quad (12)$$

derive

$$G = \frac{c^3}{4\hbar\eta_0}, \quad (13)$$

and only then write

$$\ell_P^2 = \frac{G\hbar}{c^3} = \frac{1}{4\eta_0}. \quad (14)$$

Thus the Bekenstein–Hawking coefficient is the output

$$S_H/k_B = \eta_0 A_H = \frac{A_H}{4\ell_P^2}. \quad (15)$$

3 Relative entropy and the screen modular first law

3.1 Finite relative entropy

Let σ_S be a full-rank reference state on the finite screen code subspace, and let ρ_S be a nearby screen state. If σ_S is not full rank, we restrict the local family to the common support on which the relative entropy is finite. Define the modular Hamiltonian

$$K_S = -\log \sigma_S. \quad (16)$$

The finite relative entropy is

$$D(\rho_S \| \sigma_S) = \text{Tr } \rho_S (\log \rho_S - \log \sigma_S) \quad (17)$$

$$= \Delta \langle K_S \rangle - \Delta S_S / k_B. \quad (18)$$

At the reference state, the first variation vanishes:

$$\delta D(\rho_S \| \sigma_S) \big|_{\rho=\sigma} = 0. \quad (19)$$

Therefore

$$\boxed{\delta S_S / k_B = \delta \langle K_S \rangle}. \quad (20)$$

This is the finite-screen modular first law. It is not an independent thermodynamic postulate; it is the first variation of relative entropy.

3.2 Boost modular calibration

The additional physical input is that the local causal screen reference state is boost-modular:

$$\boxed{K_S = \frac{2\pi}{\hbar} Q_\chi}, \quad (21)$$

where Q_χ is the local boost energy associated with the approximate null/Rindler generator χ . This is the finite-access analogue of local boost KMS or half-sided modular covariance. Combining (20) and (21) gives

$$\boxed{\delta S_S / k_B = \frac{2\pi}{\hbar} \delta Q_\chi}. \quad (22)$$

Equivalently, with the usual local acceleration or surface-gravity normalization,

$$\delta Q_\chi = T \delta S_S, \quad T = \frac{\hbar \kappa}{2\pi k_B c}. \quad (23)$$

Remark 1 (Where the hard problem lies). *The identity $\delta S / k_B = \delta \langle K \rangle$ is finite-dimensional relative entropy. The hard FDS lemma is (21): causal finite access must lead to a local boost modular Hamiltonian with the universal coefficient $2\pi / \hbar$.*

4 Screen modular anomaly inflow

4.1 Anomaly functional

The two local response laws are

$$\delta S_{\text{scr}} / k_B = \eta_0 \delta A, \quad (24)$$

from finite-index screen capacity, and

$$\delta S_{\text{scr}} / k_B = \frac{2\pi}{\hbar} \delta Q_\chi, \quad (25)$$

from boost modular calibration. Define the local screen modular anomaly-inflow defect

$$\delta\mathfrak{A}_{\text{scr}} = \eta_0 \delta A - \frac{2\pi}{\hbar} \delta Q_\chi. \quad (26)$$

The anomaly-cancelled sector is

$$\delta\mathfrak{A}_{\text{scr}} = 0. \quad (27)$$

This is the local equilibrium branch that will produce Einstein gravity.

4.2 Interpretation

The quantity $\delta\mathfrak{A}_{\text{scr}}$ is a calibration mismatch between finite-access screen index and bulk modular energy. If it vanishes locally, the screen ledger and the bulk boost energy can be matched by a metric response. If it does not vanish, the mismatch is a finite-screen residual. A key claim of the G1 branch is that the leading cosmological remnant is not arbitrary. It is passive, retarded, horizon-scale, and projected into the Weyl optical channel.

Principle 2 (Screen modular anomaly inflow). *The bulk metric response cancels the local finite-screen modular anomaly. The anomaly-cancelled branch is the Einstein equilibrium sector. Controlled nonzero remnants define finite-screen residual sectors.*

5 Null deformation and Einstein calibration

5.1 Raychaudhuri area variation

Assume the anomaly-free sector metricizes to a Lorentzian envelope. Then null congruence kinematics are governed by Raychaudhuri's equation

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}k^\mu k^\nu. \quad (28)$$

For a local causal screen patch in equilibrium, take

$$\theta|_p = 0, \quad \sigma_{\mu\nu}|_p = 0, \quad \omega_{\mu\nu} = 0. \quad (29)$$

To first order away from the reference point,

$$\theta(\lambda) = -\lambda R_{\mu\nu}k^\mu k^\nu + O(\lambda^2). \quad (30)$$

Thus

$$\delta A = - \int \lambda R_{\mu\nu}k^\mu k^\nu d\lambda dA, \quad (31)$$

up to the orientation convention for the local horizon generator.

5.2 Boost energy flux

The local boost energy flux is

$$\delta Q_\chi = \int \lambda T_{\mu\nu}k^\mu k^\nu d\lambda dA, \quad (32)$$

again up to the same orientation convention.

5.3 All-null response

Set $\delta\mathfrak{A}_{\text{scr}} = 0$. Using (31) and (32) in (26), one obtains

$$R_{\mu\nu}k^\mu k^\nu = \frac{2\pi}{\hbar\eta_0} T_{\mu\nu}k^\mu k^\nu \quad (33)$$

for all local null vectors k^μ . Define

$$\boxed{8\pi G = \frac{2\pi}{\hbar\eta_0}} \quad (34)$$

in units where $c = 1$, or

$$\boxed{G = \frac{c^3}{4\hbar\eta_0}} \quad (35)$$

in physical units. Then

$$R_{\mu\nu}k^\mu k^\nu = 8\pi G T_{\mu\nu}k^\mu k^\nu. \quad (36)$$

6 The Einstein calibration theorem

Theorem 1 (Screen modular anomaly inflow and Einstein calibration). *Assume:*

1. a finite-index screen entropy density $S_{\text{scr}}/k_B = \eta_0 A + o(A)$;
2. finite relative entropy and the first-variation identity $\delta S/k_B = \delta\langle K_S \rangle$;
3. boost modular calibration $K_S = (2\pi/\hbar)Q_\chi$;
4. a Lorentzian metric envelope in the anomaly-cancelled sector, so that Raychaudhuri null area variation holds;
5. local matter boost flux $\delta Q_\chi = \int \lambda T_{\mu\nu}k^\mu k^\nu d\lambda dA$;
6. all-null closure and Ward/Bianchi naturality.

Then the anomaly-cancelled sector $\delta\mathfrak{A}_{\text{scr}} = 0$ yields

$$G = \frac{c^3}{4\hbar\eta_0}, \quad (37)$$

$$S_H/k_B = \eta_0 A_H = \frac{A_H}{4\ell_P^2}, \quad (38)$$

and

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (39)$$

Proof. The relative-entropy first variation and boost calibration give

$$\delta S/k_B = \frac{2\pi}{\hbar} \delta Q_\chi. \quad (40)$$

The finite-index area response gives $\delta S/k_B = \eta_0 \delta A$. Thus

$$\eta_0 \delta A = \frac{2\pi}{\hbar} \delta Q_\chi. \quad (41)$$

Insert Raychaudhuri area variation and the boost matter flux. Since the equality holds for arbitrary sufficiently small local causal screens and all null directions, one obtains (36). Hence

$$R_{\mu\nu} - 8\pi G T_{\mu\nu} = \Phi g_{\mu\nu} \quad (42)$$

for some scalar Φ . Taking a divergence and using $\nabla^\mu T_{\mu\nu} = 0$ and $\nabla^\mu G_{\mu\nu} = 0$ gives

$$\Phi = \frac{1}{2}R - \Lambda \quad (43)$$

up to a cosmological integration constant, yielding the Einstein equation. Equation (35) gives

$$\ell_P^2 = \frac{G\hbar}{c^3} = \frac{1}{4\eta_0}, \quad (44)$$

and therefore $S_H/k_B = \eta_0 A_H = A_H/(4\ell_P^2)$. $\square$

Remark 2 (Why this is non-circular). *The derivation does not assume $S = A/(4\ell_P^2)$ at the start. It assumes an independent finite-access density η_0 , then calibrates gravitational stiffness by modular anomaly cancellation. Only afterward does $\eta_0 = 1/(4\ell_P^2)$ follow.*

7 Einstein–Hilbert inflow and uniqueness at leading order

The anomaly cancellation condition can also be read from an effective action viewpoint. The screen anomaly functional $\mathfrak{A}_{\text{scr}}$ must be cancelled by a local bulk response. Under the assumptions of locality, diffeomorphism covariance, two-derivative leading order, and a Lorentzian metric envelope, the leading bulk inflow counterterm is the Einstein–Hilbert action

$$I_{\text{EH}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} (R - 2\Lambda). \quad (45)$$

Its null variation supplies the Ricci focusing term that cancels the finite-screen modular anomaly. Higher-derivative terms can appear as subleading corrections, but the leading all-null two-derivative inflow is Einsteinian.

Principle 3 (Leading inflow uniqueness). *Within the two-derivative, local, diffeomorphism-covariant, Lorentzian metric-envelope approximation, Einstein–Hilbert gravity is the leading bulk inflow response that cancels the finite-screen modular anomaly.*

This principle is conditional. It does not exclude higher-derivative finite-size corrections, quantum corrections, nonlocal memory terms, or non-equilibrium residual sectors. It states only that the anomaly-cancelled local equilibrium limit is Einstein at leading order.

8 Optical anomaly Hessian and the 3/4 projection law

8.1 Optical tangent space

The local null screen optical deformation algebra at the identity is

$$T_I GL(2, \mathbb{R}) \simeq \mathfrak{gl}(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1). \quad (46)$$

Here A is the trace/area direction, S is the two-dimensional symmetric shear sector, and Ω is the twist/holonomy sector. A local deformation is written

$$q = q^I e_I. \quad (47)$$

8.2 Anomaly Hessian

Let $\mathfrak{A}_{\text{mod}}(q)$ denote the local modular anomaly or sufficiency loss under optical deformation. Near the reference screen,

$$\mathfrak{A}_{\text{mod}}(q) = \frac{1}{2} K_{IJ} q^I q^J + O(q^3), \quad (48)$$

where K_{IJ} is the relative-entropy/Fisher Hessian. In an $SO(2)$ -adapted basis,

$$K = K_A P_A + K_S P_S + K_\Omega P_\Omega. \quad (49)$$

The compliance is

$$C = K^{-1}. \quad (50)$$

The Weyl-active unimodular sector is

$$U = S \oplus \Omega, \quad P_U = P_S + P_\Omega. \quad (51)$$

All traces below are taken in an $SO(2)$ -adapted orthonormal optical generator basis at the reference point. Anisotropic generator normalizations are absorbed into the block weights K_A, K_S, K_Ω .

Definition 2 (Weyl compliance fraction). *The background–Weyl optical compliance fraction is*

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}. \quad (52)$$

Theorem 2 (Unbiased anomaly Hessian and the 3/4 projection law). *If the finite-access modular anomaly Hessian is unbiased over the full local optical tangent space,*

$$K_A = K_S = K_\Omega = K_0, \quad (53)$$

then

$$\kappa_{BW} = \frac{3}{4}. \quad (54)$$

If the twist/holonomy ledger is demoted, $K_\Omega^{-1} = 0$, while $K_A = K_S$, then

$$\kappa = \frac{2}{3}. \quad (55)$$

Proof. For the unbiased full optical ledger, $C = K_0^{-1} I$. The Weyl-active sector $U = S \oplus \Omega$ has dimension $2 + 1 = 3$, while the full $GL(2)$ optical tangent space has dimension 4. Hence

$$\kappa_{BW} = \frac{3K_0^{-1}}{4K_0^{-1}} = \frac{3}{4}. \quad (56)$$

If the twist/holonomy compliance is removed, only $A(1) \oplus S(2)$ remains, and the Weyl-active sector is the two-dimensional shear sector. Thus $\kappa = 2/3$. $\square$

Remark 3 (Conditional status). *The theorem does not claim that all physical screens are unbiased. It identifies the isotropic finite-access fixed point. Weighted-compliance deviations are represented by K_A, K_S, K_Ω and are tested by free- κ controls in the data-facing GIDE hierarchy.*

9 Horizon anomaly remnant and the G1DE branch

9.1 Horizon modular occupancy

The local anomaly-cancelled branch is Einsteinian. A cosmological residual can arise if finite-access modular inflow is not perfectly cancelled at the horizon scale. Let the available horizon capacity be

$$C_H(a) = \eta_0 A_H(a). \quad (57)$$

Let $C_{\text{used}}(a)$ denote the horizon-accessible finite-index recovery load: the amount of screen access capacity needed to glue or recover the horizon-accessible finite distinctions at epoch a . Define

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}. \quad (58)$$

The corresponding anomaly-load current is

$$\boxed{J_H(N) = -d_N \ln \mathcal{O}_H(N), \quad N = \ln a.} \quad (59)$$

A causal, stable, passive memory kernel gives

$$\boxed{\mathcal{R}_H(N) = \int_{-\infty}^N G_H^{\text{ret}}(N, N') J_H(N') dN'.} \quad (60)$$

The Markov limit is

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H, \quad \Gamma_H > 0. \quad (61)$$

Passivity requires

$$\int dN J_H(N) \mathcal{R}_H(N) \geq 0 \quad (62)$$

for admissible perturbations, up to sign conventions.

9.2 Weyl-normalized transfer

The production-level G1DE rule is not to add a freely normalized dark stress. Instead, the finite-screen anomaly remnant acts as a fractional correction to the GR Weyl source:

$$\mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta, \quad (63)$$

$$\boxed{\mathcal{W}_{\text{G1}}(a, k) = [1 - \kappa_{BW} \mathcal{R}_H(a)] \mathcal{W}_{\text{GR}}(a, k) + O(\mathcal{R}_H^2).} \quad (64)$$

Thus

$$\boxed{\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2).} \quad (65)$$

Minimum Ward completion gives only Ricci-channel leakage,

$$\boxed{\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).} \quad (66)$$

9.3 Low-redshift reduction

Normalize

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1. \quad (67)$$

In the G1DE-M reduction,

$$R_0 = 3 - s. \quad (68)$$

For the unbiased full optical ledger,

$$\kappa_{BW} = \frac{3}{4}. \quad (69)$$

Therefore

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a) + O((3 - s)^2), \quad \mu \simeq 1. \quad (70)$$

This is the low-redshift, Weyl-normalized, Ward-disciplined G1DE-M_{3/4} branch.

Remark 4 (Data-facing boundary). *The current G1fit-real evidence hierarchy tests the low-redshift locked reduction (70), not the full screen modular anomaly-inflow construction. The present paper supplies the foundational conditional structure behind that reduction.*

10 Why $A(a, k)$ is not a fitted amplitude

In some intermediate optical-stress notation one may write a seed amplitude $A(a, k)$ multiplying a traceless optical tensor. In the Weyl-normalized convention this is not a physical response function. It is the map that identifies a unit optical seed with a unit fractional correction to the GR Weyl source:

$$A(a, k) \longleftrightarrow \mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta. \quad (71)$$

Sampling $A(a, k)$ as an independent nuisance function would change the model class. It would convert G1DE from a finite-screen Weyl transfer model into a generic dark-stress envelope.

Implementation rule. The optical normalization $A(a, k)$ is fixed by Weyl-source normalization. It is not a G1DE parameter and must not be sampled as an independent response function. The admissible G1DE response is the fractional transfer $\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}$, with Ward-suppressed Newtonian leakage.

11 Open lemmas

The anomaly-inflow theorem isolates the remaining hard problems.

11.1 Finite-index anomaly coefficient

One must construct a natural causal finite-access algebra net

$$\mathcal{S} \mapsto \mathcal{A}(\mathcal{S}) \quad (72)$$

whose inclusions have a universal local finite-index density η_0 . This is the microscopic origin of the area valuation.

11.2 Half-sided modular boost theorem

One must show that local causal screen nesting and finite-access passivity imply boost modular flow:

$$K_{\mathcal{S}} = \frac{2\pi}{\hbar} Q_{\mathcal{X}}. \quad (73)$$

This is the deepest source of the 2π factor.

11.3 Metricization of the anomaly-free sector

One must prove that the anomaly-free finite-access modular net admits a Lorentzian metric envelope. Once this is established, Raychaudhuri coefficients and Bianchi identities are geometric consequences.

11.4 Uniqueness of leading bulk inflow

One must show that, under the G1 assumptions of locality, diffeomorphism covariance, two-derivative leading order, and metricization, Einstein–Hilbert inflow is the unique leading counterterm that cancels the screen modular anomaly.

11.5 Horizon remnant reduction

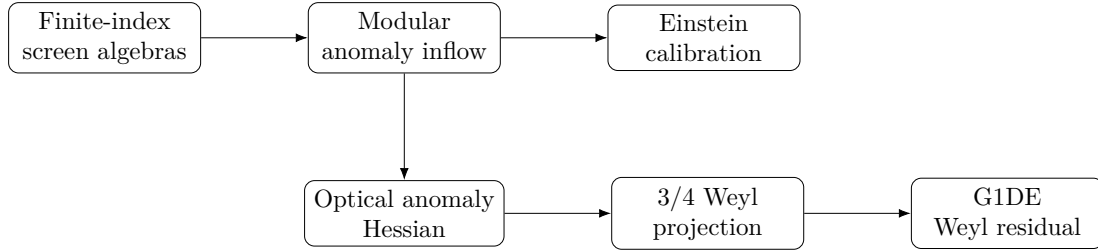
One must derive

$$\mathcal{R}_H = G_H^{\text{ret}} * J_H \quad (74)$$

from a concrete horizon-scale anomaly-load model and connect $\hat{R}_H(a)$ to a computable cosmological finite-access ensemble. The current G1DE implementation uses $\mathcal{R}_H(a) = (3 - s)\hat{R}_H(a)$ as a low-redshift reduction.

12 Relation to the G1 closure sequence

The present note should be read as a foundational calibration note. Its role in the G1 sequence is:



The logical placement is before the finite-access sheaf geometry note (D1): the present note (D0) explains the modular calibration of entropy, boost energy, and gravity; the sheaf geometry note (D1) explains how finite ledgers glue across causal screens and how the local optical Hessian yields the projection law.

13 Conclusion

This paper proposes a common modular origin for the Einstein equilibrium branch and the G1 finite-screen residual branch. The core identity is

$$\delta \mathfrak{A}_{\text{scr}} = \eta_0 \delta A - \frac{2\pi}{\hbar} \delta Q_\chi. \quad (75)$$

When the local finite-screen modular anomaly is cancelled, relative entropy first variation, boost modular calibration, Raychaudhuri null area variation, and Ward/Bianchi closure produce Einstein gravity with

$$G = \frac{c^3}{4\hbar\eta_0}, \quad S_H/k_B = \frac{A_H}{4\ell_P^2}. \quad (76)$$

When a passive horizon-scale remnant remains, its Weyl-projected low-redshift reduction gives

$$\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu - 1 \ll \Sigma - 1 \quad (77)$$

under the unbiased optical anomaly Hessian.

Thus the G1 dark-sector branch is not appended to Einstein gravity as an arbitrary dark stress. It is the controlled nonzero remnant of the same finite-screen modular anomaly-inflow structure whose zero-defect branch calibrates Einstein gravity. The remaining work is no longer a vague request to “derive gravity”. It is a set of precise lemmas: construct the finite-index causal-access algebra net, prove boost modular covariance, prove metricization of the anomaly-free sector, establish leading inflow uniqueness, and compute the horizon anomaly remnant from a concrete finite-access cosmological ensemble.

A Units and signs

Most derivations above use the dimensionless entropy S/k_B . The density η_0 has dimensions of inverse area:

$$S/k_B = \eta_0 A. \quad (78)$$

In units where $c = 1$, the calibration is

$$G = \frac{1}{4\hbar\eta_0}. \quad (79)$$

Restoring c ,

$$G = \frac{c^3}{4\hbar\eta_0}. \quad (80)$$

The sign of δA depends on whether one uses past or future local horizons and on the orientation of the affine parameter. The physical all-null response coefficient is orientation-independent once the boost flux and area variation are defined consistently.

B All-null tensor completion

Suppose

$$X_{\mu\nu}k^\mu k^\nu = 0 \quad (81)$$

for every null vector k^μ . Then

$$X_{\mu\nu} = \Phi g_{\mu\nu} \quad (82)$$

for some scalar Φ . Applying this to

$$X_{\mu\nu} = R_{\mu\nu} - 8\pi G T_{\mu\nu} \quad (83)$$

gives

$$R_{\mu\nu} - 8\pi G T_{\mu\nu} = \Phi g_{\mu\nu}. \quad (84)$$

Taking the divergence and using stress-energy conservation and the contracted Bianchi identity yields

$$\Phi = \frac{1}{2}R - \Lambda, \quad (85)$$

so that

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (86)$$

C Relation to standard entanglement-equilibrium arguments

The relative-entropy identity

$$\delta S = \delta \langle K \rangle \tag{87}$$

is standard in entanglement first-law arguments. The distinct G1 contribution is not the identity itself, but its reinterpretation inside a finite causal-screen access program with a finite-index capacity density η_0 , an optical anomaly Hessian producing the 3/4 Weyl projection, and a horizon-scale anomaly remnant producing the G1DE residual.

D Selected references

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Physical Review D* **7**, 2333–2346 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Communications in Mathematical Physics* **43**, 199–220 (1975).
- [3] W. G. Unruh, “Notes on black-hole evaporation,” *Physical Review D* **14**, 870–892 (1976).
- [4] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Physical Review Letters* **75**, 1260–1263 (1995).
- [5] R. M. Wald, “Black hole entropy is the Noether charge,” *Physical Review D* **48**, R3427–R3431 (1993).
- [6] J. J. Bisognano and E. H. Wichmann, “On the duality condition for a Hermitian scalar field,” *Journal of Mathematical Physics* **16**, 985–1007 (1975).
- [7] M. Takesaki, *Theory of Operator Algebras II*, Springer (2003).
- [8] D. Petz, “Sufficient subalgebras and the relative entropy of states of a von Neumann algebra,” *Communications in Mathematical Physics* **105**, 123–131 (1986).
- [9] R. Longo, “Notes on algebraic invariants for noncommutative dynamical systems,” *Communications in Mathematical Physics* **69**, 195–207 (1979), and subsequent developments on index and inclusions.
- [10] H. Casini, M. Huerta, and R. C. Myers, “Towards a derivation of holographic entanglement entropy,” *Journal of High Energy Physics* **2011**, 036 (2011).
- [11] N. Lashkari, M. B. McDermott, and M. Van Raamsdonk, “Gravitational dynamics from entanglement thermodynamics,” *Journal of High Energy Physics* **2014**, 195 (2014).
- [12] T. Faulkner, M. Guica, T. Hartman, R. C. Myers, and M. Van Raamsdonk, “Gravitation from entanglement in holographic CFTs,” *Journal of High Energy Physics* **2014**, 051 (2014).
- [13] O. Fawzi and R. Renner, “Quantum conditional mutual information and approximate Markov chains,” *Communications in Mathematical Physics* **340**, 575–611 (2015).
- [14] P. Hayden, O. Parrikar, and J. Sorce, “The Markov gap for geometric reflected entropy,” and related works on quantum Markov recovery and holographic codes.
- [15] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, FDS-G1 Core manuscript (2026).

- [16] Y. Wu, *Finite-Access Sheaf Geometry and the Relative-Entropy Origin of the $3/4$ Projection Law*, FDS-G1 technical note D1 (2026).
- [17] Y. Wu, *Weyl-Normalized Ward Completion and Identifiability Rules*, FDS-G1 technical note D5 (2026).
- [18] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background-Weyl Residual*, data paper (2026).

Finite-Access Sheaf Geometry and the Relative-Entropy Origin of the 3/4 Projection Law

A Conditional Closure of the G1 Finite-Screen Dark-Sector Branch

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

Draft: May 2026

Abstract

This paper develops a deeper mathematical closure for the finite-screen dark-sector branch of the FDS–G1 program. Previous G1 notes introduced a finite causal-screen entropy functional, derived a maximum-entropy optical projection law on the local null screen algebra, and implemented the resulting cosmological residual as a Weyl-normalized transfer correction. The present paper asks what mathematical object lies underneath these ingredients. The proposed answer is a finite-access sheaf: a contravariant assignment of finite, maintainable distinction ledgers to causal screens, with entropy as the coordinate-free ledger-size section and geometry as the integrable natural connection of the resulting response data.

The local optical fiber of this finite-access sheaf is the Jacobi deformation algebra

$$T_I GL(2) = A(1) \oplus S(2) \oplus \Omega(1),$$

where A is the area/trace channel, S is the two-dimensional symmetric shear sector, and Ω is the twist/holonomy sector. A finite-access screen state family

$$\rho_S(q) = \mathcal{E}_{S,q}(\rho_{\text{bulk}})$$

defines a relative-entropy Hessian

$$K_{IJ} = \partial_I \partial_J D_{\text{rel}}(\rho_S(q) \| \rho_S(0))|_{q=0},$$

which is the Fisher/Kubo–Mori metric of optical access distortions. The response compliance is $C = K^{-1}$. Projection onto the Weyl-active unimodular sector $U = S \oplus \Omega$ gives

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}.$$

The unbiased finite-access fixed point $K_A = K_S = K_\Omega$ yields $\kappa_{BW} = 3/4$, while demoting the twist/holonomy ledger gives the scalar-shear branch $\kappa_{BW} = 2/3$.

The global dark-sector residual is then interpreted as a horizon-scale gluing-load lag of the finite-access sheaf. The used horizon capacity C_{used} is the accessible distinction load required to glue horizon-accessible local ledgers; the occupancy ratio is $\mathcal{O}_H = C_{\text{used}}/C_H$ with $C_H = A_H/(4\ell_P^2)$, and the source current is $J_H = -d_N \ln \mathcal{O}_H$. A causal, stable, passive retarded response gives $\mathcal{R}_H = G_H^{\text{ret}} * J_H$. Weyl normalization and minimum Ward completion then imply

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2),$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

With $\mathcal{R}_H(a) = R_0 \widehat{R}_H(a)$, $\widehat{R}_H(1) = 1$, and $R_0 = 3 - s$, the isotropic branch recovers the G1DE-M_{3/4} reduction

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{R}_H(a), \quad \mu \simeq 1.$$

The result is a conditional closure theorem for the G1 dark-sector branch. It does not derive the Bekenstein–Hawking coefficient, the local screen first law, or the exact nonlinear Einstein equations from FDS primitives alone.

1 Scope and claim-status discipline

The FDS–G1 core program begins from a finite causal-screen entropy functional

$$S_{\text{scr}} : X_{\text{phys}} \rightarrow \mathbb{R}, \tag{1}$$

defined on a physical quotient state space of screen variables. The response one-form

$$\omega = \eta_i(X) dX^i, \quad \eta_i = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta X^i}, \tag{2}$$

contains the local area response that, under the standard horizon-entropy bridge, determines the local inverse gravitational coupling or screen stiffness. Previous G1 papers then developed screen statistics, stochastic screen dynamics, metric integrability, horizon occupancy residuals, and a likelihood-ready G1DE branch.

The present paper is not a replacement for those components. Its purpose is narrower and deeper: to provide a conditional mathematical object underneath the finite screen entropy functional, the optical response Hessian, and the 3/4 projection law. D0 supplies the local modular calibration of finite screen entropy, boost energy, and Einstein response. The present note addresses the complementary global/access question: how finite ledgers restrict, glue, and generate residual gluing defects across causal screens. The proposed object is a finite-access sheaf over causal screens.

1.1 What this paper claims

The paper makes four conditional claims.

D1 supplies the mathematical home for the Core’s response obstruction: $[\mathcal{C}_{\text{resp}}] \in H_{\text{resp}}^2(X_{\text{phys}})$. The sheaf-gluing language identifies which response non-closure survives all quotient, missing-variable, and memory-lift corrections as a true geometric residual.

The explicit correspondence is

$$\mathcal{F}_{\nabla_S} \longleftrightarrow [\mathcal{C}_{\text{resp}}] \longleftrightarrow \text{gluing defect class},$$

where $\mathcal{F}_{\nabla_S}$ is the finite-access sheaf curvature (D1, §4) and $[\mathcal{C}_{\text{resp}}]$ is the Core response obstruction class (Core, §7).

1. A finite causal-screen entropy functional can be represented as the entropy section of a finite-access sheaf over causal screens.
2. The optical-sector response Hessian is the relative-entropy/Fisher metric of local optical access distortions.

3. If the finite-access Hessian is locally unbiased over the full $GL(2)$ optical fiber, then the Weyl-active compliance share is $3/4$.
4. A horizon-scale gluing-load lag, when Weyl-normalized and minimally Ward-completed, produces the G1DE hierarchy $\Sigma - 1 = O(\mathcal{R}_H)$ and $\mu - 1 \ll \Sigma - 1$ on subhorizon linear scales.

1.2 What this paper does not claim

The result is not claimed to be an unconditional derivation of general relativity. In particular, this paper does not derive the Bekenstein–Hawking coefficient $1/(4\ell_P^2)$, the local screen first law, exact Raychaudhuri coefficients, or the full nonlinear Einstein equations from FDS primitives alone. Those remain deeper FDS-to-GR closure problems. The present result is a conditional closure of the G1 dark-sector branch, assuming the standard horizon entropy normalization and the G1 finite-screen response program.

Claim	Meaning	Status
Finite-access sheaf	causal screens carry finite maintainable distinction ledgers	programmatic FDS bridge
Entropy section	S_{scr} is the coordinate-free state-count ledger	definition/framework
Response connection	geometry is the integrable envelope of screen-response transport	conditional geometric bridge
Relative-entropy Hessian	K_{IJ} measures distinguishability of optical access distortions	information-geometric definition
$3/4$ projection law	unbiased full $GL(2)$ optical access gives $\kappa_{BW} = 3/4$	conditional theorem
$2/3$ demotion	twist/holonomy removed gives scalar-shear branch	conditional theorem
Horizon gluing lag	$\mathcal{R}_H$ is passive memory of horizon access-load drift	dynamical bridge
Weyl-normalized G1DE	$\mathcal{W}_{\text{G1}} = (1 - \kappa\mathcal{R}_H)\mathcal{W}_{\text{GR}}$	production convention

Table 1: Claim-status map for the finite-access sheaf closure.

2 From finite distinctions to finite-access sheaves

The primitive FDS argument can be stated compactly:

1. physical distinctions must be finite and maintainable;
2. maintainable distinctions require a boundary;
3. in gravity the relevant boundary is causal, namely a screen or horizon;
4. a finite causal screen has a finite state-count ledger;

5. the coordinate-free ledger of finite state-count is entropy;
6. geometry is the integrable response envelope of that ledger.

This paper sharpens the word “ledger”. A single causal screen does not stand alone. Screens refine, overlap, coarse-grain, restrict, and evolve. A finite-access theory should therefore assign ledgers not only to individual screens but consistently across a system of causal screens.

Definition 1 (Causal-screen category). *Let $\mathbf{Scr}$ denote a category whose objects are causal screen patches: local Rindler screens, causal-diamond cuts, light-sheet cross-sections, apparent-horizon patches, or observer-dependent optical screens. Morphisms*

$$f : \mathcal{S}_1 \rightarrow \mathcal{S}_2 \tag{3}$$

represent causal restriction, screen inclusion, coarse-graining, refinement, null evolution, or operational comparison between screens.

The direction of information access is naturally contravariant: a larger or later screen may restrict to a smaller ledger, while a refined screen may be coarse-grained to a lower-resolution ledger.

Definition 2 (Finite-access sheaf). *A finite-access sheaf is a contravariant assignment*

$$\mathcal{A} : \mathbf{Scr}^{\text{op}} \rightarrow \mathbf{FinLedger}, \tag{4}$$

where $\mathbf{FinLedger}$ is a category of finite maintainable distinction ledgers and ledger maps. A quantum version assigns finite-dimensional screen algebras or states,

$$\mathcal{A} : \mathbf{Scr}^{\text{op}} \rightarrow \mathbf{FinChan}, \tag{5}$$

where $\mathbf{FinChan}$ denotes finite stochastic channels or finite-dimensional quantum channels.

For a screen $\mathcal{S}$, the object $\mathcal{A}(\mathcal{S})$ is the finite ledger of distinctions accessible and maintainable on that screen. For a morphism $f : \mathcal{S}_1 \rightarrow \mathcal{S}_2$, the map $\mathcal{A}(f) : \mathcal{A}(\mathcal{S}_2) \rightarrow \mathcal{A}(\mathcal{S}_1)$ is the corresponding restriction or pullback of accessible distinctions.

Remark 1 (Why sheaf language is useful). *The word “sheaf” is used in a minimal sense. It encodes the idea that local screen ledgers must restrict and glue consistently. When they glue into a smooth metric envelope, the response looks geometric. When they fail to glue without residuals, the obstruction is a candidate dark or non-integrable response sector.*

Terminological convention. In this note, “sheaf” is used in an operational finite-access sense. Strictly, the assignment of finite ledgers to causal screens is first a *presheaf* over causal-screen contexts. When a causal-screen topology, covering families, and unique gluing conditions are imposed, the assignment satisfies a sheaf condition. Residual sectors are represented as failures of such gluing, or as obstruction data. We do not assume a full Grothendieck-topos construction in this paper.

3 Entropy as a ledger-size section

For each screen $\mathcal{S}$, the finite ledger may be represented by a classical distribution $p_{\mathcal{S}}$ or by a quantum state $\rho_{\mathcal{S}}$. The screen entropy is then the coordinate-free ledger size

$$S_{\text{scr}}(\mathcal{S}) = H(p_{\mathcal{S}}) \quad (6)$$

or

$$S_{\text{scr}}(\mathcal{S}) = S(\rho_{\mathcal{S}}) = -\text{Tr}(\rho_{\mathcal{S}} \ln \rho_{\mathcal{S}}). \quad (7)$$

When screen variables are collected as $X_{\mathcal{S}} \in X_{\text{phys}}(\mathcal{S})$, this becomes the familiar entropy functional

$$S_{\text{scr}} : X_{\text{phys}}(\mathcal{S}) \rightarrow \mathbb{R}. \quad (8)$$

Thus the G1 primitive object can be reinterpreted as a local expression of a global finite-access structure.

Principle 1 (Entropy section). *Finite screen entropy is the entropy section of the finite causal-access sheaf. It is not an arbitrary scalar field added to geometry; it is the coordinate-free state-count ledger associated with finite, maintainable, causally accessible distinctions.*

3.1 Channel representation

A useful local representation is a finite screen readout channel

$$\mathcal{E}_{\mathcal{S},X} : \mathcal{B}(\mathcal{H}_{\text{bulk}}) \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{S}}), \quad (9)$$

or a classical stochastic channel

$$p_{\mathcal{S}}(y) = \sum_x \mathcal{A}_{\mathcal{S},X}(y|x) p_{\text{bulk}}(x). \quad (10)$$

Then

$$\rho_{\mathcal{S}}(X) = \mathcal{E}_{\mathcal{S},X}(\rho_{\text{bulk}}), \quad S_{\text{scr}}[X] = S(\rho_{\mathcal{S}}(X)). \quad (11)$$

This representation explains why entropy appears before metric geometry: finite access produces a finite ledger, and entropy is the coordinate-free size of that ledger.

4 Response connection and metric integrability

The response one-form on a screen is

$$\omega_{\mathcal{S}} = dS_{\text{scr}} = \eta_i(X) dX^i. \quad (12)$$

As screens vary through Scr , response data must be transported. This motivates a response connection

$$\nabla_{\text{scr}} = d + \Gamma_{\text{scr}} \quad (13)$$

on the finite-access ledger bundle.

Principle 2 (Geometry as integrable response connection). *Metric geometry is the smooth integrable envelope of the finite-access response connection. A GR-like sector appears when local response data can be represented by a metric tensor envelope satisfying the appropriate Ward/Bianchi identities.*

Definition 3 (Finite-access response connection). *A finite-access response connection ∇_{scr} is a rule for transporting ledger states, entropy sections, or response one-forms between nearby causal-screen contexts. Locally,*

$$\nabla_{\text{scr}} = d + \Gamma_{\text{scr}}.$$

If its response curvature can be represented by a smooth metric envelope satisfying Ward/Bianchi consistency, the sector is geometrically integrable. Nonzero non-metric curvature or failure of gluing defines a residual obstruction:

$$F_{\nabla_{\text{scr}}} = 0 \Rightarrow \text{integrable ledger response}, \quad F_{\nabla_{\text{scr}}} \neq 0 \Rightarrow \text{residual/gluing defect}.$$

If response data are local and all-null compatible, they may be represented by a symmetric tensor envelope

$$F_p(k) = E_{\mu\nu}(p)k^\mu k^\nu \quad (14)$$

up to trace freedom. Ward/Bianchi closure fixes the trace integration sector. If finite-access response data fail to admit such a global envelope, the failure is a residual.

4.1 Residuals as gluing defects

Let ω_α be response one-forms on overlapping causal screen patches U_α . If the data glue globally, then on overlaps one expects

$$\omega_\alpha - \omega_\beta = d\lambda_{\alpha\beta} \quad (15)$$

with a trivial cocycle on triple overlaps,

$$\lambda_{\alpha\beta} + \lambda_{\beta\gamma} + \lambda_{\gamma\alpha} = 0. \quad (16)$$

If instead

$$\delta\lambda_{\alpha\beta\gamma} \neq 0, \quad (17)$$

then the finite-access response data do not glue into a globally integrable metric ledger without residual structure.

Remark 2 (Dark sector interpretation). *In this language, the G1 dark sector is not an arbitrary material fluid. It is a horizon-scale finite-access gluing defect, projected into the Weyl optical channel and constrained by Ward/naturality conditions.*

Residual filter. The apparent response obstruction is classified as: (1) quotient artifact – removable by gauge fixing; (2) missing variable – resolved by state-space enlargement; (3) memory projection – removed by horizontal lift; (4) true sheaf-gluing obstruction – the only category that may feed G1DE residuals.

5 The local optical fiber

A finite observer receives optical information through a two-dimensional screen transverse to a null ray. Away from caustics, the Jacobi map is an element

$$J \in GL(2, \mathbb{R}). \quad (18)$$

A local optical deformation is

$$X = \delta J J^{-1} \in \mathfrak{gl}(2, \mathbb{R}). \quad (19)$$

Under screen rotations, the deformation algebra decomposes as

$$\mathfrak{gl}(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1), \quad (20)$$

where A is the trace/area direction, S is the two-dimensional symmetric traceless shear sector, and Ω is the antisymmetric twist/holonomy sector.

Sector	Dimension	Optical meaning	G1 role
A	1	trace/determinant/area	filling and Ricci-like focusing
S	2	symmetric traceless shear	Weyl/E-mode optical shape
Ω	1	twist/holonomy	orientation/curl capacity
$U = S \oplus \Omega$	3	unimodular sector	Weyl-active optical ledger

Table 2: The local optical access fiber of a two-dimensional null screen.

The decomposition is algebraic. The coefficient $3/4$ does not follow from algebra alone. It follows when the finite-access response metric is unbiased or isotropic over the four local optical directions.

6 Relative-entropy Hessian of optical access distortions

Let

$$q^I = (q_A, q_{S_1}, q_{S_2}, q_\Omega) \quad (21)$$

be coordinates on the optical tangent fiber $T_I GL(2, \mathbb{R}) \simeq \mathfrak{gl}(2, \mathbb{R})$. We assume that $\rho_S(0)$ is full rank on the finite screen code subspace, or else restrict the local family $\rho_S(q)$ to the common support on which the relative entropy is finite. The finite-access channel under optical deformation produces a screen state family

$$\rho_S(q) = \mathcal{E}_{S,q}(\rho_{\text{bulk}}). \quad (22)$$

The distinguishability cost of the deformation is

$$D_{\text{scr}}(q) = D_{\text{rel}}(\rho_S(q) \| \rho_S(0)). \quad (23)$$

Near $q = 0$,

$$D_{\text{scr}}(q) = \frac{1}{2} K_{IJ} q^I q^J + O(q^3), \quad (24)$$

with

$$\boxed{K_{IJ} = \partial_I \partial_J D_{\text{rel}}(\rho_S(q) \| \rho_S(0))|_{q=0}}. \quad (25)$$

This is the relative-entropy Hessian of finite optical access distortions.

6.1 Exponential-family local closure

A least-biased local state family with optical constraints takes the exponential form

$$\rho_S(q) = Z(q)^{-1} \exp(\log \rho_0 + q^I \mathcal{O}_I), \quad (26)$$

where $\mathcal{O}_I$ are screen observables conjugate to the four optical generators. For such a family, the relative-entropy Hessian equals the Kubo–Mori covariance,

$$K_{IJ} = \langle \Delta \mathcal{O}_I, \Delta \mathcal{O}_J \rangle_{\text{KM}}. \quad (27)$$

In a commuting or classical limit,

$$K_{IJ} = \text{Cov}_{\rho_0}(\mathcal{O}_I, \mathcal{O}_J). \quad (28)$$

Thus the optical stiffnesses K_A, K_S, K_Ω are covariance eigenvalues of the finite-access screen channel.

Remark 3. *The exponential-family assumption is a local MaxEnt closure, not a microscopic theorem. It states that if only the optical deformation constraints are known, the least-biased finite-access state family is exponential in the corresponding optical generators.*

7 Compliance projection and the 3/4 theorem

By $SO(2)$ symmetry, the Hessian block-decomposes as

$$K = K_A P_A + K_S P_S + K_\Omega P_\Omega. \quad (29)$$

The compliance matrix is

$$C = K^{-1} = K_A^{-1} P_A + K_S^{-1} P_S + K_\Omega^{-1} P_\Omega. \quad (30)$$

The Weyl-active unimodular projector is

$$P_U = P_S + P_\Omega. \quad (31)$$

Definition 4 (Weyl-active compliance fraction). *The background–Weyl projection coefficient is*

$$\boxed{\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr}(C)}}. \quad (32)$$

All traces are taken in an $SO(2)$ -adapted orthonormal optical generator basis at the unbiased reference point. Anisotropic generator normalizations are absorbed into the block weights K_A, K_S, K_Ω . Thus the value 3/4 refers to the isotropic finite-access fixed point, not to an arbitrary coordinate normalization.

Since $\dim A = 1$, $\dim S = 2$, and $\dim \Omega = 1$,

$$\boxed{\kappa_{BW} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}}. \quad (33)$$

Equivalently, with compliances

$$\chi_A = K_A^{-1}, \quad \chi_S = K_S^{-1}, \quad \chi_\Omega = K_\Omega^{-1}, \quad (34)$$

one has

$$\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}. \quad (35)$$

Theorem 1 (Unbiased finite-access projection). *If the finite-access Hessian is locally unbiased on the full optical fiber,*

$$K_A = K_S = K_\Omega, \quad (36)$$

then

$$\kappa_{BW} = \frac{3}{4}. \quad (37)$$

Proof. At the unbiased fixed point, $C = K_0^{-1}I$ on the four-dimensional optical fiber. The Weyl-active sector $U = S \oplus \Omega$ has dimension $2 + 1 = 3$. Therefore

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr}(C)} = \frac{3K_0^{-1}}{4K_0^{-1}} = \frac{3}{4}. \quad (38)$$

□

Corollary 1 (Twist-demoted branch). *If the twist/holonomy ledger is removed or has zero compliance, $K_\Omega^{-1} = 0$, and $K_A = K_S$, then*

$$\kappa_{BW} = \frac{2}{1+2} = \frac{2}{3}. \quad (39)$$

The theorem does not claim that all physical screens are unbiased. It identifies the isotropic finite-access fixed point. Deviations from this point are represented by weighted-compliance coefficients K_A, K_S, K_Ω , and are tested by the free- κ control branch. The 3/4 law is therefore not a theorem of algebra alone. It is the isotropic MaxEnt/Fisher-compliance fraction of the full $GL(2)$ finite-access optical ledger. The 2/3 law is the demoted scalar-shear branch when the twist/holonomy ledger is absent from the accessible screen capacity.

8 Local fiber theorem and global gluing theorem

The finite-access closure naturally separates into two levels.

8.1 Local fiber theorem

The local theorem says:

$$\boxed{\text{unbiased local finite-access optical fiber} \Rightarrow \kappa_{BW} = 3/4.} \quad (40)$$

This theorem is local in screen space and does not determine the cosmological redshift kernel.

8.2 Global gluing theorem

The global theorem concerns horizon-scale access-load drift. Local ledgers must be glued over the horizon-accessible region. If this gluing has a finite-capacity lag, it produces a horizon residual $\mathcal{R}_H(a)$. This residual determines the redshift-dependent amplitude of the Weyl transfer correction.

Thus:

$$\boxed{\text{local fiber gives } 3/4, \quad \text{global access gluing gives } \mathcal{R}_H(a).} \quad (41)$$

9 Horizon gluing load and occupancy current

The used horizon capacity should not be interpreted as the entropy of the entire universe. It is the finite access load that must be maintained by the horizon ledger.

Definition 5 (Horizon access load). *Let $\mathcal{X}_a$ denote a horizon-accessible bulk distinction ensemble at epoch a . The used horizon capacity is*

$$C_{\text{used}}(a) = \sup_{\mathcal{X}_a} I_{\text{acc}}[\mathcal{X}_a; \mathcal{A}_H(\mathcal{X}_a)]. \quad (42)$$

Equivalently, a tractable relative-entropy surrogate is

$$C_{\text{used}}(a) \sim D_{\text{rel}}(\rho_{\text{bulk}}^H(a) \parallel \rho_{\text{coarse}}^H(a)). \quad (43)$$

The definition of C_{used} is structural. It identifies the horizon access-load object that a microscopic FDS model should compute. The current low-redshift G1DE implementation does not directly evaluate $C_{\text{used}}(a)$ from a cosmological distinction ensemble. Instead it uses the reduced memory response

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1, \quad R_0 = 3 - s,$$

as the data-facing low-redshift reduction. A conservative toy surrogate is

$$C_{\text{used}}^{\text{toy}}(a) = D_{\text{rel}}(\rho_{\text{LSS}}^H(a) \parallel \rho_{\text{FRW}}^H(a)),$$

where ρ_{LSS}^H denotes horizon-accessible coarse-grained large-scale-structure states and ρ_{FRW}^H is the corresponding coarse FRW reference.

The available horizon capacity is calibrated by the standard horizon bridge,

$$C_H(a) = \frac{A_H(a)}{4\ell_P^2}. \quad (44)$$

The occupancy ratio is

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}. \quad (45)$$

The horizon occupancy current is

$$\boxed{J_H(N) = -\frac{d}{dN} \ln \mathcal{O}_H(N), \quad N = \ln a.} \quad (46)$$

This current measures the scale-flow of access load relative to available horizon capacity.

10 Passive retarded gluing-lag response

The horizon residual is the retarded response of the finite-access ledger to occupancy-current drift:

$$\boxed{\mathcal{R}_H(N) = \int_{-\infty}^N G_H^{\text{ret}}(N, N') J_H(N') dN'.} \quad (47)$$

A Markov approximation gives

$$\mathcal{R}'_H(N) + \Gamma_H(N) \mathcal{R}_H(N) = J_H(N), \quad \Gamma_H > 0. \quad (48)$$

The open-system consistency conditions are:

$$G_H^{\text{ret}}(N, N') = 0 \quad (N' > N), \quad (49)$$

$$N_H(N, N') \succeq 0, \quad (50)$$

$$\int dN J_H(N) \mathcal{R}_H(N) \geq 0. \quad (51)$$

Here N_H is the noise kernel of the corresponding Schwinger–Keldysh/Feynman–Vernon influence functional. Equations (49)–(51) express causality, positive noise, and passivity.

Remark 4 (Meaning of s). *In the low-redshift G1DE reduction,*

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1, \quad R_0 = 3 - s. \quad (52)$$

Thus

$$3 - s = \mathcal{R}_H(1), \quad (53)$$

so the observable deficit $3 - s$ is the present normalized finite-access gluing-lag amplitude. The parameter s should be read as a low-redshift effective filling exponent, not as a literal spatial dimension.

11 Weyl normalization and no free optical amplitude

Define the Weyl source

$$\mathcal{W}(a, k) = -k^2(\Phi + \Psi). \quad (54)$$

In the standard Σ convention,

$$\mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta. \quad (55)$$

G1 is implemented as a finite-screen transfer correction to the GR Weyl source:

$$\boxed{\mathcal{W}_{\text{G1}}(a, k) = [1 - \kappa_{BW} \mathcal{R}_H(a)] \mathcal{W}_{\text{GR}}(a, k) + O(\mathcal{R}_H^2).} \quad (56)$$

Therefore

$$\boxed{\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2).} \quad (57)$$

Earlier screen-influence formulations used a schematic traceless optical seed

$$E_{ij, \text{TF}}^{\text{scr}} = -A(a, k) \kappa_{BW} \mathcal{R}_H(a) \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right). \quad (58)$$

The finite-access sheaf formulation clarifies that $A(a, k)$ is not a physical response function. It is the normalization map that identifies a unit optical seed with a unit fractional correction to the GR Weyl source:

$$A(a, k) \longleftrightarrow \mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta. \quad (59)$$

Thus

$$\boxed{A(a, k) \notin \Theta_{\text{G1DE}}.} \quad (60)$$

Sampling $A(a, k)$ as a nuisance parameter would change the model into a generic dark-stress source model, not the G1DE finite-screen transfer branch.

12 Minimum Ward completion and Ricci suppression

The Weyl transfer correction can be represented covariantly by an effective residual tensor

$$E_{\mu\nu}^{\text{scr}} = \mathcal{P}_W^{\text{min}}[-\kappa_{BW}\mathcal{R}_H\mathcal{W}_{\text{GR}}]_{\mu\nu}, \quad (61)$$

where $\mathcal{P}_W^{\text{min}}$ denotes the minimum Ward-completion operator that embeds a Weyl-normalized traceless seed into a conserved or controlled-exchange tensor representative.

In a gauge-invariant scalar-sector description, introduce effective screen variables

$$\pi_{\text{scr}}^{\text{GI}}, \quad q_{\text{scr}}^{\text{GI}}, \quad \delta\rho_{\text{scr}}^{\text{GI}}. \quad (62)$$

The Weyl-normalized seed is

$$k^2\pi_{\text{scr}}^{\text{GI}} = -\kappa_{BW}\mathcal{R}_H(a) (8\pi G a^2 \rho \Delta), \quad (63)$$

up to the sign and anisotropic-stress convention of the chosen Boltzmann solver. Minimum Ward completion gives the hierarchy

$$q_{\text{scr}}^{\text{GI}} = O\left(\pi_{\text{scr}}^{\text{GI}} \frac{\mathcal{H}}{k}\right), \quad (64)$$

$$\delta\rho_{\text{scr}}^{\text{GI}} = O\left(\pi_{\text{scr}}^{\text{GI}} \frac{\mathcal{H}^2}{k^2}\right) + O(\mathcal{R}_H^2). \quad (65)$$

Therefore

$$\boxed{\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right)}. \quad (66)$$

This is the mathematical origin of first-order Ricci silence: the finite-screen access lag is assigned to the Weyl optical channel with minimal Newtonian/Ricci leakage.

13 Finite-access sheaf closure theorem

We can now collect the full conditional closure.

Theorem 2 (Finite-access sheaf closure). *Assume:*

1. *physical distinctions form finite maintainable access ledgers over causal screens;*
2. *these ledgers define a finite-access sheaf $\mathcal{A} : \text{Scr}^{\text{op}} \rightarrow \text{FinLedger}$;*
3. *entropy is the coordinate-free ledger-size section S_{scr} ;*
4. *local optical access distortions span $T_I GL(2) = A \oplus S \oplus \Omega$;*
5. *the relative-entropy Hessian of the local optical fiber is unbiased, $K_A = K_S = K_\Omega$;*
6. *horizon gluing load defines $J_H = -d_N \ln(C_{\text{used}}/C_H)$;*
7. *the gluing-lag response $\mathcal{R}_H = G_H^{\text{ret}} * J_H$ is causal, stable, and passive;*
8. *the resulting residual is Weyl-normalized and minimally Ward-completed.*

Then

$$\kappa_{BW} = \frac{3}{4}, \quad (67)$$

$$\Sigma(a, k) - 1 = -\frac{3}{4}\mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (68)$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (69)$$

With $\mathcal{R}_H(a) = R_0 \hat{R}_H(a)$, $\hat{R}_H(1) = 1$, and $R_0 = 3 - s$, the locked G1DE- $M_{3/4}$ reduction is

$$\boxed{\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a), \quad \mu \simeq 1.} \quad (70)$$

Proof. The finite-access sheaf supplies finite screen states and the entropy section. The local optical fiber is $A(1) \oplus S(2) \oplus \Omega(1)$. The relative-entropy Hessian defines sector stiffnesses K_A, K_S, K_Ω and compliances $K_A^{-1}, K_S^{-1}, K_\Omega^{-1}$. By the compliance projection formula,

$$\kappa_{BW} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}. \quad (71)$$

If $K_A = K_S = K_\Omega$, this gives $3/4$. The horizon gluing load generates $\mathcal{R}_H$ by causal passive response. Weyl normalization gives $\Sigma - 1 = -\kappa_{BW}\mathcal{R}_H + O(\mathcal{R}_H^2)$. Minimum Ward completion suppresses the density/Ricci channel by $\mathcal{H}^2/k^2$ and by nonlinear powers of $\mathcal{R}_H$, giving the stated μ hierarchy. Substituting $\mathcal{R}_H = (3 - s)\hat{R}_H$ yields the low-redshift locked reduction. $\square$

14 Demotion paths and falsification registry

The closure is structured with demotion paths. Failure of one branch need not falsify the entire finite-screen program.

Condition	What fails	What may survive
$\kappa_{BW} \simeq 2/3$	full $GL(2)$ optical access ledger	scalar-shear/SPD(2) demotion
κ_{BW} far from $2/3$ and $3/4$	unbiased finite-access Hessian	weighted compliance law
Dominant first-order μ response	minimum Ward/Ricci-silent branch	finite-screen residual with density leakage
No common memory shape	horizon gluing-load kernel	separate background and optical residuals
Free $A(a, k)$ required	Weyl normalization	generic dark-stress source, not G1DE
No passive kernel for reconstructed $\mathcal{R}_H$	retarded access-lag interpretation	phenomenological Weyl transfer
No finite screen entropy response	FDS-G1 bridge	algebraic optical toy model only

Table 3: Demotion and falsification registry for the finite-access sheaf closure.

15 Relation to the G1 companion sequence

This paper is best read as a structural layer beneath the existing G1 dark-sector closure notes.

- The core FDS–G1 paper identifies finite causal-screen entropy response as the primitive gravity-side object.
- The optical-stress closure note derives the 3/4 projection law as a MaxEnt closure on the $GL(2)$ optical screen algebra.
- The non-equilibrium entropy note interprets the projection coefficient through optical-sector entropy production and response compliances.
- The screen influence functional packages entropy Hessian, retarded horizon memory, and Ward completion.
- The Weyl-normalized Ward completion note fixes the production-likelihood interface and forbids a free optical amplitude $A(a, k)$.

The present paper adds the deeper statement:

$$\boxed{\text{finite screen entropy is the entropy section of a finite causal-access sheaf.}} \quad (72)$$

It also reframes 3/4 as a relative-entropy compliance fraction of the local optical access fiber, and $\mathcal{R}_H$ as a passive retarded gluing-lag response.

16 Open problems

The present closure leaves several major tasks.

- 1. Horizon coefficient.** The capacity $C_H = A_H/(4\ell_P^2)$ is still imported through the standard horizon bridge. A stronger FDS derivation would obtain the Bekenstein–Hawking coefficient from finite-access microstructure.
- 2. Local screen first law.** The local thermodynamic bridge $\delta Q = T\delta S$ is not derived internally here. It remains a bridge from horizon thermodynamics.
- 3. Exact focusing coefficients.** Finite screen transition dynamics gives Raychaudhuri-like normal forms, but exact GR coefficients require a deeper microgeometric derivation.
- 4. Concrete computation of $C_{\text{used}}(a)$.** This paper defines C_{used} as accessible horizon gluing load. Computing it from a concrete cosmological bulk distinction ensemble remains open.
- 5. Full Boltzmann implementation.** The gauge-invariant Ward-completion hierarchy must be translated into a production Boltzmann pipeline with precise code-level anisotropic-stress conventions.

17 Conclusion

The finite-access sheaf formulation deepens the G1 dark-sector closure. Instead of beginning from an isolated entropy functional, it begins from a system of finite, maintainable, causally accessible distinction ledgers. The screen entropy functional is the entropy section of this ledger sheaf. Geometry is the integrable natural connection of the response data. Residual sectors are gluing defects or non-integrable response components.

The local optical fiber of the finite-access sheaf is $T_I GL(2) = A \oplus S \oplus \Omega$. The relative-entropy Hessian of optical access distortions supplies the sector stiffnesses. The Weyl-active compliance fraction is

$$\kappa_{BW} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}. \quad (73)$$

The unbiased finite-access fixed point gives $\kappa_{BW} = 3/4$, while twist/holonomy demotion gives $2/3$.

At the global level, horizon gluing load produces an occupancy current, its passive retarded response is $\mathcal{R}_H$, and Weyl normalization gives

$$\Sigma - 1 = -\kappa_{BW} \mathcal{R}_H + O(\mathcal{R}_H^2). \quad (74)$$

Minimum Ward completion protects the Newtonian/Ricci channel,

$$\mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (75)$$

Thus the G1DE- $M_{3/4}$ reduction is the locked branch of a finite-access, Weyl-normalized, Ward-disciplined screen-transfer class:

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{R}_H(a), \quad \mu \simeq 1.$$

The current G1fit-real evidence hierarchy tests the low-redshift locked reduction of this structure rather than the full finite-access sheaf construction:

$$M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE-2} > \text{CPL} > \Lambda\text{CDM}.$$

The slogan form is:

Gravity is the integrable natural connection of a finite causal-access ledger; G1DE is its horizon-scale Weyl gluing defect.

A Notation summary

B Cech-style gluing defect

Let $\{U_\alpha\}$ be an open cover of a causal-screen family and let ω_α be local response one-forms. If response data glue up to exact gauge transformations,

$$\omega_\alpha - \omega_\beta = d\lambda_{\alpha\beta}, \quad (76)$$

then on triple overlaps the obstruction is

$$(\delta\lambda)_{\alpha\beta\gamma} = \lambda_{\beta\gamma} - \lambda_{\alpha\gamma} + \lambda_{\alpha\beta}. \quad (77)$$

A nonzero obstruction indicates that the local finite-access response data do not integrate into a single global ledger without residual structure. In the G1DE branch, the relevant large-scale obstruction is modeled as a horizon occupancy/gluing-lag residual $\mathcal{R}_H$ and is projected into the Weyl channel.

Symbol	Meaning
Scr	category of causal screens
$\mathcal{A}$	finite-access sheaf or ledger functor
S_{scr}	screen entropy functional / entropy section
$\rho_S(q)$	screen state under optical deformation q
K_{IJ}	relative-entropy Hessian of optical access distortions
$C = K^{-1}$	compliance matrix
A, S, Ω	area, shear, and twist sectors of $\mathfrak{gl}(2)$
P_U	projector onto the unimodular Weyl-active sector $S \oplus \Omega$
κ_{BW}	background–Weyl projection coefficient
C_{used}	horizon accessible gluing load
C_H	available horizon capacity $A_H/(4\ell_P^2)$
$\mathcal{O}_H$	occupancy ratio C_{used}/C_H
J_H	occupancy current $-d_N \ln \mathcal{O}_H$
$\mathcal{R}_H$	passive retarded response of horizon gluing load
$\mathcal{W}$	Weyl source $-k^2(\Phi + \Psi)$
$A(a, k)$	Weyl-normalization map, not a fit parameter

Table 4: Notation used in the finite-access sheaf closure.

C Weighted-compliance expansion around the isotropic fixed point

Let

$$\chi_S = \chi_A(1 + \alpha), \quad \chi_\Omega = \chi_A(1 + \beta), \quad (78)$$

with $|\alpha|, |\beta| \ll 1$. Then

$$\kappa_{BW} = \frac{2(1 + \alpha) + (1 + \beta)}{1 + 2(1 + \alpha) + (1 + \beta)} \quad (79)$$

$$= \frac{3 + 2\alpha + \beta}{4 + 2\alpha + \beta} \quad (80)$$

$$= \frac{3}{4} + \frac{2\alpha + \beta}{16} + O(\alpha^2, \alpha\beta, \beta^2). \quad (81)$$

Thus deviations from $3/4$ measure anisotropy in the finite-access compliance Hessian. A coefficient near $2/3$ is not a small perturbation of the full $GL(2)$ branch; it indicates demotion of the twist/holonomy compliance.

References

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* **7**, 2333–2346 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* **43**, 199–220 (1975).

- [3] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* **98**, 1123–1126 (1955).
- [4] R. K. Sachs, “Gravitational waves in general relativity. VI. The outgoing radiation condition,” *Proc. R. Soc. Lond. A* **264**, 309–338 (1961).
- [5] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* **75**, 1260–1263 (1995).
- [6] R. Bousso, “A covariant entropy conjecture,” *JHEP* **07**, 004 (1999).
- [7] R. Kubo, “Statistical-mechanical theory of irreversible processes. I,” *J. Phys. Soc. Jpn.* **12**, 570–586 (1957).
- [8] L. V. Keldysh, “Diagram technique for nonequilibrium processes,” *Sov. Phys. JETP* **20**, 1018–1026 (1965).
- [9] R. P. Feynman and F. L. Vernon, “The theory of a general quantum system interacting with a linear dissipative system,” *Ann. Phys.* **24**, 118–173 (1963).
- [10] D. Petz, “Monotone metrics on matrix spaces,” *Linear Algebra Appl.* **244**, 81–96 (1996).
- [11] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley (2006).
- [12] S. Mac Lane and I. Moerdijk, *Sheaves in Geometry and Logic*, Springer (1992).
- [13] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, FDS–G1 Core manuscript (2026).
- [14] Y. Wu, *The 3/4 Projection Law in Finite Screen Gravity: MaxEnt Optical-Stress Closure, Ricci-Silent Residuals, and the Weyl-Locked Dark Sector*, FDS–G1 technical note D1 (2026).
- [15] Y. Wu, *Non-Equilibrium Screen Entropy and the Optical-Stress Origin of the G1 Dark Sector*, FDS–G1 technical note D2 (2026).
- [16] Y. Wu, *Screen Influence Functional for the G1 Dark Sector*, FDS–G1 technical note D3 (2026).
- [17] Y. Wu, *Weyl-Normalized Ward Completion and Identifiability Rules for G1DE*, FDS–G1 technical note D5 (2026).
- [18] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual*, FDS–G1 data paper (2026).

The 3/4 Projection Law in Finite Screen Gravity

MaxEnt Optical-Stress Closure, Ricci-Silent Residuals, and the Weyl-Locked Dark Sector

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

May 2026

Abstract

This paper gives a strengthened mathematical closure for the 3/4 projection law in Finite Distinction Systems – Gravity 1 (FDS-G1). The earlier projection-law argument observed that a finite observer receives spacetime residuals through a two-dimensional null optical screen and that the local Jacobi deformation algebra splits as

$$\mathfrak{gl}(2, \mathbb{R}) = \mathbb{R}I \oplus \mathfrak{sl}(2, \mathbb{R}),$$

namely one determinant/area/filling direction and three unimodular/shape directions. D0 embeds the same 3/4 coefficient in a modular anomaly Hessian; D1 embeds it in the relative-entropy Hessian of a finite-access sheaf. This paper isolates the optical projection law in its simplest local form: a maximum-entropy optical-stress closure on the $GL(2)$ screen algebra. The leading finite-screen residual is modeled as a local random deformation $X \in T_I GL(2, \mathbb{R}) \simeq \mathfrak{gl}(2, \mathbb{R})$ with fixed variance. In the absence of lower-order information distinguishing area, shear, and twist directions, the maximum-entropy covariance is isotropic on the optical deformation space. Projection onto the unimodular sector then gives

$$\kappa_{BW} = \frac{\text{Tr} P_{\text{uni}}}{\text{Tr} I_{\mathfrak{gl}(2)}} = \frac{3}{4}.$$

More generally, if area, shear, and twist sectors have response weights $\eta_A, \eta_S, \eta_\Omega$, the coefficient becomes

$$\kappa_{BW} = \frac{2\eta_S + \eta_\Omega}{\eta_A + 2\eta_S + \eta_\Omega},$$

with 3/4 as the isotropic fixed point and 2/3 as the scalar-shear demotion when the twist/holonomy direction is removed.

The paper then shows why the G1 residual is Ricci-silent and Weyl-active at first order. A compensating unimodular screen response is traceless by construction; through the Sachs optical equations it couples to the traceless optical tidal/Weyl channel rather than to the Ricci-density focusing channel. Consequently the local matter-growth channel is protected to first order,

$$\mu(a) = 1 + \mathcal{O}(\lambda^2),$$

while the Weyl/lensing channel receives the locked response

$$\Sigma(a) = 1 - \kappa_{BW} \lambda X_b(a) + \mathcal{O}(\lambda^2), \quad \lambda = 3 - s.$$

For isotropic lowest-order response this reduces to the G1DE-M_{3/4} relation

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s).$$

The redshift kernel $X_b(a)$ is interpreted as a retarded horizon-memory response. A minimal acceleration-gated closure, $X'_b + X_b = [1 - \epsilon_H]_+$, is stated as a candidate rather than as a theorem. This closure is the theoretical counterpart of the G1fit-real projection-locked model G1DE-M_{3/4}, whose exact-pilot

likelihood implementation fixes $\mu_0 = 0$ and $\Sigma_0 = -3(3-s)/4$ as the background–Weyl locked response. Current exact-pilot model comparisons select the locked relation at the AIC/BIC level against common dark-energy extensions, while full production evidence and expanded lensing likelihoods remain pending. The result is conditional and falsifiable: a dominant first-order μ correction, failure of a shared $X_b(a)$ shape, a measured coefficient far from the allowed optical-response-weight family, or a preference for the $2/3$ scalar-shear branch would demote the full $3/4$ projection law.

1 Scope and executive summary

This paper is a mathematical companion to the FDS-G1 finite-screen entropy-response program. The G1 core identifies gravitational coupling with finite causal-screen entropy response and interprets dark-sector residuals as failures of complete finite-screen closure rather than as ordinary material fluids. The present work addresses one narrow but important question:

Why should the leading G1 cosmological residual be a Weyl/lensing correction with amplitude $-3/4$ times the background filling defect, while the local matter-growth channel remains close to general relativity?

D2’s main purpose is not merely to derive $3/4$, but to prove its uniqueness under isotropic optical MaxEnt closure.

The answer proposed here is:

The G1 dark sector is not an additional density component. It is a traceless optical stress of finite-screen incompleteness.

This statement has three parts.

1. **Optical access.** A finite observer receives spacetime residuals through a null optical screen. The local screen deformation lives in $\mathfrak{gl}(2, \mathbb{R})$.
2. **MaxEnt projection.** The lowest-order unbiased finite-screen residual distributes over one area direction and three unimodular directions. This gives the $3/4$ projection weight.
3. **Screen stress.** The compensating unimodular residual is traceless; through the Sachs optical decomposition it is Weyl-active but Ricci-silent. Thus Σ changes while μ is first-order protected.

Let

$$\lambda = 3 - s \tag{1}$$

be the background filling defect relative to ordinary three-volume scaling. Let $X_b(a)$ be the normalized horizon-memory response kernel. The leading G1 optical-stress closure is

$$\mu(a) = 1 + \mathcal{O}(\lambda^2), \tag{2}$$

$$\Sigma(a) = 1 - \kappa_{BW} \lambda X_b(a) + \mathcal{O}(\lambda^2). \tag{3}$$

In the isotropic lowest-order optical response,

$$\kappa_{BW} = \frac{3}{4}, \tag{4}$$

so that

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s). \tag{5}$$

What this paper does not claim. The result is not claimed to be a theorem of pure algebra alone. The algebra gives the available optical sectors. The coefficient $3/4$ follows only when the leading finite-screen residual is locally maximum-entropy or isotropic on the optical deformation space. If the screen microstatistics weight area, shear, and twist differently, the coefficient shifts according to a generalized compliance-weighted formula derived below.

Table 1: Minimal map from finite-screen optical algebra to the G1DE locked response.

Object	Decomposition	G1 interpretation
$GL(2, \mathbb{R})$	$4 = 1 + 2 + 1$	finite optical screen deformation
Area sector	1 trace/determinant mode	background filling channel
Shear sector	2 symmetric traceless modes	scalar/E-mode optical shape
Twist sector	1 antisymmetric holonomy mode	orientation/curl capacity
Unimodular sector	$3 = 2 + 1$	Weyl/shape screen-stress channel
Filling defect	$\lambda = 3 - s$	effective screen-volume anomaly
G1DE lock	$\Sigma_0 = -\kappa_{BW} \lambda$	optical compensation
Isotropic branch	$\kappa_{BW} = 3/4$	full $GL(2)$ MaxEnt projection
Scalar-shear branch	$\kappa_{BW} = 2/3$	twist/holonomy demotion
Matter growth	$\mu_0 = 0$	first-order Ricci-density silence

2 Claim-status discipline

The projection law is a bridge result. It combines standard optical geometry, a finite-screen interpretation, and a lowest-order statistical assumption about residuals. Its claims are graded as follows.

Table 2: Claim status of the optical-stress closure.

Claim	Meaning	Status
Finite optical access	observable residuals are accessed through a two-dimensional null optical screen	bridge principle
Jacobi map	screen deformation is locally described by $J \in GL(2, \mathbb{R})$ away from caustics	standard optical geometry
$\mathfrak{gl}(2)$ split	$\mathfrak{gl}(2) = A(1) \oplus S(2) \oplus \Omega(1)$	algebraic theorem
MaxEnt residual	lowest-order residual covariance is isotropic in $T_l GL(2)$ absent further information	statistical bridge assumption
$3/4$ coefficient	full unimodular sector carries three of four local optical directions	conditional projection law
Weighted coefficient	$\kappa_{BW} = (2\eta_S + \eta_\Omega)/(\eta_A + 2\eta_S + \eta_\Omega)$	generalized theorem under quadratic response
Unimodular stress	shape residual is traceless on the screen	variational consequence
Ricci silence	residual does not act as first-order local density source	matter-protection bridge
Weyl activity	residual modifies lensing/optical channel	Sachs-decomposition bridge
$X_b(a)$	retarded horizon-memory response	dynamical bridge, not yet fully derived

The strongest claim in this paper is therefore not that “geometry forces $3/4$ ” without assumptions. The

stronger and safer claim is:

$$\boxed{\text{finite optical screen algebra} + \text{MaxEnt lowest-order residual} \Rightarrow \kappa_{BW} = 3/4.} \quad (6)$$

3 Finite optical screens and Sachs geometry

Consider a bundle of null geodesics with tangent k^μ reaching an observer. Let e_A^μ , $A = 1, 2$, be a screen basis orthogonal to k^μ and to the observer direction. The Jacobi map

$$J^A_B \quad (7)$$

relates an infinitesimal source-plane separation to an observed screen separation. On a non-caustic patch,

$$J \in GL(2, \mathbb{R}). \quad (8)$$

Definition 1 (Optical deformation generator). *The local optical deformation generator is*

$$X = \delta J J^{-1} \in \mathfrak{gl}(2, \mathbb{R}). \quad (9)$$

For continuous affine evolution one similarly defines the optical deformation-rate matrix

$$B^A_B = \frac{dJ^A_C}{d\lambda} (J^{-1})^C_B. \quad (10)$$

The matrix B decomposes into the Sachs optical scalars:

$$B^A_B = \frac{1}{2}\theta\delta^A_B + \sigma^A_B + \omega\epsilon^A_B, \quad (11)$$

where

$$\theta = \text{tr} B \quad (12)$$

is expansion, σ^A_B is symmetric and traceless, and $\omega\epsilon^A_B$ is the antisymmetric twist/rotation component. Thus the local optical deformation algebra has the sector count

$$\mathfrak{gl}(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1). \quad (13)$$

Equivalently,

$$\mathfrak{gl}(2, \mathbb{R}) = \mathbb{R}I \oplus \mathfrak{sl}(2, \mathbb{R}), \quad \dim \mathbb{R}I = 1, \quad \dim \mathfrak{sl}(2) = 3. \quad (14)$$

The optical tidal equation is schematically

$$\dot{B} + B^2 = \mathcal{R}, \quad (15)$$

where

$$\mathcal{R}_{AB} = -R_{\mu\nu\rho\sigma} e_A^\mu k^\nu k^\rho e_B^\sigma. \quad (16)$$

Using the four-dimensional curvature decomposition, the screen tidal matrix splits as

$$\mathcal{R}_{AB} = -\frac{1}{2}R_{\mu\nu} k^\mu k^\nu \gamma_{AB} - C_{\mu\nu\rho\sigma} e_A^\mu k^\nu k^\rho e_B^\sigma. \quad (17)$$

The first term is trace/Ricci focusing. The second term is traceless/Weyl tidal distortion. This is the geometric reason G1 can be Ricci-silent and Weyl-active at first order.

Table 3: Optical-sector counting and the G1DE locked response.

Sector	Dimension	G1 role
A	1	area / determinant / background filling
S	2	symmetric shear / lensing shape
Ω	1	twist / optical holonomy capacity
$S + \Omega$	3	unimodular Weyl-active sector
Total	4	local optical deformation space

$$\lambda = 3 - s, \quad \mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}\lambda.$$

4 Projectors on the optical deformation algebra

Use the Frobenius inner product on $\mathfrak{gl}(2)$:

$$\langle X, Y \rangle = \text{Tr}(X^T Y). \quad (18)$$

Define the normalized area generator

$$e_A = \frac{1}{\sqrt{2}}I. \quad (19)$$

Then any $X \in \mathfrak{gl}(2)$ decomposes uniquely as

$$X = x_A e_A + X_U, \quad X_U \in \mathfrak{sl}(2, \mathbb{R}), \quad (20)$$

with

$$x_A = \langle X, e_A \rangle = \frac{1}{\sqrt{2}}\text{tr}X. \quad (21)$$

Definition 2 (Area and unimodular projectors). *For $X \in \mathfrak{gl}(2)$ define*

$$P_A X = \frac{1}{2}(\text{tr}X)I, \quad P_U X = X - \frac{1}{2}(\text{tr}X)I. \quad (22)$$

Then P_A projects to the determinant/area sector and P_U projects to the unimodular sector.

To separate shear from twist, use

$$P_S X = \frac{1}{2}(X + X^T) - \frac{1}{2}(\text{tr}X)I, \quad P_\Omega X = \frac{1}{2}(X - X^T). \quad (23)$$

Then

$$P_U = P_S + P_\Omega, \quad (24)$$

with projector traces

$$\text{Tr}P_A = 1, \quad \text{Tr}P_S = 2, \quad \text{Tr}P_\Omega = 1, \quad \text{Tr}I_{\mathfrak{gl}(2)} = 4. \quad (25)$$

Proposition 1 (Optical capacity count). *The full local optical deformation space has four directions. The unimodular shape sector has three directions. Hence the unweighted unimodular projector trace ratio is*

$$\frac{\text{Tr}P_U}{\text{Tr}I_{\mathfrak{gl}(2)}} = \frac{3}{4}. \quad (26)$$

Remark 1. *Equation (26) is only a capacity count. It becomes a physical coefficient only after a response metric or residual covariance is specified. The next section supplies the minimal maximum-entropy closure.*

5 The MaxEnt optical residual lemma

One should not speak of a uniform probability distribution on the non-compact group $GL(2, \mathbb{R})$. The relevant finite-screen statement is local: near a given optical screen state, the residual lives in the tangent space $T_I GL(2) \simeq \mathfrak{gl}(2)$.

Definition 3 (Local optical residual). *A leading finite-screen optical residual is a random variable*

$$X \in \mathfrak{gl}(2, \mathbb{R}) \quad (27)$$

with

$$\mathbb{E}[X] = 0, \quad \mathbb{E} \|X\|^2 = \sigma_X^2. \quad (28)$$

Lemma 1 (MaxEnt optical residual). *Among distributions on the finite-dimensional vector space $\mathfrak{gl}(2)$ with fixed mean and fixed covariance trace, the maximum-entropy distribution with no preferred direction is the centered isotropic Gaussian. Its covariance is proportional to the identity on $\mathfrak{gl}(2)$:*

$$\mathbb{E}[X \otimes X] = \frac{\sigma_X^2}{4} I_{\mathfrak{gl}(2)}. \quad (29)$$

Consequently,

$$\frac{\mathbb{E} \|P_U X\|^2}{\mathbb{E} \|X\|^2} = \frac{\text{Tr} P_U}{\text{Tr} I_{\mathfrak{gl}(2)}} = \frac{3}{4}. \quad (30)$$

Proof. The maximum-entropy distribution on $\mathbb{R}^n$ with fixed mean and covariance is Gaussian. If no lower-order information distinguishes directions, the covariance is isotropic. For an orthogonal projector P and isotropic covariance $\sigma_X^2 I/n$,

$$\mathbb{E} \|PX\|^2 = \frac{\sigma_X^2}{n} \text{Tr} P. \quad (31)$$

Here $n = 4$ and $P = P_U$, so $\text{Tr} P_U = 3$ gives (30). $\square$

This lemma is the precise mathematical form of the phrase “unbiased finite-screen projection.” It is a lowest-order MaxEnt closure, not an unconditional geometric necessity.

6 Weighted response and the 3/4 fixed point

Finite screen microstatistics may not weight all optical directions equally. Let $\eta_A, \eta_S, \eta_\Omega > 0$ denote response weights (compliances, not stiffness penalties) in the area, symmetric-shear, and twist/holonomy sectors. Define the quadratic response cost

$$\mathcal{C}[X] = \frac{1}{2} \eta_A \|P_A X\|^2 + \frac{1}{2} \eta_S \|P_S X\|^2 + \frac{1}{2} \eta_\Omega \|P_\Omega X\|^2. \quad (32)$$

Theorem 1 (3/4 Projection Uniqueness (Theorem D2.1)). *For a finite-screen residual with sector weights $(\eta_A, \eta_S, \eta_\Omega)$, the normalized unimodular optical response coefficient is*

$$\kappa_{BW} = \frac{2\eta_S + \eta_\Omega}{\eta_A + 2\eta_S + \eta_\Omega}. \quad (33)$$

In the isotropic lowest-order case $\eta_A = \eta_S = \eta_\Omega$, this reduces to

$$\kappa_{BW} = \frac{3}{4}. \quad (34)$$

If the twist/holonomy direction is removed or has zero response weight, while $\eta_A = \eta_S$, then

$$\kappa_{BW} = \frac{2}{3}. \quad (35)$$

Proof. The weighted capacity of each sector is its dimension times its response weight. The area sector has dimension 1, the symmetric-shear sector dimension 2, and the twist sector dimension 1. The unimodular sector is shear plus twist, so its weighted share is $(2\eta_S + \eta_\Omega)/(\eta_A + 2\eta_S + \eta_\Omega)$. $\square$

Caveat 1 (What the coefficient measures). *The number $3/4$ is not merely a decorative ratio. It is a test of whether the finite screen retains the full local optical deformation algebra with isotropic lowest-order response. A measured coefficient near $2/3$ would not prove that finite-screen gravity is false; it would demote the full $GL(2)$ branch to a scalar-shear or symmetric-screen branch.*

Caveat 2 (M_κ interpretation). *M_κ is not merely a free empirical competitor. It is the anisotropic-compliance relaxation of $M_{3/4}$: the weighted coefficient $\kappa_{BW} = (2\eta_S + \eta_\Omega)/(\eta_A + 2\eta_S + \eta_\Omega)$ reduces to $3/4$ at the isotropic fixed point. A measured $\kappa_{BW} \neq 3/4$ would demote the full $GL(2)$ branch without automatically falsifying the finite-screen response program.*

7 Constrained screen closure

The $3/4$ law can be written as a constrained compensation problem. Let

$$\lambda = 3 - s \tag{36}$$

be the background filling defect. The determinant channel carries the area/filling anomaly

$$\Delta_A = \lambda X_b(a), \tag{37}$$

where $X_b(a)$ is a normalized response kernel.

Finite screen closure says that this determinant defect cannot remain as an unbalanced scalar optical ledger entry. It must be compensated by a unimodular optical response:

$$\Delta_U = -\kappa_{BW} \lambda X_b(a). \tag{38}$$

The sign is fixed by closure: the shape response is a compensation of the filling defect.

Principle 1 (Finite screen closure). *At leading order, the optical ledger residual decomposes into determinant and unimodular components,*

$$\Delta_{\text{scr}} = \Delta_A + \Delta_U + \mathcal{O}(\lambda^2), \tag{39}$$

with the compensating relation (38). The amplitude of the compensation is determined by the optical response coefficient (33).

The isotropic branch gives

$$\Delta_U = -\frac{3}{4} \lambda X_b(a). \tag{40}$$

This is the algebraic-statistical origin of the G1DE- $M_{3/4}$ amplitude lock.

8 Unimodular screen stress

The next step is to connect the unimodular optical response to physical channels. Let γ_{AB} be the screen metric and define the unimodular screen metric

$$\hat{\gamma}_{AB} = (\det \gamma)^{-1/2} \gamma_{AB}. \tag{41}$$

By construction,

$$\det \hat{\gamma} = 1. \tag{42}$$

A variation of $\hat{\gamma}$ is traceless with respect to γ :

$$\gamma^{AB} \delta \hat{\gamma}_{AB} = 0. \quad (43)$$

Suppose the shape part of the finite-screen residual action is a functional only of the unimodular metric,

$$S_{\text{scr}}^{\text{shape}} = S_{\text{scr}}^{\text{shape}}[\hat{\gamma}]. \quad (44)$$

Define the screen stress

$$T_{AB}^{\text{scr}} = -\frac{2}{\sqrt{\gamma}} \frac{\delta S_{\text{scr}}^{\text{shape}}}{\delta \gamma^{AB}}. \quad (45)$$

Lemma 2 (Unimodular screen-stress lemma). *If $S_{\text{scr}}^{\text{shape}}$ depends only on the unimodular screen metric $\hat{\gamma}_{AB}$, then the induced screen stress is traceless:*

$$\boxed{\gamma^{AB} T_{AB}^{\text{scr}} = 0.} \quad (46)$$

Sketch. A conformal variation $\delta \gamma_{AB} = 2\varphi \gamma_{AB}$ implies $\delta \gamma^{AB} = -2\varphi \gamma^{AB}$. It changes the determinant but leaves $\hat{\gamma}_{AB}$ unchanged to first order. Since $S_{\text{scr}}^{\text{shape}}$ depends only on $\hat{\gamma}$, its variation under a pure trace variation vanishes. Therefore

$$0 = \delta S_{\text{scr}}^{\text{shape}} = \int d^2 y \frac{\delta S_{\text{scr}}^{\text{shape}}}{\delta \gamma^{AB}} \delta \gamma^{AB} = -\frac{1}{2} \int d^2 y \sqrt{\gamma} T_{AB}^{\text{scr}} (-2\varphi \gamma^{AB}) = \int d^2 y \sqrt{\gamma} \varphi \gamma^{AB} T_{AB}^{\text{scr}}. \quad (47)$$

For arbitrary φ , the trace must vanish. $\square$

Thus the compensating residual is not a scalar density on the screen. It is a traceless optical stress.

9 Ricci-silent, Weyl-active residuals

The Sachs decomposition (17) shows that trace optical focusing is Ricci-like, while traceless optical tidal distortion is Weyl-like. A traceless screen stress therefore has the correct tensor character to modify the Weyl/lensing sector without introducing a leading local Ricci-density source.

Definition 4 (Ricci-silent residual). *A finite-screen residual $\mathcal{E}_{\mu\nu}^{\text{scr}}$ is Ricci-silent at first order if its density/focusing component vanishes to first order:*

$$u^\mu u^\nu \mathcal{E}_{\mu\nu}^{\text{scr}} = \mathcal{O}(\lambda^2), \quad (48)$$

while its projected traceless spatial or optical component may be first order.

Definition 5 (Weyl-active residual). *A residual is Weyl-active if it produces a nonzero traceless optical tidal correction,*

$$\mathcal{E}_{AB}^{\text{scr}} = \mathcal{E}_{\langle AB \rangle}^{\text{scr}} + \mathcal{O}(\lambda^2), \quad \gamma^{AB} \mathcal{E}_{AB}^{\text{scr}} = 0, \quad (49)$$

with amplitude proportional to the unimodular compensation (38).

The effective residual in a scalar cosmological reduction can be represented schematically as

$$\delta T_{00}^{\text{scr}} = \mathcal{O}(\lambda^2), \quad \delta T_{0i}^{\text{scr}} = \mathcal{O}(\lambda^2), \quad (50)$$

$$\delta T_{ij}^{\text{scr}} = \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right) \Pi_{\text{scr}} + \mathcal{O}(\lambda^2). \quad (51)$$

This is not an ordinary dark matter density. It is an anisotropic screen-stress residual.

10 Cosmological reduction: μ and Σ

In phenomenological modified-growth notation,

$$-k^2\Psi = 4\pi G a^2 \mu(a) \rho \Delta, \quad (52)$$

$$-k^2(\Phi + \Psi) = 8\pi G a^2 \Sigma(a) \rho \Delta. \quad (53)$$

The Newtonian channel μ is sensitive to local density/focusing. The lensing channel Σ is sensitive to the Weyl/optical combination.

Lemma 3 (Matter-channel protection). *If ordinary matter is locally conserved and the finite-screen residual is Ricci-silent to first order, then*

$$\boxed{\mu(a) = 1 + \mathcal{O}(\lambda^2)}. \quad (54)$$

Sketch. The leading Poisson equation is sourced by the matter density contrast and by any additional first-order density-like residual. Ricci silence removes the latter at order λ . Local conservation of ordinary matter keeps the standard growth source intact. Hence the first-order coefficient in μ vanishes. $\square$

Theorem 2 (MaxEnt optical-stress closure). *Let a finite causal screen have a background filling defect $\lambda = 3 - s$ carried by the determinant channel. Let the compensating residual close through the unimodular optical sector with weighted coefficient (33). Let the resulting residual be Ricci-silent and Weyl-active at first order. Then*

$$\boxed{\mu(a) = 1 + \mathcal{O}(\lambda^2)}, \quad (55)$$

$$\boxed{\Sigma(a) = 1 - \kappa_{BW} \lambda X_b(a) + \mathcal{O}(\lambda^2)}, \quad (56)$$

where

$$\boxed{\kappa_{BW} = \frac{2\eta_S + \eta_\Omega}{\eta_A + 2\eta_S + \eta_\Omega}}. \quad (57)$$

In the isotropic lowest-order branch,

$$\eta_A = \eta_S = \eta_\Omega, \quad (58)$$

so

$$\boxed{\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s)}. \quad (59)$$

Proof. The determinant channel carries the filling defect $\lambda X_b(a)$. Finite screen closure assigns a compensating unimodular optical response of amplitude $-\kappa_{BW} \lambda X_b(a)$. The coefficient is given by the weighted optical-sector share. Since the compensating residual is unimodular, it is traceless on the screen. By the Sachs decomposition, a traceless optical residual is Weyl-active rather than Ricci-density active. Therefore it corrects Σ at first order and leaves μ unchanged at first order. $\square$

Equation (59) is the desired mathematical closure of the G1DE-M_{3/4} amplitude relation.

11 The horizon-memory kernel $X_b(a)$

The projection coefficient fixes the amplitude ratio, not the full redshift kernel. The shape function $X_b(a)$ should be interpreted as a finite-screen memory response, not as an arbitrary fitting polynomial.

Let

$$N = \ln a, \quad \epsilon_H(N) = -\frac{d \ln H}{dN}. \quad (60)$$

Acceleration corresponds to $\epsilon_H < 1$. Since an event-horizon-like future boundary becomes dynamically relevant only in accelerated expansion, a minimal source for horizon-screen memory is

$$\mathcal{S}_H(N) = [1 - \epsilon_H(N)]_+, \quad (61)$$

where $[x]_+ = \max(x, 0)$.

Definition 6 (Horizon-memory closure). *A first-order finite-screen memory kernel satisfies*

$$\frac{dX_b}{dN} + \Gamma_H(N)X_b(N) = \mathcal{S}_H(N), \quad (62)$$

with formal solution

$$X_b(N) = \int_{-\infty}^N dN' \exp \left[- \int_{N'}^N \Gamma_H(u) du \right] \mathcal{S}_H(N'). \quad (63)$$

Principle 2 (Minimal Markov horizon-memory candidate). *The simplest acceleration-gated closure is*

$$\boxed{X'_b(N) + X_b(N) = [1 - \epsilon_H(N)]_+}. \quad (64)$$

It is dormant in matter and radiation eras, activates when the future horizon becomes dynamically relevant, and approaches $X_b \rightarrow 1$ in the de Sitter limit.

Caveat 3. *The present paper derives the leading amplitude coefficient κ_{BW} , not the full redshift kernel $X_b(a)$. The latter is treated as a horizon-memory response whose source and relaxation terms remain part of the next dynamical closure problem. Equation (64) is a candidate closure, not yet a theorem. The main result of this paper is the amplitude-level optical-stress lock. Deriving $\Gamma_H(N)$ and $\mathcal{S}_H(N)$ from finite-screen entropy dynamics remains an open problem.*

12 Effective filling dimension and the meaning of s

The parameter s should not be treated as a mysterious fitting exponent. It is naturally interpreted as an effective filling dimension of the finite-screen cosmological ledger.

Let $\mathcal{V}_{\text{scr}}(a)$ denote screen-accessible bulk distinction volume. Define

$$s_{\text{eff}}(a) = \frac{d \ln \mathcal{V}_{\text{scr}}}{d \ln a}. \quad (65)$$

Standard three-volume filling corresponds to $s_{\text{eff}} = 3$. The finite-screen filling defect is

$$\lambda(a) = 3 - s_{\text{eff}}(a). \quad (66)$$

In the current G1DE reduction, s should be read as a low-redshift effective filling exponent, not as a completed dynamical dimension function. A full theory would replace the constant s by $s_{\text{eff}}(a)$ derived from finite-screen occupancy dynamics.

Equivalently, define a horizon capacity

$$C_H(a) = \frac{A_H(a)}{4\ell_P^2} \quad (67)$$

and an occupancy

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}. \quad (68)$$

Then a candidate relation is

$$3 - s_{\text{eff}}(a) \sim - \frac{d \ln \mathcal{O}_H}{d \ln a}. \quad (69)$$

This equation is not used as an input in the amplitude theorem, but it indicates the direction of a deeper dynamical closure.

13 Why E_G is the natural discriminator

The statistic E_G -type observables compare lensing to growth. G1 predicts exactly this split:

$$\Sigma(a) - 1 = \mathcal{O}(\lambda), \quad \mu(a) - 1 = \mathcal{O}(\lambda^2). \quad (70)$$

Therefore the natural target is not growth alone and not distance alone, but the cross-channel ratio:

$$\frac{\text{lensing}}{\text{growth}}. \quad (71)$$

A model that adds ordinary density generally changes both growth and lensing. A generic modified-gravity parameterization allows both μ and Σ to float. G1 predicts a constrained channel split:

$$\boxed{\text{Weyl/lensing first, matter growth protected.}} \quad (72)$$

Table 4: Residual fingerprints in common alternatives.

Theory type	Growth channel	Lensing channel	Cross-channel structure
Extra matter density	$\mu \neq 1$ expected	$\Sigma \neq 1$ expected	density-driven
Generic modified gravity	free μ	free Σ	weak unless imposed
Smooth dark energy $w(a)$	mainly background	indirect lensing effects	no fixed lock
G1 finite-screen optical stress	$\mu \simeq 1$	$\Sigma < 1$ for $s < 3$	$\Sigma_0 = -\kappa_{BW}(3 - s)$

14 Demotion paths and falsification tests

The theory is deliberately structured with demotion paths.

Table 5: Demotion and falsification registry.

Condition	What fails	What may survive
Data require dominant first-order μ correction	Ricci-silent screen-stress interpretation	finite-screen residual with density component
Coefficient near 2/3	full $GL(2)$ optical capacity	scalar-shear/SPD(2) demotion
Coefficient far from 2/3 and 3/4	isotropic MaxEnt projection	weighted response-weight law (33)
No common $X_b(a)$ for background and Weyl response	background–Weyl ledger lock	separate background and optical residuals
No lensing or E_G deviation while $s < 3$ persists	optical compensation branch	background-only filling anomaly
Strong curl/rotation mean at first order	scalar isotropy assumption	full holonomy branch with nonzero mean
No acceleration-gated behavior in X_b	minimal horizon-memory kernel	alternative screen-memory dynamics
Violation of covariance/conservation closure	effective stress representation	local optical algebra only

Condition	What fails	What may survive
No finite-screen entropy response in G1 core	entire FSS/G1 bridge	none, except algebraic toy model

15 Open mathematical problems

Problem 1: derive the optical response weights. The coefficients $\eta_A, \eta_S, \eta_\Omega$ should ultimately be obtained from finite-screen microstatistics or a screen entropy Hessian. At present, isotropy is the minimal MaxEnt closure.

Problem 2: derive $X_b(a)$. The minimal equation $X'_b + X_b = [1 - \epsilon_H]_+$ is plausible but not final. A full derivation should obtain Γ_H and $\mathcal{S}_H$ from entropy production, screen capacity drift, and horizon occupancy.

Problem 3: covariant embedding. The effective residual $\mathcal{E}_{\mu\nu}^{\text{scr}}$ must be embedded in a conservation-closed covariant system satisfying

$$\nabla^\mu (G_{\mu\nu} - 8\pi G T_{\mu\nu} - \mathcal{E}_{\mu\nu}^{\text{scr}}) = 0. \quad (73)$$

Problem 4: nonlinear and small-scale limits. The present result is a linear optical-screen closure. It does not yet specify nonlinear clustering, baryonic feedback, strong lensing caustics, or black-hole interior dynamics.

Problem 5: distinguish capacity from signal. The twist/holonomy direction contributes to full optical capacity counting, but its mean signal may vanish in scalar cosmology. Future work must separate capacity covariance, parity constraints, and observable curl/B-mode signatures.

16 Conclusion

This paper strengthens the mathematical core of the G1 projection law. The 3/4 coefficient is not treated as a numerical coincidence. It is derived as the maximum-entropy projection of a finite-screen optical residual onto the unimodular sector of the null optical screen deformation algebra.

The closure chain is

$$\boxed{X \in T_I GL(2) \rightarrow A(1) \oplus S(2) \oplus \Omega(1) \rightarrow \kappa_{BW} = \frac{2\eta_S + \eta_\Omega}{\eta_A + 2\eta_S + \eta_\Omega} \rightarrow \frac{3}{4}} \quad (74)$$

under isotropic lowest-order response. The compensating residual is unimodular and therefore traceless. Through the Sachs optical decomposition, a traceless optical residual is Ricci-silent and Weyl-active at first order. Hence

$$\boxed{\mu(a) = 1 + \mathcal{O}((3-s)^2), \quad \Sigma(a) = 1 - \frac{3}{4}(3-s)X_b(a) + \mathcal{O}((3-s)^2).} \quad (75)$$

This gives the G1DE-M_{3/4} lock

$$\boxed{\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s).} \quad (76)$$

The conceptual result is sharper than a modified Poisson equation. G1 finite-screen gravity is best understood, at this level, as a Ricci-silent, Weyl-active optical-stress theory. The dark sector is the optical stress of finite-screen incompleteness.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* 7, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* 43, 199 (1975).
- [3] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* 98, 1123 (1955).
- [4] R. Sachs, “Gravitational waves in general relativity. VI. The outgoing radiation condition,” *Proc. R. Soc. Lond. A* 264, 309 (1961).
- [5] P. Schneider, J. Ehlers, and E. E. Falco, *Gravitational Lenses* (Springer, 1992).
- [6] M. Bartelmann and P. Schneider, “Weak gravitational lensing,” *Phys. Rep.* 340, 291 (2001).
- [7] R. M. Wald, *General Relativity* (University of Chicago Press, 1984).
- [8] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* 75, 1260 (1995).
- [9] J. M. Bardeen, “Gauge-invariant cosmological perturbations,” *Phys. Rev. D* 22, 1882 (1980).
- [10] C.-P. Ma and E. Bertschinger, “Cosmological perturbation theory in the synchronous and conformal Newtonian gauges,” *Astrophys. J.* 455, 7 (1995).
- [11] P. Zhang, M. Liguori, R. Bean, and S. Dodelson, “Probing gravity at cosmological scales by measurements which test the relationship between gravitational lensing and matter overdensity,” *Phys. Rev. Lett.* 99, 141302 (2007).
- [12] S. F. Daniel and E. V. Linder, “Constraining cosmic expansion and gravity with gravitational lensing,” *Phys. Rev. D* 82, 103523 (2010).
- [13] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* 07, 050 (2014).
- [14] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), doi:10.5281/zenodo.20382013.
- [15] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual*, manuscript (2026).

Non-Equilibrium Screen Entropy and the Optical-Stress Origin of the G1 Dark Sector

A Covariant Closure for Ricci-Silent, Weyl-Active Finite-Screen Residuals

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

May 2026

Abstract

This paper develops a non-equilibrium entropy closure for the dark-sector residuals of Finite Distinction Systems – Gravity 1 (FDS-G1). The companion projection-law note derived the optical amplitude lock

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s)$$

by treating the leading finite-screen residual as a maximum-entropy deformation on the local null optical screen algebra

$$\mathfrak{gl}(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1),$$

where A is the area/determinant channel, S is symmetric shear, and Ω is twist/optical holonomy. The present paper asks a deeper question: what screen dynamics produces such an optical-stress residual?

The central claim is that the G1 dark sector is not an additional density component, but non-equilibrium finite-screen entropy production projected as traceless optical stress. A finite causal screen is assigned an entropy functional whose near-equilibrium second variation has optical-sector stiffnesses

$$K_A, \quad K_S, \quad K_\Omega$$

for expansion, shear, and twist. The corresponding compliances $\chi_i = K_i^{-1}$ determine the background–Weyl projection coefficient

$$\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega},$$

with $\kappa_{BW} = 3/4$ as the isotropic entropy-Hessian fixed point and $2/3$ as the scalar-shear demotion when the twist/holonomy sector is removed. The non-equilibrium entropy production term is written in Sachs optical variables,

$$\delta S_{\text{prod}} = \frac{1}{2} \int d\lambda dA \left(K_A \theta^2 + K_S \sigma_{AB} \sigma^{AB} + K_\Omega \omega^2 \right),$$

so that the compensating residual lives in the unimodular screen sector. By construction this residual is traceless on the screen; through the Sachs curvature split it couples to the Weyl optical tidal channel rather than to the Ricci density-focusing channel.

The resulting covariant effective-stress closure predicts

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2),$$

while the matter-growth channel is protected up to second order and horizon-scale leakage,

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

Here $\mathcal{R}_H(a)$ is the horizon-screen residual response. Rather than treating the redshift kernel as a free shape, the paper defines it as a retarded memory of horizon occupancy drift,

$$\mathcal{R}'_H(N) + \Gamma_H(N)\mathcal{R}_H(N) = \mathcal{S}_H(N), \quad N = \ln a,$$

with a minimal acceleration-gated approximation $\mathcal{S}_H \simeq [1 - \epsilon_H]_+$. The constant-amplitude G1DE-M_{3/4} reduction is recovered when $\mathcal{R}_H(a) \simeq (3 - s)X_b(a)$ over the low-redshift window. The construction is conditional and falsifiable: a dominant first-order μ correction, failure of horizon-memory structure, a coefficient outside the optical-compliance family, or evidence for the 2/3 branch would demote the full non-equilibrium optical-stress closure.

1 Scope and executive summary

FDS-G1 treats gravitation as an entropy-response geometry of finite causal screens. The G1 core program identifies local gravitational stiffness with the entropy response of a finite screen, while G1DE reductions express the leading late-time residual through background and Weyl/lensing channels. The earlier optical projection law established a conditional amplitude relation

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s), \quad (1)$$

by modeling the finite-screen residual as a maximum-entropy optical deformation on the two-dimensional null screen. The present paper supplies the next layer of closure: it interprets the residual as a non-equilibrium entropy-production term and embeds it into a covariant effective-stress hierarchy.

D3 owns the thermodynamic admissibility of screen residuals. Not every residual is heat; only physically implemented irreversible record reuse, repair, synchronization, or externalization enters the entropy/resource ledger.

The central thesis is:

The G1 dark sector is not an additional density component. It is finite-screen non-equilibrium entropy production projected as Ricci-silent, Weyl-active traceless optical stress.

The effective screen stress should be read as a controlled hierarchy, not as a standalone conserved fluid with only anisotropic stress at all scales. On subhorizon scales,

$$E_{ij, \text{traceless}}^{\text{scr}} = O(\mathcal{R}_H), \quad E_{0i}^{\text{scr}} = O(\mathcal{R}_H \mathcal{H}/k), \quad E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O(\mathcal{R}_H \mathcal{H}^2/k^2). \quad (2)$$

Thus $\mu \simeq 1$ is a subhorizon first-order protection, not an absolute zero-source statement.

This statement has five components.

1. **Optical screen access.** Finite observers access spacetime residuals through a two-dimensional null optical screen. Local screen deformations live in $T_I GL(2, \mathbb{R}) \simeq \mathfrak{gl}(2, \mathbb{R})$.
2. **Entropy Hessian.** Near local screen equilibrium, non-equilibrium entropy production has a quadratic optical-sector form in expansion, shear, and twist.

3. **Compliance projection.** The response coefficient is a compliance-weighted fraction of the unimodular sector,

$$\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}. \quad (3)$$

The isotropic entropy-Hessian branch gives $\kappa_{BW} = 3/4$.

4. **Ricci silence.** The compensating residual is unimodular and therefore traceless on the optical screen. Through the Sachs split it enters the Weyl optical tidal channel before it enters the Ricci density-focusing channel.
5. **Horizon memory.** The redshift dependence is not an arbitrary polynomial. It is a retarded response of finite horizon-screen occupancy drift.

Let

$$\mathcal{R}_H(a) = (3 - s_{\text{eff}}(a))X_b(a) \quad (4)$$

be the combined horizon residual response. The leading cosmological prediction is

$$\Sigma(a, k) - 1 = -\kappa_{BW}\mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (5)$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (6)$$

Thus the G1 signature is not generic modified gravity. It is a Weyl-first, matter-growth-protected channel split.

2 Claim-status discipline

The present paper is a mathematical bridge, not a final fundamental derivation. Its claims are graded in Table 1.

The strongest safe claim is not that $3/4$ follows from algebra alone. The stronger claim is:

$$\text{finite optical algebra} + \text{isotropic non-equilibrium screen compliance} \implies \kappa_{BW} = \frac{3}{4}. \quad (7)$$

If the optical sector compliances are anisotropic, the coefficient shifts. If the twist/holonomy sector is demoted, the full $GL(2)$ branch demotes to a scalar-shear $2/3$ branch.

3 Finite causal screens and optical kinematics

Consider a null geodesic congruence with tangent k^μ reaching a finite observer. Let e_A^μ , $A = 1, 2$, be a screen basis orthogonal to k^μ and to the observer direction. The Jacobi map

$$J^A_B \quad (8)$$

relates source-plane separations to observed screen separations. Away from caustics,

$$J \in GL(2, \mathbb{R}). \quad (9)$$

Table 1: Claim status of the non-equilibrium optical-stress closure.

Claim	Meaning	Status
Finite optical access	Observable residuals are routed through two-dimensional null optical screens	bridge principle
Sachs decomposition	Optical deformation splits into expansion, shear, and twist	standard optical geometry
Entropy Hessian	Screen entropy has a quadratic near-equilibrium response in optical sectors	screen-thermodynamic bridge
Compliance projection	Sector compliances determine κ_{BW}	conditional theorem
Isotropic 3/4 branch	$K_A = K_S = K_\Omega$ implies $\kappa_{BW} = 3/4$	MaxEnt / isotropic fixed point
Unimodular stress	Shape residual is traceless on the screen	variational consequence
Ricci silence	No leading density-like source in subhorizon growth	covariant bridge with leakage terms
Horizon memory	Redshift response is retarded occupancy drift	dynamical bridge, not final theorem
G1DE-M _{3/4} reduction	$\mu_0 = 0, \Sigma_0 = -3(3 - s)/4$	low-redshift amplitude reduction

Definition 1 (Optical deformation generator). *The local residual deformation of a finite optical screen is*

$$X = \delta J J^{-1} \in \mathfrak{gl}(2, \mathbb{R}). \quad (10)$$

For continuous affine evolution, define

$$B^A{}_B = \frac{dJ^A{}_C}{d\lambda} (J^{-1})^C{}_B. \quad (11)$$

The optical deformation-rate matrix decomposes into Sachs scalars:

$$B^A{}_B = \frac{1}{2} \theta \delta^A{}_B + \sigma^A{}_B + \omega \epsilon^A{}_B, \quad (12)$$

where θ is expansion, $\sigma^A{}_B$ is symmetric traceless shear, and $\omega \epsilon^A{}_B$ is twist/rotation. Thus

$$\mathfrak{gl}(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1), \quad (13)$$

with

$$A \oplus S \oplus \Omega = \mathbb{R}I \oplus \mathfrak{sl}(2, \mathbb{R}). \quad (14)$$

The area/determinant sector has dimension 1; the unimodular sector has dimension 3.

The Sachs optical tidal equation has the schematic form

$$\dot{B} + B^2 = \mathcal{R}, \quad (15)$$

where

$$\mathcal{R}_{AB} = -R_{\mu\nu\rho\sigma} e^\mu{}_A k^\nu k^\rho e^\sigma{}_B. \quad (16)$$

Using the four-dimensional curvature decomposition,

$$\mathcal{R}_{AB} = -\frac{1}{2}R_{\mu\nu}k^\mu k^\nu \gamma_{AB} - C_{\mu\nu\rho\sigma}e_A^\mu k^\nu k^\rho e_B^\sigma. \quad (17)$$

The first term is trace/Ricci focusing. The second is traceless/Weyl tidal distortion. This split is the geometric basis of the Ricci-silent, Weyl-active G1 channel structure.

4 Equilibrium and non-equilibrium screen entropy

The equilibrium screen entropy in the G1 core is normalized by the horizon area law,

$$S_{\text{eq}} = \frac{A}{4\ell_P^2} + S_{\text{sub}}. \quad (18)$$

Equilibrium Clausius-type variations are expected to reproduce an Einstein/Jacobson-like metric closure in the local limit. The dark-sector residual arises when finite-screen closure is not exact. We therefore write a non-equilibrium entropy balance

$$\delta S_{\text{scr}} = \frac{\delta Q}{T} + \delta S_{\text{prod}}, \quad (19)$$

where δS_{prod} is the finite-screen entropy-production surplus.

The total screen entropy production decomposes into five channels:

$$\dot{S}_{\text{scr}} = \dot{S}_{\text{erase}} + \dot{S}_{\text{memory}} + \dot{S}_{\text{sync}} + \dot{S}_{\text{ext}} + \dot{S}_{\text{optical}}.$$

Only the optical channel may couple to $R_{\mu\nu}^{\text{scr}}$. The other channels represent internal finite-capacity costs that do not feed the geometric residual sector.

Principle 1 (Optical entropy production). *Near a locally isotropic screen state, the leading non-equilibrium entropy production may be expanded in the optical scalars,*

$$\delta S_{\text{prod}} = \frac{1}{2} \int d\lambda dA (K_A \theta^2 + K_S \sigma_{AB} \sigma^{AB} + K_\Omega \omega^2) + O(B^3). \quad (20)$$

Here $K_A, K_S, K_\Omega > 0$ are screen entropy stiffnesses in the expansion, shear, and twist sectors.

The sign convention is chosen so that departures from equilibrium carry a positive cost. The corresponding response compliances are

$$\chi_A = K_A^{-1}, \quad \chi_S = K_S^{-1}, \quad \chi_\Omega = K_\Omega^{-1}. \quad (21)$$

The isotropic lowest-order entropy-Hessian branch is

$$K_A = K_S = K_\Omega, \quad (22)$$

which is equivalent to

$$\chi_A = \chi_S = \chi_\Omega. \quad (23)$$

5 MaxEnt and Onsager screen response

The projection-law note used a local maximum-entropy residual on $T_1GL(2)$. The present entropy-production formulation gives a dynamical interpretation of the same assumption.

Let X_i denote the four optical deformation coordinates corresponding to A, S_1, S_2, Ω . In a linear-response approximation,

$$\dot{X}_i = - \sum_j L_{ij} \frac{\partial \mathcal{F}_{\text{scr}}}{\partial X_j} + \xi_i, \quad (24)$$

where L_{ij} is an Onsager response matrix and ξ_i is screen noise. The minimum-information, no-preferred-direction fixed point is

$$L_{ij} = L\delta_{ij}, \quad \langle \xi_i(N) \xi_j(N') \rangle = 2L\delta_{ij}\delta(N - N'). \quad (25)$$

Equivalently, the local residual covariance is isotropic on $\mathfrak{gl}(2)$.

Lemma 1 (MaxEnt-compliance projection). *Let the leading finite-screen optical residual be locally maximum-entropy on $T_1GL(2)$ and let the sector response be governed by compliances $(\chi_A, \chi_S, \chi_\Omega)$. Then the normalized unimodular response coefficient is*

$$\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}. \quad (26)$$

If $\chi_A = \chi_S = \chi_\Omega$, then

$$\kappa_{BW} = \frac{2 + 1}{1 + 2 + 1} = \frac{3}{4}. \quad (27)$$

If the twist/holonomy sector is removed while $\chi_A = \chi_S$, then

$$\kappa_{BW} = \frac{2}{1 + 2} = \frac{2}{3}. \quad (28)$$

Proof. Each sector contributes its dimension multiplied by its response compliance. The area sector has dimension 1, symmetric shear has dimension 2, and twist has dimension 1. The Weyl-active unimodular sector is $S \oplus \Omega$, so its share is (26). $\square$

For small deviations from isotropy, write

$$\chi_S = \chi_A(1 + \alpha), \quad \chi_\Omega = \chi_A(1 + \beta). \quad (29)$$

Then

$$\kappa_{BW} = \frac{3 + 2\alpha + \beta}{4 + 2\alpha + \beta} = \frac{3}{4} + \frac{2\alpha + \beta}{16} + O(\alpha^2, \alpha\beta, \beta^2). \quad (30)$$

Thus a measured offset from 3/4 directly estimates the combination $2\alpha + \beta$ of shear/twist compliance anisotropy.

6 Constrained screen closure

Let

$$\lambda(a) = 3 - s_{\text{eff}}(a) \quad (31)$$

be the finite-screen filling defect relative to ordinary three-volume scaling. The area/determinant channel carries the background filling anomaly. Screen closure requires a compensating unimodular response.

Principle 2 (Screen ledger closure). *At leading order, the finite-screen optical residual decomposes as*

$$\Delta_{\text{scr}} = \Delta_A + \Delta_U + O(\lambda^2), \quad (32)$$

with

$$\Delta_A = \lambda(a)X_b(a), \quad \Delta_U = -\kappa_{BW}\lambda(a)X_b(a). \quad (33)$$

The sign is fixed by compensation: the unimodular optical sector closes the determinant/filling defect.

It is useful to combine the amplitude defect and the memory response into a single horizon residual,

$$\mathcal{R}_H(a) = \lambda(a)X_b(a). \quad (34)$$

Then the closure relation is simply

$$\Delta_U = -\kappa_{BW}\mathcal{R}_H(a). \quad (35)$$

The low-redshift G1DE-M_{3/4} reduction corresponds to

$$\lambda(a) \simeq 3 - s = \text{constant}. \quad (36)$$

7 Unimodular optical stress

Let γ_{AB} be the screen metric and define the unimodular metric

$$\hat{\gamma}_{AB} = (\det \gamma)^{-1/2} \gamma_{AB}, \quad \det \hat{\gamma} = 1. \quad (37)$$

The shape part of the screen action depends only on $\hat{\gamma}$:

$$S_{\text{scr}}^{\text{shape}} = S_{\text{scr}}^{\text{shape}}[\hat{\gamma}]. \quad (38)$$

The induced screen stress is defined by inverse-metric variation,

$$T_{AB}^{\text{scr}} = -\frac{2}{\sqrt{\gamma}} \frac{\delta S_{\text{scr}}^{\text{shape}}}{\delta \gamma^{AB}}. \quad (39)$$

Lemma 2 (Unimodular stress). *If $S_{\text{scr}}^{\text{shape}}$ depends only on the unimodular screen metric, then*

$$\gamma^{AB} T_{AB}^{\text{scr}} = 0. \quad (40)$$

Proof. Under a pure conformal variation $\delta \gamma^{AB} = -2\phi \gamma^{AB}$, the unimodular metric $\hat{\gamma}_{AB}$ is invariant to first order. Therefore $\delta S_{\text{scr}}^{\text{shape}} = 0$ for arbitrary ϕ . But

$$\delta S_{\text{scr}}^{\text{shape}} = -\frac{1}{2} \int d^2 y \sqrt{\gamma} T_{AB}^{\text{scr}} \delta \gamma^{AB} = \int d^2 y \sqrt{\gamma} \phi \gamma^{AB} T_{AB}^{\text{scr}}. \quad (41)$$

For arbitrary ϕ , the trace vanishes. $\square$

Thus the compensating residual is not a scalar density on the screen. It is traceless optical stress.

8 Ricci-silent, Weyl-active covariant residual

Introduce an effective residual tensor

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} + E_{\mu\nu}^{\text{scr}}. \quad (42)$$

G1 does not require $E_{\mu\nu}^{\text{scr}}$ to be a material fluid. It is an effective representation of finite-screen incompleteness in the metric closure.

Definition 2 (Ricci-silent residual). *A finite-screen residual is Ricci-silent at first order if its density-like focusing component is absent in the subhorizon/quasi-static channel, up to second-order and horizon-scale leakage:*

$$E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right), \quad (43)$$

$$E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right). \quad (44)$$

Definition 3 (Weyl-active residual). *A finite-screen residual is Weyl-active if its projected traceless spatial/optical component is first order:*

$$E_{ij,\text{traceless}}^{\text{scr}} = O(\mathcal{R}_H). \quad (45)$$

Equivalently, on the optical screen,

$$E_{AB}^{\text{scr}} = E_{\langle AB \rangle}^{\text{scr}} + O(\mathcal{R}_H^2), \quad \gamma^{AB} E_{AB}^{\text{scr}} = 0. \quad (46)$$

The Sachs split routes trace optical focusing to Ricci curvature and traceless optical distortion to Weyl curvature. Therefore a unimodular screen residual is naturally Weyl-active while remaining Ricci-silent at first order.

Caveat 1 (Covariant closure). *A purely traceless spatial effective stress cannot generally be divergence-free by itself on all scales. The Ricci-silent statement is a controlled hierarchy, not an exact componentwise zero. A complete covariant embedding must satisfy either*

$$\nabla^\mu E_{\mu\nu}^{\text{scr}} = 0 \quad (47)$$

or a controlled horizon-exchange equation

$$\nabla^\mu E_{\mu\nu}^{\text{scr}} = J_\nu^H. \quad (48)$$

9 Cosmological reduction

Use the standard phenomenological functions

$$-k^2 \Psi = 4\pi G a^2 \mu(a, k) \rho \Delta, \quad (49)$$

$$-k^2 (\Phi + \Psi) = 8\pi G a^2 \Sigma(a, k) \rho \Delta. \quad (50)$$

The Newtonian channel is density/Ricci sensitive. The lensing channel is Weyl/optical sensitive.

Theorem 1 (Non-equilibrium optical-stress reduction). *Let a finite causal screen have non-equilibrium optical entropy production of the form (20). Let its horizon residual response be $\mathcal{R}_H(a)$ and its unimodular compensation coefficient be κ_{BW} as in (26). If the residual is Ricci-silent and Weyl-active in the sense above, then*

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (51)$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (52)$$

For the isotropic branch, $\kappa_{BW} = 3/4$. For a low-redshift constant-amplitude reduction $\mathcal{R}_H(a) \simeq (3-s)X_b(a)$, this gives

$$\mu(a, k) = 1 + O((3-s)^2), \quad (53)$$

$$\Sigma(a, k) = 1 - \frac{3}{4}(3-s)X_b(a) + O((3-s)^2). \quad (54)$$

Proof. The determinant/area filling defect is compensated by the unimodular sector with amplitude $-\kappa_{BW}\mathcal{R}_H$. By the unimodular-stress lemma, the compensating residual is traceless on the screen. By the Sachs Ricci/Weyl decomposition, a traceless optical residual enters the Weyl/lensing channel at first order. Since the residual is Ricci-silent, the local density-focusing source in the Poisson channel receives no leading subhorizon contribution. Bianchi consistency allows only second-order or horizon-suppressed leakage into μ . $\square$

The G1DE- $M_{3/4}$ relation follows as the low-redshift, constant-amplitude, isotropic-compliance reduction:

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s). \quad (55)$$

10 Horizon Occupancy and a Candidate Memory Kernel

The redshift shape should not be a free polynomial. It should be the retarded response of finite horizon-screen occupancy drift.

Let

$$N = \ln a, \quad \epsilon_H = -\frac{d \ln H}{dN}. \quad (56)$$

Define a horizon capacity

$$C_H(a) = \frac{A_H(a)}{4\ell_P^2}, \quad (57)$$

where A_H is an appropriate cosmological horizon screen area. Define occupancy

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}. \quad (58)$$

Definition 4 (Horizon occupancy source). *The finite-screen occupancy drift source is*

$$\mathcal{S}_H(N) = -\frac{d \ln \mathcal{O}_H}{dN}. \quad (59)$$

The horizon residual response obeys a memory equation

$$\mathcal{R}'_H(N) + \Gamma_H(N)\mathcal{R}_H(N) = \mathcal{S}_H(N), \quad (60)$$

with solution

$$\mathcal{R}_H(N) = \int_{-\infty}^N dN' \exp\left[-\int_{N'}^N \Gamma_H(u) du\right] \mathcal{S}_H(N'). \quad (61)$$

Principle 3 (Minimal acceleration-gated approximation). *A minimal low-information closure is*

$$\Gamma_H \simeq 1, \quad \mathcal{S}_H(N) \simeq [1 - \epsilon_H(N)]_+, \quad (62)$$

so that

$$\mathcal{R}'_H(N) + \mathcal{R}_H(N) = [1 - \epsilon_H(N)]_+. \quad (63)$$

This approximation is dormant during matter and radiation domination, activates when accelerated expansion makes a future-horizon-like causal boundary dynamically relevant, and approaches a constant response in the de Sitter limit.

Caveat 2 (Amplitude versus redshift kernel). *This paper derives the optical-stress amplitude coefficient. It does not complete the full redshift-kernel derivation. Equation (63) is a minimal candidate closure, while (60) is the more general horizon-memory structure. Deriving Γ_H and $\mathcal{S}_H$ from finite-screen entropy dynamics remains an open problem.*

11 Effective filling dimension

The parameter s in the low-redshift G1DE reduction should not be treated as a fundamental constant. It is an effective filling exponent.

Let $\mathcal{V}_{\text{scr}}(a)$ be the screen-accessible bulk distinction volume. Define

$$s_{\text{eff}}(a) = \frac{d \ln \mathcal{V}_{\text{scr}}}{d \ln a}. \quad (64)$$

Ordinary three-volume filling gives $s_{\text{eff}} = 3$. The finite-screen filling defect is

$$\lambda(a) = 3 - s_{\text{eff}}(a). \quad (65)$$

A low-redshift constant- s reduction corresponds to

$$s \simeq \langle s_{\text{eff}}(a) \rangle_{z \lesssim 2}. \quad (66)$$

Equivalently, if $\mathcal{O}_H = C_{\text{used}}/C_H$, then a candidate relation is

$$3 - s_{\text{eff}}(a) \sim -\frac{d \ln \mathcal{O}_H}{d \ln a}. \quad (67)$$

The deeper dynamical variable is therefore not s alone but

$$\mathcal{R}_H(a) = \lambda(a)X_b(a). \quad (68)$$

12 Null tests and observational fingerprints

The defining G1 channel split is

$$|\mu - 1| \ll |\Sigma - 1| \quad (69)$$

on subhorizon linear scales, with the precise scaling in (52)–(51).

A useful null combination is

$$\mathcal{N}_{\text{G1}}(a, k) = \Sigma(a, k) - 1 + \kappa_{BW} \mathcal{R}_H(a). \quad (70)$$

G1 predicts

$$\mathcal{N}_{\text{G1}} = O(\mathcal{R}_H^2). \quad (71)$$

For the isotropic low-redshift branch,

$$\mathcal{N}_{3/4} = \Sigma(a, k) - 1 + \frac{3}{4}(3 - s)X_b(a) = O((3 - s)^2). \quad (72)$$

Another discriminator is

$$\mathcal{Q}_{\mu\Sigma}(a, k) = \frac{\mu(a, k) - 1}{\Sigma(a, k) - 1}, \quad (73)$$

with prediction

$$\mathcal{Q}_{\mu\Sigma} = O(\mathcal{R}_H) + O(\mathcal{H}^2/k^2). \quad (74)$$

Thus E_G -type lensing/growth observables are natural probes of G1: they target precisely the ratio of Weyl-active lensing to protected matter growth.

Table 2: Residual fingerprints.

Theory class	Growth channel	Lensing channel	Channel structure
Extra matter density	$\mu \neq 1$ expected	$\Sigma \neq 1$ expected	density-driven
Smooth dark energy	mostly background	indirect lensing	no locked split
Generic modified gravity	free μ	free Σ	weak unless imposed
G1 optical stress	$\mu \simeq 1$ subhorizon	$\Sigma - 1 = O(\mathcal{R}_H)$	Weyl-first lock

13 Extensions

We list three directions but do not develop them here.

1. **Horizon shear/twist memory near black holes.** Finite-screen corrections near strong-field horizons should appear first as shear/twist membrane stress rather than as new matter density. Possible interfaces include horizon memory, ringdown, photon-sphere lensing, and scrambling.
2. **EM/GW distance-duality tests.** If the G1 residual is optical/Weyl stress, standard-siren distance ratios may carry screen-stress corrections of order $\mathcal{R}_H(a)$, though $c_T = 1$ is expected at leading order. This is an open observational direction.
3. **Finite-screen code geometry.** In an information-theoretic reading, the G1 dark sector can be interpreted as a finite-screen code-deformation residual: not a new substance, but a cost of maintaining recoverable distinction geometry under finite screen capacity. Twist/holonomy contributes to capacity counting without requiring nonzero mean rotation in scalar cosmology.

14 Demotion and falsification registry

Table 3: Demotion and falsification paths.

Condition	What fails	What may survive
Dominant first-order μ correction on subhorizon scales	Ricci-silent optical-stress closure	finite-screen residual with density component
Coefficient near 2/3	full $GL(2)$ branch	scalar-shear $SPD(2)$ branch
Coefficient far from optical-compliance family	MaxEnt / compliance projection	unrelated phenomenological lock
No common horizon residual $\mathcal{R}_H(a)$ for background and Weyl	background–Weyl ledger lock	separate residual channels
No lensing or E_G deviation while filling defect persists	optical compensation branch	background-only filling anomaly
Large first-order curl mean	scalar isotropy assumption	holonomy branch with parity breaking
No acceleration-gated or occupancy-memory behavior	minimal horizon-memory closure	alternative screen-memory dynamics
Violation of any covariant conservation or controlled exchange	effective-stress embedding	local optical algebra only
Failure of finite-screen entropy response in G1 core	full FSS/G1 bridge	algebraic toy model only

15 Open mathematical problems

Problem 1: derive the entropy Hessian. The stiffnesses K_A, K_S, K_Ω should ultimately be obtained from a finite-screen entropy functional rather than postulated.

Problem 2: derive the horizon source. The source $\mathcal{S}_H = -d_N \ln \mathcal{O}_H$ must be computed from finite-screen occupancy dynamics. The acceleration-gated form $[1 - \epsilon_H]_+$ is only a minimal approximation.

Problem 3: covariant embedding. The effective residual $E_{\mu\nu}^{\text{scr}}$ must be embedded in a conservation-closed system satisfying

$$\nabla^\mu (G_{\mu\nu} - 8\pi G T_{\mu\nu} - E_{\mu\nu}^{\text{scr}}) = 0 \quad (75)$$

or a controlled exchange equation with a horizon bath.

Problem 4: nonlinear and strong-field limits. The present analysis is linear and optical. Nonlinear structure, baryonic feedback, caustics, strong lensing, black holes, and ringdown remain outside its scope.

Problem 5: twist covariance and observability. The twist/holonomy direction contributes to full optical capacity, but its mean may vanish in scalar cosmology. Future work must separate capacity covariance, parity constraints, and observable curl/B-mode signals.

Problem 6: relation to G1fit-real production evidence. The present paper supplies a theoretical closure. Its observational counterpart requires production likelihoods, expanded weak-lensing and E_G data, and model comparison against flexible μ - Σ alternatives.

16 Relation to the G1fit-real evidence tests

The observational counterpart of this closure is the G1fit-real program, where the low-redshift reduction G1DE-M_{3/4} is tested against SN, BAO, RSD, and E_G data. The present paper supplies the non-equilibrium screen-entropy closure behind that model; it does not use the likelihood results as an input. Current exact-pilot and medium-prior nested-evidence comparisons select the projection-locked relation at the AIC/BIC and evidence level against common dark-energy extensions, while full production evidence, expanded weak-lensing likelihoods, and redshift-kernel discrimination refinement remain pending.

17 Conclusion

The optical projection law showed how a finite screen can produce the amplitude relation

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s). \quad (76)$$

The present paper embeds that result into a deeper non-equilibrium screen-entropy closure.

The core chain is

$$\begin{aligned} \boxed{\delta S_{\text{prod}}} &\implies \boxed{K_A, K_S, K_\Omega} \implies \boxed{\chi_i = K_i^{-1}} \implies \boxed{\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}} \\ &\implies \boxed{\text{traceless optical stress}} \implies \boxed{\Sigma - 1 = -\kappa_{BW}\mathcal{R}_H + O(\mathcal{R}_H^2), \mu - 1 = O(\mathcal{R}_H^2, \mathcal{R}_H\mathcal{H}^2/k^2)}. \end{aligned}$$

In the isotropic entropy-Hessian branch,

$$\chi_A = \chi_S = \chi_\Omega, \quad (77)$$

so

$$\kappa_{BW} = \frac{3}{4}. \quad (78)$$

In the low-redshift constant-amplitude reduction,

$$\mathcal{R}_H(a) \simeq (3 - s)X_b(a), \quad (79)$$

recovering the G1DE-M_{3/4} lock.

The conceptual result is that the G1 dark sector is not an ordinary density fluid and not a generic modified Poisson term. It is finite-screen non-equilibrium entropy production projected as Ricci-silent, Weyl-active traceless optical stress. The redshift dependence is a horizon-memory problem; the coefficient is an entropy-compliance problem; the observational signature is a lensing/growth split.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* **7**, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* **43**, 199 (1975).
- [3] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* **98**, 1123 (1955).
- [4] R. Sachs, “Gravitational waves in general relativity. VI. The outgoing radiation condition,” *Proc. R. Soc. Lond. A* **264**, 309 (1961).
- [5] R. M. Wald, *General Relativity* (University of Chicago Press, 1984).
- [6] P. Schneider, J. Ehlers, and E. E. Falco, *Gravitational Lenses* (Springer, 1992).
- [7] M. Bartelmann and P. Schneider, “Weak gravitational lensing,” *Phys. Rep.* **340**, 291 (2001).
- [8] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* **75**, 1260 (1995).
- [9] C. Eling, R. Guedens, and T. Jacobson, “Non-equilibrium thermodynamics of spacetime,” *Phys. Rev. Lett.* **96**, 121301 (2006).
- [10] J. M. Bardeen, “Gauge-invariant cosmological perturbations,” *Phys. Rev. D* **22**, 1882 (1980).
- [11] C.-P. Ma and E. Bertschinger, “Cosmological perturbation theory in the synchronous and conformal Newtonian gauges,” *Astrophys. J.* **455**, 7 (1995).
- [12] P. Zhang, M. Liguori, R. Bean, and S. Dodelson, “Probing gravity at cosmological scales by measurements which test the relationship between gravitational lensing and matter overdensity,” *Phys. Rev. Lett.* **99**, 141302 (2007).
- [13] S. F. Daniel and E. V. Linder, “Constraining cosmic expansion and gravity with gravitational lensing,” *Phys. Rev. D* **82**, 103523 (2010).
- [14] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* **07**, 050 (2014).
- [15] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), doi:10.5281/zenodo.20382013.
- [16] Y. Wu, *The 3/4 Projection Law in Finite Screen Gravity: MaxEnt Optical-Stress Closure, Ricci-Silent Residuals, and the Weyl-Locked Dark Sector*, technical note (2026).
- [17] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual*, manuscript / data paper (2026).

Screen Influence Functional for the G1 Dark Sector: Entropy Hessian, Ward Closure, and Retarded Horizon Memory

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

May 2026

Abstract

This paper formulates a screen influence functional for the G1 dark sector in the Finite Distinction Systems–Gravity 1 (FDS-G1) program. Previous G1 notes derived a 3/4 optical projection law and a non-equilibrium screen-entropy origin for Ricci-silent, Weyl-active residuals. The present paper packages those ingredients into an effective influence framework: finite-screen entropy incompleteness produces a retarded horizon-memory residue $\mathcal{R}_H$, and the leading operator by which that residue enters cosmological perturbation theory is a Ward-closed unimodular optical stress.

The formal core is a quadratic non-equilibrium screen-entropy Hessian on the local optical deformation algebra,

$$I_{\text{scr}}^{\text{neq}} = \frac{1}{2} \int_{\mathcal{N}} d\lambda dA (K_A \theta^2 + K_S \sigma_{AB} \sigma^{AB} + K_\Omega \omega^2),$$

with compliance matrix $C = K^{-1}$. The Weyl-active projection coefficient is

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr} C} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}.$$

The isotropic entropy-Hessian fixed point $K_A = K_S = K_\Omega$ gives $\kappa_{BW} = 3/4$, while demotion of the twist/holonomy sector gives the scalar-shear branch $\kappa_{BW} = 2/3$.

The horizon residue is treated as an auxiliary retarded memory field obeying

$$\mathcal{R}'_H(N) + \Gamma_H(N) \mathcal{R}_H(N) = -\frac{d \ln \mathcal{O}_H}{dN}, \quad N = \ln a,$$

where $\mathcal{O}_H = C_{\text{used}}/C_H$ is the finite-screen occupancy of the horizon capacity $C_H = A_H/(4\ell_P^2)$. A Ward-completion theorem then gives the subhorizon hierarchy

$$E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{R}_H), \quad E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right),$$

so that

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad \mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

The low-redshift reduction $\mathcal{R}_H(a) \simeq (3-s)X_b(a)$ recovers the projection-locked G1DE relation $\mu_0 = 0$, $\Sigma_0 = -3(3-s)/4$. The resulting framework identifies G1DE-M_{3/4} as the constant-amplitude, low-redshift, isotropic-compliance reduction of a retarded screen influence functional.

The influence functional is retarded and open-system in the Schwinger–Keldysh sense, not a conservative local action. A technical appendix sharpens the remaining closure by defining the screen entropy Hessian as a relative-entropy Hessian, the horizon occupancy as horizon-accessible distinguishability, and the effective stress as a minimum-norm Ward completion of the optical seed.

1 Scope and executive summary

This paper is the fourth theory note in the G1 dark-sector closure sequence. Its purpose is not to introduce a new phenomenological parameterization. Its purpose is to assemble the previously derived optical projection and non-equilibrium screen-entropy structure into a single effective object: a *screen influence functional* for the G1 dark sector.

D4 mediates between microscopic realization and output $R_H(a)$, but must not claim unique source–kernel identification. The factorization $R_H = K_H * J_H$ carries a quotient equivalence class: $(K_H, J_H) \sim (K'_H, J'_H)$ iff $K_H * J_H = K'_H * J'_H$. D4 classifies possible realizations inside this class but avoids uniqueness.

The guiding statement is:

The G1 dark sector is a Ward-closed retarded screen influence whose leading operator is unimodular optical stress.

Equivalently,

$$\begin{aligned} \text{finite-screen entropy incompleteness} &\longrightarrow \text{retarded horizon residue } \mathcal{R}_H \\ &\longrightarrow \text{Ward-closed optical stress } E_{\mu\nu}^{\text{scr}}. \end{aligned} \tag{1}$$

This framing is designed to answer three questions left open by the shorter projection-law and non-equilibrium notes:

1. What is the effective object that generates the optical-stress residual?
2. Why is the leading residual Weyl-active but Ricci-density silent at first order?
3. How does the low-redshift model G1DE-M_{3/4} arise as a reduction of a retarded horizon-memory response?

The central result is the following schematic chain:

$$\begin{aligned} \boxed{I_{\text{scr}}^{\text{neq}}} &\Rightarrow \boxed{K_A, K_S, K_\Omega} \Rightarrow \boxed{C = K^{-1}} \Rightarrow \boxed{\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C}} \searrow \\ &\quad \boxed{\mathcal{O}_H} \Rightarrow \boxed{\mathcal{R}_H} \swarrow \\ &\Rightarrow \boxed{E_{ij, \text{TF}}^{\text{scr}}} \Rightarrow \boxed{\Sigma \neq 1, \quad \mu \simeq 1}. \end{aligned}$$

Equivalently, the operator chain is

$$-\text{d}_N \ln \mathcal{O}_H \xrightarrow{G_H^{\text{ret}}} \mathcal{R}_H \xrightarrow{C=K^{-1}, P_U} -\kappa_{BW} \mathcal{R}_H \xrightarrow{\mathcal{P}_W^{\text{ret}}} E_{\mu\nu}^{\text{scr}} \xrightarrow{\mathcal{W}_\Sigma, \mathcal{W}_\mu} (\Sigma, \mu). \tag{2}$$

The effective stress is a hierarchy, not a standalone conserved fluid with only anisotropic stress at all scales. Ward/Bianchi closure completes the leading traceless spatial stress by suppressed momentum and density components. In the subhorizon linear regime, the hierarchy is

$$E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{R}_H), \quad E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (3)$$

This is the mathematical meaning of Ricci silence in G1: it is not literal absence of all Ricci-sector components, but Ward-suppressed density leakage.

2 Relation to the G1 closure sequence

The present note should be read as a companion to three earlier layers:

Layer	Question answered
<i>D1</i> : 3/4 projection law	Why does the finite optical screen give a 3/4 Weyl-active projection under a MaxEnt / isotropic residual assumption?
<i>D2</i> : non-equilibrium entropy	What screen entropy dynamics produces optical stress?
<i>D3</i> : Ward-closed dark sector	How does optical stress enter covariant perturbation theory while keeping $\mu \simeq 1$ and $\Sigma \neq 1$?
This paper	What effective influence functional unifies entropy Hessian, retarded horizon memory, and Ward completion?
G1fit-real	Does the low-redshift reduction G1DE-M _{3/4} receive exact-likelihood support?

Thus this paper is not a data paper. It is a theory-side effective closure. It uses the G1fit-real model only as a low-redshift reduction target, not as an input to the derivation. A sharper conditional theorem collecting the relative-entropy screen Hessian, passive horizon-memory response, and minimum Ward completion is stated in Appendix A.

3 Claim-status discipline

The language of an influence functional is useful, but it must be handled with care. The screen residual is retarded and dissipative. Therefore the appropriate object is closer to an open-system influence functional than to a conservative local action. This paper uses “influence functional” in that effective-theory sense.

Table 1: Claim status of the screen influence closure.

Claim	Meaning	Status
Finite optical screen	observable residuals are projected through a two-dimensional null screen	bridge principle
$SO(2)$ sector split	$\mathfrak{gl}(2) = A(1) \oplus S(2) \oplus \Omega(1)$	algebraic / optical geometry
Entropy Hessian	K_A, K_S, K_Ω are second variations of screen entropy production	closure ansatz
Compliance projection	$\kappa_{BW} = \text{Tr}(P_U K^{-1}) / \text{Tr}(K^{-1})$	conditional theorem
Retarded memory field	$\mathcal{R}_H$ responds causally to horizon occupancy drift	dynamical bridge
Ward completion	leading optical stress requires suppressed $0i$ and 00 components	covariance / power counting
μ suppression	$\mu - 1 = O(\mathcal{R}_H^2) + O(\mathcal{R}_H \mathcal{H}^2 / k^2)$	subhorizon prediction
Weyl lock	$\Sigma - 1 = -\kappa_{BW} \mathcal{R}_H + O(\mathcal{R}_H^2)$	data-facing reduction
Low-redshift G1DE-M _{3/4}	$\mathcal{R}_H \simeq (3 - s)X_b$ and $\kappa = 3/4$	reduced model

The strongest safe statement is:

screen entropy Hessian + retarded memory + Ward closure

$$\implies \quad \Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2),$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

4 Finite optical screen sectors

A finite observer receives spacetime residuals through a null optical screen. On a non-caustic patch, the Jacobi map is

$$J_B^A \in GL(2, \mathbb{R}), \quad A, B = 1, 2. \quad (4)$$

The local deformation generator is

$$X = \delta J J^{-1} \in \mathfrak{gl}(2, \mathbb{R}). \quad (5)$$

Equivalently, along an affine parameter λ one may use

$$B_B^A = \frac{dJ_C^A}{d\lambda} (J^{-1})^C_B. \quad (6)$$

The Sachs decomposition is

$$B_B^A = \frac{1}{2} \theta \delta_B^A + \sigma_B^A + \omega \epsilon_B^A. \quad (7)$$

Thus

$$\mathfrak{gl}(2) = A(1) \oplus S(2) \oplus \Omega(1). \quad (8)$$

Here A is the area/trace channel, S is the symmetric traceless shear channel, and Ω is the twist/holonomy channel.

Under the local screen rotation group $SO(2)$, the sectors may be described as

$$A : \mathbf{0}^+, \quad S : \mathbf{2}, \quad \Omega : \mathbf{0}^-. \quad (9)$$

In an isotropic and parity-respecting screen background, the leading Hessian is block diagonal on these representations.

Let P_A, P_S, P_Ω denote the orthogonal projectors under the Frobenius inner product. The unimodular projector is

$$P_U = P_S + P_\Omega, \quad (10)$$

with traces

$$\text{Tr } P_A = 1, \quad \text{Tr } P_S = 2, \quad \text{Tr } P_\Omega = 1, \quad \text{Tr } P_U = 3. \quad (11)$$

5 Screen entropy Hessian and compliance projection

The non-equilibrium screen-entropy production is taken to be quadratic at leading order:

$$I_{\text{scr}}^{\text{neq}} = \frac{1}{2} \int_{\mathcal{N}} d\lambda dA (K_A \theta^2 + K_S \sigma_{AB} \sigma^{AB} + K_\Omega \omega^2). \quad (12)$$

Here K_A, K_S, K_Ω are entropy cost coefficients: they are eigenvalues of the screen entropy Hessian on the A, S, Ω sectors. The response compliances are their inverses:

$$\chi_A = K_A^{-1}, \quad \chi_S = K_S^{-1}, \quad \chi_\Omega = K_\Omega^{-1}. \quad (13)$$

Define the Hessian operator

$$K = K_A P_A + K_S P_S + K_\Omega P_\Omega, \quad (14)$$

so that the compliance/covariance operator is

$$C = K^{-1} = K_A^{-1} P_A + K_S^{-1} P_S + K_\Omega^{-1} P_\Omega. \quad (15)$$

Definition 1 (Weyl-active compliance projection). *The $G1$ background–Weyl projection coefficient is the fraction of residual compliance carried by the unimodular sector:*

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C}. \quad (16)$$

Using the sector traces gives

$$\kappa_{BW} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}. \quad (17)$$

Equivalently,

$$\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}. \quad (18)$$

Proposition 1 (Isotropic optical fixed point). *If the screen entropy Hessian is isotropic at leading order,*

$$K_A = K_S = K_\Omega, \quad (19)$$

then

$$\kappa_{BW} = \frac{2+1}{1+2+1} = \frac{3}{4}. \quad (20)$$

If the twist/holonomy sector is demoted, $K_\Omega^{-1} = 0$, and $K_A = K_S$, then

$$\kappa_{BW} = \frac{2}{1+2} = \frac{2}{3}. \quad (21)$$

Thus $3/4$ is not a bare numerological count. It is the compliance projection of an isotropic screen entropy Hessian. Deviations from $3/4$ measure anisotropy in the sector compliances. Since $C = K^{-1}$ is a positive compliance/covariance operator in the stable branch,

$$C \succeq 0, \quad (22)$$

one obtains the consistency bound

$$0 \leq \kappa_{BW} \leq 1. \quad (23)$$

A fitted value outside this interval would falsify the positive-compliance interpretation of the screen influence functional.

Let

$$\chi_S = \chi_A(1 + \alpha), \quad \chi_\Omega = \chi_A(1 + \beta). \quad (24)$$

Then

$$\kappa_{BW} = \frac{3 + 2\alpha + \beta}{4 + 2\alpha + \beta} = \frac{3}{4} + \frac{2\alpha + \beta}{16} + O(\alpha^2, \beta^2). \quad (25)$$

Hence a measured deviation from 0.75 can be read as a constraint on screen response anisotropy.

6 Retarded horizon-response output

The G1 dark-sector amplitude should not be treated as an arbitrary redshift function. The basic object is a retarded horizon-occupancy residue.

Define horizon capacity

$$C_H(a) = \frac{A_H(a)}{4\ell_P^2}, \quad (26)$$

and finite-screen occupancy

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}. \quad (27)$$

Let $N = \ln a$. The source for screen memory is the occupancy drift

$$\mathcal{S}_H(N) = -\frac{d \ln \mathcal{O}_H}{dN}. \quad (28)$$

Definition 2 (Retarded horizon residue). *The G1 horizon residue $\mathcal{R}_H$ is an auxiliary memory field obeying*

$$\boxed{\mathcal{R}'_H(N) + \Gamma_H(N)\mathcal{R}_H(N) = \mathcal{S}_H(N)}. \quad (29)$$

The retarded solution is

$$\mathcal{R}_H(N) = \int_{-\infty}^N dN' G_H(N, N') \mathcal{S}_H(N'), \quad (30)$$

where

$$G_H(N, N') = \exp \left[- \int_{N'}^N \Gamma_H(u) du \right]. \quad (31)$$

If $\Gamma_H > 0$, the memory is stable and decays in the absence of source. A future microscopic closure should also impose passivity or entropy-production positivity. In time-domain language this means that the response cannot extract net work from an unforced screen bath; schematically,

$$\int dN \mathcal{S}_H(N) \mathcal{R}_H(N) \geq 0 \quad (32)$$

for the physical sign convention. In frequency-domain language, an equivalent condition is positivity of the dissipative part of the retarded response function. These are not extra data-fitting assumptions; they are consistency conditions on the allowed memory kernels.

The earlier low-redshift parameterization is recovered by writing

$$\mathcal{R}_H(a) \simeq (3 - s) X_b(a), \quad (33)$$

where s is a low-redshift effective filling exponent and $X_b(a)$ is a normalized response shape. The minimal acceleration-gated candidate

$$X'_b(N) + X_b(N) = [1 - \epsilon_H(N)]_+, \quad \epsilon_H = - \frac{d \ln H}{dN}, \quad (34)$$

is the special approximation $\Gamma_H \simeq 1$ and $\mathcal{S}_H \simeq [1 - \epsilon_H]_+$. It is not the full dynamical derivation.

7 The screen influence functional

The dark-sector residual is dissipative and retarded. Therefore it is better described by an influence functional than by an ordinary conservative action. A schematic decomposition is

$$I_{G1}^{\text{eff}} = I_{\text{EH}} + I_{\text{matter}} + I_{\text{scr}}^{\text{eq}} + I_{\text{scr}}^{\text{neq}} + I_{\text{mem}}^{\text{ret}}. \quad (35)$$

Here $I_{\text{scr}}^{\text{eq}}$ belongs to the equilibrium entropy-response geometry of the G1 core, while $I_{\text{scr}}^{\text{neq}}$ is the quadratic optical entropy production in Eq. (12).

The memory part can be represented locally with an auxiliary field constraint,

$$I_{\text{mem}}^{\text{aux}} = \int dN \Lambda_H(N) [\mathcal{R}'_H(N) + \Gamma_H(N) \mathcal{R}_H(N) - \mathcal{S}_H(N)]. \quad (36)$$

Because the memory response is dissipative and retarded, it should not be read as an ordinary symmetric quadratic action. A more appropriate schematic representation is an open-system or Schwinger–Keldysh influence term,

$$I_{\text{mem}}^{\text{ret}} \sim \int dN dN' \mathcal{S}_H^-(N) G_H^{\text{ret}}(N, N') \mathcal{S}_H^+(N'), \quad (37)$$

possibly accompanied by a noise kernel,

$$I_{\text{noise}} \sim \frac{i}{2} \int dN dN' \mathcal{S}_H^-(N) N_H(N, N') \mathcal{S}_H^-(N'). \quad (38)$$

For the present purpose, the essential point is causal structure. The screen residual at time N depends on past horizon occupancy drift, not on future data.

The leading operator by which the memory field enters cosmological perturbation theory is unimodular optical stress:

$$E_{ij, \text{TF}}^{\text{scr}} = -\mathcal{A}(a, k) \kappa_{BW} \mathcal{R}_H(a) \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right) + O(\mathcal{R}_H^2), \quad (39)$$

where $\mathcal{A}(a, k)$ is a normalization that maps dimensionless screen residue to the chosen effective-stress convention. In the reduced μ, Σ notation, this normalization is absorbed into $\Sigma - 1$.

8 Ward projector and minimum completion

The constitutive law in Eq. (39) defines only the leading spatial traceless seed. It does not by itself define a covariantly closed effective stress. The physical residual should therefore be read as a retarded Ward completion of this seed:

$$E_{\mu\nu}^{\text{scr}} = \mathcal{P}_W^{\text{ret}}[Q]_{\mu\nu}, \quad (40)$$

where

$$Q_{ij} = -\kappa_{BW} \mathcal{R}_H(a) \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right) + O(\mathcal{R}_H^2) \quad (41)$$

is the unimodular optical-stress seed, and $\mathcal{P}_W^{\text{ret}}$ denotes the minimal retarded completion satisfying the Ward/Bianchi constraint.

In a local flat-patch analogy, the idea is analogous to completing a source with a retarded transverse projector. Schematically one may use

$$P_{\mu\nu}^{\perp} = \eta_{\mu\nu} - \frac{\partial_\mu \partial_\nu}{\square_{\text{ret}}}, \quad (42)$$

and a spin-two transverse projector

$$\mathcal{P}_{\mu\nu\alpha\beta}^{TT} = \frac{1}{2} \left(P_{\mu\alpha}^{\perp} P_{\nu\beta}^{\perp} + P_{\mu\beta}^{\perp} P_{\nu\alpha}^{\perp} \right) - \frac{1}{3} P_{\mu\nu}^{\perp} P_{\alpha\beta}^{\perp}. \quad (43)$$

This expression is not meant as a complete FRW gauge-invariant construction. It states the local principle: the completion is retarded, transverse, and minimal. A production implementation should replace it by a fully gauge-invariant cosmological perturbation operator.

Equivalently, the Ward completion may be characterized variationally as the minimum-norm symmetric tensor compatible with the optical-stress seed:

$$E_{\mu\nu}^{\text{scr}} = \arg \min_{E_{\mu\nu}} \|E_{\mu\nu}\|_{\mathcal{G}}^2 \quad (44)$$

subject to

$$E_{ij, \text{TF}}^{\text{scr}} = Q_{ij}, \quad \nabla^\mu E_{\mu\nu}^{\text{scr}} = J_\nu^H, \quad (45)$$

with $J_\nu^H = 0$ in the closed limit. This minimum-completion formulation makes explicit why the density and momentum components are not independent dark-fluid variables: they are the smallest Ward-required completion of the leading optical stress.

9 Ward completion of optical stress

The effective field equation is written as

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} + E_{\mu\nu}^{\text{scr}}. \quad (46)$$

If ordinary matter is conserved, Bianchi closure requires

$$\nabla^\mu E_{\mu\nu}^{\text{scr}} = 0. \quad (47)$$

If the finite screen exchanges with a horizon bath, one may instead allow controlled exchange

$$\nabla^\mu E_{\mu\nu}^{\text{scr}} = J_\nu^H, \quad (48)$$

where J_ν^H is horizon-scale and hence subhorizon suppressed.

Theorem 1 (Ward completion of traceless optical stress). *Assume that the leading screen residual in the scalar sector is a traceless spatial optical stress,*

$$E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{R}_H), \quad (49)$$

and that $E_{\mu\nu}^{\text{scr}}$ is Ward closed, up to horizon-scale exchange. In the subhorizon linear regime $k \gg \mathcal{H}$, the minimal completion satisfies

$$E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad (50)$$

$$E_{00}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right) + O(\mathcal{R}_H^2). \quad (51)$$

Power-counting proof. The divergence of the leading traceless spatial stress carries one spatial derivative, hence scales as $k E_{ij,\text{TF}}^{\text{scr}}$. The Ward identity can be balanced by a momentum component whose time-derivative/expansion contribution carries a scale $\mathcal{H}$; therefore $E_{0i}^{\text{scr}}/E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{H}/k)$. Applying the constraint once more gives the density-like component suppressed by a second power, $E_{00}^{\text{scr}}/E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{H}^2/k^2)$. Nonlinear self-sourcing contributes $O(\mathcal{R}_H^2)$. $\square$

This is the Ward-closed meaning of Ricci silence. The density channel is not exactly absent at all scales. It is suppressed relative to the optical stress by powers of $\mathcal{H}/k$ and by nonlinear order.

The minimum-completion version of this Ward hierarchy, including the precision gauge-invariant scalar-sector route, is stated as a conditional theorem in Appendix A.

Caveat 1 (Gauge and quasi-static scope). *This hierarchy is a subhorizon, quasi-static power-counting statement. A fully gauge-invariant formulation with a complete covariant embedding of the screen influence functional is still required for superhorizon scales and for a production-level perturbation implementation. Appendix A gives a technical minimum-completion formulation and a gauge-invariant scalar-sector route toward that implementation.*

10 Cosmological reduction

Use the standard phenomenological definitions

$$-k^2\Psi = 4\pi G a^2 \mu(a, k) \rho \Delta, \quad (52)$$

$$-k^2(\Phi + \Psi) = 8\pi G a^2 \Sigma(a, k) \rho \Delta. \quad (53)$$

The Newtonian channel μ is sensitive to density-like Ricci sourcing. The Weyl/lensing channel Σ is sensitive to optical traceless stress.

Combining the constitutive law with Ward completion gives

$$\boxed{\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2),} \quad (54)$$

$$\boxed{\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).} \quad (55)$$

These are stronger than the shorthand $\mu_0 = 0$. They state that matter growth is first-order protected only in the subhorizon linear regime, with controlled horizon-scale leakage.

In the low-redshift constant-amplitude reduction,

$$\mathcal{R}_H(a) \simeq (3 - s) X_b(a), \quad (56)$$

and for isotropic screen entropy Hessian,

$$\kappa_{BW} = \frac{3}{4}. \quad (57)$$

Thus one obtains

$$\mu(a, k) = 1 + O((3 - s)^2) + O\left((3 - s) \frac{\mathcal{H}^2}{k^2}\right), \quad (58)$$

$$\Sigma(a, k) = 1 - \frac{3}{4}(3 - s) X_b(a) + O((3 - s)^2). \quad (59)$$

At the amplitude level, this is the G1DE-M_{3/4} lock

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s). \quad (60)$$

11 Compensation sign and Weyl suppression

The sign of the Weyl response is fixed by finite-screen ledger closure. A positive horizon occupancy lag corresponds to

$$\mathcal{R}_H > 0. \quad (61)$$

The determinant/filling channel then carries a positive residual. The unimodular optical sector compensates it:

$$\Delta_U = -\kappa_{BW} \Delta_A. \quad (62)$$

Since the unimodular sector is Weyl-active, the leading lensing response has opposite sign:

$$\boxed{\mathcal{R}_H > 0 \quad \Rightarrow \quad \Sigma - 1 < 0.} \quad (63)$$

Thus negative Weyl amplitude is not inserted by hand. It is the sign of optical compensation for a positive filling/occupancy residual. If future data robustly require $\Sigma > 1$ for positive $\mathcal{R}_H$, the compensation orientation or the identification of $\mathcal{R}_H$ would have to be revised.

12 G1 null tests

The influence-functional closure suggests two direct diagnostics.

12.1 Weyl-lock null test

Define

$$\mathcal{N}_{G1}(a, k) = \Sigma(a, k) - 1 + \kappa_{BW} \mathcal{R}_H(a). \quad (64)$$

G1 predicts

$$\boxed{\mathcal{N}_{G1}(a, k) = O(\mathcal{R}_H^2)}. \quad (65)$$

A statistically significant nonzero $\mathcal{N}_{G1}$ at first order would falsify the single-residue background–Weyl lock.

12.2 Channel-split suppression test

Define

$$\mathcal{C}_{G1}(a, k) = \frac{\mu(a, k) - 1}{\Sigma(a, k) - 1}. \quad (66)$$

Using Eqs. (54)–(55),

$$\boxed{\mathcal{C}_{G1} = O(\mathcal{R}_H) + O\left(\frac{\mathcal{H}^2}{k^2}\right)}. \quad (67)$$

Therefore the true data-facing signature is not merely $\Sigma < 1$, but

$$\boxed{|\mu - 1| \ll |\Sigma - 1|} \quad (68)$$

for linear subhorizon modes.

13 Branch tree and demotion paths

The influence functional is intentionally structured with demotion branches.

Table 2: Branch tree and demotion registry.

Outcome	What fails or demotes	What may survive
$\kappa_{BW} \simeq 3/4$	none	full $GL(2)$ optical branch
$\kappa_{BW} \simeq 2/3$	twist/holonomy sector	symmetric $SPD(2)$ scalar-shear branch
Intermediate κ	isotropic Hessian fixed point	anisotropic compliance branch
Strong first-order μ correction	Ricci-silent Ward hierarchy	density-leakage finite-screen model

Outcome	What fails or demotes	What may survive
Constant- Σ favored	resolved horizon-memory shape	amplitude lock may survive; output-shape or source-kernel lift demoted
No shared $\mathcal{R}_H(a)$ for background and Weyl	single-residue ledger lock	separate background and optical residuals
No acceleration/horizon gating	minimal source model	alternative occupancy current
$\Sigma > 1$ for positive $\mathcal{R}_H$	compensation sign convention	reversed ledger orientation
No finite-screen entropy Hessian	influence-functional closure	algebraic toy model only

This tree is important. G1 does not rest on a single brittle number. It gives a hierarchy of more general branches and specifies what would be demoted by data.

14 Relation to G1fit-real

The observational counterpart of this influence-functional closure is the G1fit-real program. The low-redshift reduction G1DE- $M_{3/4}$ is the data-facing model obtained by taking

$$\kappa_{BW} = \frac{3}{4}, \quad \mathcal{R}_H(a) \simeq (3-s)X_b(a), \quad (69)$$

and by retaining the subhorizon leading-order relation

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s). \quad (70)$$

The present paper supplies the theory-side influence closure behind that model. It does not use likelihood results as an input. Current exact-pilot comparisons select the projection-locked relation at the AIC/BIC level. As of the G1fit-real v1.3 medium-prior pass, the six-model nested-evidence hierarchy is $M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE-2} > \text{CPL} > \Lambda\text{CDM}$, with margins $\Delta \log Z \simeq 0.97, 1.92, 6.70, 9.43,$ and 11.96 against the five controls. Top-control wide-prior sensitivity passed ($M_{3/4} > M_\kappa > \text{const-}\Sigma$ with $\Delta \log Z \simeq 0.85, 1.96$). This paragraph is version-locked to the v1.3 pass; consult the latest G1fit-real and archive front matter for newer numbers.

The distinction between amplitude and shape is crucial. If a constant- Σ control were preferred in future runs, the κ_{BW} amplitude branch could survive while the specific horizon-response output shape, or its chosen source-kernel lift, would be demoted.

15 Extensions

The influence-functional viewpoint suggests several extensions, but they are not developed here.

15.1 Horizon shear/twist memory near black holes

Near black-hole horizons, finite-screen corrections should first appear as shear/twist memory rather than density hair:

$$\delta S_{\text{hor}} = \frac{\delta A}{4\ell_P^2} + \frac{1}{2} \int (K_S \sigma^2 + K_\Omega \omega^2) d\lambda dA. \quad (71)$$

This may connect to horizon memory, tidal response, ringdown, photon-ring optics, and scrambling.

15.2 Standard sirens and optical distance duality

The conservative expectation is that local tensor speed remains protected,

$$c_T = 1, \quad (72)$$

but finite-screen optical stress may affect amplitude transport. A possible branch is

$$\frac{d_L^{\text{GW}}}{d_L^{\text{EM}}} = 1 + \alpha_{\text{GW}} \mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (73)$$

where $\alpha_{\text{GW}} = 0$ if EM and GW amplitudes couple identically to the screen influence. This is a future standard-siren test, not a result of this paper.

15.3 Screen code deformation

In a quantum-information interpretation, the area sector measures code capacity, while the uni-modular sector measures recoverability geometry or code deformation. The G1 dark sector may then be interpreted as a finite-screen code-deformation residual projected into spacetime optics.

16 Open mathematical problems

1. Derive K_A, K_S, K_Ω from a microscopic finite-screen entropy functional rather than postulating a quadratic Hessian.
2. Derive $\mathcal{O}_H(a)$ and $C_{\text{used}}(a)$ from screen-accessible bulk distinction accounting.
3. Embed the retarded influence functional in a fully covariant open-system formalism, including the corresponding noise kernel and passivity conditions.
4. Compute the exact Ward projector beyond subhorizon power counting, including gauge-invariant scalar perturbation variables.
5. Determine whether the twist/holonomy sector contributes only covariance or also measurable second-order curl/B-mode signatures.
6. Fix the constitutive mapping normalization $\mathcal{A}(a, k)$ between the optical-stress amplitude and the phenomenological $\Sigma(a, k)$ convention, including its relation to the standard $8\pi G a^2 \rho \Delta / k^2$ normalization.
7. Test the null combinations $\mathcal{N}_{G1}$ and $\mathcal{C}_{G1}$ in production likelihood pipelines.

17 Conclusion

This paper formulated a screen influence functional for the G1 dark sector. The central object is a retarded horizon residue $\mathcal{R}_H$ sourced by finite-screen occupancy drift. The leading operator generated by this residue is a unimodular optical stress. The response coefficient is the compliance projection

$$\kappa_{BW} = \frac{\text{Tr}(P_U K^{-1})}{\text{Tr}(K^{-1})}, \quad (74)$$

which gives 3/4 at the isotropic entropy-Hessian fixed point and 2/3 when the twist/holonomy sector is demoted.

Ward completion turns the leading optical stress into a covariant hierarchy:

$$E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{R}_H), \quad E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (75)$$

Therefore

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (76)$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (77)$$

The low-redshift reduction $\mathcal{R}_H(a) \simeq (3-s)X_b(a)$ recovers the G1DE-M_{3/4} lock

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s). \quad (78)$$

Equivalently, G1DE-M_{3/4} is the constant-amplitude, low-redshift, isotropic-compliance reduction of this screen influence functional.

The conceptual result is that the G1 dark sector is not an ordinary density component and not an arbitrary modified-gravity function. It is a Ward-projected, Ward-closed retarded influence of finite-screen entropy incompleteness on the optical Weyl channel.

A Technical Appendix: Relative-Entropy Screen Hessian and Minimum Ward Completion

This appendix records a sharper mathematical formulation of the two least-closed objects in the main text. First, the entropy Hessian K is defined as the relative-entropy Hessian of a finite-screen state manifold. Second, the effective stress $E_{\mu\nu}^{\text{scr}}$ is defined as the minimum-norm Ward completion of a leading unimodular optical-stress seed. The appendix is a conditional closure: it does not yet construct the microscopic screen Hilbert space, but it specifies what such a construction must reproduce.

A.1 Finite-screen state manifold

Let q^I denote local optical deformation coordinates on the tangent space of the finite optical screen,

$$q^I = (A, S_1, S_2, \Omega), \quad I = 1, \dots, 4. \quad (79)$$

Here A is the area/trace channel, S_1, S_2 are the two symmetric shear channels, and Ω is the twist/holonomy channel. Let

$$\rho_S(q) \quad (80)$$

be the finite-screen state associated with a small optical deformation q , and let $\rho_S(0)$ be the locally isotropic reference screen state.

Definition 3 (Screen relative-entropy distinguishability). *The distinguishability cost of the deformation q is*

$$D_{\text{scr}}(q) = S_{\text{rel}}(\rho_S(q) \parallel \rho_S(0)), \quad (81)$$

where $S_{\text{rel}}(\rho \parallel \sigma)$ denotes relative entropy.

For a smooth state family, the expansion near $q = 0$ is

$$D_{\text{scr}}(q) = \frac{1}{2} K_{IJ} q^I q^J + O(q^3). \quad (82)$$

Thus the screen entropy Hessian is

$$K_{IJ} = \left. \frac{\partial^2}{\partial q^I \partial q^J} S_{\text{rel}}(\rho_S(q) \parallel \rho_S(0)) \right|_{q=0}. \quad (83)$$

This identifies K with the information-geometric metric of the local screen state manifold. In this language, the coefficients K_A, K_S, K_Ω in the main text are not arbitrary response weights. They are eigenvalues of the relative-entropy Hessian on irreducible optical sectors.

A.2 $SO(2)$ block structure and compliance fraction

The local screen rotation group decomposes the optical deformation algebra as

$$\mathfrak{gl}(2) = A(1) \oplus S(2) \oplus \Omega(1), \quad (84)$$

with representation labels

$$A : \mathbf{0}^+, \quad S : \mathbf{2}, \quad \Omega : \mathbf{0}^-. \quad (85)$$

If $\rho_S(0)$ is isotropic and parity respecting, different $SO(2)$ sectors do not mix at quadratic order. Therefore

$$K = K_A P_A + K_S P_S + K_\Omega P_\Omega, \quad (86)$$

where P_A, P_S, P_Ω are the corresponding orthogonal projectors. The compliance matrix is

$$C = K^{-1} = K_A^{-1} P_A + K_S^{-1} P_S + K_\Omega^{-1} P_\Omega. \quad (87)$$

The Weyl-active unimodular projector is

$$P_U = P_S + P_\Omega. \quad (88)$$

Thus

$$\kappa_{BW} = \frac{\text{Tr}(P_U K^{-1})}{\text{Tr}(K^{-1})} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}. \quad (89)$$

Because K is a positive relative-entropy Hessian on the stable branch, $C = K^{-1}$ is positive semidefinite on the responding subspace. Hence

$$0 \leq \kappa_{BW} \leq 1. \quad (90)$$

At the isotropic fixed point $K_A = K_S = K_\Omega$, one obtains $\kappa_{BW} = 3/4$. If the twist/holonomy channel is projected out, $K_\Omega^{-1} = 0$, and $K_A = K_S$, one obtains $\kappa_{BW} = 2/3$.

A.3 Horizon occupancy as bulk distinguishability pressure

The main text defines

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}, \quad C_H(a) = \frac{A_H(a)}{4\ell_P^2}. \quad (91)$$

A microscopic completion must define C_{used} . The information-geometric proposal is that C_{used} is horizon-accessible bulk distinguishability demand, not ordinary thermodynamic entropy:

$$\boxed{C_{\text{used}}(a) = S_{\text{rel}}(\rho_{\text{bulk}}^H(a) \parallel \rho_{\text{coarse}}^H(a))}. \quad (92)$$

Here $\rho_{\text{bulk}}^H(a)$ is the horizon-accessible bulk state and $\rho_{\text{coarse}}^H(a)$ is the corresponding coarse FRW or screen-compressed reference state. The source for the horizon residue is then

$$\mathcal{S}_H(N) = -\frac{d}{dN} \ln \left[\frac{S_{\text{rel}}(\rho_{\text{bulk}}^H \parallel \rho_{\text{coarse}}^H)}{A_H/(4\ell_P^2)} \right], \quad N = \ln a. \quad (93)$$

This definition makes the G1 source a finite-screen distinguishability pressure: the bulk state demands a certain distinguishability budget, while the horizon supplies only finite screen capacity.

A.4 Passive retarded memory

The retarded memory field is

$$\mathcal{R}_H(N) = \int_{-\infty}^N dN' G_H^{\text{ret}}(N, N') \mathcal{S}_H(N'). \quad (94)$$

The admissible kernels are restricted by three conditions.

1. **Causality:**

$$G_H^{\text{ret}}(N, N') = 0 \quad \text{for } N' > N. \quad (95)$$

2. **Stability:** in a Markov reduction,

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = \mathcal{S}_H, \quad \Gamma_H > 0, \quad (96)$$

or, more generally, the poles of the response lie in the stable half-plane.

3. **Passivity / entropy positivity:** for the physical sign convention,

$$\int dN \mathcal{S}_H(N) \mathcal{R}_H(N) \geq 0. \quad (97)$$

Equivalently, in a frequency-domain description the dissipative part of the response has the sign required for positive entropy production.

A full open-system treatment should also include a stochastic screen-memory fluctuation,

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = \mathcal{S}_H + \xi_H, \quad \langle \xi_H(N) \xi_H(N') \rangle = N_H(N, N'), \quad (98)$$

with $N_H \succeq 0$. The deterministic reduction used in G1DE-M_{3/4} concerns the mean response $\langle \mathcal{R}_H \rangle$.

A.5 Minimum Ward completion

Let the leading optical seed be

$$Q_{ij}^{\text{TF}} = -\mathcal{A}(a, k) \kappa_{BW} \mathcal{R}_H(a) \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right) + O(\mathcal{R}_H^2). \quad (99)$$

This seed is not yet a covariantly closed effective stress. Define the physical screen residual as the minimum-norm Ward completion:

$$\boxed{E_{\mu\nu}^{\text{scr}} = \arg \min_{E_{\mu\nu}} \|E\|_{\mathcal{G}}^2} \quad (100)$$

subject to

$$E_{ij}^{\text{TF}} = Q_{ij}^{\text{TF}}, \quad \nabla^\mu E_{\mu\nu} = J_\nu^H, \quad E_{\mu\nu} = E_{\nu\mu}. \quad (101)$$

The norm $\|\cdot\|_{\mathcal{G}}$ may be chosen as a DeWitt-like quadratic norm or a screen-response norm. Different reasonable norms can change higher-order details, but the subhorizon scaling follows from the Ward constraint and the fact that the seed is spatial and traceless.

Theorem 2 (Minimum Ward completion hierarchy). *Assume $Q_{ij}^{\text{TF}} = O(\mathcal{R}_H)$ and that J_ν^H is horizon-scale. In the subhorizon linear regime $k \gg \mathcal{H}$, the minimum Ward completion obeys*

$$E_{ij, \text{TF}}^{\text{scr}} = O(\mathcal{R}_H), \quad (102)$$

$$E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad (103)$$

$$E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (104)$$

Sketch. The spatial divergence of Q_{ij}^{TF} contributes a term of order $k\mathcal{R}_H$ to the Ward equation. A momentum component E_{0i} with one time/expansion scale $\mathcal{H}$ balances this term at order $\mathcal{R}_H \mathcal{H}/k$. Applying the constraint once more gives a density-like component suppressed by another factor $\mathcal{H}/k$. Nonlinear self-coupling of the screen residual contributes $O(\mathcal{R}_H^2)$. $\square$

Consequently the Weyl/lensing response is first order in $\mathcal{R}_H$, while the Ricci-density response is suppressed:

$$\Sigma - 1 = -\kappa_{BW} \mathcal{R}_H + O(\mathcal{R}_H^2), \quad (105)$$

$$\mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (106)$$

A.6 Gauge-invariant scalar-sector route

A production cosmology implementation should not rely on a local Minkowski projector. Instead, introduce gauge-invariant effective screen variables

$$\pi_E^{\text{GI}}, \quad q_E^{\text{GI}}, \quad \delta E_{\text{GI}}, \quad (107)$$

corresponding to anisotropic stress, momentum, and density-like screen components. The G1 constitutive input is

$$\pi_E^{\text{GI}} = O(\mathcal{R}_H). \quad (108)$$

The gauge-invariant Ward completion should then determine

$$q_E^{\text{GI}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad (109)$$

$$\delta E_{\text{GI}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (110)$$

This is the production-ready version of the minimum Ward completion theorem: build the completion directly in the scalar perturbation variables used by a Boltzmann or likelihood pipeline.

A.7 Twist/holonomy as covariance sector

The twist/holonomy coordinate Ω is part of the full $GL(2)$ optical capacity branch, but this does not require a nonzero first-order scalar rotation mean. The correct interpretation is

$$\langle \Omega \rangle = 0, \quad \langle \Omega^2 \rangle = O(\mathcal{R}_H^2) \quad (111)$$

for scalar cosmology. Thus Ω contributes to the screen covariance and the compliance fraction, while direct lensing curl or rotation observables are expected to be second-order or covariance-level tests. The branch distinction is

$$GL(2) : \quad A(1) + S(2) + \Omega(1) \Rightarrow \kappa_{BW} = 3/4, \quad (112)$$

$$SPD(2) : \quad A(1) + S(2) \Rightarrow \kappa_{BW} = 2/3. \quad (113)$$

Absence of a first-order curl mean therefore does not falsify the 3/4 branch; a measured demotion of κ_{BW} toward 2/3 would.

A.8 Appendix theorem

Combining the definitions above yields the compact closure statement:

Theorem 3 (Relative-entropy screen Hessian and minimum Ward completion). *Let finite screen states form a smooth family $\rho_S(q)$ over local optical deformations $q \in T_I GL(2)$, and let K be the relative-entropy Hessian in Eq. (83). Let the horizon occupancy source be Eq. (93), and let $\mathcal{R}_H$ be its causal, stable, passive retarded response. Let $E_{\mu\nu}^{\text{scr}}$ be the minimum Ward completion of the unimodular optical seed. Then*

$$\kappa_{BW} = \frac{\text{Tr}(P_U K^{-1})}{\text{Tr}(K^{-1})}, \quad (114)$$

$$\Sigma - 1 = -\kappa_{BW} \mathcal{R}_H + O(\mathcal{R}_H^2), \quad (115)$$

$$\mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (116)$$

If $K \propto I$ on $T_I GL(2)$, then $\kappa_{BW} = 3/4$. If the twist/holonomy sector is demoted, then $\kappa_{BW} = 2/3$.

This theorem is conditional. It specifies the mathematical target that a microscopic FDS-to-G1 derivation must realize: a finite screen state manifold with relative-entropy Hessian, a horizon distinguishability occupancy current, a passive retarded memory kernel, and a minimum Ward completion of the unimodular optical seed.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* 7, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* 43, 199 (1975).
- [3] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* 98, 1123 (1955).
- [4] R. Sachs, “Gravitational waves in general relativity. VI. The outgoing radiation condition,” *Proc. R. Soc. Lond. A* 264, 309 (1961).
- [5] R. M. Wald, *General Relativity*, University of Chicago Press (1984).
- [6] P. Schneider, J. Ehlers, and E. E. Falco, *Gravitational Lenses*, Springer (1992).
- [7] M. Bartelmann and P. Schneider, “Weak gravitational lensing,” *Phys. Rep.* 340, 291 (2001).
- [8] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* 75, 1260 (1995).
- [9] R. P. Feynman and F. L. Vernon, “The theory of a general quantum system interacting with a linear dissipative system,” *Ann. Phys.* 24, 118 (1963).
- [10] J. Schwinger, “Brownian motion of a quantum oscillator,” *J. Math. Phys.* 2, 407 (1961).
- [11] L. V. Keldysh, “Diagram technique for nonequilibrium processes,” *Sov. Phys. JETP* 20, 1018 (1965).
- [12] S. Kullback and R. A. Leibler, “On information and sufficiency,” *Ann. Math. Statist.* 22, 79 (1951).
- [13] R. A. Fisher, “Theory of statistical estimation,” *Proc. Cambridge Philos. Soc.* 22, 700 (1925).
- [14] D. Petz, “Monotone metrics on matrix spaces,” *Linear Algebra Appl.* 244, 81 (1996).
- [15] L. Onsager, “Reciprocal relations in irreversible processes. I,” *Phys. Rev.* 37, 405 (1931).
- [16] J. M. Bardeen, “Gauge-invariant cosmological perturbations,” *Phys. Rev. D* 22, 1882 (1980).
- [17] C.-P. Ma and E. Bertschinger, “Cosmological perturbation theory in the synchronous and conformal Newtonian gauges,” *Astrophys. J.* 455, 7 (1995).
- [18] P. Zhang, M. Liguori, R. Bean, and S. Dodelson, “Probing gravity at cosmological scales by measurements which test the relationship between gravitational lensing and matter overdensity,” *Phys. Rev. Lett.* 99, 141302 (2007).
- [19] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* 07, 050 (2014).
- [20] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), doi:10.5281/zenodo.20382013.
- [21] Y. Wu, *The 3/4 Projection Law in Finite Screen Gravity: MaxEnt Optical-Stress Closure, Ricci-Silent Residuals, and the Weyl-Locked Dark Sector*, technical note (2026).

- [22] Y. Wu, *Non-Equilibrium Screen Entropy and the Optical-Stress Origin of the G1 Dark Sector*, technical note (2026).
- [23] Y. Wu, *Ward-Closed Finite-Screen Dark Sector: Retarded Optical Stress, Horizon Memory, and G1 Null Tests*, technical note (2026).
- [24] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual*, Zenodo / manuscript (2026).

Weyl-Normalized Ward Completion and Identifiability Rules for G1DE

Finite-Screen Memory as a Transfer Operator, Not an Independent Dark Stress Source

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

Draft: May 2026

Abstract

This implementation note fixes the production-level normalization and identifiability rules for the G1DE screen-influence reduction. The central point is that G1DE does not add an independently normalized dark stress amplitude. Instead, it applies a finite-screen fractional optical projection correction to the general-relativistic Weyl source. Defining

$$\mathcal{W}(a, k) \equiv -k^2(\Phi + \Psi), \quad \mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta,$$

the Weyl-normalized implementation is

$$\mathcal{W}_{\text{G1}}(a, k) = [1 - \kappa_{BW} \mathcal{R}_H(a)] \mathcal{W}_{\text{GR}}(a, k) + O(\mathcal{R}_H^2).$$

Thus

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2),$$

while the Newtonian channel receives only Ward-suppressed leakage,

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

The note makes explicit that the optical normalization factor $A(a, k)$ is not a fitted response function; it is fixed by the Weyl-source convention. It also introduces a memory-shape normalization $\mathcal{R}_H(a) = R_0 \hat{R}_H(a)$, $\hat{R}_H(1) = 1$, so that the projection coefficient κ_{BW} , the present-day residual amplitude R_0 , and the redshift-memory shape $\hat{R}_H(a)$ are identifiable. The locked model G1DE-M_{3/4} is recovered by $\kappa_{BW} = 3/4$, $R_0 = 3 - s$, $\hat{R}_H = X_b$, and $\mu = 1$.

1 Purpose and scope

This note is a production-interface companion to the G1 finite-screen dark-sector sequence. Its task is not to rederive the optical projection coefficient κ_{BW} , the entropy-Hessian origin of the screen influence, or the retarded horizon-memory kernel. Upstream of this production convention, D0 identifies $\mathcal{R}_H$ as a passive horizon-scale remnant of incomplete modular anomaly inflow, while D1 interprets the same quantity as a finite-access gluing-lag residual. Those ingredients belong to the companion theory notes on the 3/4 projection law, non-equilibrium screen entropy, and the screen influence functional. The purpose here is narrower and more implementation-facing:

1. fix the normalization map between the optical-stress seed and the standard phenomenological Weyl-source convention;
2. state which quantities may be fitted in G1DE production likelihoods and which quantities must not be fitted;

3. separate projection coefficient, residual amplitude, and redshift-memory shape;
4. specify Ward-suppressed leakage into the Newtonian growth channel;
5. define null tests and demotion paths for production analyses.

D5 owns: (1) Ward/Bianchi closure; (2) Weyl normalization; (3) no-free- $A(a, k)$ enforcement; (4) identifiability rules; (5) production likelihood conventions.

The guiding statement is:

Implementation thesis. G1DE is not an independently normalized dark-stress model. It is a Weyl-normalized, Ward-disciplined finite-screen transfer class. Its primary action is a fractional finite-screen correction to the GR Weyl source.

This distinction matters because an implementation that samples an arbitrary optical amplitude $A(a, k)$ no longer tests the G1DE projection-locking claim. It tests a free dark-stress envelope instead.

2 Standard modified-growth convention

We use the scalar perturbation convention

$$-k^2\Psi = 4\pi G a^2 \mu(a, k) \rho \Delta, \quad (1)$$

$$-k^2(\Phi + \Psi) = 8\pi G a^2 \Sigma(a, k) \rho \Delta. \quad (2)$$

Here Δ is the comoving matter density contrast used in the likelihood implementation. We define the Weyl source

$$\mathcal{W}(a, k) \equiv -k^2(\Phi + \Psi), \quad (3)$$

and the GR Weyl source

$$\mathcal{W}_{\text{GR}}(a, k) \equiv 8\pi G a^2 \rho \Delta. \quad (4)$$

In GR,

$$\mathcal{W} = \mathcal{W}_{\text{GR}}, \quad \Sigma = 1. \quad (5)$$

A production implementation of G1DE should modify the transfer from $\rho \Delta$ to Weyl curvature by a finite-screen factor. It should not introduce a new freely normalized dark-stress source.

3 Three production principles

3.1 Weyl Normalization Principle

Principle 1 (Weyl Normalization). *The finite-screen residual is normalized as a fractional correction to the GR Weyl source. To first order in the screen residual,*

$$\mathcal{W}_{\text{G1}}(a, k) = [1 - \kappa \mathcal{R}_H(a)] \mathcal{W}_{\text{GR}}(a, k) + O(\mathcal{R}_H^2). \quad (6)$$

Here κ is the optical projection/compliance coefficient. In the isotropic finite-screen branch $\kappa = \kappa_{BW} = 3/4$.

Equation (6) immediately gives

$$\Sigma(a, k) - 1 = -\kappa \mathcal{R}_H(a) + O(\mathcal{R}_H^2). \quad (7)$$

Thus the symbol $A(a, k)$, when used to represent the amplitude of an optical-stress seed, is fixed by the Weyl-source convention:

$$A(a, k) \longleftrightarrow \mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta. \quad (8)$$

It is not an additional physical response function.

3.2 No Free Optical-Amplitude Principle

Principle 2 (No Free Optical-Amplitude). *The normalization $A(a, k)$ must not be sampled as an independent nuisance parameter or response function in a G1DE production likelihood. It is fixed by the Weyl-source convention (4). Sampling $A(a, k)$ turns the model into a free dark-stress source and removes the Weyl-normalized projection-locking claim being tested.*

Equivalently, if Θ denotes the fitted parameter set of a production model, then

$$\boxed{A(a, k) \notin \Theta_{\text{G1DE}}.} \quad (9)$$

A model that allows an arbitrary function multiplying the optical seed should be labeled as an adversarial free-response or dark-stress envelope, not as a primary G1DE branch.

The Weyl-normalized production convention $W_{G1} = (1 - \kappa R_H)W_{GR}$ is admissible only if $A(a, k)$ is not sampled independently. If data require free $A(a, k)$, D5 must force demotion to generic dark-stress.

3.3 Memory-Shape Normalization Principle

A second identifiability issue concerns the product

$$\Sigma - 1 = -\kappa \mathcal{R}_H. \quad (10)$$

If the overall amplitude of $\mathcal{R}_H(a)$ is not fixed, then κ and $\mathcal{R}_H$ are rescaling-degenerate. Therefore the memory response should be factorized as

$$\boxed{\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1.} \quad (11)$$

Here $R_0 = \mathcal{R}_H(1)$ is the present-day finite-screen residual amplitude, while $\hat{R}_H(a)$ is a normalized redshift-memory shape.

For the current G1DE-M reduction,

$$R_0 = 3 - s, \quad \hat{R}_H(a) = X_b(a), \quad X_b(1) = 1. \quad (12)$$

Therefore

$$\boxed{\Sigma(a, k) - 1 = -\kappa(3 - s)X_b(a) + O((3 - s)^2).} \quad (13)$$

For the locked branch,

$$\boxed{\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)X_b(a) + O((3 - s)^2).} \quad (14)$$

Remark 1 (Identifiability of κ). *The free- κ branch is interpretable only after a memory-shape normalization has been imposed. With $X_b(1) = 1$, κ is a projection/compliance coefficient. Without this normalization, κ can be absorbed into the arbitrary amplitude of $\mathcal{R}_H$.*

Remark 2 (Output response versus source-kernel lift). *The normalized memory shape $\hat{R}_H(a)$ is a data-facing output response. A deeper realization may factor it as $\mathcal{R}_H = K_H * J_H$, but this factorization is not unique without a lift choice. D6 formulates this ambiguity as a memory quotient and proposes a maximum-caliber horizontal lift. Therefore the production likelihood should sample R_0 and $\hat{R}_H(a)$, not a separately normalized K_H , J_H , or optical amplitude $A(a, k)$.*

4 Factorized G1DE production theorem

The previous principles can be summarized as a production-level factorization theorem.

Theorem 1 (Weyl-normalized G1DE factorization). *Assume:*

1. the finite-screen residual is normalized as a fractional correction to the GR Weyl source;
2. the optical projection coefficient is κ ;
3. the screen residual factorizes as $\mathcal{R}_H(a) = R_0 \hat{R}_H(a)$, $\hat{R}_H(1) = 1$;
4. Ricci leakage is controlled by the minimum Ward completion of the optical seed.

Then, on subhorizon linear scales,

$$\boxed{\Sigma(a, k) - 1 = -\kappa R_0 \hat{R}_H(a) + O(R_0^2),} \quad (15)$$

while

$$\boxed{\mu(a, k) - 1 = O(R_0^2) + O\left(R_0 \hat{R}_H(a) \frac{\mathcal{H}^2}{k^2}\right).} \quad (16)$$

For the locked isotropic-compliance branch, $\kappa = 3/4$. For the current low-redshift G1DE-M reduction, $R_0 = 3 - s$ and $\hat{R}_H = X_b$.

This theorem separates the three quantities that are otherwise easy to confuse:

$$\boxed{\Sigma - 1 = - \underbrace{\kappa}_{\text{projection}} \underbrace{R_0}_{\text{screen amplitude}} \underbrace{\hat{R}_H(a)}_{\text{memory shape}} + O(R_0^2).} \quad (17)$$

The Weyl-source normalization is not a fourth physical factor; it is the convention that identifies the unit optical seed with a unit fractional correction to $\mathcal{W}_{\text{GR}}$.

5 Interpretation of s , amplitude, and sign

Under the current low-redshift reduction,

$$R_0 = 3 - s. \quad (18)$$

Therefore s is not merely a symbolic filling dimension in the production likelihood. It is the present-day normalized residual amplitude written in the G1DE-M convention:

$$\boxed{s = 3 - R_0.} \quad (19)$$

The sign chain is then

$$\boxed{s < 3 \Rightarrow R_0 > 0 \Rightarrow \mathcal{R}_H > 0 \Rightarrow \Sigma - 1 < 0 \quad (\kappa > 0).} \quad (20)$$

For $\kappa = 3/4$,

$$\boxed{\Sigma_0 = -\frac{3}{4}(3 - s).} \quad (21)$$

This present-day amplitude relation is independent of detailed redshift-shape discrimination. Hence amplitude support and memory-shape support should be interpreted separately in model comparison.

6 Why this is not an independent dark-stress model

A conventional dark-fluid or stress-source model begins with additional components in an effective stress tensor and asks how they change the potentials. G1DE, in the production convention advocated here, begins with a finite-screen correction to the Weyl transfer:

$$\mathcal{W}_{\text{GR}} \longrightarrow (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}. \quad (22)$$

The effective screen stress tensor, when used, is a representation of this Weyl-transfer correction under minimum Ward completion. It is not a freely normalized physical fluid.

Schematically,

$$\delta \mathcal{W}_{\text{scr}} = -\kappa \mathcal{R}_H \mathcal{W}_{\text{GR}}, \quad (23)$$

and the covariant representation is

$$E_{\mu\nu}^{\text{scr}} = \mathcal{P}_W^{\text{min}} [-\kappa \mathcal{R}_H \mathcal{W}_{\text{GR}}]_{\mu\nu}, \quad (24)$$

where $\mathcal{P}_W^{\text{min}}$ denotes the minimum Ward completion. The density and momentum components of this representation are therefore not independent first-order sources. They are Ward-suppressed completion terms.

7 Gauge-invariant Ward-completed interface

7.1 Weyl-normalized anisotropic seed

For a production-level scalar-sector implementation, introduce effective gauge-invariant screen variables

$$\pi_{\text{scr}}^{\text{GI}}, \quad q_{\text{scr}}^{\text{GI}}, \quad \delta \rho_{\text{scr}}^{\text{GI}}. \quad (25)$$

The leading G1 screen seed is anisotropic/Weyl-active. In the Weyl-source convention one may write schematically

$$\boxed{k^2 \pi_{\text{scr}}^{\text{GI}} = -\kappa \mathcal{R}_H(a) (8\pi G a^2 \rho \Delta)}. \quad (26)$$

This equation fixes the Weyl-source convention. Code-level implementations must translate it into the sign and anisotropic-stress normalization used by the chosen Boltzmann solver. Different codes may differ by signs or numerical factors in their definitions of π , σ , or $\Phi - \Psi$.

7.2 Ward-completed leakage hierarchy

The Ward-completed momentum and density components obey the subhorizon hierarchy

$$q_{\text{scr}}^{\text{GI}} = O\left(\pi_{\text{scr}}^{\text{GI}} \frac{\mathcal{H}}{k}\right), \quad (27)$$

$$\delta \rho_{\text{scr}}^{\text{GI}} = O\left(\pi_{\text{scr}}^{\text{GI}} \frac{\mathcal{H}^2}{k^2}\right) + O(\mathcal{R}_H^2). \quad (28)$$

Consequently,

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right), \quad (29)$$

while

$$\Sigma(a, k) - 1 = -\kappa \mathcal{R}_H(a) + O(\mathcal{R}_H^2). \quad (30)$$

Thus the core data-facing signature is not simply $\Sigma < 1$, but the channel split

$$\boxed{|\mu - 1| \ll |\Sigma - 1| \quad \text{on subhorizon linear scales.}} \quad (31)$$

8 Allowed and forbidden production branches

The following registry should be used to keep production likelihoods interpretable.

Branch	Status	Response	Interpretation
Locked $M_{3/4}$	Primary branch	$\Sigma = 1 - \frac{3}{4}R_0\hat{R}_H, \mu = 1$	Isotropic compliance, normalized memory shape, no leading Ricci leakage.
Free- κ	Compliance control	$\Sigma = 1 - \kappa R_0\hat{R}_H, \mu = 1$	Tests whether the 3/4 isotropic-compliance value is selected. Identifiable only if $\hat{R}_H(1) = 1$.
Constant- Σ	Memory-shape control	$\Sigma = 1 + \Sigma_c, \mu = 1$	Tests whether the redshift-memory shape is resolved beyond a constant Weyl offset.
Ward leakage	Ricci-leakage control	$\Sigma = 1 - \kappa R_0\hat{R}_H, \mu = 1 + \nu_2 R_0^2 \hat{R}_H^2 + \nu_H R_0 \hat{R}_H \mathcal{H}^2 / k^2$	Tests whether the Newtonian channel requires Ward-disciplined leakage.
Free shape	Shape reconstruction	$\hat{R}_H(a) = \sum_i c_i B_i(a), \hat{R}_H(1) = 1$	Tests memory shape without losing amplitude identifiability.
G1DE-2	Adversarial envelope	Flexible response	Tests whether data require freedom beyond the Weyl-normalized branch.
Free $A(a, k)$	Forbidden / not G1DE	Arbitrary amplitude map	Turns the model into a free dark-stress source and destroys Weyl normalization.

Implementation warning. The normalization $A(a, k)$ must not be sampled as an independent nuisance parameter or response function. It is fixed by the Weyl-source convention $\mathcal{W}_{\text{GR}} = 8\pi G a^2 \rho \Delta$. Sampling $A(a, k)$ converts the G1DE reduction into a free dark-stress model.

9 Identifiability conditions

A G1DE production fit is identifiable only if the following conditions are satisfied:

1. **Weyl normalization is fixed.** The optical seed is normalized relative to $\mathcal{W}_{\text{GR}} = 8\pi G a^2 \rho \Delta$.
2. **Memory shape is normalized.** Write $\mathcal{R}_H(a) = R_0 \hat{R}_H(a)$, with $\hat{R}_H(1) = 1$ or an equivalent fixed normalization.
3. **κ is varied only relative to that fixed shape.** Otherwise κ and the amplitude of $\mathcal{R}_H$ are degenerate.
4. **$A(a, k)$ is never sampled.** It is a convention map, not a physical response.

5. **Free-shape reconstructions must impose a normalization.** If $\mathcal{R}_H(a)$ is reconstructed in bins or basis functions, either κ is fixed or a shape normalization is imposed.

These conditions separate amplitude evidence from shape evidence. The present-day amplitude is

$$\Sigma_0 - 1 = -\kappa R_0. \quad (32)$$

The locked branch predicts

$$\Sigma_0 - 1 = -\frac{3}{4}(3 - s). \quad (33)$$

This relation can be tested even when redshift-shape discrimination remains weak.

10 Production null tests

10.1 Weyl-transfer residual

Define

$$\boxed{\mathcal{N}_\Sigma(a, k) = \Sigma(a, k) - 1 + \kappa \mathcal{R}_H(a).} \quad (34)$$

For the primary G1DE branch,

$$\mathcal{N}_\Sigma = O(\mathcal{R}_H^2). \quad (35)$$

In the locked branch one sets $\kappa = 3/4$.

10.2 Ward-leakage residual

Define

$$\boxed{\mathcal{N}_\mu(a, k) = \mu(a, k) - 1.} \quad (36)$$

G1DE predicts

$$\mathcal{N}_\mu = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (37)$$

The ratio diagnostic

$$\mathcal{C}_{G1}(a, k) = \frac{\mu(a, k) - 1}{\Sigma(a, k) - 1} \quad (38)$$

may be useful for interpretation, but additive residuals $\mathcal{N}_\Sigma$ and $\mathcal{N}_\mu$ are numerically safer in likelihoods because the denominator of $\mathcal{C}_{G1}$ can become small.

11 G1DE-2 as an adversarial envelope

G1DE-2 should be interpreted as an adversarial envelope around the Weyl-normalized transfer branch. It asks whether the data require response freedom beyond the constrained structure

$$\Sigma - 1 = -\kappa R_0 \hat{R}_H, \quad \mu - 1 = O(R_0^2, R_0 \mathcal{H}^2 / k^2). \quad (39)$$

If $M_{3/4}$ beats G1DE-2 in evidence, then projection locking and Ward discipline are evidence-selected over flexible response freedom. If they tie, the free-response envelope is compatible but unnecessary. If G1DE-2 wins strongly, the broader finite-screen hypothesis may survive, but the Weyl-normalized locked reduction is incomplete.

G1DE-2 should not be allowed to hide a free $A(a, k)$ unless it is explicitly labeled as a non-G1DE dark-stress envelope. Otherwise, a comparison with $M_{3/4}$ would not test projection locking fairly.

12 Demotion and falsification hierarchy

Because the production response is factorized, different failures have different meanings.

Outcome	Interpretation
Constant- Σ favored over $X_b(a)$	Memory-shape closure is demoted or unresolved; the Weyl-amplitude branch may survive.
Free- κ favored with $0 < \kappa < 1$	Isotropic 3/4 compliance branch is demoted to a weighted-compliance branch.
Free- κ favored with $\kappa \notin [0, 1]$	Positive-compliance screen interpretation fails.
Large subhorizon $ \mu - 1 \sim \Sigma - 1 $	Ward-closed optical-stress branch fails.
Free $A(a, k)$ required	Weyl normalization fails; the model is not a G1DE production branch.
G1DE-2 strongly favored	The locked reduction is incomplete; diagnose whether the failure is compliance, shape, leakage, or normalization.

This demotion hierarchy prevents overclaiming. The amplitude relation $\Sigma_0 = -\kappa R_0$ is lower-level than the detailed redshift-memory shape $\hat{R}_H(a)$. Evidence for amplitude suppression can therefore be stronger than evidence for a particular memory kernel.

13 Implementation recipe

13.1 Minimal locked implementation

Use

$$R_0 = 3 - s, \quad \hat{R}_H(a) = X_b(a), \quad X_b(1) = 1, \quad \mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \mu(a, k) = 1, \quad \Sigma(a, k) = 1 - \frac{3}{4} R_0 \hat{R}_H(a). \quad (40)$$

The fitted parameter is s . Neither κ nor $A(a, k)$ is sampled.

13.2 Free- κ control

Use

$$\mathcal{R}_H(a) = (3 - s)X_b(a), \quad X_b(1) = 1, \quad \mu(a, k) = 1, \quad \Sigma(a, k) = 1 - \kappa(3 - s)X_b(a). \quad (41)$$

The parameters s and κ are sampled. This control tests the isotropic-compliance value $\kappa = 3/4$.

13.3 Constant- Σ control

Use

$$\mu(a, k) = 1, \quad \Sigma(a, k) = 1 + \Sigma_c. \quad (42)$$

This is a memory-shape control. It asks whether the data require the redshift dependence $X_b(a)$, rather than only a constant Weyl offset.

13.4 Ward-leakage extension

Use

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \Sigma(a, k) = 1 - \kappa R_0 \hat{R}_H(a), \quad \mu(a, k) = 1 + \nu_2 R_0^2 \hat{R}_H^2(a) + \nu_H R_0 \hat{R}_H(a) \frac{\mathcal{H}^2}{k^2}. \quad (43)$$

The coefficients ν_2 and ν_H test whether the Newtonian growth channel requires Ward-disciplined leakage. G1DE predicts that these terms are unnecessary in the subhorizon exact-pilot limit.

13.5 Free-shape reconstruction

If one reconstructs the memory shape,

$$\hat{R}_H(a) = \sum_i c_i B_i(a), \quad (44)$$

one must impose a normalization such as

$$\hat{R}_H(1) = 1. \quad (45)$$

Otherwise κ , R_0 , and the shape coefficients are not separately identifiable.

14 Relation to evidence interpretation

The factorized production doctrine clarifies how evidence comparisons should be read:

- $M_{3/4} > \text{CPL}$ means that a constrained Weyl-transfer response is favored over a background-only equation-of-state deformation.
- $M_{3/4} > M_\kappa$ means that exact 3/4 locking is not penalized and may be selected under the fixed memory-shape normalization.
- $M_{3/4} > \text{constant-}\Sigma$ means that the proposed memory shape helps relative to a constant Weyl offset.
- $M_{3/4} > \text{G1DE-2}$ means that the locked finite-screen transfer rule is preferred over an adversarial free-response envelope.

Conversely, failure modes should be mapped to the relevant factor. A failure of $X_b(a)$ is not the same as a failure of $\kappa = 3/4$, and a failure of $\kappa = 3/4$ is not the same as a failure of Weyl normalization or Ward suppression.

In the current G1fit-real medium-prior matched evidence pass, the locked model $M_{3/4}$ ranks first among the tested six-model set:

$$M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE2} > \text{CPL} > \Lambda\text{CDM}.$$

This hierarchy supports the Weyl-normalized projection-locking interpretation. The amplitude-level lock ($\kappa \sim 3/4$) is selected over free- κ , constant- Σ , and free-response G1DE-2, while the full evidence numbers and prior-sensitivity checks are reported in the G1fit-real data paper.

15 Conclusion

The essential implementation point is that G1DE is not an independently normalized dark-stress model. It is a finite-screen Weyl-transfer correction whose normalization is fixed by the GR Weyl source. The data-facing response is

$$\boxed{\mathcal{W}_{\text{G1}} = (1 - \kappa R_0 \widehat{R}_H) \mathcal{W}_{\text{GR}} + O(R_0^2),} \quad (46)$$

with

$$\boxed{\Sigma - 1 = -\kappa R_0 \widehat{R}_H + O(R_0^2),} \quad (47)$$

and

$$\boxed{\mu - 1 = O(R_0^2) + O\left(R_0 \widehat{R}_H \frac{\mathcal{H}^2}{k^2}\right).} \quad (48)$$

For the locked G1DE-M branch,

$$\boxed{\kappa = \frac{3}{4}, \quad R_0 = 3 - s, \quad \widehat{R}_H = X_b, \quad X_b(1) = 1, \quad \mu = 1.} \quad (49)$$

The normalization $A(a, k)$ is not a parameter. The memory shape must be normalized before κ is interpreted. The Newtonian channel is not independently modified at first order; it receives only Ward-suppressed leakage. These rules define G1DE as a Weyl-normalized, Ward-disciplined finite-screen transfer class.

A Code-facing equation blocks

A.1 Locked response block

$$R_0 = 3 - s, \quad X_b(1) = 1, \quad R(a) = R_0 X_b(a), \quad \mu(a, k) = 1, \quad \Sigma(a, k) = 1 - \frac{3}{4} R(a). \quad (50)$$

A.2 Free- κ response block

$$R_0 = 3 - s, \quad X_b(1) = 1, \quad R(a) = R_0 X_b(a), \quad \mu(a, k) = 1, \quad \Sigma(a, k) = 1 - \kappa R(a). \quad (51)$$

A.3 Ward-leakage response block

$$R(a) = R_0 \widehat{R}_H(a), \quad \widehat{R}_H(1) = 1, \quad \mu(a, k) = 1 + \nu_2 R^2(a) + \nu_H R(a) \frac{\mathcal{H}^2}{k^2}, \quad \Sigma(a, k) = 1 - \kappa R(a). \quad (52)$$

B Notation dictionary

Symbol	Meaning
$\mathcal{W}$	Weyl source, $-k^2(\Phi + \Psi)$
$\mathcal{W}_{\text{GR}}$	GR Weyl source, $8\pi G a^2 \rho \Delta$
$\mathcal{T}_{\text{G1}}^{\mathcal{W}}$	G1 Weyl-transfer operator
$A(a, k)$	convention-fixing map from optical seed to unit fractional Weyl correction; not a fit parameter

Symbol	Meaning
$\mathcal{R}_H(a)$	finite-screen horizon-memory residual
R_0	present-day residual amplitude, $R_0 = \mathcal{R}_H(1)$
$\hat{R}_H(a)$	normalized memory shape, $\hat{R}_H(1) = 1$
$X_b(a)$	specific normalized active-filling response window used in G1DE-M
s	low-redshift amplitude parameter, $R_0 = 3 - s$
κ	optical projection/compliance coefficient
κ_{BW}	locked finite-screen projection coefficient, $\kappa_{BW} = 3/4$ in the isotropic branch
$\pi_{\text{scr}}^{\text{GI}}$	gauge-invariant screen anisotropic seed
$q_{\text{scr}}^{\text{GI}}$	Ward-completed momentum leakage
$\delta\rho_{\text{scr}}^{\text{GI}}$	Ward-completed density/Ricci leakage
ν_2, ν_H	Ward-leakage control parameters

References

- [1] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, FDS-G1 Core manuscript (2026).
- [2] Y. Wu, *The 3/4 Projection Law in Finite Screen Gravity: MaxEnt Optical-Stress Closure, Ricci-Silent Residuals, and the Weyl-Locked Dark Sector*, manuscript / technical note, 2026.
- [3] Y. Wu, *Screen Influence Functional for the G1 Dark Sector: Entropy Hessian, Ward Closure, and Retarded Horizon Memory*, manuscript / technical note, 2026.
- [4] Y. Wu, *G1DE-M_{3/4}: A Projection-Locked Background–Weyl Residual*, manuscript / data paper, 2026.
- [5] R. P. Feynman and F. L. Vernon, The theory of a general quantum system interacting with a linear dissipative system, *Annals of Physics* **24**, 118–173 (1963).
- [6] J. Schwinger, Brownian motion of a quantum oscillator, *Journal of Mathematical Physics* **2**, 407–432 (1961).
- [7] L. V. Keldysh, Diagram technique for nonequilibrium processes, *Soviet Physics JETP* **20**, 1018–1026 (1965).
- [8] J. M. Bardeen, Gauge-invariant cosmological perturbations, *Physical Review D* **22**, 1882–1905 (1980).
- [9] C.-P. Ma and E. Bertschinger, Cosmological perturbation theory in the synchronous and conformal Newtonian gauges, *The Astrophysical Journal* **455**, 7–25 (1995).
- [10] P. Zhang, M. Liguori, R. Bean, and S. Dodelson, Probing gravity at cosmological scales by measurements which test the relationship between gravitational lensing and matter overdensity, *Physical Review Letters* **99**, 141302 (2007).
- [11] E. Bellini and I. Sawicki, Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity, *Journal of Cosmology and Astroparticle Physics* **07**, 050 (2014).
- [12] S. F. Daniel and E. V. Linder, Constraining cosmic expansion and gravity with gravitational lensing, *Physical Review D* **82**, 103523 (2010).

- [13] S. Weinberg, Photons and gravitons in perturbation theory: Derivation of Maxwell's and Einstein's equations, *Physical Review* **138**, B988–B1002 (1965).

Maximum-Caliber Horizontal Lift for the G1 Memory Quotient

Path Relative Entropy, Passive Horizon Kernels, and the Rank-One Origin of G1DE-M_{3/4}

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

Draft: May 2026

Abstract

This technical note isolates the memory-shape problem in the G1 finite-screen dark-sector branch. The data-facing object is not a unique microscopic source–kernel pair (K_H, J_H) , but the quotient output

$$\mathcal{R}_H(a) = (K_H * J_H)(a),$$

which enters the scalar perturbation sector through the Weyl-normalized normal form

$$\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

The factorization $\mathcal{R}_H = K_H * J_H$ has a gauge-like ambiguity: different passive kernels and occupancy sources can generate the same observable output. We propose a canonical resolution: the source–kernel representative is selected by a maximum-caliber horizontal lift. Given a passive reference path measure $\mathbb{P}_0[x]$ for hidden horizon memory variables and an output constraint $\mathbb{E}_{\mathbb{P}}[O(N)] = \mathcal{R}_H(N)$, the canonical lift is the path-space information projection

$$\mathbb{P}_* = \arg \min_{\mathbb{P}: \mathbb{E}_{\mathbb{P}}[O] = \mathcal{R}_H} D_{\text{KL}}(\mathbb{P} \parallel \mathbb{P}_0).$$

The dual source J_H is the Lagrange multiplier of this constraint, and the linear response kernel is the Hessian/covariance of the passive reference process. If the reference dynamics is stable and overdamped, the response is Stieltjes:

$$\chi_H(s) = \int_0^\infty \frac{d\mu_H(\gamma)}{s + \gamma}, \quad d\mu_H \geq 0.$$

The rank-one spectral truncation gives the one-state passive memory equation

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H,$$

and, after the fixed optical compliance fraction 3/4, yields G1DE-M_{3/4}. Higher-rank positive Stieltjes or Jacobi realizations are controlled memory-shape extensions; unconstrained free-response models are adversarial envelopes outside the passive lift class. The note clarifies the status of the commonly used shape $X_b(a)$: it is a normalized output response $\widehat{\mathcal{R}_H}(a)$, not a uniquely derived microscopic kernel. The remaining microscopic task is to derive the passive reference path measure and occupancy source convention from finite-screen modular-index dynamics.

1 Introduction

The G1 finite-screen dark-sector program has developed a data-facing low-redshift reduction in which a horizon-scale finite-screen memory output modifies the Weyl source while leaving the Newtonian/Ricci response approximately unchanged. In its projection-locked branch the scalar response is

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{\mathcal{R}}_H(a), \quad \mu(a, k) \simeq 1, \quad (1)$$

where $\widehat{\mathcal{R}}_H(1) = 1$ and $3 - s$ is the present amplitude of the low-redshift horizon memory output. Previous closure notes isolate three structural elements behind this expression:

1. a finite-screen optical compliance fraction $\kappa_{BW} = 3/4$ at the unbiased $GL(2)$ optical fixed point;
2. a Ward-completed Weyl transfer in which the leading correction enters Σ , while $\mu - 1$ is higher-order or subhorizon-suppressed;
3. a horizon memory output $\mathcal{R}_H(a)$ representing a passive finite-screen remnant.

D6 does not make (K_H, J_H) identifiable. It selects canonical representatives inside an output equivalence class. The horizontal lift $[R_H] \mapsto (K_H, J_H)_{\text{can}}$ uses maximum-caliber selection under the criteria: (1) passive realization; (2) minimal memory production; (3) detailed-balance compatibility; (4) no additional free output degrees; (5) consistency with D7 Markov realization.

The weakest link is not the projection law, but the redshift shape of $\mathcal{R}_H(a)$. A common schematic factorization is

$$\mathcal{R}_H(N) = \int_{-\infty}^N K_H(N, N') J_H(N') \, dN', \quad N = \ln a, \quad (2)$$

where K_H is a causal memory kernel and J_H is a horizon occupancy current, often written formally as

$$J_H = -\frac{d}{dN} \ln \mathcal{O}_H, \quad \mathcal{O}_H = \frac{C_{\text{used}}}{C_H}. \quad (3)$$

However, data do not separately observe K_H and J_H . They observe the output $\mathcal{R}_H(a)$ through its Weyl projection. Therefore a correct implementation note must distinguish three layers:

$$\boxed{\text{output } \mathcal{R}_H} \longleftarrow \boxed{\text{source-kernel lift } (K_H, J_H)} \longleftarrow \boxed{\text{microscopic finite-screen dynamics}}. \quad (4)$$

This note gives a precise mathematical formulation of that distinction. We first identify the observable memory quotient. We then introduce a memory factorization bundle whose base is the output $\mathcal{R}_H$ and whose fibers are all source-kernel factorizations producing that output. The central proposal is that a canonical representative of the fiber is selected by a maximum-caliber horizontal lift: among all path measures on hidden horizon states that reproduce the output, choose the one of least path relative entropy with respect to a passive reference process. This information projection yields the dual source, the Onsager kernel, the Stieltjes response class, and the rank-one model as the lowest spectral truncation.

The note is deliberately technical and modest. It does not claim to derive the microscopic finite-screen state family or the occupancy functional C_{used} . It supplies an interface between the data-facing G1DE model and the deeper finite-screen theory. Its main purpose is to prevent overclaiming: $X_b(a)$ or $\widehat{\mathcal{R}}_H(a)$ is a normalized output shape, not a uniquely derived microscopic kernel.

2 Claim status and notation

We use the following conventions.

- The time variable is $N = \ln a$. A prime denotes d/dN .
- $\mathcal{R}_H(N)$ is the scalar horizon memory output. Its normalized shape is $\widehat{\mathcal{R}}_H(N) = \mathcal{R}_H(N)/\mathcal{R}_H(0)$, with $a = 1$ corresponding to $N = 0$.
- The present amplitude is written

$$R_0 = \mathcal{R}_H(0) = 3 - s. \quad (5)$$

The parameter s is a low-redshift effective filling exponent; its deficit $3 - s$ is not assumed to be a literal spatial dimension.

- The Weyl-normalized observable map is

$$\Sigma - 1 = -\kappa_{BW}\mathcal{R}_H + O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right), \quad (6)$$

$$\mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (7)$$

At the isotropic optical fixed point $\kappa_{BW} = 3/4$.

- The notation $K_H * J_H$ denotes a causal convolution or, more generally, a causal Volterra response. In a stationary low-redshift approximation,

$$\mathcal{R}_H(N) = \int_0^\infty K_H(\tau) J_H(N - \tau) d\tau. \quad (8)$$

Claim-status discipline. This note proves conditional statements. If a passive reference path process is given, then a maximum-caliber lift yields a convex response potential and a positive kernel. If the passive generator is overdamped/self-adjoint in the relevant slow sector, then the kernel is Stieltjes. If the leading spectral mode dominates, then the rank-one model follows. The note does not prove from bare FDS primitives that the required reference process exists.

3 The G1 output normal form

The primary data-facing object is $\mathcal{R}_H(a)$. To leading order, its scalar perturbation imprint is the Weyl transfer

$$\mathcal{W}_{G1} = (1 - \kappa_{BW}\mathcal{R}_H) \mathcal{W}_{GR} + O(\mathcal{R}_H^2), \quad (9)$$

with minimum Ward completion suppressing the Ricci/Newtonian leakage. Hence

$$\Sigma - 1 = -\kappa_{BW}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (10)$$

At the unbiased optical fixed point,

$$T_I GL(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1), \quad (11)$$

where A is the area/trace direction, S is the two-dimensional shear sector, and Ω is twist/holonomy. The Weyl-active unimodular sector is

$$U = S \oplus \Omega, \quad (12)$$

with $\dim U = 3$, while the full optical ledger has dimension 4. If the optical Hessian or compliance metric is isotropic at the fixed point,

$$\kappa_{BW} = \frac{\dim U}{\dim GL(2)} = \frac{3}{4}. \quad (13)$$

Thus the leading data-facing normal form is

$$\boxed{\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu - 1 = O\left(\mathcal{R}_H^2, \mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).} \quad (14)$$

With $\mathcal{R}_H(a) = (3-s)\widehat{\mathcal{R}_H}(a)$, this becomes

$$\boxed{\Sigma(a, k) - 1 = -\frac{3}{4}(3-s)\widehat{\mathcal{R}_H}(a), \quad \mu(a, k) \simeq 1,} \quad (15)$$

up to the stated higher-order corrections.

4 The memory quotient

4.1 Source–kernel factorization is not unique

In a stationary representation,

$$\widetilde{\mathcal{R}_H}(s) = \chi_H(s)\tilde{J}_H(s), \quad (16)$$

where χ_H is the transfer function of the memory kernel. If $Q(s)$ is an admissible invertible filter, then

$$\chi_H(s) \mapsto \chi_H(s)Q(s), \quad \tilde{J}_H(s) \mapsto Q(s)^{-1}\tilde{J}_H(s) \quad (17)$$

leaves $\widetilde{\mathcal{R}_H}(s)$ invariant. Thus the pair (K_H, J_H) is not directly identifiable from $\mathcal{R}_H$ alone.

Definition 1 (Memory factorization bundle). *Let $\mathcal{M}_{\text{fact}}$ be the set of admissible source–kernel pairs (K, J) , where K is causal, stable, and passive, and J is an admissible horizon source. Define*

$$\pi : \mathcal{M}_{\text{fact}} \rightarrow \mathcal{M}_{\text{out}}, \quad \pi(K, J) = K * J = \mathcal{R}. \quad (18)$$

The base $\mathcal{M}_{\text{out}}$ contains observable outputs $\mathcal{R}$. The fiber over $\mathcal{R}$ is

$$\pi^{-1}(\mathcal{R}) = \{(K, J) : K * J = \mathcal{R}\}. \quad (19)$$

The production likelihood lives primarily on $\mathcal{M}_{\text{out}}$, because scalar observables at leading order depend on $\mathcal{R}_H$, not on a unique representative (K_H, J_H) . A microscopic theory or implementation convention selects a horizontal lift from output space to the factorization bundle.

4.2 Gauge-invariant output quantities

Factorization-invariant quantities include

$$R_0 = \mathcal{R}_H(0), \quad \alpha_H(N) = \frac{d}{dN} \ln \mathcal{R}_H(N), \quad \beta_H(N) = \frac{d^2}{dN^2} \ln \mathcal{R}_H(N), \quad (20)$$

and normalized ratios $\mathcal{R}_H(N_i)/\mathcal{R}_H(N_j)$. These are output observables. By contrast, pole locations Γ_i , spectral weights Z_i , or occupancy currents J_H require a factorization gauge or external information.

5 Maximum-caliber horizontal lift

5.1 Path-space information projection

Let $x(N)$ denote hidden horizon memory variables, and let

$$O(N) = Cx(N) \quad (21)$$

be the slow observable whose expectation is the output $\mathcal{R}_H(N)$. Let $\mathbb{P}_0[x]$ be a passive reference path measure representing equilibrium or unforced horizon modular-index dynamics.

Definition 2 (Maximum-caliber lift). *Given an output trajectory $\mathcal{R}_H(N)$, the canonical path measure $\mathbb{P}_*$ is*

$$\boxed{\mathbb{P}_* = \arg \min_{\mathbb{P}: \mathbb{E}_{\mathbb{P}}[O(N)] = \mathcal{R}_H(N)} D_{\text{KL}}(\mathbb{P} \parallel \mathbb{P}_0).} \quad (22)$$

The Lagrange multiplier enforcing the output constraint is the dual source $J_H(N)$. Standard variational calculus gives

$$\frac{d\mathbb{P}_*}{d\mathbb{P}_0} = \frac{1}{Z[J_H]} \exp \left(\int J_H(N) O(N) \, dN \right), \quad (23)$$

where

$$\Psi[J_H] = \log Z[J_H]. \quad (24)$$

The output is the functional derivative

$$\boxed{\mathcal{R}_H(N) = \frac{\delta \Psi}{\delta J_H(N)}}. \quad (25)$$

Equation (25) is the convex-dual source gauge. In this framework J_H is not first introduced as a unique microscopic source; it is the dual coordinate of the maximum-caliber output constraint. A microscopic occupancy formula such as (3) is a further claim: it identifies this dual coordinate with a finite-screen occupancy current.

5.2 Dissipation potential and convexity

Expanding Ψ near $J_H = 0$,

$$\Psi[J_H] = \Psi[0] + \int \bar{O}(N) J_H(N) \, dN + \frac{1}{2} \int J_H(N) K_H(N, N') J_H(N') \, dN \, dN' + \dots \quad (26)$$

For perturbations around a centered reference process $\bar{O} = 0$,

$$\mathcal{R}_H(N) = \int K_H(N, N') J_H(N') \, dN' + O(J_H^2), \quad (27)$$

where

$$K_H(N, N') = \langle O(N) O(N') \rangle_{0,c}. \quad (28)$$

Since K_H is a covariance operator, it is positive semidefinite. The quadratic dissipation potential is

$$\Psi_2[J_H] = \frac{1}{2} \int J_H K_H J_H \geq 0. \quad (29)$$

Thus linear passivity is the convexity of the path relative-entropy projection.

6 Passive kernels and Stieltjes representation

If the passive reference process is stationary in N , then $K_H(N, N') = K_H(N - N')$ for $N \geq N'$. If the unresolved horizon dynamics is overdamped and generated by a nonnegative self-adjoint operator L_H , then

$$K_H(\tau) = \langle O, e^{-\tau L_H} O \rangle_0, \quad \tau \geq 0. \quad (30)$$

The spectral theorem gives

$$\boxed{K_H(\tau) = \int_0^\infty e^{-\gamma\tau} d\mu_H(\gamma), \quad d\mu_H(\gamma) \geq 0.} \quad (31)$$

The transfer function is the Stieltjes function

$$\boxed{\chi_H(s) = \int_0^\infty \frac{d\mu_H(\gamma)}{s + \gamma}.} \quad (32)$$

The class (32) is causal, stable, and passive. It is also completely monotone in the time domain:

$$(-1)^n \frac{d^n}{d\tau^n} K_H(\tau) \geq 0, \quad n = 0, 1, 2, \dots \quad (33)$$

Thus a G1 passive horizon kernel is not a general oscillator. It is an overdamped relaxation spectrum.

7 Rank-one lift and G1DE-M_{3/4}

If the spectral measure is dominated by one slow mode,

$$d\mu_H(\gamma) = Z_H \delta(\gamma - \Gamma_H) d\gamma, \quad Z_H > 0, \quad \Gamma_H > 0, \quad (34)$$

then

$$K_H(\tau) = Z_H e^{-\Gamma_H \tau}. \quad (35)$$

Equivalently,

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = Z_H J_H. \quad (36)$$

Absorbing Z_H into the amplitude convention gives the rank-one memory equation

$$\boxed{\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H.} \quad (37)$$

The normalized output is

$$\widehat{\mathcal{R}_H}(a) = \frac{\mathcal{R}_H(a)}{\mathcal{R}_H(1)}, \quad \mathcal{R}_H(a) = (3 - s) \widehat{\mathcal{R}_H}(a). \quad (38)$$

Combining (37) with the fixed optical projection gives the minimal output model:

$$\boxed{\text{G1DE-M}_{3/4} : \quad \Sigma - 1 = -\frac{3}{4} (3 - s) \widehat{\mathcal{R}_H}(a), \quad \mu \simeq 1.} \quad (39)$$

Thus G1DE-M_{3/4} is the rank-one maximum-caliber horizontal lift of the G1 memory quotient, followed by the fixed 3/4 Weyl projection.

8 Moment, Hankel, and Jacobi interpretations

The Stieltjes transfer has a low-frequency expansion

$$\chi_H(s) = m_0 - sm_1 + s^2 m_2 - \dots, \quad (40)$$

where

$$m_n = \int_0^\infty \gamma^{-(n+1)} d\mu_H(\gamma). \quad (41)$$

The moment sequence must satisfy Hankel positivity:

$$H_{ij}^{(r)} = m_{i+j} \succeq 0. \quad (42)$$

The rank of the Hankel moment matrix equals the minimal number of discrete relaxation bands needed to realize the response.

For rank one,

$$\boxed{m_0 m_2 - m_1^2 = 0}. \quad (43)$$

For rank two or higher,

$$m_0 m_2 - m_1^2 > 0. \quad (44)$$

Violation of Hankel positivity indicates failure of the passive Stieltjes class.

Equivalently, the Stieltjes response admits a canonical Jacobi representation

$$\chi_H(s) = \mu_0 \langle e_0, (sI + J_{\text{Jac}})^{-1} e_0 \rangle, \quad (45)$$

where $J_{\text{Jac}} \geq 0$ is a Jacobi matrix. Its continued fraction is

$$\chi_H(s) = \frac{\mu_0}{s + b_0 - \frac{a_1^2}{s + b_1 - \frac{a_2^2}{s + b_2 - \dots}}}, \quad a_i > 0. \quad (46)$$

The m -step truncation defines a controlled passive memory extension $M_{3/4}^{(m)}$.

9 Noise and fluctuation-dissipation information

The maximum-caliber construction naturally supplies both the mean response and the fluctuation covariance of the hidden horizon process. In an open-system description one may write schematically

$$x' = Ax + bJ_H + \xi_H, \quad \mathcal{R}_H = c^T x, \quad (47)$$

with noise covariance

$$\langle \xi_H(N) \xi_H(N')^T \rangle = N_H(N, N'). \quad (48)$$

Near equilibrium the response and noise are not independent. A fluctuation-dissipation-type relation has the schematic form

$$N_H(\omega) \sim \coth\left(\frac{\beta\omega}{2}\right) \text{Im} \chi_H(\omega), \quad (49)$$

with the appropriate modular inverse temperature β . The precise cosmological stochastic prediction requires a fully covariant open-system formalism and is left as an extension. The important point is structural: a stochastic Weyl or lensing residual covariance, if present, should not be modeled independently of the response kernel.

10 Model hierarchy and identifiability

The memory quotient formulation suggests the following disciplined hierarchy.

Model or control	Interpretation
constant- Σ	Output amplitude / DC-gain only; tests whether m_0 is enough.
$M_{3/4}^{(1)}$	Rank-one maximum-caliber/Stieltjes lift plus 3/4 Weyl projection; this is G1DE-M _{3/4} .
$M_{3/4}^{(2)}$	Rank-two positive memory; tests whether a second hidden relaxation band is resolved.
$M_{\kappa}^{(m)}$	Passive memory of order m , but optical coefficient free; tests optical isotropy.
G1DE-2 / free response	Unconstrained response envelope; adversarial control outside the passive-lift class.
free $A(a, k)$	Forbidden in G1: changes the fixed normalization rather than a memory deformation.

A key identifiability point follows: current scalar data should first be interpreted as constraining $\mathcal{R}_H(a)$ and its output shape moments. Inferring K_H and J_H separately requires a gauge choice, a maximum-caliber lift, stochastic covariance information, or an independent finite-screen occupancy model.

11 Main closure theorem

Theorem 1 (Maximum-caliber horizontal-lift closure). *Assume:*

1. the G1 scalar dark-sector branch is described by an output $\mathcal{R}_H(N)$;
2. the leading observable map is Weyl-normalized,

$$\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu - 1 = O\left(\mathcal{R}_H^2, \mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right); \quad (50)$$

3. hidden horizon memory variables admit a passive reference path measure $\mathbb{P}_0$;
4. the canonical lift is selected by minimizing path relative entropy subject to $\mathbb{E}[O] = \mathcal{R}_H$;
5. the passive reference generator is stable and overdamped in the relevant slow sector;
6. the dominant spectral response is rank one.

Then the canonical lift is linearized by a positive kernel with Stieltjes transfer function. Its rank-one truncation satisfies

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H, \quad (51)$$

and its leading scalar cosmological reduction is

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{\mathcal{R}_H}(a), \quad \mu(a, k) \simeq 1. \quad (52)$$

Proof. The maximum-caliber variational problem gives the exponentially tilted path measure (23). Its log-partition functional $\Psi[J_H]$ is convex. Hence its second variation around the passive reference process is a positive kernel K_H , and the linear response is $\mathcal{R}_H = K_H * J_H$. Stability and overdamped passive dynamics imply the spectral representation (32). Rank-one spectral dominance gives $\chi_H(s) = Z_H/(s + \Gamma_H)$, which is equivalent to $\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = Z_H J_H$. After absorbing Z_H into the amplitude convention and applying the fixed 3/4 Weyl projection with Ward-suppressed μ leakage, the stated G1DE- $M_{3/4}$ form follows. $\square$

Corollary 1 (Output status of $X_b(a)$). *A redshift shape $X_b(a)$ used in the low-redshift G1DE branch should be interpreted as*

$$X_b(a) = \widehat{\mathcal{R}_H}(a), \quad (53)$$

namely a normalized output response. It is not, by itself, a unique microscopic kernel or source.

Corollary 2 (Controlled memory extensions). *If data require additional memory moments while preserving Stieltjes positivity and Ward suppression, the correct controlled extensions are $M_{3/4}^{(m)}$, the m -rank passive memory realizations, rather than an unconstrained free-response surface.*

12 Null tests

The maximum-caliber quotient/lift picture implies the following tests.

Output projection test. On linear subhorizon scales,

$$\frac{\mu - 1}{\Sigma - 1} = O(\mathcal{R}_H) + O\left(\frac{\mathcal{H}^2}{k^2}\right). \quad (54)$$

An order-one ratio would indicate failure of Ward/Ricci slaving.

Rank-one memory test. If output moments can be reconstructed, rank-one memory predicts

$$\Delta_H = m_0 m_2 - m_1^2 \simeq 0. \quad (55)$$

Positive Δ_H suggests higher memory rank; negative Δ_H violates passive Stieltjes positivity.

Kernel positivity test. Any inferred kernel in a passive gauge should be completely monotone. Oscillatory or negative-residue kernels are outside the overdamped G1 memory class.

Source-gauge test. If a finite-screen occupancy proxy becomes available, the dual source from the maximum-caliber lift should be compatible with

$$J_H = -\frac{d}{dN} \ln(C_{\text{used}}/C_H). \quad (56)$$

Failure here would challenge the microscopic occupancy model, not necessarily the output normal form.

13 Open problems

The note reduces the memory-shape problem to several sharper open tasks:

1. **Reference path measure.** Derive the passive reference process $\mathbb{P}_0[x]$ from finite-screen modular-index dynamics.
2. **Occupancy source.** Derive $C_{\text{used}}(a)$ and hence $J_H = -d_N \ln(C_{\text{used}}/C_H)$ from a microscopic finite-screen state family.
3. **Covariant open-system completion.** Promote the scalar maximum-caliber lift to a gauge-invariant covariant Schwinger-Keldysh or influence-functional framework.
4. **Noise kernel.** Work out the stochastic Weyl/lensing covariance implied by the same passive reference process.
5. **Memory-rank evidence.** Implement $M_{3/4}^{(2)}$ and higher positive-rank memory models before interpreting free-response envelopes as fundamental alternatives.
6. **Optical coefficient controls.** Continue testing the fixed 3/4 coefficient against free- κ branches and twist-demoted alternatives.

14 Conclusion

The G1 memory problem is best formulated as a quotient-and-lift problem. The scalar data-facing object is the horizon output $\mathcal{R}_H(a)$. Source–kernel factorizations (K_H, J_H) are representatives in a fiber over that output, and a unique representative requires an additional lift principle. The proposed canonical lift is the maximum-caliber path-relative-entropy projection onto a passive reference horizon process. Its dual source is the Lagrange multiplier enforcing the output, its linear kernel is a covariance/Onsager Hessian, and stable overdamped reference dynamics gives a Stieltjes memory class.

The rank-one spectral truncation is precisely the memory closure behind G1DE-M_{3/4}. Thus

$$\boxed{\text{G1DE-M}_{3/4} = \text{rank-one maximum-caliber horizontal lift} + \text{fixed } 3/4 \text{ Weyl projection.}} \quad (57)$$

The formulation preserves the main leading prediction,

$$\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H, \quad \mu \simeq 1, \quad (58)$$

while clarifying that $X_b(a)$ is a normalized output shape, not a uniquely derived microscopic kernel. The deepest remaining task is to derive the passive reference path measure and the occupancy source convention from finite-screen modular-index microphysics.

A Relative entropy and maximum caliber

For probability measures $\mathbb{P}$ and $\mathbb{P}_0$ on path space, the path relative entropy is

$$D_{\text{KL}}(\mathbb{P}||\mathbb{P}_0) = \int \log \left(\frac{d\mathbb{P}}{d\mathbb{P}_0} \right) d\mathbb{P}. \quad (59)$$

Minimizing this quantity subject to linear constraints

$$\mathbb{E}_{\mathbb{P}}[O_i] = r_i \quad (60)$$

leads to the exponential family

$$\frac{d\mathbb{P}_{\lambda}}{d\mathbb{P}_0} = \frac{1}{Z(\lambda)} \exp \left(\sum_i \lambda_i O_i \right). \quad (61)$$

The log-partition function $\Psi(\lambda) = \log Z(\lambda)$ satisfies

$$\frac{\partial \Psi}{\partial \lambda_i} = \mathbb{E}_{\mathbb{P}_{\lambda}}[O_i], \quad \frac{\partial^2 \Psi}{\partial \lambda_i \partial \lambda_j} = \text{Cov}_{\mathbb{P}_{\lambda}}(O_i, O_j). \quad (62)$$

The continuous-time formulas used in the main text are the functional version of this result.

B Stieltjes functions and complete monotonicity

A function $\chi(s)$ of the form

$$\chi(s) = \int_0^{\infty} \frac{d\mu(\gamma)}{s + \gamma}, \quad d\mu \geq 0, \quad (63)$$

is a Stieltjes function. Its time-domain kernel is

$$K(\tau) = \int_0^{\infty} e^{-\gamma\tau} d\mu(\gamma), \quad (64)$$

which is completely monotone:

$$(-1)^n K^{(n)}(\tau) = \int_0^{\infty} \gamma^n e^{-\gamma\tau} d\mu(\gamma) \geq 0. \quad (65)$$

This is the mathematical expression of an overdamped passive relaxation spectrum.

C Hankel rank and memory order

For a positive discrete measure

$$d\mu(\gamma) = \sum_{i=1}^m Z_i \delta(\gamma - \Gamma_i) d\gamma, \quad Z_i > 0, \quad (66)$$

the moments are

$$m_n = \sum_{i=1}^m Z_i \Gamma_i^{-(n+1)}. \quad (67)$$

The infinite Hankel matrix $H_{ij} = m_{i+j}$ has rank m if the Γ_i are distinct. Thus moment rank equals minimal memory order. Rank one implies $m_0 m_2 - m_1^2 = 0$; rank two or higher gives $m_0 m_2 - m_1^2 > 0$ under nondegenerate positive weights.

D Implementation recipe

A production implementation can separate layers as follows.

1. Fit the output normal form:

$$\Sigma - 1 = -\frac{3}{4}(3-s)\widehat{\mathcal{R}_H}(a), \quad \mu = 1. \quad (68)$$

2. Test constant- Σ as an amplitude-only control.
3. Test a rank-one passive lift, i.e. a one-pole $\widehat{\mathcal{R}_H}(a)$.
4. If shape residuals remain, test rank-two positive Stieltjes memory before moving to unconstrained response.
5. Test free κ only as an optical-isotropy control, not as a replacement for the memory hierarchy.
6. Treat free $A(a, k)$ as outside the G1 class, because it changes the fixed normalization.

References

- [1] L. Onsager, Reciprocal relations in irreversible processes I and II, *Phys. Rev.* **37**, 405 (1931); **38**, 2265 (1931).
- [2] E. T. Jaynes, Information theory and statistical mechanics, *Phys. Rev.* **106**, 620 (1957).
- [3] R. Kubo, The fluctuation-dissipation theorem, *Rep. Prog. Phys.* **29**, 255 (1966).
- [4] H. Mori, Transport, collective motion, and Brownian motion, *Prog. Theor. Phys.* **33**, 423 (1965).
- [5] R. Zwanzig, Memory effects in irreversible thermodynamics, *Phys. Rev.* **124**, 983 (1961).
- [6] R. P. Feynman and F. L. Vernon, The theory of a general quantum system interacting with a linear dissipative system, *Ann. Phys.* **24**, 118 (1963).
- [7] D. Petz, Sufficient subalgebras and the relative entropy of states of a von Neumann algebra, *Commun. Math. Phys.* **105**, 123 (1986).
- [8] M. G. Krein and A. A. Nudelman, *The Markov Moment Problem and Extremal Problems*, AMS, 1977.
- [9] Y. Wu, *Screen Modular Anomaly Inflow and the Einstein Calibration of Finite Access Entropy*, G1 technical manuscript, 2026.
- [10] Y. Wu, *Weyl-Normalized Ward Completion and Identifiability Rules*, FDS–G1 technical note D5 (2026).
- [11] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual*, data-facing manuscript (2026).

Finite Markov-Screen Realization of the G1DE-M_{3/4} Normal Form

A Detailed-Balance Prototype for the 3/4 Weyl Projection, Rank-One Horizon Output, and Ward-Suppressed Ricci Leakage

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

Draft: May 2026

Abstract

This note gives a finite-state microscopic prototype for the G1DE normal form. Previous G1 notes separate the data-facing horizon response output $\mathcal{R}_H(a)$ from a non-unique source-kernel factorization $K_H * J_H$, and implement the dark-sector residual as a Weyl-normalized transfer correction rather than as an independently normalized dark stress source. The purpose of the present note is narrower and more constructive: to show that the formal ingredients used by the G1DE-M_{3/4} branch can be realized by a finite detailed-balance Markov screen.

For each screen size N we introduce a finite state space Ω_N , a passive reference measure σ_N , reversible Markov dynamics with generator Q_N , and centered macro-port observables

$$F_a, \quad a \in \{A, S_1, S_2, T, H, R\},$$

representing the trace optical, two shear, twist/holonomy, horizon-memory, and Ricci-leakage channels. With the nonnegative relaxation operator $\mathcal{L}_N = -Q_N$, the response matrix is

$$\chi_{ab}^{(N)}(s) = \langle F_a, (s + \mathcal{L}_N)^{-1} F_b \rangle_{\sigma_N}.$$

Reversibility makes $\mathcal{L}_N$ self-adjoint and nonnegative, hence $\chi^{(N)}$ has a positive matrix-valued Stieltjes representation. If the four optical ports are unbiased in both covariance and dissipation, the Weyl-active fraction of the optical response is

$$\kappa_{BW} = \frac{\dim(S_1 \oplus S_2)}{\dim(A \oplus S_1 \oplus S_2)} = \frac{3}{4}.$$

If the horizon port is approximately lumpable and dominated by one slow eigenmode, its output satisfies the rank-one equation

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H.$$

If Ward conditioning makes the Ricci port entropy-stiff, then the Newtonian response satisfies

$$\mu - 1 = O(\mathcal{R}_H^2) + O(\mathcal{R}_H \mathcal{H}^2 / k^2).$$

Consequently, the low-redshift Weyl-normalized output is

$$\Sigma(a, k) - 1 = -\frac{3}{4} \mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad \mu(a, k) \simeq 1.$$

With $\mathcal{R}_H(a) = (3 - s) \hat{\mathcal{R}}_H(a)$ and $\hat{\mathcal{R}}_H(1) = 1$, this gives the production normal form

$$\Sigma(a, k) - 1 = -\frac{3}{4} (3 - s) \hat{\mathcal{R}}_H(a), \quad \mu(a, k) \simeq 1.$$

The result is conditional and constructive: it does not derive the true finite-screen microphysics, but it gives a concrete finite-state realization target for the G1DE-M_{3/4} interface. This realization is not unique; it is a constructive existence target showing that the G1DE-M_{3/4} interface is compatible with finite passive Markov dynamics.

1 Scope and claim status

The G1 dark-sector sequence has three implementation-facing claims that must be kept distinct:

1. the likelihood primarily sees an output response $\mathcal{R}_H(a)$;
2. a factorization $\mathcal{R}_H = K_H * J_H$ is a source–kernel lift, not a uniquely observed microscopic object;
3. the leading G1DE branch modifies the Weyl source by a normalized finite-screen transfer factor, not by an independently normalized dark-stress source.

This note adds a finite-state realization prototype underneath those claims.

Non-uniqueness. This paper proves existence of at least one admissible finite-state passive lift, not uniqueness of the microscopic screen dynamics. If two generators L_N, L'_N produce the same $R_H(a)$, they are observationally equivalent at the G1fit-real level unless additional observables are specified.

The note is not a derivation of General Relativity and not a proof that Nature uses the particular Markov chain written below. It proves a conditional realizability statement: if a finite reversible Markov screen has optical unbiasedness, a slow horizon mode, and Ward-stiff Ricci leakage, then its low-spectrum response realizes the G1DE-M_{3/4} normal form.

Layer	Object	Status in this note
Data-facing output	$\mathcal{R}_H(a)$	retained as observable response
Lift layer	(K_H, J_H)	induced by reversible Markov response
Microscopic prototype	$(\Omega_N, \sigma_N, Q_N, F_a)$	constructed conditionally
Optical projection	$\kappa_{BW} = 3/4$	follows from optical unbiasedness
Horizon shape	$\hat{\mathcal{R}}_H(a)$	rank-one slow-mode output
Ward leakage	$\mu - 1$	Schur/conditioning suppressed

2 Finite Markov screens

2.1 State space and ports

For each finite screen size N , let

$$\Omega_N = \Omega_{\text{cell}}^N$$

be a finite product state space. A microscopic screen state is

$$\omega = (\omega_1, \dots, \omega_N) \in \Omega_N.$$

Each cell carries port variables

$$f_a : \Omega_{\text{cell}} \rightarrow \mathbb{R}, \quad a \in \mathcal{P} := \{A, S_1, S_2, T, H, R\}.$$

The port labels are interpreted as follows:

Port	Meaning
A	trace/area optical channel
S_1, S_2	two shear optical channels
T	twist/holonomy optical channel
H	horizon memory or occupancy output channel
R	Ricci/density leakage channel

The centered macroscopic port observables are

$$F_{a,N}(\omega) = \frac{1}{\sqrt{N}} \sum_{i=1}^N (f_a(\omega_i) - \mathbb{E}_{\sigma_1} f_a). \quad (1)$$

The $N^{-1/2}$ scaling is convenient for response theory. For large-deviation statements one may instead use $N^{-1} \sum_i f_a$; the two conventions differ only by the fluctuation scale.

2.2 Passive reference state

Let σ_N be a faithful probability measure on Ω_N . In the simplest independent-cell case,

$$\sigma_N = \sigma_1^{\otimes N}.$$

More generally, σ_N may include finite-range interactions, provided the macro-variables obey an area-extensive large-deviation principle.

Definition 2.1 (Static macro-quasipotential). *Let $x = (x_a)_{a \in \mathcal{P}}$ denote the empirical macro-port vector. A static quasipotential $\Phi_N(x)$ is defined by*

$$\sigma_N\{x_N \simeq x\} \asymp \exp[-N\Phi_N(x)].$$

Near the passive fixed point $x = 0$,

$$\Phi_N(x) = \frac{1}{2} x^T K_N x + O(|x|^3), \quad K_N = \nabla^2 \Phi_N(0). \quad (2)$$

For independent cells, K_N is the inverse covariance matrix of the cell port vector. This is the finite-state version of the relative-entropy Hessian used in the continuum G1 notes.

3 Reversible Markov screen dynamics

3.1 Detailed-balance generator

Let Q_N be a continuous-time Markov generator on Ω_N :

$$(Q_N g)(\omega) = \sum_{\omega' \in \Omega_N} r_N(\omega, \omega') [g(\omega') - g(\omega)]. \quad (3)$$

Assume detailed balance with respect to σ_N :

$$\sigma_N(\omega) r_N(\omega, \omega') = \sigma_N(\omega') r_N(\omega', \omega). \quad (4)$$

Then Q_N is self-adjoint on $L^2(\sigma_N)$ and has nonpositive spectrum. Define the nonnegative relaxation operator

$$\mathcal{L}_N = -Q_N \succeq 0. \quad (5)$$

The Dirichlet form is

$$\mathcal{E}_N(g, h) = \langle g, \mathcal{L}_N h \rangle_{\sigma_N} = \frac{1}{2} \sum_{\omega, \omega'} \sigma_N(\omega) r_N(\omega, \omega') [g(\omega) - g(\omega')] [h(\omega) - h(\omega')]. \quad (6)$$

In particular,

$$\mathcal{E}_N(g, g) \geq 0.$$

3.2 Matrix-valued response

For $s > 0$, define the port response matrix

$$\chi_{ab}^{(N)}(s) = \langle F_{a,N}, (s + \mathcal{L}_N)^{-1} F_{b,N} \rangle_{\sigma_N}. \quad (7)$$

This is the finite Markov-screen impedance.

Lemma 3.1 (Matrix-Stieltjes representation). *For every finite reversible Markov screen, there exists a positive semidefinite matrix-valued measure $dM_N(\gamma)$ on $[0, \infty)$ such that*

$$\chi^{(N)}(s) = \int_0^\infty \frac{dM_N(\gamma)}{s + \gamma}. \quad (8)$$

Proof. Because $\mathcal{L}_N$ is self-adjoint and nonnegative in $L^2(\sigma_N)$, the spectral theorem gives

$$(s + \mathcal{L}_N)^{-1} = \int_0^\infty \frac{dE_N(\gamma)}{s + \gamma},$$

where E_N is the spectral projection measure. Define

$$dM_{N,ab}(\gamma) = \langle F_{a,N}, dE_N(\gamma) F_{b,N} \rangle_{\sigma_N}.$$

For any vector $u = (u_a)$,

$$u^\top dM_N(\gamma) u = \left\langle \sum_a u_a F_{a,N}, dE_N(\gamma) \sum_b u_b F_{b,N} \right\rangle_{\sigma_N} \geq 0.$$

Thus $dM_N \succeq 0$, and (8) follows. $\square$

4 Optical unbiasedness and the 3/4 projection

4.1 Optical port space

Define the four-dimensional optical port space

$$\mathcal{P}_{\text{opt}} = A(1) \oplus S_1(1) \oplus S_2(1)(1). \quad (9)$$

The Weyl-active unimodular subspace is

$$U = S_1 \oplus S_2, \quad \dim U = 3. \quad (10)$$

Let P_U be the projector onto U .

Definition 4.1 (Optical unbiasedness). *The finite Markov screen is optically unbiased if, on $\mathcal{P}_{\text{opt}}$,*

$$\text{Cov}_{\sigma_N}(F_{a,N}, F_{b,N}) = c_N \delta_{ab}, \quad (11)$$

and

$$\mathcal{E}_N(F_{a,N}, F_{b,N}) = d_N \delta_{ab}, \quad (12)$$

for $a, b \in \{A, S_1, S_2, T\}$.

Under this condition, the optical response block is approximately proportional to the identity:

$$\chi_{\text{opt}}^{(N)}(s) = \chi_N^{\text{opt}}(s)I_4 + O(\epsilon_{\text{aniso}}). \quad (13)$$

This holds exactly if the four optical observables span an invariant eigenspace of $\mathcal{L}_N$ with a common spectral measure, and it is the leading fixed-point form if optical anisotropies are irrelevant under coarse-graining.

Proposition 4.1 (Finite-screen optical projection fraction). *If the optical response block is proportional to the identity, then*

$$\kappa_N(s) = \frac{\text{Tr}(P_U \chi_{\text{opt}}^{(N)}(s))}{\text{Tr}(\chi_{\text{opt}}^{(N)}(s))} = \frac{3}{4}. \quad (14)$$

Proof. Using (13),

$$\text{Tr}(P_U \chi_{\text{opt}}^{(N)}) = \chi_N^{\text{opt}} \text{Tr}(P_U) = 3\chi_N^{\text{opt}},$$

while

$$\text{Tr}(\chi_{\text{opt}}^{(N)}) = 4\chi_N^{\text{opt}}.$$

Their ratio is 3/4. □

Remark 4.1 (Weighted-compliance branch). *If optical unbiasedness fails, the coefficient becomes*

$$\kappa = \frac{C_{S_1} + C_{S_2} + C_T}{C_A + C_{S_1} + C_{S_2} + C_T}. \quad (15)$$

For the shear-degenerate case $C_{S_1} = C_{S_2} = C_S$, this is

$$\kappa = \frac{2C_S + C_T}{C_A + 2C_S + C_T}.$$

This is the microscopic meaning of the free- κ compliance control.

5 Horizon port and rank-one output

5.1 Horizon response

The scalar horizon-memory response is the H -port projection

$$\chi_H^{(N)}(s) = \langle F_{H,N}, (s + \mathcal{L}_N)^{-1} F_{H,N} \rangle_{\sigma_N} = \int_0^\infty \frac{d\mu_H^{(N)}(\gamma)}{s + \gamma}. \quad (16)$$

A source–kernel representation is

$$\mathcal{R}_H(N) = \int_{-\infty}^N K_H(N - N') J_H(N') dN'. \quad (17)$$

At this level J_H is a dual source or drive; identifying it with an occupancy current is an additional source-gauge claim.

5.2 Approximate lumpability and spectral gap

Definition 5.1 (Rank-one horizon dominance). *The horizon port is rank-one over the low-redshift window if*

$$d\mu_H^{(N)}(\gamma) = Z_1^{(N)} \delta(\gamma - \Gamma_H^{(N)}) d\gamma + d\mu_\perp^{(N)}(\gamma), \quad (18)$$

with

$$0 < \Gamma_H^{(N)} \ll \gamma \quad \text{for } \gamma \in \text{supp } d\mu_\perp^{(N)}, \quad (19)$$

and $d\mu_\perp^{(N)}$ weak over the observational window.

Then

$$K_H(\tau) = Z_1 e^{-\Gamma_H \tau} + K_\perp(\tau), \quad |K_\perp(\tau)| \lesssim Z_\perp e^{-\Gamma_2 \tau}. \quad (20)$$

Keeping the first band gives

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = Z_1 J_H. \quad (21)$$

Absorbing Z_1 into the source or amplitude convention gives the one-state memory equation

$$\boxed{\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H.} \quad (22)$$

The production normalization is

$$\mathcal{R}_H(a) = R_0 \widehat{\mathcal{R}}_H(a), \quad R_0 = \mathcal{R}_H(1) = 3 - s, \quad \widehat{\mathcal{R}}_H(1) = 1. \quad (23)$$

The commonly used shape is the output alias

$$X_b(a) \equiv \widehat{\mathcal{R}}_H(a). \quad (24)$$

It is not, by itself, a uniquely reconstructed microscopic kernel.

6 Ward conditioning and Ricci-port suppression

6.1 Conditioning

Let $C_\alpha(\omega)$ be finite-screen Ward-constraint observables. The physical reference state is the entropy projection

$$\sigma_N^{\text{phys}} = \arg \min_{\rho: \mathbb{E}_\rho C_\alpha = 0} D_{\text{KL}}(\rho \| \sigma_N). \quad (25)$$

Equivalently, one may work with a conditional expectation

$$\mathbb{E}_{\text{Ward}, N} : L^2(\sigma_N) \rightarrow L^2(\sigma_N^{\text{phys}}) \quad (26)$$

and a constrained relaxation operator

$$\mathcal{L}_N^{\text{phys}} = \mathbb{E}_{\text{Ward}, N} \mathcal{L}_N \mathbb{E}_{\text{Ward}, N}. \quad (27)$$

This is an information projection, not necessarily a literal projection onto a hard constraint subspace.

6.2 Schur-complement suppression

Let W denote the Weyl/optical output block and R the Ricci/density leakage block. The quadratic covariance or stiffness matrix may be written as

$$K = \begin{pmatrix} K_{WW} & K_{WR} \\ K_{RW} & K_{RR} \end{pmatrix}. \quad (28)$$

Contraction over Ward-constrained Ricci variables gives the effective Weyl block

$$K_{WW}^{\text{eff}} = K_{WW} - K_{WR} K_{RR}^{-1} K_{RW}. \quad (29)$$

If K_{RR} is stiff, the Ricci response is suppressed. In the cosmological scalar response this becomes

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (30)$$

The Weyl response remains first order:

$$\Sigma(a, k) - 1 = -\kappa \mathcal{R}_H(a) + O(\mathcal{R}_H^2). \quad (31)$$

At the optically unbiased fixed point, $\kappa = 3/4$.

The Markov-screen model supplies a stiff Ricci port that suppresses density-like response. The explicit $\mathcal{H}^2/k^2$ scaling in cosmological reduction is a Ward-to-Boltzmann scalar-sector bridge used in the production G1DE likelihood, not a property derived directly from the finite Markov chain.

7 Finite Markov-Screen Realization Theorem

Theorem 7.1 (Finite Markov-Screen Realization of the G1DE normal form). *Assume a sequence of finite Markov screens*

$$(\Omega_N, \sigma_N, Q_N, F_{a,N}), \quad a \in \{A, S_1, S_2, T, H, R\},$$

satisfying:

1. Q_N is reversible with respect to σ_N ;
2. the macro-port quasipotential has a positive Hessian at the passive fixed point;
3. the four optical ports A, S_1, S_2, T are unbiased in covariance and dissipation, and the optical port subspace is invariant under the reversible generator with a common low-spectrum Stieltjes measure,

$$\chi_{\text{opt}}(s) = \chi_{\text{opt}}^0(s) I_4 + O(\epsilon_{\text{aniso}});$$

4. the horizon port H is approximately lumpable and dominated by one slow spectral band;
5. Ward conditioning makes the Ricci port entropy-stiff;
6. the Weyl-source normalization is fixed, so no independent optical amplitude $A(a, k)$ is fitted.

Then the finite-screen response is matrix-Stieltjes, the optical projection coefficient is $\kappa_{BW} = 3/4$, the horizon output is rank-one to leading order, and the low-redshift Weyl-normalized scalar response is

$$\Sigma(a, k) - 1 = -\frac{3}{4}\mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (32)$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (33)$$

With $\mathcal{R}_H(a) = (3-s)\widehat{\mathcal{R}}_H(a)$ and $\widehat{\mathcal{R}}_H(1) = 1$, this gives

$$\boxed{\Sigma(a, k) - 1 = -\frac{3}{4}(3-s)\widehat{\mathcal{R}}_H(a), \quad \mu(a, k) \simeq 1.} \quad (34)$$

Proof sketch. Reversibility gives self-adjointness and nonnegativity of $\mathcal{L}_N = -Q_N$. The spectral theorem gives the matrix-Stieltjes representation (8). Optical unbiasedness makes the four-dimensional optical block proportional to the identity, so the Weyl-active trace fraction is 3/4. Rank-one horizon dominance reduces the horizon Stieltjes response to the one-pole equation (22). Ward conditioning contracts or Schur-complements the Ricci port, producing (30). Weyl normalization identifies the optical seed with a fractional correction to the GR Weyl source, producing (31). Combining these statements gives (34). $\square$

8 An Explicit Minimal Toy Model

(Existence Proof-of-Concept, Not a Spacetime Microstate Proposal)

This section gives a deliberately simple finite-state model realizing the assumptions. It is a proof-of-concept, not a proposed microscopic theory of spacetime.

8.1 Cell state

Let a cell state be

$$\omega_i = (o_i, h_i, r_i), \quad (35)$$

where

$$o_i = (o_A, o_{S_1}, o_{S_2}, o_T) \in \{-1, +1\}^4, \quad h_i \in \{-1, +1\}, \quad r_i \in \{-1, 0, +1\}. \quad (36)$$

The optical observables are the four Rademacher components

$$f_A = o_A, \quad f_{S_1} = o_{S_1}, \quad f_{S_2} = o_{S_2}, \quad f_T = o_T. \quad (37)$$

The horizon and Ricci ports are

$$f_H = h_i, \quad f_R = r_i. \quad (38)$$

8.2 Reference state

Take the single-cell reference measure

$$\sigma_1(o, h, r) = \frac{1}{16} \frac{e^{-\lambda_H V_H(h)}}{Z_H} \frac{e^{-\lambda_R r^2}}{Z_R}. \quad (39)$$

The optical part is exactly unbiased. The parameter $\lambda_R \gg 1$ makes Ricci leakage rare or stiff. The product reference state is

$$\sigma_N = \sigma_1^{\otimes N}. \quad (40)$$

8.3 Reversible jumps

Let the Markov dynamics flip one coordinate at a time. Optical flips all have the same rate r_o :

$$o_a \mapsto -o_a, \quad a \in \{A, S_1, S_2, T\}, \quad \text{rate } r_o. \quad (41)$$

Horizon flips are slow:

$$h \mapsto -h, \quad \text{rate } \epsilon r_H, \quad 0 < \epsilon \ll 1. \quad (42)$$

Ricci moves are either strongly penalized by σ_1 or rapidly relaxed with rate $r_R \gg r_o$ and detailed-balance Metropolis factors:

$$r(\omega, \omega') = r_0 \min \left(1, \frac{\sigma_1(\omega')}{\sigma_1(\omega)} \right) \quad (43)$$

for r -moves. This ensures detailed balance.

The spectrum has:

- an optically degenerate band at the common optical relaxation scale;
- a slow horizon band at $\Gamma_H \sim \epsilon r_H$;
- a stiff or weak Ricci leakage sector controlled by λ_R and r_R .

Thus the toy model realizes the hierarchy

$$\text{optical } 3/4, \quad \text{rank-one } \mathcal{R}_H, \quad \text{Ward-stiff } \mu \text{ leakage}. \quad (44)$$

8.4 Explicit two-state flip spectrum

For a symmetric two-state flip variable $y \in \{-1, +1\}$ with transition rate r ,

$$Q = \begin{pmatrix} -r & r \\ r & -r \end{pmatrix}. \quad (45)$$

The eigenvectors are the constant mode 1 and the contrast mode y , with eigenvalues

$$\lambda_0 = 0, \quad \lambda_1 = -2r. \quad (46)$$

Equivalently, the positive relaxation generator $\mathcal{L} = -Q$ has spectrum

$$\gamma_0 = 0, \quad \gamma_1 = 2r. \quad (47)$$

Thus, for the minimal product-chain toy,

$$\gamma_{\text{opt}} = 2r_o, \quad \Gamma_H = 2\epsilon r_H, \quad \gamma_R \simeq 2r_R, \quad (48)$$

with possible additional Ricci stiffness supplied by the penalty scale λ_R in the conditioned ensemble.

If

$$\Gamma_H \ll \gamma_{\text{opt}}, \gamma_R, \quad (49)$$

then the horizon port is the unique slow output mode. Its scalar response is

$$\chi_H(s) \simeq \frac{Z_H}{s + \Gamma_H}, \quad (50)$$

and in time domain

$$K_H(\tau) \simeq Z_H e^{-\Gamma_H \tau}. \quad (51)$$

After absorbing Z_H into the source normalization, the rank-one output obeys

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H. \quad (52)$$

For a single-pole Stieltjes response,

$$m_n = \frac{Z_H}{\Gamma_H^{n+1}}, \quad (53)$$

so the rank-one Hankel diagnostic satisfies

$$m_0 m_2 - m_1^2 = 0. \quad (54)$$

A positive value of this determinant in future output reconstructions would indicate a multiband passive horizon response rather than failure of the finite Markov-screen class.

9 Model hierarchy implied by the realization

The finite Markov-screen construction gives the following interpretation of production branches.

Branch	Markov-screen meaning	Failure/demotion interpretation
$M_{3/4}^{\text{rank1}}$	optical-unbiased, rank-one horizon mode	primary low-spectrum branch
$M_{3/4}^{\text{multiband}}$	optical-unbiased, two or more horizon bands	broader passive memory spectrum
M_κ	optical covariance/dissipation not isotropic	weighted compliance fixed point
const- Σ	only DC Weyl offset resolved	output shape not resolved
Ward leakage	Ricci port not infinitely stiff	controlled constrained-port correction
G1DE-2	free response envelope	outside passive locked realization
free $A(a, k)$	arbitrary Weyl-port normalization	leaves G1DE class

10 Null tests

10.1 Optical isotropy

The optical covariance and dissipation matrices should be approximately proportional to identity:

$$\text{Cov}(F_a, F_b) \propto \delta_{ab}, \quad \mathcal{E}(F_a, F_b) \propto \delta_{ab}, \quad a, b \in \{A, S_1, S_2, T\}. \quad (55)$$

A persistent evidence preference for free κ means this symmetry is broken.

10.2 Horizon Stieltjes positivity

The reconstructed horizon memory moments must lie in a positive Stieltjes cone:

$$H_{ij} = m_{i+j} \succeq 0. \quad (56)$$

Violation means no passive reversible Markov-screen realization in this class.

10.3 Rank-one consistency

Rank-one predicts

$$m_0 m_2 - m_1^2 \simeq 0. \quad (57)$$

A positive deviation indicates a multi-band passive memory extension rather than immediate failure of the G1 class.

10.4 Ward leakage

The subhorizon diagnostic should satisfy

$$\frac{\mu - 1}{\Sigma - 1} = O(\mathcal{R}_H) + O\left(\frac{\mathcal{H}^2}{k^2}\right). \quad (58)$$

If it is $O(1)$, the Ward-stiff Ricci port hypothesis fails.

10.5 Normalization

Sampling a free optical amplitude $A(a, k)$ corresponds to rescaling the Weyl port measure itself. That changes the Markov-screen impedance normalization and is not a G1DE production branch.

11 Open microscopic tasks

The construction reduces the remaining microscopic work to precise tasks:

1. construct physically motivated finite screen state spaces Ω_N or finite screen algebras $\mathcal{A}_N$;
2. derive the passive reference measures σ_N from finite-access screen ledgers;
3. derive reversible or CP-detailed-balance dynamics Q_N from finite-screen update rules;
4. prove optical unbiasedness or classify its weighted-compliance deformations;
5. prove approximate lumpability and spectral gap for the horizon port;
6. derive Ward constraints as entropy projections or conditional expectations;
7. prove convergence of the finite response matrices to a stable continuum screen impedance.

These are hard problems. The value of the Markov-screen realization is that it turns them into explicit mathematical targets rather than informal gaps.

Generator checklist. An admissible Markov-screen generator must demonstrate: (1) finite state space; (2) detailed balance or controlled non-equilibrium breaking; (3) slow horizon mode; (4) Ward-stiff Ricci leakage suppression; (5) output $R_H(a)$ matching; (6) no hidden free $A(a, k)$.

12 Conclusion

The finite Markov-screen construction gives a concrete microscopic prototype for the G1DE normal form. A reversible finite-state screen automatically produces a positive matrix-valued Stieltjes response. Optical unbiasedness gives the Weyl-active fraction $3/4$. A slow lumpable horizon port

gives the rank-one output equation $\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H$. Ward conditioning makes Ricci leakage higher-order or subhorizon-suppressed. Weyl normalization then gives

$$\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu \simeq 1.$$

With $\mathcal{R}_H(a) = (3-s)\widehat{\mathcal{R}}_H(a)$, the G1DE-M_{3/4} production branch follows.

The most compact statement is:

In the D7 realization class, G1DE-M_{3/4} is the rank-one Weyl-port output of a finite reversible Markov screen.

This note does not complete the microscopic derivation of the true finite-screen ensemble. It supplies a minimal realization target showing that the G1DE-M_{3/4} interface is mathematically implementable by finite-state passive dynamics.

A Relation between response and large deviations

For independent cells, the empirical macro-port vector

$$x_N = \frac{1}{N} \sum_{i=1}^N f(\omega_i)$$

satisfies a large-deviation principle

$$\sigma_N\{x_N \simeq x\} \asymp e^{-NI(x)}.$$

The rate function is the Legendre dual of the cumulant generating function

$$\Lambda(\lambda) = \log \mathbb{E}_{\sigma_1} e^{\lambda \cdot f}.$$

Near the fixed point,

$$I(x) = \frac{1}{2} x^T \text{Cov}(f)^{-1} x + O(|x|^3).$$

Thus the static Hessian is the inverse covariance. In reversible dynamics, the dynamic response is controlled by the same covariance together with the relaxation operator $\mathcal{L}_N = -Q_N$.

B Sign conventions for generators

The Markov generator Q_N has nonpositive spectrum and satisfies

$$\frac{d}{dt} \mathbb{E}[g(\omega_t)] = \mathbb{E}[Q_N g(\omega_t)].$$

For response theory it is convenient to use the nonnegative relaxation operator

$$\mathcal{L}_N = -Q_N.$$

The resolvent is then $(s + \mathcal{L}_N)^{-1}$, and the response is Stieltjes in the variable s .

C Notation dictionary

Symbol	Meaning
$\mathcal{R}_H(a)$	data-facing scalar horizon output
$\widehat{\mathcal{R}}_H(a)$	normalized output shape, $\widehat{\mathcal{R}}_H(1) = 1$
$X_b(a)$	production alias for $\widehat{\mathcal{R}}_H(a)$
R_0	present amplitude, $R_0 = \mathcal{R}_H(1) = 3 - s$
K_H, J_H	source–kernel lift representatives
F_a	finite-screen port observables
Q_N	reversible Markov generator
$\mathcal{L}_N = -Q_N$	nonnegative relaxation operator
$\chi_{ab}(s)$	matrix-valued screen impedance
κ_{BW}	optical Weyl-active compliance fraction
$A(a, k)$	forbidden free optical amplitude; fixed by Weyl normalization

Closing Comparative Structural Matrix

This closing matrix repeats and expands the front-matter positioning table so that the structural comparison remains visible independently of the front matter. It is not a scorecard; it is an attribute map. Standard-limit structure should be distinguished from production inheritance: recovering a GR/ Λ CDM-like limit is not the same as already possessing the full numerical, nonlinear, CMB, or community infrastructure of the standard model. All frameworks require primitives, calibration assumptions, and observational interfaces; the relevant question is whether those bridges are established, open, flexible, or prediction-locking.

Framework	Primitive	Bridge / calibration	Dark-sector specificity	Prediction lock	Flexibility	Falsification clarity	Maturity / stress
GR+ Λ CDM	Metric geometry, stress-energy, constant Λ , CDM	Equivalence principle, smooth manifold, Einstein-Hilbert, conservation, initial conditions, bias/baryonic modeling, unexplained Λ	Low: Λ is a constant residual, not derived dynamics	High for standard cosmology; low for Λ origin	Low baseline freedom; unexplained dark sectors	High but tested in bulk; Λ origin still open	Established production baseline; stressed by Λ problem, late-time tensions, quantum/gravity interface
CPL / $w_0 w_a$	Effective dark-energy fluid	Smooth $w(a)$ expansion plus perturbation/closure priors	Medium phenomenological specificity; low microscopic mechanism	Low–medium	Moderate flexibility	Benchmark-level: useful comparative test, not a fundamental prediction	Useful evolving-DE benchmark; weak physical mechanism
Horndeski / EFT of DE	Scalar-tensor / EFT operator basis	Field content, stability, screening, local/GW constraints, EFT truncation	Broad class; not a unique dark-sector mechanism	Usually low unless restricted	High without strong priors	Depends on chosen model; class-level falsification is weak	Mature formalism; often underconstrained and flexibility-limited
$f(R), f(Q)$, related MG	Modified geometric action / connection sector	Chosen function, screening, local tests, growth/GW constraints	Model-dependent	Medium if fixed	Medium–high	Model-level: specific $f(X)$ can be rejected; class is broad	Strong literature; many regions constrained by local/growth/GW tests
Thermodynamic / emergent gravity	Entropy, horizon thermodynamics, information principles	Horizon entropy, local first law, Raychaudhuri bridge, metric/Ward closure	Usually indirect for late-time dark sector	Often low unless residual law added	Model-dependent	Weak without concrete locked residual + Known stress / limitation	Conceptually adjacent; lacks locked data-facing residual fingerprint
Quantum-spacetime programs	String/AdS-CFT/LQG/causal-set primitives	Compactification/landscape, holographic dictionary, semiclassical/continuum recovery	Usually indirect	Low near-term dark-sector lock	Varies; often high model multiplicity	Framework-level: long path to cosmological prediction + Known stress / limitation	High ambition; long empirical path for late-time cosmology
FDS-G1 / $M_{3/4}$	Finite causal-screen ledger, entropy response, integrable geometry, gluing defects	Finite distinguishability, causal-screen localization, entropy ledger, stiffness bridge, D0 modular calibration lemmas, Ward/Bianchi, Weyl normalization, optical projection, likelihood implementation	High: structured background–Weyl residual with near-GR growth	High: $\mu(a, k) = 1$, $\Sigma - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a)$	Low in $M_{3/4}$; controlled by free- κ , const- Σ , G1DE-2	Explicit: 6 demotion paths including free- κ , const- Σ , $ \mu - 1 \sim \Sigma - 1 $, free $A(a, k)$, CPL/ Λ CDM, lensing failure	Exact-pilot evidence-selected[†] ; top-control wide-prior stable; pending production refinement, expanded lensing, full baseline sensitivity, independent replication

Production maturity note. This matrix evaluates theoretical structure and empirical testability, not current infrastructure. GR+ Λ CDM currently dominates production breadth: numerical relativity, nonlinear N -body simulations, CMB/BBN pipelines, and community code ecosystems. FDS–G1 does not yet replace that infrastructure. These are implementation-readiness facts, not direct measures of primitive depth, prediction lock, or explanatory compression.

[†]Medium-prior exact-pilot status; production evidence, expanded lensing, and independent replication pending.

Comparative Notes on Major Competing Frameworks

The following notes summarize the structural comparison behind the matrix above. They are not total-score rankings. Each framework is strongest on a different axis: production maturity, formal generality, microscopic ambition, or prediction-locked late-time residual structure.

GR+ Λ CDM. GR+ Λ CDM remains the established production baseline for relativistic astrophysics and precision cosmology. It has mature infrastructure for strong-field waveforms, nonlinear structure formation, CMB inference, BBN consistency, and large-scale survey modeling. FDS–G1 does not currently replace that production apparatus. Its claim is different: the tested $M_{3/4}$ branch targets the late-time dark sector, where Λ CDM uses an unexplained constant residual. G1 replaces a free constant-like dark-sector term with a finite-screen background–Weyl residual whose output is locked by

$$\mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a).$$

Thus GR+ Λ CDM leads in production maturity, while FDS–G1 claims a more specific and falsifiable late-time residual mechanism. Full confirmation still requires production evidence refinement, expanded lensing, and independent replication.

CPL / w_0w_a . CPL is a useful phenomenological benchmark for evolving dark energy because it compresses a broad class of background deviations into two equation-of-state parameters. Its weakness is that it does not supply a microscopic or geometric mechanism linking expansion history to lensing response. FDS–G1 addresses this gap by treating the dark-sector signal as a finite-screen residual rather than as an arbitrary fluid. In the $M_{3/4}$ branch, the background deviation and Weyl response are tied to the same output shape $\hat{R}_H(a)$, with the growth channel kept near GR. The comparison is therefore not merely one of fit quality: CPL is a flexible benchmark, whereas $M_{3/4}$ is a prediction-locked residual model. If CPL wins under production evidence, the G1DE observational branch is demoted.

Scalar-tensor theories and EFT of dark energy. Horndeski and EFT-of-dark-energy frameworks provide a mature covariant language for departures from GR. Their strength is formal generality: many modified-gravity or dark-energy scenarios can be embedded in

an operator or function basis. Their risk is the same generality: without strong priors, the model space can become too flexible to yield a sharp prediction before data. FDS–G1 uses compatible EFT interfaces where needed, but its $M_{3/4}$ branch is much narrower. It predicts a Ricci-suppressed, Weyl-active residual pattern, with $\mu \simeq 1$ and a locked Σ -channel response. This makes the branch easier to falsify than generic scalar-tensor freedom. If data require a free $A(a, k)$ or $|\mu - 1| \sim |\Sigma - 1|$, the G1DE identity is demoted.

Quantum-spacetime programs. String theory, AdS/CFT, loop quantum gravity, and causal-set approaches pursue microscopic or quantum-spacetime foundations. Their ambition and mathematical depth are high, and some provide explicit microscopic degrees of freedom or holographic dictionaries. Their weakness for the present purpose is the long bridge from microscopic structure to a unique, near-term, late-time cosmological residual fingerprint. FDS–G1 takes a different route: it remains open at the microscopic calibration level, but it offers an immediate macroscopic prediction lock for the late-time survey regime. Its finite-screen entropy-response structure is therefore not a replacement for UV-complete quantum gravity; it is a data-facing finite-access framework whose first empirical test is the background–Weyl residual. Its burden is to survive lensing, production, and replication tests.

End of FDS–G1 Complete Series Archive.